

Germanic syntax from a comparative perspective

Diachronic and synchronic issues

Edited by

Ermenegildo Bidese

Federica Cognola

 Open Germanic Linguistics

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Contents

1	Germanic syntax from a comparative perspective	1
	Ermenegildo Bidese & Federica Cognola	
2	Complementation and relativization: A new perspective	11
	Katrin Axel-Tober & Ellen Brandner	
3	Determinacy and the CP-TP Bottleneck	69
	Elly van Gelderen	
4	A comparative investigation of verbal mood in Old High German, Old Saxon and Old English adverbial clauses	101
	Marco Coniglio, Chiara De Bastiani, Roland Hinterhölzl & Svetlana Petrova	
5	OCP effects in Germanic possession	131
	Alexandra Rehn & Hannah Booth	
6	Some remarks on the properties and analysis of expletive <i>es</i> in German	181
	Eric Fuß	
7	An extraction restriction with complement-less prepositions in British English but not dialectal German	235
	Richard Stockwell, Anke Himmelreich & Carson T. Schütze	
8	Pseudo-resumptives in Pennsylvania Dutch: A preliminary analysis	295
	Michael T. Putnam	
9	The phenomenon of Complementizer Agreement: New data from South Tyrol	323
	Ermenegildo Bidese & Helmut Weiß	

Contents

10 Parasitic participles and ellipsis in VP-focus pseudoclefts Jan-Wouter Zwart	353
Index	386

Chapter 1

Germanic syntax from a comparative perspective

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Comparative Germanic syntax is one of the most productive areas of research in linguistics. In fact, Germanic varieties exhibit a collection of syntactic phenomena (verb-second, CP- and TP-expletives, complementizer agreement, that-trace effect, subordination, relativization, scrambling, SVO vs. SOV, and many others) that have inspired substantial work on the architecture of the language faculty. The *Comparative Germanic Syntax Workshop* (CGSW) is the forum in which all these topics have been discussed, both from a comparative synchronic and diachronic perspective, since the first conference's edition back in 1984 (Trondheim, Norway <https://www.let.rug.nl/~zwart/cgsw/history.html>). In 2021 the University of Trento hosted the 35th edition of the CGSW, which was held for the first time in Italy. This volume collects a selection of papers presented and discussed during the conference in Trento, all of which employ a comparative approach.

With a large number of examples from many dialects and earlier stages of Germanic languages, Axel and Brandner's paper offer a new perspective on the integration of embedded clauses and relative clauses into a "generalized correlative construction" for clause embedding, which turns out to follow the model of bipartite equative constructions. Indeed, the conjunction that introduces both relative and complement clauses in many Germanic languages is an equative particle or goes back to a lexical item that originally derived from an equative particle (e.g., *as* in English and in many German dialects, *som* in Scandinavian). Equation itself involves the idea that two entities are in a formal relationship, not only

Ermenegildo Bidese & Federica Cognola

semantically but also syntactically. Based on this observation Axel and Brandner's analysis proposes a completely new view of clause embedding in which the CP layer is understood as a Small Clause with the equative particle as its head. The subject of the Small Clause, i.e. the Specifier position, is occupied in the complement clauses by the 'that' complementizer as 'double' of the correlative clause, in the relative clauses by the d-pronoun as 'double' of the externally merged correlative clause. This solution offers an innovative explanation both for the emergence of complementation and its different types from a correlative construction, and for the nature of clausal embedding.

In her programmatic paper, Elly van Gelderen explores how a determinacy approach (Chomsky 2019; Chomsky et al. 2019; Goto & Ishii 2019) can contribute to the analysis of phenomena concerning the C-T complex (V2, C-deletion, TP-expletives, that-trace, -subject-less relatives), from both a synchronic and a diachronic perspective. The principle of determinacy - a typical third factor constraint - states that there must be only one object X in a workspace, otherwise it cannot be sent to transfer (spell-out). In a phase approach, however, certain workspaces (i.e., phases) will contain more than one instance of an item X when the complement of the phase head is sent to Transfer. For instance, a subject in the specifier of v*P that moves to the specifier of TP will have two copies in the workspace that is the complement of C and therefore cannot move to the specifier of CP. This movement makes the derivation indefinite. CP and TP duplicate many of each other's functions. Indeed, there are many other cases in which the interrelationship between CP and TP in terms of feature inheritance leads to determinacy violations. The paper explores various cross-linguistic ways of avoiding indeterminacy by working out a typology of choices for the C-T complex in Old English, Modern English, and Dutch. The paper by Coniglio & De Bastiani & Hinterhölzl & Petrova is devoted to the investigation of mood alternations between indicative and subjunctive in adverbial clauses in Old High German (OHG), Old Saxon (OS) and Old English (OE). The authors show that these three Early Germanic varieties share striking quantitative (subjunctive in around 30% of sentences in the three varieties, cf. Table 1, p.1) and qualitative (subjunctive with nonfactual or counterfactual utterances like wishes, commands or hypothetical clauses, in which, though, also the indicative is documented) distribution of mood forms and claim that the phenomenon should receive the same analysis. Key in this common account is that mood selection in adverbial clauses is not fed by the main clause type, which means that the triggers for the mood alternations should be searched for within the embedded clause. Moreover, there is variation in the distribution of mood across the semantic classes of adverbial clauses, with concessive and purpose clauses featuring the highest rate of subjunctive forms

1 *Introductory remarks*

(cf. Coniglio et al. 2018). The presence of subjunctive is favored in this specific semantic context by the same conditions ruling the distribution of subjunctive in modern Greek and Romance (see Giannakidou 1998 and later, Farkas 1992), i.e. subjunctive is favored in non-veridical contexts. However, the semantic property of non-veridicality is still not sufficient to fully account for the distribution of mood forms in Early Germanic, given that subjunctive alternates with indicative in the relevant contexts. By relying on novel data on OHG, OS and OE and on cross-linguistic comparison, Coniglio & De Bastiani & Hinterhölzl & Petrova propose that the lexical subordinators of the adverbial clause might play a role in the selection of subjunctive or indicative moods. Specifically, they show that different subordinating elements are hosted in three different FPs in the left periphery which they suggest correlates with different mood selectional properties based on whether the adverbial clause is anchored to the utterance situation or to the situation denoted by the matrix verb which in turn largely correlates with (non-)veridicality.

In their paper, Hannah Booth and Alexandra Rehn propose a novel account for possessive constructions in German, by also considering the so-called Possessor Linking Construction (PLC), which is the most common construction for the expression of a possessive relationship in spoken, informal and dialectal varieties of German. In this construction, the possessor carries dative case and precedes the possessum, which must be preceded by a possessive element (*sein* ‘his’, in this case) (1a). In German, the possessor can also be realized through a complex DP bearing the genitive case (adnominal genitive, (1b)) or be introduced by the preposition *von* ‘of’ (prepositional possessive construction, (1c)):

- (1) a. Dem Mann sein Haus / *Sein Haus dem Mann
 the.dat man his house his house the.dat man
 b. Das Haus des Mannes / Des Mannes das Haus
 the house of.the.gen man.gen of.the.gen man.gen the house
 c. Das Haus von dem Mann
 the house of the man
 ‘The man’s house.’

By focusing on synchronic data from German varieties and dialects, which mostly exhibit a three-case system due to the loss of genitive case, and synchronic and diachronic data from Low German which was characterized by a three-case system in contrast to present-day varieties exhibiting now a two-case system (nominative, non-nominative), Booth and Rehn analyze the morphosyntax of PLC and other possessive construction as a case of Obligatory Contour Principle

Ermenegildo Bidese & Federica Cognola

(OCP) applied to morphosyntactic phenomena (Nevins 2012). They propose that the possessive element in PLC and the genitive case or the prepositional marker in other constructions function as a linker, whose functions is preventing an OCP violation which would otherwise arise from two adjacent DPs within a single constituent, cf. (2a).

- (2) a. * Possessor DP Possessum DP
- b. Possessor DP sein (his) Possessum DP
- c. Possessor DP gen Possessum DP
- d. Possessum DP gen Possessor DP
- e. Possessum DP von (of) Possessor DP

Through this hypothesis, they account for the fact that a feminine DP bearing genitive case but lacking genitive marking on the noun cannot precede the possessum in German - something which, they show, was though possible in low older stages of Low German which had genitive marking on feminine nouns.

- (3) a. * Der Frau Haus
 the.gen woman house
- b. Des Mannes Haus
 the.gen man.gen house
 ‘The woman’s /man’s house.’

In their account, the linker also marks the possessive relation. Eric Fuß’s work on expletives offers also new insights into another prototypical phenomenon of Germanic syntax. In fact, it deals with the prefinite expletive insertion, a phenomenon, which usually shows up inthetic clauses, when the clause lacks a topical subject and conveys a sentence focus interpretation. Syntactically, the presence of the expletive element is required by the edge-feature linked to C. From the perspective of information structure, C is taken to host a discourse-related δ -feature in clauses with prefinite expletive insertion, that has not been valued by any topical phrases in the course of the syntactic derivation. In clauses with a topical subject the δ -feature is donated to T and valued by the subject phrase in SpecTP (and then eventually transferred to SpecCP). In clauses with prefinite expletive, Fuß assumes that the δ -feature is donated to T as well, but then re-transferred to C via T-to-C movement.

Richard Stockwell, Anke Himmelreich & Carson T. Schütze deal with prepositional object gaps (POG) in English, i.e. the possibility of dropping the object pronoun after the preposition typically in sentences featuring the verb *to have*.

1 Introductory remarks

- (4) The box has papers in (it).

They claim that a parallel phenomenon is also found in German, specifically with so-called pronominal adverbial *da*-constructions. In this language, neuter pronouns referring to inanimates selected by a preposition must be realized through the element *da* (pronominal adverb):

- (5) Fritz hat an ein Auto gedacht. → daran / *an es
 Fritz has of a car thought at it
 ‘Fritz thought of a car.’

In standard German, *da* cannot be separated from the selecting preposition - an option which is though grammatical in Northern German varieties/dialects, cf. the so-called *Spaltungskonstruktion* (Fleischer 2002). In this construction, *da* can be fronted (6a) or moved to the middle field whereas the preposition is stranded. In the varieties in which this split construction is admitted, the pronominal *da* element can be dropped (6b).

- (6) a. Da habe ich keine Zeit für / Dafür habe ich keine Zeit.
 Da have I neg time for / Da-for have I net time
 ‘I do not have time for that.’
 b. Im Sommer sollte man auch gelegentlich einen Wassernapf
 in summer should one also occasionally a water.bowl
 (da)neben stellen.
 da.next put
 ‘In summer, one should occasionally put a bowl of water next to it.’

In the paper it is claimed that the *da*-drop construction replicates the prepositional object gaps (POG) found in English and a parallel between them is drawn at different linguistic levels. Novel data collected for German are discussed in the paper: while a previous study (Stockwell & Schütze 2019) showed that POG constructions involving object fronting are less felicitous than other involving other arguments in English, such as effect is not found in German, where *da*-dropping appears to be less felicitous in all contexts. According to the authors, the characteristics of the POG construction in the two languages can be accounted for within a theory of A-movement in which object extraction in English is marginal due to the fact that the object must cross an object - a configuration which is circumvented in German due to the availability of scrambling in this language.

Michal Putnam’s paper addresses an understudied, but very vital diasporic variety of German, Pennsylvania Dutch, specifically, the lack of wo-compounds

in this variety in comparison to German and Dutch. Interestingly, Pennsylvania Dutch has developed a pseudo-resumptive structure, in which the w-element represented by the w-word *was* 'what' is combined with a complex, in which the preposition is lexicalized by a d-element:

- (7) Pennsylvania Dutch (default)
- a. *Was* sin(d) die Schtanne g'macht *devun*?
 what are the stars made it-from
 'What are the stars made out of/from?'
 - b. *Was* bischt sell n duh *defor*?
 what are(you) that PROG do it-for
 'What are you doing that for?'

Starting from an extended projection consisting of a DP embedded in a CP, Putnam analyzes the specific properties of pseudo-resumptives in Pennsylvania Dutch and argues that pseudo-resumptives are a repair strategy related to the fact that in this German variety there is no English-like r-stranding in most environments. Through this strategy, pseudo-resumptives avoid the appearance of English-style r-stranding hyperextending the structure of *da*-compounds. Since the main difference between *wo*- and *da*-compounds lies in their feature content, i.e. whether (or not) the topmost project contains a d or w feature, Pennsylvania Dutch compensates for this lack with the above strategy. Since the seminal work of Bayer (1984), inflected complementizers is one of the specific morphosyntactic phenomena of the Germanic languages (although it has been discovered also in non-Germanic varieties) and one of the most studied. Several analyses have been proposed to explain both the distribution of the phenomenon and its function. Its origin is still a matter of debate. A recent proposal (see Weiß 2018) argues that it results from the reanalysis of subject clitics as part of the inflection. This initially produces a double inflection – in a first diachronic step on the verb in second position and then also on the verb in final position – secondly, the new morpheme also appears on the complementizer. Bidese and Weiß presents new data from a vital Germanic variety, namely South Tyrolean, in which Complementizer Agreement has hardly ever been investigated. The data collection provides a differentiated picture of the situation in South Tyrol. On the one hand, they confirm Weiß' (2018) hypothesis on the diachronic path of the phenomenon, on the other they suggest another possible developmental path: the new morphology can be transferred first on complementizers and then on verbs in the end position – as it happened in the Upper Vinschgau variety – or first on verbs in the end position and afterward on complementizers – as it is going on in Meran.

1 *Introductory remarks*

Jan Wouter Zwart's paper deals with Dutch pseudocleft constructions featuring either an infinitive or a past participle (8). In this construction the participle in the lower VP is parasitic on the past participle in the antecedent VP, i.e. the participle within the pseudo-cleft is only possible if a licensing auxiliary appears in the preceding VP.

(8) Dutch

Wat ik heb ge-daan is [keihard werk-en / ge-werk-t].
 what I AUX:1SG GE-do:PTCP be:3SG real.hard work-INF GE-work-PTCP
 'What I did is work real hard.'

The author focusses on VP-focus pseudocleft constructions, i.e. construction in which the B element in the copular pseudocleft construction [A cop B] is a focussed VP (and in which A is a *wh*-element). In this construction both the specificational and predicational readings are possible. When the cleft is specificational, the element in B (a book) provides value to an empty position within the clause, therefore (9a) can be paraphrased as (9b).

(9) Dutch

- a. [Wat ik <e> lees] is [een boek].
 what I read:1SG BE:3SG INDF book
 'What I'm reading is a book.'
- b. Ik lees een boek.
 I read:1SG INDF book
 'I'm reading a book.'

In predicational pseudocleft, B introduces a specific property for the element in the empty position in the main clause: therefore, a paraphrase like (10b) would not be possible.

(10) Dutch

- [Wat ik <e> lees] is [interessant].
 what I read:1SG BE:3SG interesting
 'What I'm reading is interesting.'
- a. *I'm reading interesting.
 - b. I'm reading *x*, and *x* is interesting.

In VP-pseudoclefts, the past participle in B is parasitic on the past participle in A, and only the specificational reading is possible. Zwart discusses constructions similar to Dutch pseudocleft cross-linguistically, but no construction fully

Ermenegildo Bidese & Federica Cognola

comparable to the Dutch one is found. In order to account for the facts, Zwarts proposes that in VP-pseudoclefts with parasitic participle, the antecedent clause in A contains a functional head mediated by a dummy verb which is responsible for the licensing of a [+anterior] features. Since the focus VP in B is associated with the variable in the clause in A, past participle morphology can be assigned, analogously to (9a).

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1 *Introductory remarks*

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Chapter 2

Complementation and relativization: A new perspective

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The paper argues that complement and relative clauses are both slightly different instantiations of a general correlative structure which in turn follows the model of equative/comparative constructions with its necessarily bipartite structure. As such, it is in line with recent semantic treatments of clausal embedding as being predication with a nominal in the matrix clause. The starting point is the observation that in many Germanic languages (mostly dialects and earlier stages), the equative particle or lexical items derived from it (e.g. *as* in English, *som* in Scandinavian) is found in relative and complement clauses alike and that d-pronouns/demonstrative pronouns seem to be rather ‘an addition’ – which however eventually succeeded as complementizers in the standardized languages. Some consequences of such an analysis for ‘long wh-extraction’ are discussed and it is shown that the correlative structure is superior to accounts in terms of successive cyclic movement.

Keywords: relative clauses, complement clauses, equative particle, correlative, wh-extraction

1 Introduction

In recent semantic work on clausal complementation, it is suggested to take complement clauses (henceforth: CCs) semantically as predicates of (possibly empty) nominals or of content nouns (*rumour* etc.) cf. Kratzer (2006, 2013, 2016), Moulton (2009), Elliott (2017, 2020), Simeonova (2018, 2020). Likewise, Moltmann (2020) works out a predicative analysis of attitude verbs within a truth-maker

based semantics. Under such predicative analyses, clausal complements are never directly selected by a lexical verb (or noun) as such, rather they modify and thus deliver the content for a nominal in the matrix clause which in turn is the ‘real’ argument of the verb.

From the syntactic side, it has been suggested by many researchers to take the complementizers in argument clauses as what they look like, namely as pronouns, more specifically as relative pronouns, [Manzini & Savoia \(2003, 2011\)](#), [Kayne \(2014\)](#), [Arsenijević \(2009\)](#), [Roussou \(2020\)](#) among many others. This research has focussed on English and the Romance languages where we indeed find complementizers based on demonstratives (Engl. *that*, Germ. *dass*), on *d*-pronouns, or on interrogative, i.e. *w*-pronouns (French *que*, Italian *che*), as they occur in relative clauses (henceforth: RCs) respectively.

It is a well-known fact, however, that many languages use different complementizers for the various types of embedded clauses (e.g. [Franco 2012](#)). RCs, in particular, often show special ‘particles’. In a recent paper, [De Cuba \(2017\)](#) concentrates on languages that show such differences and discusses the interpretation of the embedded clauses (referential vs. non-referential,) but he does not fully account for the correlation between complementizer type and interpretation and, in particular, he does not work out how RCs should then be analysed.

This is the point where we will step in by looking closely at the lexical sources and the meanings of the different types of complementizers – whether they occur in CCs, in RCs or in both alike. We will thereby focus on a type of complementizer that is as yet under-researched, even though it is attested in many languages – namely the complementizer corresponding to English *as* (and its *w*-counterpart *how*). In other words, we are interested in a type of complementizer that otherwise is found in equative/comparative constructions and will thus be referred to in the following as ‘equative particle’. Equative particles are well-known to occur in RCs, cf. the (dialectal) English *as*-RCs ([Herrmann 2005](#)) and the *som*-complementizers in the Scandinavian languages ([Stroh-Wollin 2002](#)).

In section 2 of the paper, we will present data from several Germanic languages (including dialectal and historical varieties) which show that equative particles as complementizers are much more wide-spread than previously thought. Thus, the complementation-as-relativization analysis, beginning with [Arsenijević \(2009\)](#) and [Kayne \(2014\)](#), will gain further support – but this time from the perspective of a different type of complementizer, namely the equative particle.

Based on these data, we will show how the equation semantics perfectly fits the semantic requirements for a complementizer, namely the “equation” of a nominal in the matrix with the propositional content of the embedded clause (cf. [Elliott 2020](#)), be it an RC or an CC. In the case of RCs, we will propose

2 Complementation and relativization

that the equation applies to the two nominals in the respective clause, i.e. the head noun and the “missing”, i.e. elided argument in the RC such that there is one individual referred to with two different predicates. In case of a clausal complement (traditionally assumed to be selected as such by the matrix verb), we will argue that there is always a nominal present in the matrix, crucially a pronoun. Due to this, there is no nominal predicate, i.e. no lexical content, and therefore the equation results in a ‘complete transfer’ of the content of the embedded clause. In this case, predicate modification “runs empty” as there is no further predicate. Consequently, there is no ellipsis in the embedded clause. This derives the crucial difference between RCs and CCs while allowing them to have the same type of complementizer.

The embedded clause itself will be analysed as a Small Clause with the TP as the predicate, the equative particle as the head of the Small Clause, cf. [Bowers \(1993\)](#), and the pronominal complementizers (if present) as the subject of this Small Clause. It will be shown that due to the anaphoricity of these *d*-pronouns (aka complementizers), essentially the same effect can be obtained as with the equative particle alone. This leads one to expect that there is a certain amount of variation concerning the choice of the complementizers – and this is actually the empirical situation that we find, including the co-occurrence of a pronominal and an equative complementizer.

The general structure that we propose for RCs and CCs alike is the following:

- (1) ... [head noun/correlate]... [_{SC} (*d*-pronoun) AS [_{TP} predicate]]

Thus, the novel analytical perspective we shall propose here is that CCs and RCs are slightly different instantiations of this basic correlative structure rather than arguing that CCs *are* actually RCs.

The rest of the paper is structured as follows: In [Section 2](#), we will present data from Germanic languages (including some dialectal and historical varieties) that provide evidence for a (sometimes hidden) correlative nominal in the matrix clause of a CC. [Section 4](#), then spells out the analysis sketched above. [Section 5](#), briefly discusses the consequences for long *wh*-extraction that the suggested analysis has. Mainly based on data from Alemannic, it will be shown that a base-generation analysis, see [Adger & Gillian \(2005\)](#), [Brandner & Berger \(2018\)](#), in terms of prolepsis, [Salzmann \(2006\)](#), is able to capture the properties of long *wh*-extraction in an adequate way. [Section 6](#) concludes.

Katrin Axel-Tober & Ellen Brandner

2 Complementizer types

2.1 The THAT-type complementizer and its competitors in early Germanic

Before going into the discussion of the equative particle, we will briefly discuss the role of the THAT-type complementizer in early Germanic. In discussions of Germanic complementizers, the THAT-type complementizer has been in the focus of attention both in synchronic (Kayne 2014) and diachronic or comparative accounts (e.g. Roberts & Roussou 2003, Baunaz 2015, Nordström & Boye 2016). It is generally assumed that the “Germanic languages are united in employing reflexes of the demonstrative pronoun **tod* for this purpose” (Harbert 2007: 415). The Romance languages, by contrast, derive their finite-clause complementizers from the interrogative pronoun **k^{wo}*. In Standard German, the complementizer in argument clauses is *dass*, a cognate of English *that* and as in English, it is homonymous with the neuter form of the demonstrative pronoun. According to an often-cited scenario, the THAT-type complementizer goes back to the demonstrative correlate that shifted from the superordinate clause into the C-position of the emerging CC: “Most of the type 1 complementizers [= complementizers that express epistemic neutrality; AUTHORS] probably developed from a demonstrative pronoun (Old English *þæt*, Old High German *daz*, Old Norse *þat*, Gothic *þata*; Oxford English Dictionary: “that, conj.”) which was part of the matrix clause and referred to a following apposed main clause [...]” (Nordström & Boye 2016: 134). Lühr (2004) and Axel-Tober (2017), however, have raised a number of serious empirical and conceptual objections against this diachronic account. Axel-Tober (2017) argues that the declarative THAT-type complementizer in West Germanic does not directly go back to the demonstrative pronoun, but rather to a THAT-type relative complementizer that in turn evolved from the neuter form of the demonstrative-based relative pronoun, again much in line with a complementation-as-relativization analysis.

It should be noted that in the quotation above, Nordström & Boye (2016) also refer to putative THAT-type complementizers in North and East Germanic. However, contrary to what is often assumed in cross-linguistic overviews (cf. also Heine & Kuteva 2002, Harbert 2007: 415–417), the Nordic *at*-complementizer as in (2) is probably not derived from the demonstrative *þat*, but from the local preposition *at* via an intermediate step that persisted into Old Norse, where *at* was an infinitival complementizer (Heltoft 1995: 130V. Faarlund 2004, 2015, Norde 2009: 196), exemplified in (3):

2 *Complementation and relativization*

(2) Modern Norwegian

Eg kan ikkje lova at eg aldri skal gjera det igjen.

I can not promise that I never shall do it again.

‘I cannot promise that I will never do it again.’

(Faarlund 2015: 1)

(3) Old Norwegian

Kjartan kaus heldr at vera með konungi en fara til Íslands.

Kjartan chose rather to be with king than go to Iceland.

‘Kjartan chose to stay with the king rather than go to Iceland.’

(Laxd 129.17, cited after Faarlund 2015: 6)

Note that in Old Norse, subject or object clauses could alternatively occur with the complementizer *er/es*, which otherwise introduced RCs and was “by far the most common relative complementizer in all the earliest sources” (Wagener 2017: 260).¹

(4) Old Norse

fannsk þat á ollu, er hon þóttisk vargefin found.rfl.

that.n on all that she seemed.rfl ill-matched

‘It showed in everything that she thought she was marrying beneath her.’

(Nj 25.24, Faarlund 2004: 256: 26a)

Both *er-* and *at*-clauses were frequently associated with a demonstrative correlate in the superordinate clauses (= *þat* in (4)); cf. section 3. for further discussion.

In the Nordic languages *er/es* was largely replaced by the equative-based complementizer *sem/sum/som* in RCs. The earliest attestations of this type of complementizer are in Swedish runes from the 11th century and in a Norwegian inscription from 1180 (Wagener 2017: 263). In the modern Scandinavian languages, *som/sem* serves as the unmarked relative complementizer, see also section 3. below. Thus, for the North Germanic languages, the lexical source of the complementizer are not *d*-pronouns but, rather particles that introduced RCs (*er/es*), the preposition *at*, and the equative particle.

Turning to East Germanic, again, contrary to a widely-held conviction, there is no evidence for the THAT-type complementizer that we know from West Germanic. In Gothic, *ei* and *patei* (*pata*-Nom/Acc + *ei*, also as *pamma*-Dat+*ei* = *pammei* and *piz*-Gen+*ei* = *pizei*; cf. Miller 2019) are the two major complementizers.

¹According to Sapp 2019: c36, Old Norse *er* was an “all-purpose complementizer, serving also as a marker of comparison (as)”. Generally, *er* is assumed to be a locative adverb, see the discussion in Franco & Boef (2015).

Katrin Axel-Tober & Ellen Brandner

Contrary to what is claimed in the quotation above, *thata* alone is not attested as a complementizer, only the compound complementizer with (enclitic) *ei*. *Ei* is primarily a relative complementizer.² The compound complementizer typically occurs with *verba sentiendi* or *dicendi* in indicative clauses, but often only *ei* appears (Harbert 2007, Miller 2019, Longobardi 1979, 2019).³ The *d*-pronoun may additionally show up as a matrix correlate, cf. Section 4. The etymology of *ei* is not entirely clear. It has been argued to go back to the Indo-European pronominal stem **e-*, **i-* (Lehmann 1986).

(5) Gothic

- a. qap̃ du imma þatei broþar þeins qam
said to him THATEI brother your came
‘told him that your brother came’ (Lk 15:27)
(Miller 2019: 445, ex. 223)
- b. Ni hugjaip̃ ei qemjau gatairan witop̃ ...
NEG think.2PL.PT EI came.1SG.OP abolish.INF law
‘Do not think that I have come to abolish the law ...’ (Mt 5:17)
(adapted from Miller 2019: 430, ex. 170)

To sum up, in our view, the relevance of the THAT-type complementizer has been overestimated in the literature. Early West Germanic, Old Norse and Gothic show different types of complementizers – not only complementizers based on the neuter *d*-pronoun, but also complementizers derived from other types of pronouns, from a preposition or from the equative particle. In earlier Germanic, the overlap between complementizers used in RCs on the one hand, and in ‘complement’ clauses (i.e. object or subject clauses) on the other hand, was even larger than in the modern (standard) varieties.

Considering now the West Germanic *d*-complementizers, recall that in English *that* is the unmarked complementizer both in ‘complement’ clauses and in RCs, whereas (Standard) German has largely lost this neuter *d*-complementizer in relatives. Instead, RCs are introduced by *d*-pronouns with agreement and case features:⁴

²In RCs, enclitic *-ei* combines with a *d*-pronoun (e.g. masc *saei* = *sa* + *ei*), a *d*-adverb (e.g. *þarei* (*þar* + *-ei*) ‘there where’), or a personal pronoun (e.g. 1. Sg. *ikei* = *ik* + *ei*).

³Subject clauses only show *þatei* (Longobardi 1979).

⁴In the standard language the (uninflected) *d*-complementizer only survives in RCs with temporal antecedents (Harbert 2007, Axel-Tober 2017) and in cleft-like constructions with clefted adverbials (Müller 2022).

2 *Complementation and relativization*

(6) German

Die Lehrerin fragte nach den Mädchen,
 the teacher enquired about the.DAT.PL girls,
 die/*das krank waren.
 d-pron.nom.pl/*d-pron.neuter.sg sick were
 ‘The teacher enquired about the girls that were sick.’

Crucially, the phi-features of the *d*-pronoun agree with the head noun. The case marking of this pronoun, however, is governed by its grammatical function in the embedded clause. In the example above, it is the subject of the embedded clause and thus carries nominative case, whereas the head noun in the matrix clause is marked with dative governed by the preposition *nach*. The agreement phenomena (together with the *wh*-pronouns⁵ in English relatives) seem to support the traditional analysis of RCs in which the relative pronoun moves out of the embedded clause to a left-peripheral position. As already indicated, in the remainder of this paper, we will cast doubt on this straightforward analysis⁶ by establishing on a broad empirical basis that the use of an equative particle seems to be a very common and widespread strategy to build RCs. This leads to the proposal that embedded clauses are indeed Small Clauses with equative particles as heads. In consequence of this, the *d*-pronoun will be argued to be (optionally) base-generated in the subject position of this Small Clause, being in a correlative relation with the nominal in the matrix clause.

In order to establish this analysis, we will now discuss the role of the equative particle or, more generally speaking, of AS-type complementizers in Germanic and beyond.

-
- (i) Das letzte Mal, **dass** wir uns getroffen haben, war am Bodensee.
 ‘The last time that we met was at the Lake of Constance.’

- (ii) Es war am Bodensee, **dass** wir uns getroffen haben.
 ‘It was at the Lake of Constance that we met.’

⁵Relative *wh*-pronouns (*who*, *which*) only began to spread in the Middle English period. Their introduction has been ascribed to the influence of Latin *qui*, *quae*, *quod* and/or to the influence of French *qui*, *que* in the literature. Dekeyser (1997), by contrast, hypothesises that the *wh*-strategy itself must have been rooted in the grammar of Late Old English, but its quantitative rise in particular in complex registers occurred under Romance influence. Nikolas & Truswell (2017) highlight the fact that headed RCs with openly filled specifiers are typologically rare. They argue that this feature is nevertheless quite common in Indo-European languages as a stable set of lexical items was inherited from Proto-Indo-European. We will not be able to address this issue in the present paper.

⁶Note that according to the WALS atlas, the use of pronouns in RC formation is a rather marked strategy, essentially confined to Germanic and Romance languages. cf., <https://wals.info/feature/122A#2/23.1/151.9>.

Katrin Axel-Tober & Ellen Brandner

2.2 AS-type complementizers in Germanic

2.2.1 Comparative/equative and complement clauses

To get a first impression about the type of construction that is our empirical point of departure, consider the data below from (modern) Alemannic, an Upper German dialect spoken mainly in Southwest Germany and in Switzerland.⁷ In all three constructions below, we find the particle *ass*. The first one is a garden variety of a comparative construction in which we would find *als* in Standard German and *than* in English. The second one is a so-called hypothetical clause (English *as if*), and finally we see a CC, embedded under *say*:

(7) Alemannic

- a. die isch etz scho größer ass ich comparative
she is now already taller AS I
'She is already now taller than me.'
- b. der tuet grad so ass ob er nünt gwisst het hypothetical
he acts PRT so AS if he nothing known had
'He acts as if he hadn't known anything.'
- c. %er het gseet ass er schpöter kunnt CC
he has said AS he later comes
'He said that he will come later.'

The *ass*-complementizer also occurs with so-called content nouns (i.e. nouns like *belief*, *hope*, *conviction* etc.):

(8) Alemannic

- S git no Hoffnig, ass mer das Spiil no gwinne.
it gives still hope AS we the match still win
'There is still hope that we will the match after all.' (SynAlm FB5_6a)

Most of the variants of Alemannic would use the more familiar complementizer *dass* in (7c) and (8), however some sub-variants⁸ (as indicated by %) regularly

⁷The Alemannic data discussed in the paper come from the SynAlm project. SynAlm was a DFG-funded project (2011–2016), led by Ellen Brandner at the University of Konstanz that investigated the syntax and morphology of Alemannic. The results can be found under <https://ilg-server.ling.uni-stuttgart.de/synalm/html/>.

⁸The variant where the complementizer *ass* is attested, according to the data in SynAlm, is located in the Low Alemannic area, Southwest Germany, northwest of Freiburg and some instances in Alsace, see below for examples. This is the same area in the Wenker-Atlas, where

2 *Complementation and relativization*

show *ass* in such syntactic environments. Interestingly, these variants use this particle also in equatives, cf. (9a), whereas again the other variants show *wie* in this environment – as it is the case in Standard German, cf. (9b):

(9) Alemannic

- a. % die isch etz scho so groß ass ich.
she is now already as tall AS I
- b. die isch etz scho so groß wie ich.
she is now already as tall HOW I
'She is already now as tall as me.'

Before going into more data with AS as a complementizer, let us first consider its origin. The particle *ass* (and its cognate form *as* in English) stems from the complex form *al-sô*, as attested in the Old High German example in (10):

(10) Old High German

- thia fart sie al so gisitotun
the journey they AL SO carried.out
'the journey which they carried out in this way'
(Otfrid V 10,14)

Later, the final vowel was weakened to *al-se* and the schwa was dropped, cf. modern German *als*. In the Alemannic variants the /l/ was elided, just like in English. *Al* itself is an intensifying element that developed from the adverb/adjective *al* with the meaning whole/complete⁹ and which is found today e.g. in the quantifier *all*. The *so*-part goes back to *sô* and is dubbed as a correlative particle in the DWB¹⁰. It is often described as an adverb of manner¹¹ derived from ie. **sue-*, **se-* 'so' (in ablaut and case variants) and which is related to the pronominal stems ie. **se-*, **seue-* 'aside, separate, on its own' (Köbler 2018). *Als* (and its variants) can be found regularly as an equative particle – it was in competition with its *wh*-counterpart *wie* already from Early New High German on.¹²

the complementizer *dass* is marked as occurring without an onset, i.e. *ass* instead of *dass*, see also fn. 14. On this map, there is a second area with *ass* as the prevalent version. This area is located Northwest of Nürnberg – an area which was not covered in SynAlm. See map 189 in 'Sprachatlas des Deutschen Reichs': <https://www.regionalsprache.de/SprachGis/RasterMap/WA/32>.

⁹Bd.1,Sp.214 (DWB)

¹⁰Bd.1,Sp.260 (DWB)

¹¹Concerning the manner-adverb reading, see the discussion in Section 4.

¹²See Jäger (2018, 2019, 2021) for a thorough discussion of the 'replacement' of *als/ass* by *wie*. Her work covers the diachronic development under a syntactic perspective as well as dialectal variation.

Katrin Axel-Tober & Ellen Brandner

Thus, the version in (9a) could be seen as an archaic form that has survived in some dialects.

As shown in (7c) and (8), some variants of Alemannic can use this particle as a complementizer in CCs. In order to exclude the possibility that such data are merely some single occurrences¹³, we checked for this construction in further dialects, see also *Axel-Tober & Brandner (2019)*. And indeed, it turned out that it is much more widespread than previously thought.¹⁴

The following data are from Alsatian, an Alemannic dialect spoken in Alsace (a region in northeast France, nowadays):

(11) Alsatian

- a. I weiss, *ass* I stërwe muess.
I know AS I die must
'I know that I have to die.'
- b. ... *ass* I dr etz sa, *ass* d heim ges.
AS I you-DAT now say AS you home go
'... that I am saying to you now that you go home.'

These examples have been taken from a dictionary (= *ElsWb*) which was composed during the end of the 19th century – but it also contains examples that are much older. According to this source, *ass* was not only used in declarative CCs

¹³One reviewer pointed out to us that Mòcheno (Felsentalerisch), see *Rowley (2003)*, *Cognola 2013*), exhibits the same phenomenon:

- (i) hait is khelter AS gester
today is colder AS yesterday
- (ii) an hunt AS pèllt, paist nia
a dog AS barks, doesn't bite
- (iii) de lait AS i khenn
the people AS I know

¹⁴*Fleischer (2005: fn.10)* in his survey on relative complementizers in the German dialects, cites some examples of this type but rejects the idea that *ass* has anything to do with the equative particle. He points to the lack of further examples in German dialects – according to his sources. This gap is filled with our data. *Leu (2015)* suggests that *ass* an instance of a “*d*-drop” (cf. the definite article *d-er* vs. the simple pronoun *er*) and cites Yiddish and the North-Germanic languages as exhibiting the same phenomenon, as these typically show a complementizer without a *d*-component. However, as discussed already above, the complementizers in Scandinavian languages evolved out of the preposition corresponding to English *at*. Yiddish employs the equation particle as well, see below.

2 *Complementation and relativization*

with verbal or adjectival predicates, but also in clausal complements of prepositions, in combination with relative or interrogative phrases (cf. *who that, which that* in dialectal/colloquial English) or in multi-word adverbial subordinators (cf. *when/if that* in dialectal/colloquial English).¹⁵

(12) Alsatian

Bis **asz** du kasch Adschwang wèrde, Schlof ich gwiss scho in
 until AS you can ... become, sleep I certainly already in
 der Erde.
 the earth
 ‘Until you can become A., I will sleep certainly already in the earth.’
 Mäder Mü. Ged. 39.

(13) Ich weiss nit für wene **ass** es is, **ass** ich schaff.¹⁶

I know not for whom AS it is AS I work
 I don’t know for whom it is that I work.’

(14) Kann-s komme, wënn **ass** de wit.

Can-2sg come when AS you want
 ‘(You) can come whenever you want.’

According to the example material given in the Palatine dictionary (PfWB), *as/aß* once was found in the same set of syntactic environments as in Alsatian. Here, it is explicitly given as an alternative to the THAT-type complementizer *daß* in the entry under the lemma *daß*. *As/aß* is, however, the older form that has been driven back by *daß*. By 1960, *as/aß* had already been largely replaced by *daß*.

In Pennsylvania German (also called Pennsylvania Dutch), which partly descends from Palatine variants, we still find a continuation of this “older stage”

¹⁵The following examples are given under the lemma *ass* with superscript ‘d’ = *dass*. (The superscripts have been inserted to allow the reader to identify the corresponding lexical item in Standard German). There is a further lemma *as* (with superscript l: *als*), under which examples of comparative and equative clauses are given. The graphematic/transcriptional distinction between *ass* and *as* in the dictionary does not seem to correspond with a phonemic difference in the dialect, however. This is also the reason why dialect speakers have difficulties in distinguishing between *dass* and *als* when they use Standard German: “Both *ass*-items (= ‘that’ and = ‘as’) are often mixed up when speaking or writing in Standard German.”

¹⁶In this example, the relative phrase *für wene* ‘for whom’ has been extracted out of a further clause with *ass* that is a part of a cleft construction.

Katrin Axel-Tober & Ellen Brandner

(again alongside with *dass*-complementation¹⁷). In this variety as well, *ass* is not only used as a complementizer in comparative clauses, (17, 18), and in equative clauses or hypothetical conditional clauses, (17), but again also in CCs of verbs, adjectives and prepositions, cf. (18) and (19).

(15) Pennsylvania German

Es iss so alt *ass* Mensche alt sin.
it is as old AS men old are
'It is as old as mankind.'
(Haag 1988: 138)

(16) (Wu aerscht die golde Frucht

So schee geschtanne hot.)
Iss nix meh *ass* mer sucht un nix meh *ass* mer wott.
is nothing more AS one searches and nothing more AS one wants
'...is nothing more than one searches and nothing more than one wants'
'(How beautifully the golden fruit stood there! It is nothing more than
one looks for, and nothing more than one could wish for.'
(Haag 1988: 84)

(17) Der *Lawyer* hot gshwetz d *ass* wann er arrig bees waar.¹⁸

the lawyer has talked AS when he very angry were
'The lawyer talked as if he were very angry.'

¹⁷The Pennsylvania German examples are largely cited from the anthology compiled by Earl C. Haag 1988. The anthology contains literature (prose and poetry) from the 19th and early 20th centuries. We cited the Pennsylvania German examples from here as this is a published and permanently accessible source. In present-day Pennsylvania Dutch the use of *dass* instead of *ass* is probably more frequent, but all the uses of *ass* that we discuss in the paper are still active in contemporary Pennsylvania German as can be easily verified by e.g. reading through the newspaper texts published online at <https://hiwwewiedriww.wordpress.com/>. The anthology only contains seven examples with *dass*, as opposed to 96 with *ass*. In five instances, *dass* is part of the consecutive multi-word subordinator *so dass* 'so that'. There are only two cases where *dass* introduces a noun or verbal subject clause (*mei Hoffning* - *dass* 'my hope that' (p. 127); *Noh folgt's, dass aer ken friede hot* 'then it will also follow that he does not have peace' (p. 258). The 96 *ass*-cases cover the range of constructions presented above (comparative/equative constructions, hypothetical conditionals, CCs of verbs, nouns and prepositions) as well RCSs, cf. Section 2.2.2 below. Furthermore, *ass* figures in complex/multi-word adverbial subordinators (*zidder ass* 'since/from that', *nochdem ass* 'after that that').

¹⁸The example is taken from Rauchs Pennsylvania Dutch hand book, 1879 (entry under *bees* (=angry), in the edited version of the 'The Dutch dialect project': http://www.ancest4.com/Tim/dutch_rauch2.html

2 Complementation and relativization

- (18) a. Katzlich hot ein Mann zu mir gsaat, **ass** die Weibslait nix
recently has a man to me said, AS the women.people nothing
waert waere.
worth were
'Recently, a man told me that women are unworthy.'
(Haag 1988: 115)
- b. Kannscht schier net glaawe, **ass** du fliegscht Sex hunnert Meil die
Can.you hardly not believe AS you fly six hundred miles the
Schtund.
hour
'You can hardly believe that you fly six hundred miles an hour.'
(Haag 1988: 304)
- (19) a. Ich bin schtols, **ass** ich en Pennsilfaanisch Deitscher bin.
I am proud AS I a Pennsylvania German am
'I am proud that I am a Pennsylvania German man.'
(Haag 1988: 130)
- b. Der Wasch waar so froh, **ass's** net der Kiwwler waar.
the Wasch was so happy AS=it not the condenser was
'Wasch was so happy that it was not the condenser.'
(Haag 1988: 109)
- (20) Wann's aa yuscht dauere deet **Bis ass's** gut Sach all gesse iss.
when=it also just last does until AS=the good thing all eaten is'
'Even if it takes a while until a good thing is finished.'
(Haag 1988: 98)

The *ass*-complementizer in Pennsylvania German also occurs in embedded clauses after so-called content nouns:

- (21) Un's iss so oft der Fall, **ass** yuscht enwennich iww'rich iss.
and=it is so often the case AS just a.little left.over is
'And it is often the case that only a little is left over.'
(Haag 1988: 98)
- (22) Liewe Leit, sell iss yuscht en Bekanntmachung, **ass** es widder en
dear people this is just an announcement AS is again a
pennsylvanisch-deitscher Gottesdienscht gebt in April.
pennsylvanish-german church service gives in April
'Dear people, this is just an announcement that there will be a

Katrin Axel-Tober & Ellen Brandner

Pennsylvanian German church service in April.’

<https://hiwwewiedriwwe.wordpress.com/category/deutsche-versammlinge/>

22.06.2021

A further West Germanic variety that uses a complementizer based on the equative particle is Yiddish (Jacobs 2005, Taube 1994). The following examples are taken from Taube (1994):¹⁹

(23) Yiddish

- a. di mame zogt as der gan-eydn iz dortn...
the mother says AS the paradise is there...
‘Mother says that paradise is over there.’
- b. s’iz meglex as mit der cayt zol er nemen dray toler a wox.
it’s possible AS with the time shall he take three dollar a week
‘It is possible that with time he will be earning three dollars a week.’
- c. bay ayx, lemishl, kumt aroys as ale gent zenen glayx.
by you, e.g., comes ou AS all geese are equal
‘According to your view, for instance, it follows that all geese are equal.’

Outside West Germanic,²⁰ Scandinavian *som/sem* is a further example of a complementizer that derives from a comparative/predicative particle. *Som/sem* is primarily a relative complementizer, but it also occurs in special types of ‘complement’ clauses, namely in embedded interrogatives, where it may co-occur with the *wh*-pronoun (Taraldsen 1986, Harbert 2007: 426–427, Larsson 2014):

(24) Norwegian

¹⁹Taube (1994) is mainly concerned with the alternation between *as* and *wos* ($\approx$ what), used in relative clauses. He isolates the semanto-pragmatic notions ‘factivity’ and ‘emotivity’, cf. Kiparsky & Kiparsky (1970) as the relevant factors for the possibility of choosing *wos*. In this paper, we are not (yet) concerned with factors like that, as we aim to establish the widespread use of the equative particle as a complementizer.

²⁰Sam Featherston (personal communication) has informed us that in his English dialect *as* can be used as a complementizer under verbs like *say*, *think* etc. Unfortunately, we have not been able to find further (written) evidence for this construction. However, in the light of the dialectal data from (Continental) West Germanic, that we illustrated in this section, it would not come as a surprise that it is found in dialectal/spoken English as well. An investigation of contemporary English dialects would be of merit – especially in the light of the fact that *as* is rather common in English RCs – which will be discussed in the next section.

2 *Complementation and relativization*

Vi vet hvem **som** har gjort dette.
 we know who SOM has done this
 'We know who has done this.'
 (Taraldsen 1986: 11, ex. 27)

As the use of *som* in CCs seems restricted to interrogatives with (extracted) subjects (Westergaard et al. 2012, also Franco & Boef 2015), we will not go further into this here, but see Section 5 for some remarks.²¹

Finally, in the creole language Tok Pisin, an English (German) based Creole, spoken in Papua Neuguinea, the same lexical item that is found in equatives occurs in CCs:

- (25) Tok Pisin
 Em tok **olsem** mi mas skul na kisim gutpela save.
 He said like I must school and get good knowledge
 He said I must go to school and acquire good knowledge.
 (example from APICS²² , nr. 22-164)

Example (26) illustrates the use of *olsem* as an equative particle. It has been grammaticalized from an expression meaning 'all the same' (see e.g. Meyerhoff 2002):

- (26) Tok Pisin
 Emi tok planti West Papua papal iwok **olsem** ol dokta, ...
 3P.PM say many West Papua people PM-work as PL doctor
 'He says that many Western Papuans work as doctors, ...
 (cited from Kornacki 2015: 54)

Outside Germanic, in the West African language Twi, the verb *se* with the meaning of 'resemble', 'be like' is used as a complementizer (Heine & Kuteva 2002: 257):

- (27) Twi
 a. kofi **se** amma.
 Kofi be:like Amma
 'Kofi is like Amma.'

²¹Cf. Larsson (2014) for a survey of the acceptability of this construction in the contemporary North Germanic dialects: <http://www.tekstlab.uio.no/nals/#/chapter/62>.

²²Apics: The Atlas of Pidgin and Creole language structures online (<https://apics-online.info/>), the example can be found under <https://apics-online.info/valuesets/22-95>

Katrin Axel-Tober & Ellen Brandner

- b. na ama nim se kofi yɛɛ adwuma no.
 past Ama know (that) Kofi did work the
 'Ama knew that Kofi had done the work.'

A whole list of lexical items that occur as complementizers and which are based on "conjunctions expressing similarity" is presented and discussed at length in [Lefebvre & Loranger \(2008\)](#). We will refrain here from further illustrations and instead refer the reader to this comprehensive work.

Beside the above cases of the equative particle used as the complementizer, we find many languages where the equative particle shows up in its *wh*-form. This is the case for the Hungarian complementizer *hogy*, which evolved out of an equative particle meaning 'how', [Kiss \(2021\)](#). A comparable source has been attributed to Greek *pos*. In Ancient Greek, the major complementizer was *hóti*, which goes back to the neuter of the relativiser *hóstis*, cf. [Nicholas \(1998: 211\)](#). In Middle Greek several competitors evolved, in particular also *pô:s* (*pos*) 'how', which is the interrogative counterpart to *hô:s*, the instrumental case of Proto-Indo-European **yo*.

This phenomenon is also found in colloquial uses of English *how* ([Legate 2010](#), [Nye 2013](#)). There is, however, no free alternation between the *how*- and the *that*-complementizer. We will briefly come back to this issue in Section 6.

- (28) He told me **how** she hadn't seen her husband for 3 months and...
 ([Nye 2013: 2](#))

Note finally that in nearly all the languages discussed in this section, the equative particle may introduce the clausal complements of perception verbs ([Zimmermann 1993](#), [Liefke 2021](#)), exemplified here merely with Modern German:

- (29) Modern Standard German
 Ich sah **wie** er die Straße überquerte.
 I saw WIE he the road crossed
 'I saw how he crossed the road.'

In sum, the use of an equative particle as the complementizer of an embedded argumental clause is by no means an exception and should thus be considered as one strategy to realise the 'marking' embeddedness.

2.2.2 Relative clauses

Until now we have mainly discussed instances where an equative particle occurs in embedded clauses that are undoubtedly CCs – with matrix predicates

2 *Complementation and relativization*

like *believe, say, know, see* etc. that are usually taken to ‘select’ for clausal complements. A further well-known area where equative particles are employed as complementizers is RCs. Again concentrating on Germanic, these are attested in some contemporary dialects and especially in older stages of (West-) Germanic.

In some sub-variants of Alemannic *ass* may introduce not only equative clauses and CCs, but also relative clauses, (30a). Even the “plain” equative particle *so* is used in this environment, (30b):²³

(30) Alemannic

- a. % des huus, **ass** mo etz kaufe kaa
the house AS one now buy can
 - b. % des huus, **so** mo etz kaufe kaa
the house so one now buy can
- both: ‘The house which is on sale now.’

Similarly, in Pennsylvania German the *ass*-complementizer is used as a complementizer in RCs:²⁴

(31) Pennsylvania German

- Sie schwetze vun Sache **ass** Yaahre zerick ghaeppent sin.
they talk of things AS years ago happened have
‘They are talking of things that happened years ago.’
(Haag 1988: 317)

- (32) Hab ... gsehne... en Mischthaufe **ass** geroche hot wie Blumme.
have ...seen ... a dungheap AS smelt has like flowers
‘I have seen a dungheap that smelt like flowers.’
(Haag 1988: 309)

²³Concerning the Alemannic data, note that the example is marked with %, reflecting the fact that not all variants use the equative particle. The reason is that the most common particle in RCs is the *w*-version of the locative adverb (= *wo*): (i) des huus, wo mo etz kaufe kaa Locative adverbs as RC-complementizers are fairly widespread not only in Germanic but occur in a wide variety of languages, e.g. (spoken) French, Greek, among many others. In the following, we will ignore these cases. In the history of German, the use of a locative adverb was/is one of the many strategies to introduce RCs and in Alemannic and this strategy has won out in most of the variants. Cf. Bräuning (2020) for an extensive overview and discussion about the various strategies to form RCs in Alemannic, see also Brandner & Bräuning (2013) for some earlier speculations about the role of equatives in RC-formation.

²⁴In the examples in (31) and (32) the relative heads have feminine (*Sache*) and masculine (*Mischthaufe*) gender. This rules out the possibility to analyse *ass* as the neuter relative *d*-pronoun *das* with a dropped initial *d*.

Katrin Axel-Tober & Ellen Brandner

In historical German, *so* is attested as a relative particle from Old High German onwards (e.g. Behaghel 1928: 285–288, Erdmann 1874 56f., Ebert et al. 2013: 447 § S 268; cf. also Jäger 2018: 62, fn. 67, and p. 512 for an overview of the older literature). As is observed in Jäger (2018: 62, fn. 67), *so* is already used in relative-like function as a translation of the Latin relative pronoun *quod* in the Old High German *Isidor* translation:²⁵

(33) Old High German

Endi ist sin namo, **so** sie inan nemnant.
and is his name SO they him call
‘and it is his name with which they call him.’
Lat. et hoc est nomen, quod uocabunt eum
(Isidor 9,6; cited from Jäger 2018: 62, fn. 67)

In Middle High German, (34), and in Early New High German, in particular, relative *so* becomes more frequent (Ebert et al. 2013: 447 § S 268), cf. (35).

(34) Middle High German

- a. daz uns müeze werden alles des gebetes ein teil **so** hiute
that us must become all the prayer-GEN a part SO today
wirt gesprochen.
becomes spoken
‘that we must receive all of the prayer that is spoken today.’
(Reinmar von Zw 11,2f.; adapted from Paul et al. 2007: 405))²⁶
- b. von priesterlichem ampte **sô** wir haben von gote
by priestly office SO we have of God
‘by the priestly office that we have been assigned by God’
(cited from BMZ/lemma *sô* <https://woerterbuchnetz.de/>)

(35) Early New High German

²⁵In this example, as in many OHG examples discussed in the literature, a modal/manner reading of *so* is also possible: ‘and it is his name as they call him’. *So* would then be a modal relative adverb. A similar example from the Old Saxon *Heliand* is discussed by Dal (2014: 244)

- (i) nāhor gengun sulike gesiðos, so he im selbo gecōs.
nearer went such companions SO he himself chose

‘There neared such companions which/as he had chosen for himself.’ (Heliand 1279)

²⁶*So* occurs in the manuscript Paris, Bibl. Nat. fonds allem. 32, 14. Jh. [= C]. Other manuscripts have *des* or *daz*, i.e. the neuter form of the relative *d*-pronoun in genitive or accusative case. Cf. Ebert et al. (2013: 447 § S 268).

2 Complementation and relativization

das geleyd, **so** Johan. huß und Hieronymo geben war
the escort SO Johannes Hus and Hieronymo given was
‘the escort that was provided to Jan Hus and Hieronymos’
(Luther, Adel 62; adapted from [Ebert et al. 2013](#): 447 § S 268)

The online Middle High German dictionary also gives examples of relative *als* (< *al* + *sô*; cf. above) under the lemma *alsô*, *alse*, *als* (in paragraph 1.2.7), (36). Parallel examples also occur in Early New High German, (37).

- (36) Middle High German
diu mir her nâch die nôt klaget, / als ir durh mich
who me there after the harm laments ALS you through me have
habt gedolt/.
suffered
‘who then laments the harm that you have suffered through me’.
(Wh 231,11; adapted from
http://www.mhdwb-online.de/wb.php?link_id=4695000)

- (37) Early New High German
und damit die zwene gulden, als sie yme ... geben han
and there.with the two guldens ALS they him given has
‘and with that the two guldens that they have given to him’
(Frankfurter Urk. 138; cited from [Ebert et al. 2013](#): 447 § S 268)

The use of *so* and *als* as relative particles is not restricted to High German, but is also e.g. attested in Middle Low German:

- (38) Middle Low German
dyt weren de word, so ik en lerede.
this were the words SO I them taught
‘This were the words that I had taught them.’
(Reinke 6006, cited from [Ihden 2020](#): 61, ex. 169, who refers to [Kock 1904](#): 2)

In MHG and Early New High German a further relative particle is attested, namely *und(e)*, which is homonymous with the coordinating conjunction ‘and’; cf. (39a). While it seems to be unexpected that a coordinating conjunction can be used as a subordinator, i.e. a C°-element, [Ferraresi & Weiß \(2011\)](#) point to the fact that *und* is semantically maximally underspecified, which has arguably enabled the development of ‘modal’ uses in equative constructions as in (39b). They put

Katrin Axel-Tober & Ellen Brandner

forward the diachronic hypothesis that the relative particle developed from the equative use (cf. already Paul et al. 2007: 427). If this scenario is correct, *und(e)* in historical German would be a further example of an equative-based (relative) complementizer.

(39) Middle High German

- a. wann mit dem urteil **und** ir urteilt, wert ir geurteilt...
because with the judgement UND you judge will.be you judged
‘because you will be judged with the judgement with which you will be judged.’
(Mentelbibel, Mt. 7,2; cited from Ebert et al. 2013: 447 § S 268)
- b. zuo also starken leiden **unt** ich von iu hân.
to such strong suffering UND I of you have
‘to such a strong suffering as I have it experienced through you.’
(NL1787,1–3, (C); cited from Paul et al. 2007: 427)

2.2.3 Interim conclusion

From the data discussed above, we can rather safely conclude that the usage of the equative particle in CCs and RCs is a robust phenomenon within Germanic – and obviously also outside of Germanic. In most varieties of Germanic, the *d*-complementizer is equally a possibility. In Palatine German it seems to have been somehow “stronger” as it has largely replaced the older and more archaic complementizer *ass* in the 20th century. Nevertheless, we find varieties like Pennsylvania German or subvariants of Alemannic that show AS throughout all types of clausal embedding. On the other hand, there are also languages like Modern (Standard) English that use the *d*-type complementizer (*that*) in both CCs and RCs. Many languages choose one type for CCs and the other type for RCs (De Cuba 2017).

The following table shows some of the attested combinations – not to give a complete overview, but rather to illustrate the range of variation:

We find languages with complete uniformity, others with different items in all three surroundings, and then those which cluster two of them with the exclusion of the third. It should be noted that some languages also show ‘doubled forms’ of complementizers. For example, Gothic does not only use the simple *swa*-complementizer (cf. (40) from an example from Miller 2019: 552), but also *swase* as in (41):

2 *Complementation and relativization*

Table 1: Usage of equative particle in CCs and RCs within Germanic

	complement	relative	equative
Pennsylvania German	ass	ass	ass
Alemannic %	ass	ass	ass
Standard English	that	that	as
dialectal English	that	as	as
Alemannic	dass	(d-pronoun) + wo	+ wie
Gothic	(d-pronoun) + ei	(d-pronoun) + ei	swa
Yiddish	az	wos	az

- (40) (tau)jaina izwis mans **swa** jah jūs taujaip̃ im
 ‘let men do to you **as** you would do to them’
 (Mt. 7:12; cited from [Miller 2019](#): 552)

- (41) ni **swaswe** fifjand ina rahnjaip̃
 ‘do not count him **as** an enemy’ (2 Thess 3:15A/B)

In modern German dialects, we find these doubled forms of the equative/comparative particle as well, see also [Jäger \(2018\)](#):

- (42) die isch etz scho größer/so gross **als wie** ich
 she is now already taller/as tall as how I (SynAlm, FB1.1-5, FB7.13-2)

As was pointed out above, some variants employ indeed two lexical items in the same cell, but with different meanings concerning e.g. factivity or temporal overlap (cf. the perception verbs). What we have presented so far does thus not do justice to the overall situation. In light of the fact, however, that there are languages that use the equative particle in all three surroundings, we will concentrate in the next section on how this might come about – both from a semantic and a syntactic perspective. We will further suggest a structure that provides a position for the optional *d*-pronouns and offer some speculations about how the current situation e.g. in English might have arisen. Before going into that in more detail we will, however, first look at the role of nominal elements in the matrix clause. A major difference between CCs and RCs is that the former, especially after *say/think*-type verbs, often do not show a ‘head noun’ (correlate) and thus

Katrin Axel-Tober & Ellen Brandner

give rise to the impression that these CCs are directly selected by the governing predicates as it has been advocated in the traditional analysis of clausal complementation. In the next section, we will discuss the historical development of a nominal element in the matrix clause in the form of a so-called ‘correlative’ pronoun. This is to argue that our proposal given in (1) and repeated here in (43), is also justified when it comes to the nominal in the matrix.

(43) ... [head noun/correlate]... [_{SC} (d-pronoun) AS [_{TP} predicate]]

3 The correlate construction

Proto-Indo-European did not show the type of finite CCs that we know from e.g. modern Germanic and Romance. According to Lühr (2008) the evolution of relative conjunctions (i.e. relative particles/complementizers) from relative pronouns must have already happened in Proto-Indo-European. However, the precursors of the modern type of CCs “were apparently originally RCs connected to a reference word” (Lühr 2008: 156). The predicates did not subcategorize for finite clauses, but for DPs that could then be combined with an ‘explicative’ clause that was syntactically not a complement, but an adjunct. This pattern (= ‘nominal element + explicative clause’) is abundantly attested in the older Indo-European languages.

As to Germanic (early and modern), an instantiation of this pattern is the widely-discussed ‘correlate’ or ‘proleptic’ construction. It is a well-known fact that in the Germanic OV languages, clausal and non-clausal complements have a different syntactic distribution. Whereas non-clausal direct objects occur to the left of the governing verb in the so-called ‘middle field’, clausal ‘objects’ do not typically show up in this position. The unmarked position for a clausal ‘complement’ is to the right of the governing verb, i.e. in the ‘post/final field’. In many cases, the ‘post-posed’ clause is associated with a pronominal element in the middle field.

(44) Modern German

- a. Ama hat es bereut, dass sie zurückgetreten ist.
Ama has it regretted that she resigned is
‘Ama has regretted it that she has resigned.’
- b. Modern Dutch
dat Jan het betreurt dat Marie zwanger is.
that Jan it regretted that Mary pregnant is
‘that Jan regrets it that Mary is pregnant.’

2 *Complementation and relativization*

(adapted from [Den Dikken 2017](#))

The type of clause combining, where a dependent clause occurs at the right (or left) periphery of the matrix/superordinate clause and is associated with a correlate/ proleptic proform, has been argued to be an important cross-linguistic strategy to avoid direct clausal embedding ([Den Dikken 2017](#)).

3.1 **The Germanic correlate construction**

The correlate construction, in particular of the type with a *d*-proform in the matrix, was very wide-spread in early Germanic. As [Lühr \(2004\)](#) argues, the use of the *d*-correlate is a hallmark feature of early West Germanic clause combining. She mainly presents material from the Old Saxon Heliand. It is, however, a well-known fact that the correlate construction was also a typical of Old High German ([Müller & Frings 1959](#), [Johnk 1979](#), [Axel-Tober 2012](#), [Axel-Tober 2017](#)) and Old English ([Hopper & Traugott 1993](#), [Roberts & Roussou 2003](#); [Los 2015](#)).

- (45) Old Saxon
 ef he **that** giquâði, **that** sie sie quica lêtin (Heliand 3857f.)
 if he DEM said THAT they them alive leave
 ‘If he said (it), that they should leave them alive.’
 ([Lühr 2004](#): 165, her (12))
- (46) Old High German
 Er tháhta odowila **tház**,\ **thaz** er ther dúriwart wás.
 he thought maybe DEM THAT he the gatekeeper was
 ‘Maybe he thought (it), that he was the gate-keeper.’
 ([Axel-Tober 2017](#): 34, her (17))
- (47) Old English
 þæt gefremede Diulius hiora consul, þæt þæt wearð tidlice
 DEM arranged Diulius their consul, ÐÆT DEM beginning in-time
 þurhtogen.
 achieved
 ‘Their consul Diulius arranged (it) that it was started on time.’
 (c. 880, Orosius 46.172.2; from [Hopper & Traugott 1993](#): 186)

Regarding North Germanic, Old Norse clauses introduced by the *at*-complementizer (object and subject clauses, adverbial clauses and complements of prepositions)

Katrin Axel-Tober & Ellen Brandner

are indeed often attested with a demonstrative in the superordinate clause (Faarlund 2004). Instead of the neuter demonstrative in the respective case governed by the matrix predicate (e.g. genitive in (48)), the correlate can also be the adverb *svá* ‘thus’, cf. (48b) below).

(48) Old Norse

- a. ok beiddi Þorfinn þess ins sama, at hann skyldi játa
and asked Thorfinn THAT.gen the same, that he should grant
konungi þeim hluta landa, er hann átti áðr.
king.dat that part lands.gen ER he owned before
‘and asked the same of Thorfinn, that he should grant the king that
part of the lands which he had owned before.’
(Hkr II.209.11; Faarlund 2004: 87, ex. 17b)
- b. hygg nú svá fyrir hag þínum at þar liggur við líf
think now THUS for condition your THAT there lies by life
þitt.
your
‘Now remember for your own sake that your life depends on it.’
(Nj 160.8; Faarlund 2004: 156, ex. 100)

The demonstrative correlate was not only attested in combination with *at*-clauses, but also with subject or object clauses introduced by the complementizer *er*, (cf. Section 2 above example (4)).

In East Germanic, the correlate construction seems to have been pertinent as well. Both the simple complementizer *ei* and the compound one (inflected *thata + ei*) are attested in clauses that are associated with a neuter demonstrative pronoun in the superordinate clause (Longobardi 1979, Harbert 1992, Miller 2019).

(49) Gothic

- a. þamma nifaginop, ei þai ahmans izwis ufhausjand.
that.DAT not-rejoice EI the spirits to.you are.subject
‘Do not rejoice in this that the demons are subject to you.’
(Gothic bible Luke 10:20)
- b. domjandans pata patei ains faur allans gaswalt ...
thinking that.ACC THAT.EI one for all dies
‘thinking about this that one died for all’
(Gothic bible 2. Cor. V:15)

2 Complementation and relativization

So the use of the neuter demonstrative pronoun as a correlate was probably a Pan-Germanic phenomenon, but as we have demonstrated above, its homonymous use as a complementizer is absent from Old Norse and also from Gothic, where *pata* had to combine with *ei*, a generalized complementizer of different origin. In sum, it seems quite obvious that in older stages, a correlate in the matrix was the usual way to ‘connect’ an embedded clause to the matrix clause. Note that in many examples, the correlate and the complementizer are not identical in form. This casts doubts on the common diachronic analysis that this correlate is the source of the complementizer which ‘entered’ the embedded clause via a shift of the clause boundary (e.g. Müller & Frings 1959, cf. also Roberts & Rousou 2003 for formal syntactic account). The word order in the languages with asymmetric verb placement (verb-final in subordinate clauses, verb-second/first in main clauses) is a further problem, as a simple shift of the correlate from the matrix into the embedded clause won’t allow it to occur in the C-position of the embedded clause and block verb movement (Axel-Tober 2017), see also Weiß (2020). Thus, the data actually do not support a scenario in which the emerging ‘complement’ clause was a separate main clause in a paratactic sentential sequence, but rather point to a source structure in which the correlate and the embedded clause built one constituent whereby the *d*-pronoun was assigned the theta-role from the matrix verb and the embedded clause was first merged in an adjoined position to it and then subsequently extraposed. In the next section we will shortly discuss some analyses of this type that have been proposed for the correlate construction in modern German.

3.2 The correlate construction in modern German

In German (both in the standard language and in the dialects), the unmarked correlate is now *es*, which is based on the neuter 3. person singular form of the person pronoun.

(50) Modern Standard German

Petra wird (es) bedauern, dass sie ihre Hilfe zugesagt hat.
 Petra will it regret THAT she her help offered has
 ‘Petra will regret it that she has offered her assistance.’

- (51) Mir ist (es) klar, dass das viel Arbeit macht.
 me.DAT is it.NOM clear, that that much work makes
 ‘To me, it is clear that this will take a lot of work.’

Katrin Axel-Tober & Ellen Brandner

Es is only used with object clauses of transitive verbs and with subject clauses, it is restricted to nominative and accusative case. With a correlate in the dative/genitive case, the DAS-proform, based on the neuter form of the demonstrative pronoun has to be used:²⁷

(52) Modern Standard German

- a. Petra hat **dem** widersprochen, **dass** sie ihre Hilfe zugesagt
 Petra has THAT.DAT objected THAT she her help offered
 hat.
 has
 ‘Petra has objected to the fact that she has offered her assistance.’
- b. Petra ist sich **dessen** bewusst, **dass** sie Hilfe zugesagt hat.
 Petra is REFL THAT.GEN aware that she help offered has
 ‘Petra is aware of the fact that she has offered her assistance.’

Diachronically, there is a general tendency in German(ic) verbal/adjectival complementation to replace genitive or dative case by prepositional government.²⁸ This is why so-called ‘prepositional object clauses’ have become a major type of clausal ‘complementation’ both in the Standard language and in the dialects. When the object of a preposition-governing verb is realized as a clause, a so-called ‘pronominal adverb’ is used in the matrix clause. In this case, the ‘correlate’ thus consists of the *d*-pronominal part *da(r)* + the governed preposition (= *darüber* ‘there.over’, *darauf* ‘there.on’, *daran* ‘there.at’, *damit* ‘there.with’ ..., or

²⁷In structures as in (50) to (52), where the associated clause occurs in the post field, the use of the correlate is optional (e.g. Zifonun et al. 1997:1478–1483). When the subordinate clause occurs in the middle field or in the prefield/SpecC position of the matrix clause, oblique correlates (dative, genitive, prepositional (cf. below)) are obligatory:

- (i) Petra hat ***(dem)**, dass sie ihre Hilfe zugesagt hat, widersprochen.
 Petra has that-DAT that she her help offered has objected
- (ii) Petra ist sich ***(dessen)**, dass sie ihre Hilfe zugesagt hat, bewusst.
 Petra is REFL that.GEN that she her help offered has aware

²⁸(Standard) German still has a relatively rich case system and the neuter form of the demonstrative has three different forms (*das*-NOM/ACC, *dessen*-GEN, *dem*-DAT). This may also be a reason why the *da(r)*+P-correlate, even though it has gained much ground in the last centuries has not fully replaced the dative and genitive *d*-proforms. Cf. also Gunkel & Hartmann (2020) on ‘prepositional object clauses’ in further Continental West Germanic languages and in North Germanic. In their account, accented forms of prepositional adverbs (*DARüber*, *DARan*) give rise to different structures than those with weak (*damit*, *daran*) or reduced prepositional adverbs (*dran*). Only these weak forms take CP-complements.

2 Complementation and relativization

reduced forms: *drüber, drauf, dran...*), cf. the following examples from Standard and Pennsylvania German:

- (53) a. Modern Standard German
 Petra hat sich **darüber** geärgert, **dass** sie ihre Hilfe zugesagt hat.
 Petra has REFL there.about annoyed that she her help offered has
 ‘Petra is annoyed about the fact that she has offered her help.’
 b. Pennsylvania German
 Hend ihr schunn **draagedenkt** **ass** net all die Monnkis uff
 have you already there.on/at.thought AS not all the monkeys on
 vier Bee laafe duhne?
 four legs walk do
 ‘Have you already thought about the fact that not all monkeys walk
 on four legs?’
 (Haag 1988: 131)

It has been discussed in the literature whether correlate-taking predicates sub-categorize for an NP/DP (or PP) instead of a CP (e.g. Müller 1995, Sudhoff 2003, Sudhoff 2016, Zimmermann 2016). Zimmermann (2016) proposes the following unified structure for all types of ‘correlates’ in (Standard) German:

- (54) $[XP\ X\alpha\ ([_{PP}\ P)_{\beta}\ [_{DP}\ [_{D'}\ [_D\ \{\{es/das\}, \emptyset\}]]\ CP]\ ([)]_{\beta}\ X_{-\alpha}]$
 (Zimmermann 2016: 399, ex. 10a modified)

In this analysis, X is the governing predicate which takes either a DP- or PP-complement. The cataphoric correlate is an overt or zero determiner which hosts an additional position for an explicative modifier in the form of a CP. (In the case of prepositional government the adverbial form *da(r)* is base-generated in D and then raised to P.) In Zimmermann’s account, the embedded CP occupies SpecDP and is thus accessible for government by the predicate with its c-selectional properties (e.g. the case of the DP, the specific preposition, +/- finiteness of the CP).²⁹ The DP-shell can also be realized by a zero head, this is why the correlate *es* is facultative (with most predicates). Sudhoff (2016), by contrast, proposes an analysis similar to one that has been used for restrictive RCs (Streube 1992, Sternefeld 2006), where correlate is in D° and takes the CP as its complement:

²⁹If the CP is extraposed, the idiosyncratic PPs and DPs with lexical cases can undergo deletion at PF. Zimmermann does not provide an explanation for the fact that extraposition is obligatory in sentences with correlate *es*. Sudhoff (2003, 2016) attributes this to the fact that (overt) *es* is phonologically weak and cannot be accented, it cannot constitute a phonological head (Pittner 1995). Since the syntactic head of the DP-shell cannot be stressed/accented, clauses associated with correlate-*es* have to undergo extraposition.

Katrin Axel-Tober & Ellen Brandner

- (55) [CP [C' dass [VP [VP Fred_j [VP vielleicht [VP t_j [V' [DP [D' (es) t_k] bedauert]]]]] [CP dass p]_k]]]]
 'that F. (it) perhaps regrets that p'
 (adapted with slight simplifications from Sudhoff 2003: 94, ex. 60b)

In the following section, we also adopt an analysis in which the CP in a correlate construction is a complement of the correlate-D°, but will put forward a structure that is based on a recent proposal by Cinque (2020).

In most accounts of modern German, the correlate construction is taken to be predicate-dependent. It is generally argued that some classes of verbs, in particular those which allow long *wh*-extraction ('bridge verbs'), do not allow a correlate (neither silent or overt). These classes, which e.g. contain the *verba dicendi* and some *verba sentiendi*, are taken to subcategorize for a CP (and not a DP), which they 'directly' take as a complement. It should be noted, however, that even these verb classes are often used with a *es* or *das* in the matrix clause. This kind of uses are then not taken as instantiations of the 'true' correlate construction, but as anaphoric sentential proform to which the CP is adjoined in an apposition-like structure (Pütz 1986², Sudhoff 2003, 2016, Zimmermann 2016). By contrast, our unified analysis of RCs and CCs, rests on the assumption that there is always a nominal present in the matrix clause. In other words, even in bridge-verb constructions the embedded CP is not a complement of the verb. Apparent long 'extractions' will thus have to receive a totally different analysis than traditionally assumed. We will discuss this in Section 5.

4 'Equality'

As was demonstrated in Section 3, there is robust evidence for the use of complementizers based on equative particles in Germanic and beyond. It cannot be an accident that so many different languages came up with a similar strategy for clausal embedding, namely to invoke a particle to express somehow 'equality'. That equality or rather simultaneity plays a role with perception verbs, cf. (30) above, seems intuitively obvious, because here the event of 'seeing' and the observed event happens at the same time and thus the use of the equative particle can be plausibly brought together with the temporal simultaneity of the two events. But what about the cases where the particle occurs as the complementizer of embedded clauses, selected by matrix predicates like *say*, *think*, *believe* as in the cases shown in Section 2? Besides this, the widespread use of the equative particle in RCs is also in need of a principled explanation.

2 Complementation and relativization

Elliott (2020) recently proposed that the relationship between a content nominal in the matrix and the embedded clause is one of equality rather than of modification as assumed in the work starting with Kratzer (2006). His crucial observation is that these content nouns may occur with all types of determiners and/or quantifiers if they are used without an embedding clause. This also holds if the embedded clause is a RC. However, as soon as the content noun is accompanied by an embedded clause that delivers the content, only the definite determiner is possible:

- (56) a. Several/some facts speak against this.
 b. Several/some facts that should be considered
 c. The/*a fact that it is raining

Elliot derives this contrast by appealing to the Maximize presupposition! constraint, cf. Heim (1991), which requires that in case the uniqueness of an entity is already given, other determiners than the definite article are ruled out. If the content of the respective noun is equated with the content of the clause, uniqueness is entailed as there is already one single referent in the discourse, namely the content of the embedded clause. Only the definite determiner is thus possible in this case. This argumentation can also be extended to the correlatives, discussed in Section 3, since *d*-pronouns are definite per se.

As our focus is on the morpho-syntax, we will defer from a semantic discussion and just follow Elliott in his argumentation that there are good reasons to invoke the notion of equality in the realm of clausal embedding. The fact that so many languages indeed bear this semantics “on their sleeves” lends further support to such an approach. To make the idea of ‘equality’ work on a morpho-syntactic basis, it is necessary to reconsider the categorial status of the embedded C-projection. As we will argue below, a Small Clause structure will allow us to capture the equality semantics of clausal embedding as well as the variation found at the ‘left periphery’. Specifically concerning the German *ass/dass*-variation, we will suggest that the equative particle forms the head of the Small Clause and that the *d*-pronoun occupies the subject position of Small Clause.

4.1 The equation construction

Before going into clausal complementation with the equative particle, we will outline some general observations on ‘equation’. Note first that an equative/comparative construction is “correlative” by its very nature, as two entities must always be involved in order to equate or compare them. Whether the relation between these two entities is characterised as being equal (equative) in some

Katrin Axel-Tober & Ellen Brandner

respect or different (comparative) is secondary and not necessarily encoded in the particle itself.³⁰ Already on this very general level of thought, equative particles are very well suited to connect two elements and we will suggest that grammars exploit this inherent ‘dualness’ to connect viz. equate a nominal in the matrix with the content of an embedded clause – just as it is proposed in Elliott (2020). The question to be addressed in this section is how to implement this idea into the syntax.

As a starting point, we will introduce the terminology used in Haspelmath & Buchholz (1998) in their survey of equative/comparative constructions in various languages. Let us consider a ‘typical’ equation structure like in (57):

- (57) a. John is as tall/smart as Mary.
 b. Hans ist so groß/klug wie Maria.
 [so/as (property of) X] [so/as/wie ... (property of) Y]
 P-marker Parameter S-marker Standard

The Parameter introduces the property to what the equation/comparison will apply, i.e. height, smartness etc. The marker before the adjective (*so/as*) induces a specific interpretation: there are several (at least two) degrees or extents to which John can be tall/smart. Thus a scale is evoked and John’s height/smartness will eventually be situated on this scale, i.e. it will be given a value. The Standard delivers this value; in our example the height/smartness of Mary about which the speaker is supposed to know – otherwise the equation/comparison would be vacuous. It doesn’t matter whether or not we know the exact height/smartness of Mary in terms of meters/centimeters or IQ-values. What is important is that Mary has such a value and in this sense it can be treated as a fixed value³¹, which is eventually determined in the context/world knowledge. John – or better to say John’s height/smartness – will now get this very same value via the equation relation (or a higher/lower one in case of a comparative). Thus, the marker of

³⁰See e.g. Southern German dialects that use the particle *wie* in the equative and the comparative, whereby the latter is uniquely identified by the morphology on the adjective:

(i) (so) groß wie... ((as) tall as)
 größ-er wie (taller than)

French also uses *que* both in comparatives and equatives (*plus grand que* vs. *aussi grand que*). Here as well the decision between equative/comparative is relegated to further lexical items.

³¹We are aware of the fact that the semantics of comparatives/equatives is usually treated in quantificational terms, cf. Rett (2008) for an overview, see Hohaus & Zimmermann (2021) for a recent discussion who extend the quantificational analysis to equation/comparison of properties.

2 *Complementation and relativization*

the parameter induces a sort of kind-reading of the predicate (Parameter): the set of all possible values that may apply to the given property are evoked and then, when equating it with the value given by the Standard, one value is picked out and will then apply to John.

In addition to inducing this kind-reading, the parameter-marker can be seen as cataphoric pronoun/element. This can be illustrated nicely with the following (anecdotic child language) data:

- (58) German
 wenn ich [wie David gross] bin,...
 when I how David tall am
 intended meaning: ‘when I have reached the height (or age) of David.’

The Standard with its marker *wie* is situated directly in a position within the adjectival projection. It is in this case treated like a numerical measure phrase of the type ‘he is 1.80 meters high’. Let us assume that adjectives (of this type) have something like a ‘degree argument’ within their projection – for the sake of concreteness we posit it in the specifier without further discussion. In the tree on the right, we can see this ‘shortened’ version with the standard directly inserted in the Degree-phrase. In the tree on the right, one which represents the usual equative construction, the standard is ‘extraposed’, i.e. right-adjoined to the AP, and the original degree position is occupied by the parameter marker. We will give explicit trees for the embedded clauses below where we also address the issue of deletion, which will become crucial especially in the context of relative clauses. The trees in Figure 1 are merely meant to illustrate the intuition behind a typical³² equative/comparative construction with its bi-partite structure:

The Parameter marker (*so* in our case) could then be seen in addition as a placeholder for an expression yet to come which will then deliver the content with which it can be replaced. As such, we are already very close to what we find in clausal embedding as Elliott (2020) describes it.

Before going into the syntax, a brief remark on the status of degrees and properties: degrees have often been seen as a semantic primitive and being restricted essentially to gradable adjectives as in (57). Anderson & Marcin (2015) however

³²Numerical measure phrases resist usually this kind of extraposition. We will not go into a deeper discussion of this issue here, but it seems to us that this is not a matter to be handled strictly within the grammar. Rather, the *so...wie* construction gives us an instruction to reach eventually a value by a “detour” in that it introduces something of which the speaker assumes that the value is known to the hearer. But if the numerical value is known, it can be directly inserted into the degree phrase. It is thus more a matter of economy, viz. subject to the Blocking Principle.

Katrin Axel-Tober & Ellen Brandner

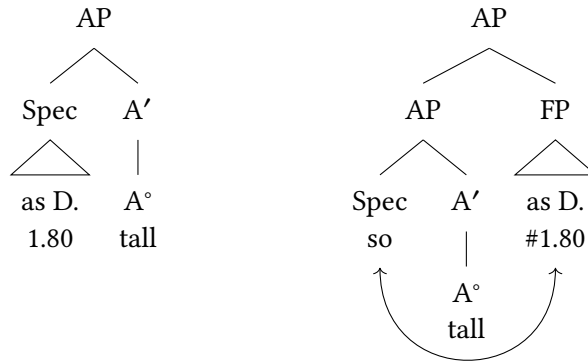


Figure 1: Structure of an equation

argue that degrees should be seen merely as a special type of kinds, namely kinds of states. This means that having a property to a certain degree is nothing more than being in a certain state. Their claim is based on the similarity (or identity) of the lexical items used in these three constructions, as illustrated for German in (59) (their example 5):

- (59) modern Standard German
so groß wie degree → kind of states
as tall how
so tanzt wie manner → kind of events
as dances how
so ein Hund wie kind → kind of nominals
as a dog how

Under this perspective, manner-predicates can be seen as kinds of events (e.g. dancing with the arms above the head, dancing with the arms on the hip, dancing slowly, etc.) and nominals are of the usual kind. Regardless of whether the conception of kinds is adequate in this context (see [Moltmann 2015](#) for a critical discussion), the important point for us is that the equative may apply to all types of predicates in this analysis, i.e. also to properties that describe an event which in turn means also to embedded clauses – if taken as predicates. One important point is that the equation relation necessarily consists of the cataphoric element (the Parameter marker in Haspelmath & Buchholz’s terms) and the Standard, also marked with such a particle, i.e. the equative particle. Thus, the abstract structure of clausal embedding and of RCs looks like the following:

2 Complementation and relativization

- (60) Parameter marker Standard marker predicate
 CCs: correlate equals predicate
 RCs: head noun equals predicate with gap

In the following, this will be made more precise.

4.2 Relative clauses with equative semantics

For RCs, we will assume an equative structure. However, there is of course an important difference in interpretation: In clausal embedding, the correlate/placeholder is devoid of any descriptive content and thus the modification does not lead to an intersection of properties, rather equality can apply directly, giving rise to the obligatory definiteness of the correlate, resp. content nouns, cf. [Elliott's](#) (2020) observation and our brief discussion above.

This is different in RCs: here, the intersection of properties is exactly what we want to achieve such that a given discourse referent is described by "more" properties³³: those of its nominal predicate, the predicate in the matrix clause and the predicate in the RC. The suggestion we want to make is that in case of RCs, the equation targets the nominal predicate of the head noun and the nominal expression in the RC while they 'share' the D-position, see below.

To illustrate, take the following example:

- (61) [The cat [as ___ is sitting over there] is hungry]

The paraphrase of such a sentence would be something like: 'The cat which is sitting over there is the same cat which is hungry'. Applied to a head noun with the quantifier *all* for example, the same type of paraphrase can be used, as the quantifier applies to both instances:

- (62) [All cats [as ___ do have stripes] do not eat mice]

The paraphrase of this sentence is: 'the set of cats that do have stripes equals the set of cats that do not eat mice'. Similar considerations apply to a head noun with an indefinite determiner as in:

- (63) [A cat [as ___ does have stripes] does not eat mice]

³³Recall that the conjunction *und* 'and' could be found as the RC introducer in historical German, cf. Section 3.2.2.

Katrin Axel-Tober & Ellen Brandner

The indefinite determiner here is not interpreted as an existential quantifier, but it is bound by an implicit universal quantifier. A paraphrase would be: ‘if *x* is a cat and if *x* has stripes, then *x* does not eat mice’. We have an immediate explanation why the determiner in RCs is not restricted to the definite determiner as it is with content nouns, resp. pronouns (correlates), the latter being inherently definite. The reason is that only the nominal predicate is equated but that the (clausal) predicate of the RC still need something to predicate over. Thus what is known, resp. presupposed is that we are talking about a cat, but it is not known what the further properties are. Therefore “Maximise Presupposition!” does not apply.

Anticipating the discussion about the (nearly³⁴) obligatory gap in RCs – which is one of the main differences between CCs and RCs – note that the idea of using the properties of equative/comparative constructions gives us immediately an idea why the gap occurs in RCs only: in this case there are two identical instances of the same phrase and thus a deletion operation takes place – as it is known from equation/comparative constructions, cf. [Bacskai-Atkari \(2018\)](#) for a recent proposal.

- (64) [the [cat]]ⁱ [as [[cat]]ⁱ is sitting over there]], is hungry
 Parameter Standard marker predicate1 predicate2
 nominal equals nominal

As there are no identical lexical items in the case of CCs, the deletion operation does not apply and we do not have a gap in CCs.

Note that this conception differs slightly from the more common predicate modification approach as in [Heim & Kratzer \(1998\)](#) which is usually syntactically implemented as an adjunction to the NP (in a head-external analysis). Rather, the representation in (64) is more in line with [Cinque’s \(2020\)](#) ‘double headed’ analysis. Cinque proposes that the different type of RCs (restrictive, appositive, maximalizing, etc.) all have the same underlying structure: the RC is situated in a position between the determiner of the ‘head noun’ (the external head) and the ‘internal head’, which is generated as a nominal projection without a (referential) D-layer within the embedded clause, see below for further elaboration.

To summarize, our observations concerning the role of equative particles in clausal embedding has prompted us to propose the following (abstract) structures:

³⁴We will not go into a discussion of resumptive pronouns, see Section 5 for some considerations.

2 *Complementation and relativization*

Table 2: Different environments for the equative particle

	Parameter	Standard Marker	Standard
Equation	property of X	as	property of Y
RC	[... [DP[NP]]...]	as	[... NP ...]
CC	[... it ...]	as	[clausal predicate]

4.3 **Application to complement clauses**

In the degree construction above, we posited the placeholder/correlate in the specifier of the AP. But as we have seen, this ‘typical’ degree construction is but one instance of a general correlative construction with a parameter and a standard. We shall claim now that essentially all kinds of phrases may serve as the parameter. In the case of RCs, it is the head noun, and in CCs, the nominal (or prepositional) correlate in the matrix.

Concerning the embedded clauses themselves, we will suggest – again based on the data with the equative particle – that they can be captured best if we take them generally as an instance of a Small Clause, cf. [Bowers \(1993\)](#). We take the TP as the predicate and the head of the Small Clause is lexicalized by the equative particle. Its subject may either be empty or realized by a pronominal, i.e. the ‘doubled’ correlate as in the data illustrated in [Section 3](#). This configuration then led to the development of the ‘traditional’ *that*-complementizer. What is commonly called the CP-layer or the Left Periphery is thus actually the subject and head of a Small Clause:

The next questions are now: How is this Small Clause introduced into the structure? How can the different outcomes that we saw above be captured? And what do the elements with their varying lexical sources (equative particle and/or *d*-pronoun) in this left periphery contribute when they ‘connect’ with the correlate?

We will start with the RC. Departing from the analysis by [Cinque \(2020\)](#), where all types of RCs are based on a double-headed structure, we will assume that the RC, i.e. the Small Clause in our understanding, is inserted under a functional projection (YP) above the (external) nominal (dP) with the D-head external to both of them. Simplifying somewhat Cinque’s proposal for ease of exposition, we will suggest the structure in [Figure 3](#):

Now the very same structure can be used for CCs. The sole difference is that in this case, the external dP is empty as there is no lexical nominal restriction

Katrin Axel-Tober & Ellen Brandner

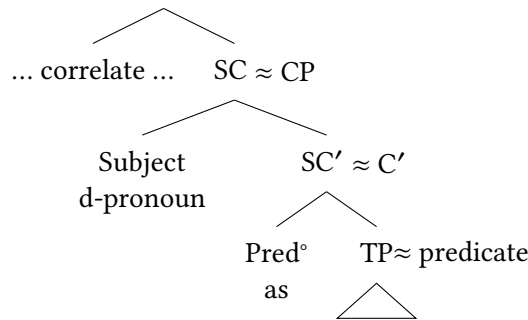


Figure 2: General structure of RCs and CCs

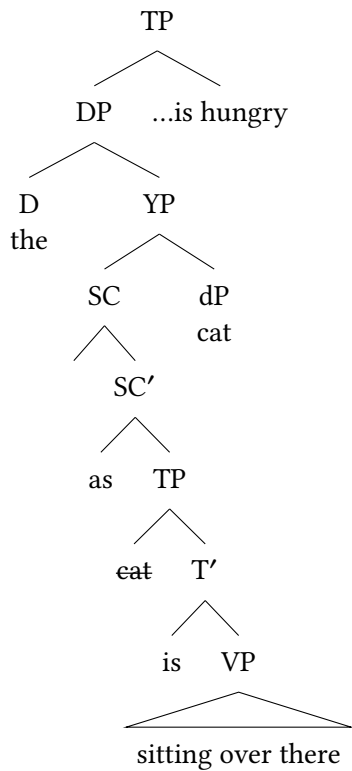


Figure 3: Structure of a RC

2 Complementation and relativization

in case the D-position is occupied by the (potentially silent) correlative pronoun (Figure 4).

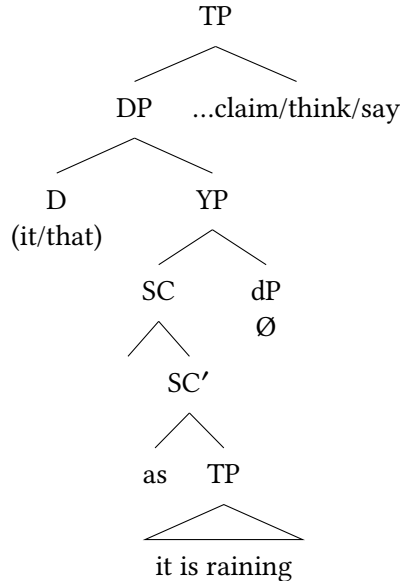


Figure 4: Structure of a CC

A deletion operation does not take place as the structure does not involve two identical lexical items.

For the sake of completeness, let us illustrate how this general structure can be applied to ‘standard’ equatives/comparatives as well. The only difference is that here, the SC is not situated in the functional extension of a nominal, but rather of a predication adjective, respectively an understood manner component of a verb. For reasons of concreteness³⁵, we take this position of the parameter marker in the matrix as the head of the functional extension (YP) of the AP/VP:

These would be the structures for a language like Pennsylvania German or some variants of Alemannic, as we illustrated in Section 2, with the head of

³⁵It is probably more adequate to posit it in the head of the specifier of YP – just as it is generally assumed for the DegreePhrase in adjectival projections, cf. also the structural sketches in Figure 1. If the parallelism between degree and manner holds, then VP should be able to host such a “degree”- phrase as well, hosting then a manner adverbial in case of a lexical verb. Importantly, the mechanism that this position is filled with the cataphoric element *so* and subsequent ‘extraposition’ applies in the same way. The question concerning the exact position of *so* in these cases will be left open to further research.

Katrin Axel-Tober & Ellen Brandner

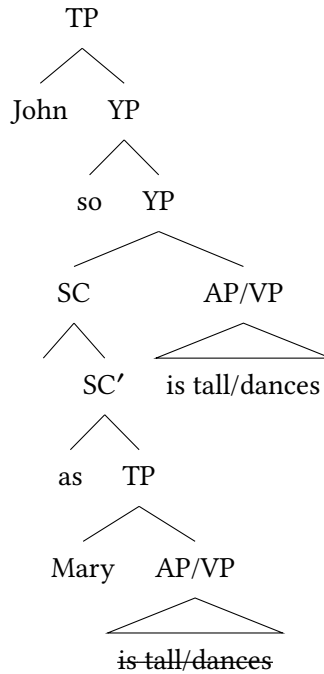


Figure 5: Structure of an equative construction

the Small Clause spelled with the equative particle. But as we have seen, many Germanic languages use the *d*-pronoun as the ‘complementizer’. We will suggest that it is the subject position of the Small Clause that may host such a *d*-pronoun, cf. the structure in Figure 2.

We have argued above that the equative particle seems to be an optimal candidate for fulfilling the task to ‘connect’ an embedded clause, i.e. a predicate with a nominal (and argumental) element in the matrix clause. Nevertheless, one very common strategy in Germanic is the use of a *d*-pronoun instead of the particle. The question is thus, how come that a *d*-pronoun may fulfil this function as well? And furthermore: can we develop a plausible scenario why the *d*-pronoun strategy superseded the equative particle – at least in the standardized, written variants?

As discussed in Section 2.1, we follow the analysis that the *d*-pronoun complementizer – at least in West Germanic – developed out of a RC with the neuter *that/das* demonstrative. So the question boils down to how RCs with *d*-pronouns evolved at all. Again, we can only provide a first sketch of a putative analysis,

2 *Complementation and relativization*

which may serve as a point of departure for future accounts of *d*-pronoun relatives in Germanic.

Recall that in contemporary German, the *d*-pronoun in RCs bears the phi-features and crucially the case of the nominal within the RC. This used to be different in earlier stages where this pronoun could share the case with the nominal in the matrix. This phenomenon is traditionally called ‘case attraction’, cf. Erdmann (1874) and Sapp (2019) for a recent discussion. A typical example from Old High German is given in (65), but we find the same pattern in Old English (66), Old Icelandic (67) among others (data are taken from Sapp; his (38, 39, 3 respectively, glosses adapted).

(65) Old High German

annuzi min-es fater thes dar in himile ist
face my-gen father d-gen there in heaven is
‘The face of my father who is in heaven’
(Tatian 96:1, cited in Harbert 1992:110)

(66) Middle High German

ere ... des dienst-es des iu wirt getan
honor ... the.gen service-gen d-gen you.dat is done
‘... the honor of the service, that is done to you’
(Karl der Gr. 9666, Harbert 1992:110)

(67) Old Icelandic

Hann setti jarl í hverju fylki, þann er doema skyldi lög
he set earl.acc in each district d.acc ER judge should law
‘He placed an earl in each district, who should judge the law.’
(Heimskringla 98)

Is this
Old En-
glish
[cfr.
text]
or Mid-
dle High
German?

Thus, the *d*-pronoun obviously does not originate within the RC (nor is it moved to a left peripheral position, leaving a trace which gives rise to the gap, cf. our explanation of the gap in terms of deletion under identity). Rather, what seems to be the case is that it occurs as a doubled version of the head noun. As a demonstrative, it is free to take its reference outside of its clause and thus may share it with the head noun. Now if – as we suggest – this pronoun is situated in the subject position of the Small Clause, we can understand how this construction delivers the same function: as the subject of the Small Clause it will be predicated over by the SC-predicate, i.e. the content of this predicate now holds of this pronoun. Due to co-reference with the head noun in the matrix

clause, the same goal is achieved as with the equative particle, namely that the property expressed by the TP is ‘transferred’ to the nominal in the matrix.

Interestingly, as Sapp shows in great detail, the common development seems to be that these case-attracted nominals appear together with a relative particle (*er* in Old Norse, *ei* in Gothic, see the discussion in Section 2.1), which in turn seems to be the earliest strategy. Later the particle disappears and in the final state, we find either an invariant form of the *d*-pronoun (English *that*) or a *d*-pronoun that bears the case from within the RC, see Sapp (2019: his (62)). As for the latter point, we can only offer a speculation right now: the case-attracted demonstrative clearly builds a constituent with the RC (cf. Sapp 2019 for extensive discussion). In our proposal, this is equally the case. This leaves us with an ‘additional’ argument-like element within one clausal domain, but for which there is no appropriate case-assigner within this domain, cf. the genitive in (65) within a clause with only a copula. And furthermore, due to the deletion operation in the RC, the ‘real’ subject is not expressed. However, given that the case that would be assigned otherwise to the gap, is free³⁶ to be overtly realized, we tentatively suggest that the case assignment to the relative pronoun is a surface repair-strategy that will yield a ‘proper’ clause with all the nominals bearing the ‘correct’ case – at least on the surface. As discussed in Sapp (2019), relying on Pittner (1995), the loss of case-attraction and replacement of it with the local (or invariant) case goes hand in hand with the loss of the particle. In our scenario, this would mean that the head of the Small Clause is no more spelled out and the underlying configuration is thus blurred. This then could be the input for a reanalysis such that the *d*-pronoun is re-analysed as a head, as in Van Gelderen’s (2004) Head-preference principle and we find the typical behaviour of a ‘complementizer’. See also Section 5, for some speculations on the doubly filled Comp Phenomenon in the context of question formation.

What we have not addressed until now is the surface position of this Small Clause. Again, following Cinque, we assume that it is extraposed in languages with externally headed RCs and adjoined quite low in the structure, at a projection of the matrix VP. According to our analysis, this would then hold for CCs as well. The consequences of this proposal for the well-known extraction asymmetries concerning CCs and RCs will be addressed in Section 5.

³⁶There is one fact that might bolster this claim: Alemannic uses the particle strategy to form RCs but it has optionally relative pronouns of the Standard German type. In addition, it may use resumptive pronouns, e.g. with datives (or obliques). However, as soon as the relative pronoun occurs, a resumptive is completely ruled out. In the scenario sketched, this would follow automatically as then this dative would have to be assigned twice.

5 Extraction

The unified structural analysis of RCs and CCs that we proposed above is of course immediately confronted with the problem of the well-known extraction asymmetries between cross-clausal extractions. Indeed, one of the main arguments for the idea that CCs are directly selected by the respective matrix verb is the fact that only these are transparent for long *wh*-extraction – in contrast to adjunct clauses and RCs. According to standard assumptions, only CCs are in the right structural configuration such that the gap (or trace or copy) can be ‘identified’³⁷ by the moved *wh*-phrase. This is not the case in relative and adjunct clauses: be it because the intermediate landing site is already occupied by an operator (as in RCs) or the relevant c-command relations cannot be established (adjunct clauses). In the proposal advocated here, all clauses are extraposed alike and thus in an adjunct position from which extraction should be strictly banned in all cases – contrary to fact. Equally, since we suggested above that RCs and CCs share all relevant properties in terms of structure, the impossibility to extract out of a RC seems to provide direct counter evidence to the here suggested analysis of clausal embedding.

However, the traditional analysis of long *wh*-extraction in terms of successive cyclic movement has been challenged in recent years on both empirical and conceptual grounds, cf. Adger & Gillian (2005), Den Dikken (2009, 2018), McDaniel et al. (2015), Schippers et al. (2020), Brandner & Berger (2018). One alternative analysis to cyclic movement involves base-generation of the *wh*-phrase in the matrix clause with only local movement to the matrix Spec-CP. The embedded clause is then taken as a kind of RC³⁸ and acts as a predicate for the extracted phrase. The tradition term for such a construction is “prolepsis”, cf. Salzmann (2006, 2017) for an implementation in generative grammar. For reasons of space, we will sketch how prolepsis is completely in accordance with our proposal for clausal embedding.

5.1 The proleptic construction as the basis for ‘long extraction’

Example (68) illustrates a typical proleptic construction from Modern German:

³⁷We do not commit ourselves to any of the current implementations of long *wh*-extraction. As will become clear immediately, we will argue for a base-generation (proleptic) analysis.

³⁸In some languages, which have different complementizers for CCs and RCs, the complementizer “switches” to the RC-complementizer in case of extraction. This is the case in some Celtic languages, see Adger & Gillian (2005) and also in some varieties of Alemannic, see Brandner & Berger (2018).

Katrin Axel-Tober & Ellen Brandner

- (68) Von wem hast du gesagt/gedacht/behauptet [dass du *(ihn) nicht
of whom have you said/thought/claimed [that you him not
kennst]?
know]
‘Of whom did you say that you don’t know him, viz. Who did you say
that you don’t know?’

The important properties are (i) the *wh*-phrase in the matrix is assigned case via the preposition *von* (‘of’) and is thus not identical to the (supposed) *wh*-phrase in the embedded clause and (ii) there is no gap in the embedded clause – a resumptive pronoun must be used instead.

Again, a look into diachrony can give us important hints. First, in former stages, the *wh*-phrase in the matrix could also bear accusative, instead of being case-marked by a ‘dummy preposition’. Second, a resumptive was not obligatory. Both properties are illustrated by the example in (69a). The proleptic construction could also be formed by extracted relative pronouns instead of interrogative pronouns. In this case as well, the extracted pronoun showed accusative case, cf. (69b):

- (69) 18th/19th century New High German
- a. wen lesen wir dass __ dem Moses erschienen ist?
whom-ACC read we that the-DAT Moses appeared is
‘Of whom do we read that he appeared to Moses?’
 - b. den dritten, welchen er gehört, daß er gerne Bier trinke
the third which-ACC he heard that he gladly beer drinks
‘the third of whom he has heard that he likes drinking beer.’
(Lesebuch III, 760, 31; Behaghel 1928: 550)

Thus, this construction is somehow the opposite to what we saw in the case attraction cases discussed above, where there is an ‘additional’ nominal in the embedded clause, bearing a case which is obviously assigned by the matrix verb. According to Behaghel (1928: 551), the version with the preposition replaced the old structure in the 19th century due to the “...seemingly completely unlogic form of the matrix clause...” with its additional argument – obviously not selected by the matrix verb – but nearing a verbal structural case (accusative).

In the structure that we propose in Figure 6, the natural place for this proleptic argument is within the complex DP, i.e. where otherwise the correlate is situated. We suggest that the external dP moves to the specifier of the DP, due

2 Complementation and relativization

to the *wh*-feature present on the whole DP which then will be checked via spec-head-agreement. For theta-theoretic reasons, this configuration then excludes the lexicalization of the D^0 position for which we assumed that it comes along with an empty dP, bearing the theta-role itself as a pronoun. Thus, proleptic argument and correlate exclude each other, see already [Bennis \(1986\)](#) for this observation. Regarding the optionality/obligatoriness of the resumptives, see next section.

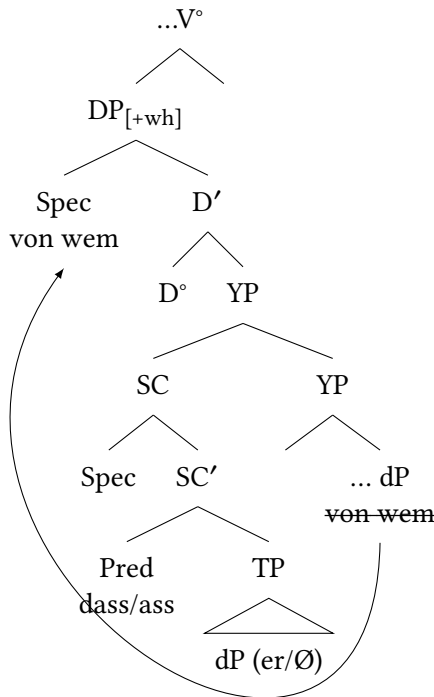


Figure 6: Structure with complex DP.

Thus far, the traditional and our approach make the same predictions. Concerning the general impossibility of extraction out of RCs, again, the same predictions are made, independently of whether the impossibility of extraction is

Katrin Axel-Tober & Ellen Brandner

derived from (a modern version) of the Complex NP constraint, attributed to the impossibility to cross a (relative) operator or simply to the fact that the clause is in an adjoined position (in a traditional external head analysis). In our proposal, there is simply no room for a proleptic argument as the head noun of the RC is ‘the correlate’ and there is thus no possibility to build a further proleptic construction. The same reasoning applies to content nouns: in the case of ‘real’ relativization, extraction is of course impossible and in the content-reading, again, the correlate is overt and thus no further proleptic argument can be generated within this DP.

In sum, an analysis of long *wh*-extraction in terms of a proleptic construction as yet another instance of our generalized “correlate-embedded clause” analysis can capture the basic asymmetries between argument and adjunct clauses as well.

The question then is how the ‘standard’ cases of long *wh*-extraction, as in *Who do you think that John will invite* can be accounted for. The most pressing problem for our account is the case assignment, respectively the ‘formal identity’ of the *wh*-phrase in matrix and in its base position. We can only sketch some speculations here. As is well-known, in case of long extraction, (direct) objects seem to be much easier to extract than subjects (McDaniel et al. 2015, Schippers et al. 2020). In our scenario, this could be explained by assuming that object case can be delivered by the matrix verb, as in the early proleptic construction in (69). In both cases, we have a subject-extraction, but the ‘extracted’ *wh*-phrase bears accusative – obviously assigned by the matrix verb. This is possible because the correlate can never get an overt realization in this configuration. Given that nominative is already ‘consumed’ by the matrix subject,³⁹ the asymmetry between object and subject extraction (and its various escape strategies like scope-marking, see Section 5.3, prolepsis, parenthesis, resumptives etc.) could be accounted for in our approach. That some languages like German and Dutch allow quite readily for long distance subject extraction, is nevertheless a fact that still awaits an explanation.⁴⁰

³⁹Note that even if the matrix verb is passivized, i.e. when nominative is free (at least in German – which allows impersonal passives), subject extraction is highly marked:

(i) ?? wer wurde gesagt [dass __ nicht eingeladen wurde]?
 who was said that not invited was
 ‘Of whom it was said that he was not invited?’

⁴⁰We refer the reader to the DFG funded project by Ankelien Schippers at the University of Oldenburg: <http://www.ankelienschippers.com/research/>

2 *Complementation and relativization*5.2 **Resumptive pronouns**

Concerning the occurrence of the resumptive, again only a short remark on a possible solution: recall that we suggested that the gap in RCs is due to ellipsis, i.e. deletion under identity. Given now that the proleptic argument in a ‘long *wh*-extraction’ does not only have a different surface form (today a PP) but in addition a different feature composition (*wh*-feature), it might seem plausible that this lack of parallelism is responsible for the strong preference of a resumptive.

As for the possibility to have a gap in former stages, cf. (69a), it has to be shown in future work what the conditions exactly are. Note that in this example, we have a simple *w*-pronoun and the gap corresponds to the subject of the embedded clause – and a missing (non-overt) subject can quite easily be identified, due to the syntactic surrounding. It would be interesting to see whether the (im-)possibility of having a gap correlates with the Keenan/Comrie hierarchy such that the lower position within the hierarchy is, the more likely it is to require a resumptive – as it is the case in relativization, see also Brandner (2020) for a more detailed treatment of resumptive pronouns in Alemannic long *wh*-extractions.

5.3 **Scope marking and copy construction**

As is well known, especially in German, there are further possibilities to build ‘long *wh*-extraction’:

- (70) a. was hast du gedacht wen er einladen will? scope marking
 what have you thought who he invite will
 b. wen hast du gedacht wen er einladen will? copy construction
 who have you thought who he invite will
 both: ‘Who do you think that he will invite?’

The scope marking construction was analysed already by Dayal (1993) as being based on a correlative construction, the so-called ‘indirect approach’. This analysis gave rise to an extensive discussion – which is still going on – on whether this analysis can be applied to all kinds of scope marking constructions. However, in light of the structure of clausal embedding that we suggested here, an analysis where the correlate is *wh*-moved inside its clause and the embedded question being predicated over this correlate would follow immediately. The only thing that is different from a more typical correlative construction is the fact that there is also *wh*-movement inside the embedded clause – however, this movement as such is completely independent from the higher movement – as it would be the case in simple embedded questions.

Katrin Axel-Tober & Ellen Brandner

Concerning the copy-construction, recall that this construction is often taken as the hallmark empirical evidence for successive cyclic movement: here we can see overtly that the *wh*-phrase lands in the specifier of the embedded CP. However, recent analyses, cf. Pankau (2014) and most recently Rugna (2020), cast doubt on such a straightforward analysis in terms of successive cyclic movement. Note that there are many instances where we either do not find copying “proper” (77b and c) or where exact copying leads to ungrammaticality (71a):

(71) modern Standard German

- a. * **Welchen Artikel** hast du gesagt, [**welchen Artikel** man lesen
which article have you said which article one read
sollte]?
should
- b. **Welchen Artikel** hast du gesagt [**den** man lesen sollte]?
which article have you said d- one read should
both: ‘which article did you say that one should read?’
- c. **Wen** denkst du [[**wen von den Studenten**] man einladen sollte]?
who think you who.acc oft he students one invite should
,Which of the students should one invite?’

In (a), the *which-wh*-phrase cannot be ‘repeated’ in the embedded clause, casting doubt on successive cyclic movement. On the other hand, as (b) shows, a *d*-pronoun is perfectly fine, indicating that the embedded clause can or should be analysed as a RC. (c) finally is again an instance of a non-perfect copy, again casting doubt on ‘literal’ movement.

In our approach to ‘long *wh*-extraction’, i.e. as one instantiation of the correlate construction, these non-matching versions of a copy construction do not pose a problem. The proleptic argument of the matrix clause is base generated there and thus can take the shape it wants. The harder question to solve is why there are constraints in the embedded clause, in particular the ban on *which*-phrases.⁴¹ We cannot go in a deeper discussion here, but as Rugna (2020) suggests, it may have to do with the *d*-linked status of these phrases which require access to the discourse and are therefore restricted to matrix clauses. Under this perspective this ban is thus more an issue that has to be solved in the pragmatics. Note that in a simple embedded question, e.g. under a verb like *wonder*, a *which*-phrase is

⁴¹As discussed in Rugna (2020: 189) this ban does not apply in all languages. Afrikaans is said to allow such constructions.

2 Complementation and relativization

perfectly fine. Thus it cannot be due a narrow syntactic reason, why (a) is out – or at least very awkward.

These considerations lead us to the final issue we want to discuss in this section, namely the landing site of the *wh*-phrase in the embedded clause. Recall that we proposed that what is traditionally taken to be the ‘Left Periphery’ of a clause is essentially the functional layer of a Small Clause with the equative particle as the head. The *d*-pronoun complementizers are situated in its specifier position (subject of the Small Clause).

It is well known that in embedded questions, *d*-pronoun-complementizers may co-occur with the fronted *wh*-phrase, a typical instance of the Doubly filled Comp Filter:

(72) Alemannic

I muss wüsse [[weller Artiggel] dass ich lese sott].

I must know which article that I read should

‘I must know which article I have to read.’

But if the *d*-pronoun complementizer is in the specifier position, as we have argued so far, what could be then the landing site of the *wh*-phrase?

There are two possibilities: either we allow the left periphery in the embedded clause to ‘expand’ such that several layers in the sense of Rizzi (1997) can be projected and the *wh*-phrase moves into the Foc-position, as is standardly assumed.

Or we assume that the *d*-pronoun complementizers may have grammaticalized into heads, following the well-known Spec-to-head reanalysis, as suggested and to be shown operative in many environments by Van Gelderen (2004). In this case, the spec-position would be free and could serve as a landing site for the *wh*-phrase, yielding the DFC-phenomena illustrated in (72). Those variants that did not grammaticalize the *d*-pronoun would then be those which do not allow DFCs. It seems, however, hard to underpin this distinction empirically, as there seem to be no independent diagnostics despite the DFC-phenomenon regarding the head-status of the *d*-pronoun. But note that this scenario would make a clear prediction, namely that those variants that employ the equative particle should allow DFCs, since in this case, the spec-position is always free to host a *wh*-phrase. We have seen already some instances of this in Section 2 – but we have to leave the final empirical justification to future work.

6 Conclusion

We started with the observation that many variants (not only) of Germanic employ the equative particle to introduce embedded clauses – relative as well as

verbal ‘complement’ clauses. This uniformity lends further support to the idea that CCs and RCs share relevant properties. Based on this data, we suggested to analyse clausal embedding as being based on the notion of equation – not only semantically, but also syntactically. Since an equation always involves two entities being in a formal relationship, we proposed a “generalized correlative construction” for clausal embedding – much in line with recent considerations from the semantic side. We then showed on an empirical basis that there are good reasons to always posit a nominal in the matrix – even if phonetically empty in some cases. The crucial distinction between CCs and RCs was traced back to the presence of a nominal predicate in RCs (the head noun). This forces the equation relation to target not the whole nominal expression but rather merely the entity which is described by the nominal predicate – while sharing the (referential) DP-layer. This was implemented by recourse to Cinque’s (2020) double-headed RC structure. If there is no nominal predicate (correlative pronoun), a direct ‘replacement’ is possible, i.e. a verbal CC.

As for the embedded clause itself, we proposed to analyse it as a Small Clause – again inspired by the equative particle used as a complementizer. We then sketched a possible way to integrate the *d*-pronoun complementizer as an externally merged ‘double’ of the correlate in the specifier position of the Small Clause. Finally, we sketched how this analysis may also solve some long-standing problems in the realm of long *wh*-extraction. In particular, the scope marking and the copy construction with two occurrences of a *wh*-element in each clause give further support for the generalized correlative construction.

As we proposed a completely different view on clausal embedding, necessarily many issues must be left open. For example, a much debated issue that we did not address at all is factivity – which is in particular a prominent issue in the semantic-oriented discussions of complementation. We noted only in passing that the complementizer choice might be relevant for the interpretation, cf. the *how*-complementizer in (spoken) English.⁴² Still, we have to keep in mind that there are languages that use the same particle/strategy in all embedding environments and also do not distinguish between factive or non-factive – while others do. Thus, there seems to be no universal requirement that languages use distinct complementizers. If there are several possibilities, however, then languages seem to exploit these different possibilities to mark additional distinctions.⁴³

⁴²But see the work by Roussou, e.g. (Roussou 2010) on Greek, where this difference is clearly encoded in the grammar.

⁴³We just mention the study by D’Arcy & Tagliamonte (2015) who show for Toronto English that the choice between *wh*-pronoun and *that* is governed by animateness, see also Cheshire et al.

2 *Complementation and relativization*

Our focus was to provide the syntactic basis how different types of complementizers may arise. In the Small Clause structure that we suggested, there is a natural place for additional (pro)nominal items. Note that in this scenario, the wh-type complementizers as they are familiar from the Romance languages⁴⁴ could be captured in a similar way. What has to be answered in this context is to which extent demonstratives and wh-pronouns overlap in their semantic contribution. It has to be shown by future research in which way these complementizers enter a language community (via language contact or internal language change?) and how they are used to indicate further grammatical distinctions and whether there are natural pathways of development to detect. We hope to have started a discussion which will lead ultimately to new insights about the nature of clausal embedding.

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(2013) who show that topicality may also be a factor governing the distribution of the various complementizers.

⁴⁴See for example [Poletto & Sanfelici \(2021\)](#) whose structural proposal is very similar to ours, especially they also rely on a Cinque-style double headed analysis. They suggest that the difference between wh-pronouns (Romance) and demonstratives (Germanic) correlates with the 'inner' and the 'outer' position within this structure. As we concentrated in this paper on Germanic, we will refrain from an explicit proposal for Romance here, see also [Kayne \(2021\)](#).

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Chapter 3

Determinacy and the CP-TP Bottleneck

Elly van Gelderen

In the last 25 years, generative grammar has undergone a paradigm shift from relying on principles in Universal Grammar to principles that are less specific to the faculty of language. These are usually referred to as Third Factors. In this paper, I take one such non-language specific principle, namely determinacy, and apply it to variation and change in the CP-TP layer, focusing on English and Dutch. I show that variation surrounding Verb-second, TP-expletives, *that*-trace, C-deletion, and subject-less relatives is due to the need for an efficient computation of the CP-TP boundary.

1

Key-words: C-deletion, CP-TP boundary, determinacy, Subject-less relatives, *that*-trace, TP-expletives, Verb-second

1 Introduction

Universal Grammar served as the foundation of generative grammar for a long time before the latter underwent a shift from relying on principles in Universal Grammar to principles that are less specific to the faculty of language. These are usually referred to as Third Factors (Chomsky 2005) and include Minimal Search, Inclusiveness, and No Tampering. In this paper, I take one such non-language specific principle, namely determinacy, and apply it to variation and change in the CP-TP layer. I show that variation surrounding Verb-second, TP-expletives,

¹This article focuses on the CP-TP layer, which I also discuss in van Gelderen (2022). Examples and discussions overlap with that work but the focus is different. I'd like to thank the audience at CGSW 35, William Kruger, Johanna Wood, Stefanie Bode, and two anonymous reviewers for suggestions on the presentation and content.

Elly van Gelderen

that-trace, C-deletion, and subject-less relatives is due to the need for an efficient computation of the CP-TP boundary.

The outline is as follows. In Section 2, I discuss the paradigm shift, from UG to Third Factors, and provide examples of how determinacy works. In Section 3, I first show some overlap in the functions of C and T and then provide examples of variation that are due to determinacy resolutions around the CP/TP ‘bottle neck’, namely in *that*-trace, C-deletion, and subject-less relatives. In Section 4, I show how Verb-second (V2) is a problem for determinacy if both CP and TP occur. When, in the history of English, the TP is introduced around 1400, the V2 option disappears. The phenomena in Section 3 and Section 4 follow if CP can be deleted (in English) or TP isn’t present (in Old English and Dutch).

2 From UG to third factors

UG of the 1950s to 1970s includes a lot of language-specific instructions and rules. It initially includes rules for e.g. antecedent-reflexive relations, for passives, and for *wh*-movement. When the model shifts to the Principles and Parameters’ one, the parameters help to account for cross-linguistic differences while still keeping language acquisition central. In that model, children acquiring their language(s) choose a setting (compatible with the data) for language-specific phenomena, such as *wh*-movement and null-subjects. UG remains rich in the Government and Binding framework of the 1980s (Chomsky 1981: 3).

However, if the Faculty of Language developed in humans only 100,000 to 200,000 years ago, it makes sense to attribute less to language specific principles. Because of their specificity, principles of UG wouldn’t have had enough time to evolve and, if there had been time, the changes would possibly have made languages very different from each other. Chomsky et al. (2019: 230) put it this way, “UG must meet a criterion of evolvability”. That’s why UG is currently assumed to contain just a simple operation merge that combines two elements into an unordered set, as in (1).

(1) Merge

Select two lexical items α and β and form the set $\{\alpha, \beta\}$ in a workspace.

(Chomsky et al. 2019)

More important than principles specific to the language faculty, as in (1), are “general properties of organic systems” (Chomsky 2004: 105). Three factors are considered crucial in the development of language and they are listed below.

Three Factors

3 *Determinacy and the CP-TP Bottleneck*

“1. Genetic endowment, apparently nearly uniform for the species, which interprets part of the environment as linguistic experience, a nontrivial task that the infant carries out reflexively, and which determines the general course of the development of the language faculty. Among the genetic elements, some may impose computational limitations that disappear in a regular way through genetically timed maturation ...;

2. Experience, which leads to variation, within a fairly narrow range, as in the case of other subsystems of the human capacity and the organism generally.

3. Principles not specific to the faculty of language”. “(a) principles of data analysis that might be used in language acquisition and other domains; (b) principles of structural architecture and developmental constraints that enter into canalization, organic form, and action over a wide range, including principles of efficient computation, which would be expected to be of particular significance for computational systems such as language”. (Chomsky 2005: 6)

The first factor principle, the “genetic endowment” comprises a greatly reduced UG which Chomsky (2005: 11–12) argues includes just merge, i.e. (1). Take two syntactic objects and combine them to make another object. Since merge can continue indefinitely, it also derives the recursive nature of language.

The second factor is the learner’s need to be exposed to one or more languages (spoken or signed) to build the lexicon and to become familiar with interface constraints. Lexical differences are responsible for all cross-linguistic variation and parameters are now only relevant to that domain: speakers have to learn which features a lexical item has and which grammatical categories are included.

Third factor principles are not specific to language and include Minimal Search, Determinacy, and Structural Economy and take over some of what used to be attributed to first factor, i.e. UG principles, such as Subjacency, Locality, and Anti-Locality. In what follows, I’ll focus on the Determinacy Principle. In the spirit of attributing as much as possible to non-linguistic or third factors, Chomsky et al. (2019: 246) argue that determinacy is such a principle that is needed to account for the locality of movement. In this section, because it is the most complete account, I provide Goto & Ishii’s (2019) formulation of determinacy and some examples of input that must be unambiguous.

Within a Workspace -read phase-, there can be only one object X, “avoiding in determinate rule application.” Putting this positively gets us (2) and this rule, combined with a derivation by phase model, also accounts for islands.

Elly van Gelderen

(2) **The Principle of Determinacy**

If Structural Description (SD) for a rule holds for some [Workspace], then Structural Change (SC) must be unique. (Goto & Ishii 2019: 91)

Boeckx (2012: 4) points out that Chomsky (1964) already writes that “transformations must be unambiguous”.

If CP, v*P, and DP are phases and the complement of the phase head is sent to Transfer, certain workspaces will contain more than one instance of an item. For instance, a subject in the specifier of the v*P that moves to the specifier of the TP will have two copies in the workspace that is the complement of C and therefore cannot move to the specifier of the CP. That movement makes the derivation indeterminate. Note that (2) resembles (3), which is usually seen as a UG Principle not a third factor one.

(3) **Anti-Locality**

- a. Movement must not be too local. (Grohmann 2003: 26)
- b. Movement of a phrase from the specifier of XP must cross a maximal projection other than XP. (cf. Erlewine 2016: 431 and Deal 2019: 408)

As an example of a determinacy violation (or an anti-locality one), English shows a subject island effect where extraction of a *wh*-element from the specifier of the TP in (4a) into the CP in (4b) results in an ungrammatical sentence (Goto & Ishii 2019: 94).

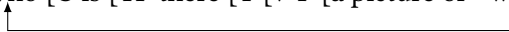
- (4) a. A picture of Mary pleased you?
 - b. * Who did a picture of please you?
 - c. [CP who [C-did [TP [a picture of <who>] [T [v*P [<a pictures of who>] [v* [please you]]]]]]].
- Workspace: not allowed by determinacy Transfer 1

In (4c), a derivation of (4b), it is shown that the complement of v* is transferred to the interface and the next workspace is the complement of C. The latter contains two copies of *who* and is therefore not unambiguous, as required by (2).

If the subject stays in the specifier of the v*P, as in (5a), there is no violation when *who* moves to the CP, as shown in (5c), because the specifier of TP is filled with an expletive and there would not be two copies of *who* in the workspace.

- (5) a. There is a picture of who on the wall?
- b. Who is there a picture of on the wall?

3 Determinacy and the CP-TP Bottleneck

- c. [CP who [C is [TP there [T [v*P [a picture of <who>] [v* ...
- 

Note that accounting for the grammaticality of (5b) would be difficult under Subjacency and related constraints because only one TP is crossed.

As argued in detail by Goto & Ishii (2019), in addition to banning extraction from subject position and movement of the subject to a topic position, the requirement for determinate input also accounts for the *that*-trace effect, as in (6a), and the impossibility of further-raising, as in (7a). The violating two copies are shown in (6b) and (7b), respectively.

- (6) a. * Who do you think that read the book?
 b. Who do you think that <who> T <who> read the book
 workspace Transfer
- (7) a. * John seems that reads a book.
 b. John T <John> seems that <John> T <John> reads a book.
 workspace Transfer

In (6b), the workspace contains two copies and is therefore indeterminate and ungrammatical. Deleting C, as in (8), makes it determinate because the C transfers its features to T which means that T becomes the phase head and the number of copies of *who* is reduced to one.

- (8) a. Who do you think read the book?
 b. Who do you think <who> T <who> read the book
 workspace Transfer

The ungrammaticality of further-raising, as in (7a), earlier referred to as the Tensed S Condition (TSC) because a DP cannot move out of a tensed clause, is likewise accounted for by determinacy, as is seen from the multiple copies of *John* in the workspace in (7b).

The C-deletion approach to *that*-trace violations is one adopted in Chomsky (2015: 10–11) but it violates the No Tampering Condition because something is deleted from the derivation. An alternative is not to include the C in the derivation and to choose a T as phase head but this requires the selection from the lexicon to ‘look ahead’. It seems to me violating the No Tampering Condition is preferable.

In this section, I have outlined the shift in generative grammar from UG to Third factors and given an example of a Third Factor, i.e. the Determinacy Principle, one that will feature in what follows next.

Elly van Gelderen

3 The CP/TP ‘bottle neck’

In this section, I first outline how CP and TP duplicate many of each other’s functions. In many cases, having both present results in violations of determinacy. Cross-linguistic variation shows ways to resolve these violations.

A clear indication that C and T fulfill similar functions is that, in the current generative framework, T inherits tense and agreement features from the phase head C. This Feature Inheritance proposal (Chomsky 2001) captures the dependence of mood in the C and finiteness in the T. For instance, indicative mood is marked through *that* in the CP layer (or in the ForceP) and requires a finite verb in its clause. This is achieved by means of tense and phi-features on T. Similarly, if the mood is irrealis, the C is *for* and the T in the accompanying clause will be non-finite. The CP and TP thus work together but, as I will show, throughout the history of generative grammar, only one or the other has played a role: sometimes TP (as with Subjacency in English) and sometimes CP (as with phases). This dependence is also shown when the C head is deleted, the phase is cancelled, and phasehood is transferred to T, as in (8) above.

In the 1970s and 1980s, the clause boundary was relevant for Subjacency and for identifying islands. Once the clause is seen as a combination of a TP and a CP, it is the TP that counts as the bounding node (in English) which can only be crossed once. In (9), the first movement crosses one TP and the second another TP and this results in a grammatical sentence.

- (9) Who did TP[Mary <did> think CP[<who> | that TP[Peter met <who>]]].
-

Crossing two TPs, as in (10) where the intermediate specifier of the CP is filled, results in ungrammaticality.

- (10) *Who did TP[Mary <did> wonder CP[when C TP[Peter met <who>]]].
-

If both CP and TP counted as bounding nodes, the second movement in (9) would not have a grammatical result. This means that, even though there are two phrases, only one counts for Subjacency/locality. At the time, it was also argued that bounding nodes were parametrized, either TP, as in English, or CP, as in Italian (see e.g. Rizzi 1980), again suggesting that the CP and TP have a differing degree of importance, but that it is either CP or TP that counts, and not both in the same language.

3 Determinacy and the CP-TP Bottleneck

As an alternative to Subjacency, Chomsky (1986) formulates the concept of barriers that block government and hence disallow movement across because the empty position would not be properly governed. Here too, we see the redundancy of the TP. A barrier depends on the notion of Blocking Category (BC), as in (11), where L-marking means a theta and agreement relation with a sister, e.g. a verb L-marks its theta-marked object.

- (11) γ is a BC for β iff γ is not L(exically)-marked and γ dominates β .

A barrier is defined in (12) if either there is a BC category in between it and β or it is itself a BC. As with the bounding nodes of Subjacency, movement can't cross more than one barrier.

- (12) γ is a barrier for β iff (a) or (b):
- a. γ immediately dominates δ , δ a BC for β = barrier by inheritance
 - b. γ is a BC for β , γ is not IP [i.e. TP] (Chomsky 1986: 14) = inherent barrier

What is important for this paper is that by (12b), the TP cannot by itself be a BC but has to inherit barrierhood from the CP through (12a). The complexity of the definition in (12) is due to the CP/TP area.

In the Minimalist Program, starting around Chomsky (2001), phases replace bounding nodes and barriers. Phases restrict the “computation of expressions [...] to a single cyclic/compositional process” (Chomsky 2008: 142). Once a phase is complete, its complement is transferred to the interfaces and no longer remains accessible to the computation. There are two clausal domains that Chomsky recognizes as phases, the CP and v^*P . The TP is not a phase although it is the locus of the φ -features. The T “manifests the basic tense features if and only if it is selected by C” and it therefore “makes sense to assume that Agree- and Tense-features are inherited from C, the phase head” (Chomsky 2008: 143–144). Under phase theory, the Subjacency violation in (10) is accounted for because, after *who* adjoins to [met <who>], *who* is not extractable because the C containing *when* cannot be deleted. There are two more instances where C can be deleted, which I turn to now. These are C deletion in declarative clauses and C deletion in certain relatives.

In a declarative subordinate clause with both a CP and TP, as in (13), the complementizer has tense, checks agreement with the subject, and assigns nominative case to this subject. In English, that holds whether an overt complementizer *that* is present or not. When C deletes, T inherits tense and agreement features, as seen in (13b).

Elly van Gelderen

- (13) a. I know (that) she left.
 b. I know CP [C TP [she T vP [<she> left]]]
 [u-phi: 3S]/[i-tense/Case] [i-3S]/[u-Case]
-

Two analyses are possible: either the CP is not lexically filled but present or the CP is not present. The latter option is explored by Doherty (2000: chapter 2). He argues that sentences without the *that* complementizer are in fact just TPs (IPs for him). The differences in topicalization, shown in (14) and (15), provide a major piece of evidence. Assuming that topics have to be positioned in the expanded CP, there is simply no such position. Hence the ungrammaticality of (15); see also Rizzi & Shlonsky (2007: 151) who cite Grimshaw (1997).

- (14) I hope that [this book] you will read.

- (15) *I hope [this book] you will read.
 (both from Doherty 2000: 13)

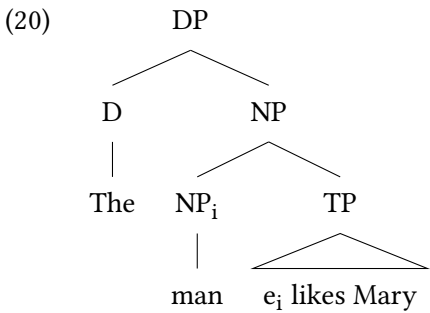
Subject-less relative clauses, as in (16) to (18), occur in many present-day varieties of English, but they are no longer considered part of the standard language. Jespersen (1997: 135) uses the term ‘contact clause’ for all relative clauses that leave out the pronoun and complementizer and that term is still current, e.g. it is used in Doherty (2000) and Radford (2019).

- (16) *Anybody does that ought to be locked up.*
 ‘Anyone **who** does that ought to be locked up.’ (from Quirk et al. 1985: 1250)
 (17) *It’s Simon did it.*
 ‘It is Simon **who** did it.’ (from Quirk et al. 1985: 1250)
 (18) *He’s brought 4 lads in are explosive.*
 ‘He has brought four lads in **who** are explosive.’ (Radford 2019: 14)

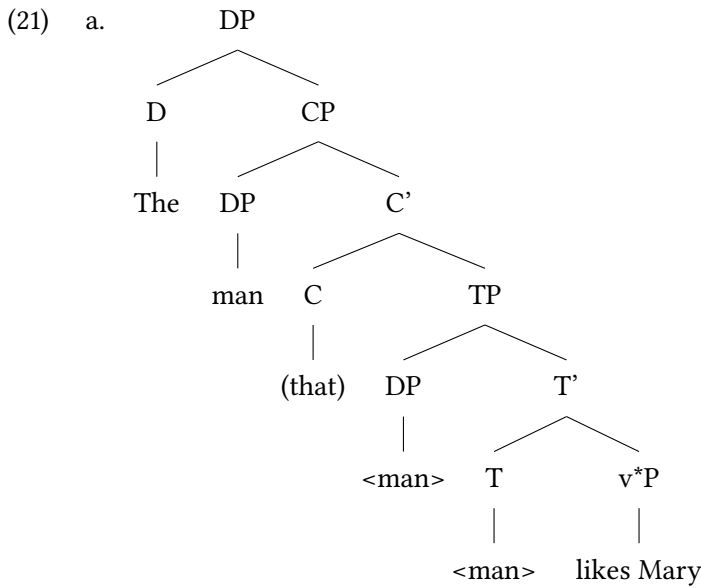
Syntactically, I will argue that these constructions just use a TP, in a way very similar to the strategy on how to avoid *that*-trace violations. Doherty’s (2000: 66) structure for the grammatical subject-less relatives in (19), is also one of a CP-less structure with a direct link between the external head N, i.e. *man*, and the subject position, as shown in the slightly adapted (20).

- (19) *The man likes Mary lives next door.*
 ‘The man who likes Mary lives next door’.

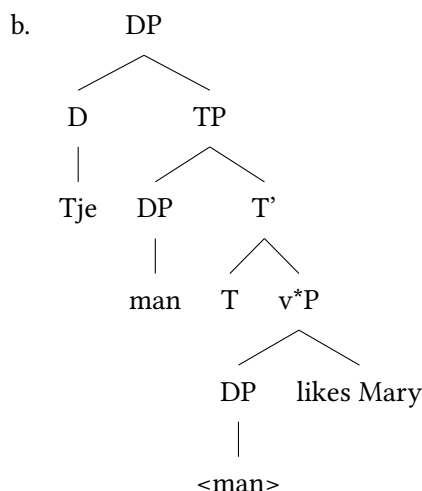
3 Determinacy and the CP-TP Bottleneck



Doherty assumes externally-headed relative clauses. With a raising analysis for relative clauses, as in [Kayne \(1994\)](#), the data can also be accounted for using the same options available in the functional lexicon: some languages have CP relative clauses (and can't have subject-less relatives) and others TP ones (and can have these). The trees are shown in (21), where (21a) is the indeterminate structure because *man* appears in the specifier of TP and of v*P and (21b) the determinate one, with only one instance of *man*.



Elly van Gelderen



If C has all the features to start with, why is T needed? Richards (2007: 565) suggests that C “would actually be *unable* to do all of the work ... without the mediation of proxy nonphase heads”. If feature valuation is part of transfer, C can’t hold on to its [u-phi] features because these couldn’t be valued once TP is transferred and, hence, “u-phi must spread from edge to non-edge” (Richards 2007: 569). In this section and the next, I argue that it is in fact possible to do without one of the two functional heads and will show there how that would work.

There are some questions with Feature Inheritance, as Chomsky (2008: 144) notes. “The mechanism is a narrow violation of [the No Tampering Condition]. The usual question therefore arises: does it violate [the Strong Minimalist Thesis]? If it does, then the device belongs to UG (perhaps parametrized), lacking a principled explanation”. Chomsky elaborates a little later in the same paper on the universality of Feature Inheritance: “[a]nother question is whether inheritance is obligatory or optional. For C-T, that raises familiar questions about universality of EPP and about mechanisms of agreement” (Chomsky 2008: 149).

In short, the CP-TP complex functions as a boundary for locality and as a holder of agreement and nominative case features. In English, the features from C are generally transferred to T but it also has the option of doing without C. Other languages opt for different solutions, as I now show for Dutch and Old English.

If the reason for the ungrammaticality of *that*-trace sentences is due to indeterminacy (a subject in both the specifier of the TP and v*P), then that effect is not expected in languages without or with optional movement to the specifier

3 Determinacy and the CP-TP Bottleneck

of the TP. This is borne out in Dutch (22), as has been known since Maling & Zaenen (1978).

- (22) a. Dutch
 Wie denk je dat weggegaan is?
 who think you that left has
 ‘Who do you think left?’
 b. Wie denk je CP[<wie> dat v*P[<wie> weggegaan is]].

The same *that*-trace violation can be avoided, as Goto & Ishii (2019) show, in a variety of languages that lack a subject in the specifier of the TP. For instance, Japanese has been argued since Fukui (1986) and Kuroda (1988) to not move the subject outside of the v*P and to not show a *that*-trace effect. Likewise, many varieties of Arabic allow *that*-trace sequences (Halila 1992: 166) and can leave their subjects in the v*P.

Old and Middle English, like Dutch (and German), can be argued to have a CP but not a TP, up to around 1400 when the situation changes. Here, I will show that *that*-trace sequences are indeed common in the earlier stages of English and that they can be accounted for as in Dutch, namely without a TP. This discussion will also help set the stage for the arguments used in the next section for the loss of T to C movement.

Allen (1977: 122) says that *wh*-extractions from a clause “are quite common” in Old English but she provides only one instance of *that* followed by an extracted subject, namely (23).

- (23) Ac hwæt saegst ðu ðonne ðæt__ sie forcuðre ðonne sio
 But what say you then that be.SUBJ more.wicked than be
 ungesceadwisnes?
 foolishness
 ‘But what do you say is more wicked than foolishness.’
 (Alfred, *Boethius* XXXVI.8, from Allen 1977: 122)

A search of the YCOE (Prose) (of all *that* followed by zero) reveals the selection of complementizers not followed by a subject in e.g. (24) to (26). There are, however, hundreds of these. I have marked the absence of the subject with underlining but that subject arguably never moved through that position.

Elly van Gelderen

- (24) & of þam ilcan bocum tyn capitulas, þa ic geond stowe awrat
and of those same books ten chapters those I throughout place wrote
& ic wiste **þæt** __ **swiðost nedðearflecū wæron**.
and I knew that most needful were
'and ten chapters of the same books which I had transcribed and which I
knew were most necessary.' (Bede 278.1-2)
- (25) & he geornlice þa ongan leornian & don þa þe he ongeat
and he gladly then began learn and do those REL he perceived
þætte __ **belumpun** to þære mynsterlican clænsunge & arfæstnesse.
that belonged to that monastic purity and goodness
'and he then gladly began to learn and carry out those things which he
perceived to belong to monastic purity and goodness.' (Bede
450.30-541.1-2)
- (26) Hwæt, we genog georne witon ðæt nanne mon þæs ne twooð þæt se
Indeed we very well know that no man that NEG doubts that that
sie strong on his mægene þe mon gesihð **þæt**__ stronglic weorc **wyrð**.
be strong in his virtue REL man sees that strongly work works
'We know very well that no man doubts that he is strong in his virtue
who people see to be doing laborious work.' (Boethius 38.2-4)

Middle English *that*-trace sequences, as in (27), have been documented by Bergh & Seppänen (1994). These sequences become less frequent throughout the Middle English period and cease to exist in the Early Modern period.

- (27) Grymbert who wolde ye **that**__ sholde goo and daye hym to come.
Grymbert who want you that should go and dare him to come
'Grymbert, who do you want that should go and dare him to come.'
(Caxton's Reynard the Fox, from Bergh & Seppänen 1994: 132)

Bergh and Seppänen argue that the "general decrease of *that* was a major factor" in the loss of *that*-trace sequences "between the early 15th century and the late 17th century" (Bergh & Seppänen 1994: 138). This means the optionality of C is connected to the presence of a T: if T is present, C can be optional and, if T is not, C has to be there.

So far, I have provided examples of complementizers followed by subject gaps. If Dutch, (German,) and older stages of English lack a TP, there is no gap. That explains the grammaticality of (22) and (23) to (26). If the TP is optional but the

3 Determinacy and the CP-TP Bottleneck

CP obligatory, *that*-deletion in complements is not expected either and that is borne out.

In languages such as Dutch, the complementizer is not typically deletable. With a verb like *beloven* ‘promise’, the only possibility is a V-final subordinate with a complementizer, as in (28a), and not one without, either with V-last or Verb-second, as in (28b) and (28c), respectively.

- (28) a. Dutch
 Hij beloofde **dat** hij haar ging opzoeken.
 he promised that he her went visit
- b. V-last
 * Hij beloofde hij haar **ging opzoeken**.
 he promised he her went visit
- c. V-second
 * Hij beloofde hij **ging** haar opzoeken.
 he promised he went her visit
 ‘He promised he’d go visit her.’

That-deletion occurs in those cases where the embedded object is a direct quote as with verbs like Dutch *zeggen* ‘to say’ in (29). Note that the word order in the embedded clause is V2 which is typical for a main clause not an embedded one. That makes the connection between these clauses very loose.

- (29) Ik zei hij **moet** dat maar doen. V2 subordinate
 I said he must that PRT do
 ‘I said that he should do that.’

The semantics of the matrix verbs is responsible for the appearance of an embedded clause that either has the characteristics of a V2 main clause, as in (29), or of a V-final embedded one with the complementizer, as in (28a). I will therefore argue that the embedded CP cannot delete its C head.

Turning to older stages of English, *that*-deletion doesn’t occur after verbs like *(ge)hyran* ‘hear’, as (30) shows. According to Mitchell (1985: 25–34), complementizers are only left out for direct speech in Old English (31).

- (30) Ic ... secgan hyrde **þæt** hie gesawon ...
 I say heard that they saw ...
 ‘I heard it said that they saw ..’ (Beowulf, 1345-7)

Elly van Gelderen

- (31) Ic secge eow. manega **cweðað** to me on ðam micclum dæge. drihten
 I tell you many say to me on that great day Lord
 drihten. la hu ne witegode we on þinum naman.
 Lord hear how NEG prophesied we in your name.
 ‘I’ll tell you: Many will say to me to me on that great day “Lord, Lord, lo!
 have we not prophesied in thy name ...”’
 (*Catholic Homilies*, first series, Clemoes, 351.183, from [Mitchell 1985](#), II. 29)

So, complements of ‘say’ verbs act like main clauses and that means they are CPs but without a complementizer. It is therefore also possible to say that verbs such as *say* in these languages always have a CP complement but with an optional complementizer: if the complementizer is not present, Verb-movement occurs (cf. [Haider 1986](#): 53).

In early Middle English, e.g. the 13th century *Wohunge* (see [van Gelderen 1993](#)), the only verbs that allow *that*-deletion are *seggen* ‘to say’, as in (32), and *cwid-den* ‘to say’. These are verbs whose complements are more like main clauses in Dutch (29) and Old English (31) as well. These clauses are not verb-final as typical subordinate clauses are.

- (32) he **seide** ne dred tu nawt for ...
 he said NEG dread you not for
 ‘He said do not fear because ...’ (*Wohunge*, 412)

None of the other verbs in this early Middle English text, e.g. *praie* ‘to pray’, *þunche* ‘seem’, *understonde* ‘understand’, *wearnen* ‘refuse’, *see*, *leue* ‘to grant’, have *that*-deletion, as (33) shows.

- (33) A iesu swete iesu leue þ te luue of þe beo...
 Oh Jesus sweet Jesus grant that the love of you will be
 ‘Oh, Jesus, sweet Jesus, grant that the love of you will be’ (*Wohunge*, 55–56)

In Layamon, another early Middle English text, *that*-deletion does not occur either. Sentences (24) to (36) are typical instances where *that* is present, taken from the Caligula version.

- (34) & swa heo sculden under-stonden. þat ich am king of Brut-londe.
 ‘and so they must understand that I am king of Brittany.’ (Layamon, *Brut*, Cal. 3065)

3 Determinacy and the CP-TP Bottleneck

- (35) Nu þu wult under-uon. þat nauer ær nes idon.
 ‘Now you will undertake that (it) was never before done.’ (Layamon, *Brut*, Cal. 15118)
- (36) and saide þat he wolde. aquellen heom a londe.
 ‘and said that he wanted to destroy him on land.’ (Layamon, *Brut*, Cal. 14874)

The fact that *that* must be present can be seen as additional evidence that tense features occupy C and that they must be lexicalized. This means that, in this period, Verb-movement occurs in main clauses and the complementizer *that* must be present in subordinate clauses.

If subject-less relatives are CP-less clauses, languages like Old English and Dutch, with CPs but without TPs, will be expected not to have them and this is borne out by the non-appearance of subject-less relatives in Old English, and the ungrammaticality of them in Dutch (37), which is the translation of (16).

- (37) Dutch
 *Iemand [___ dat doet] moet opgesloten worden.
 Someone that does must locked.up be
 ‘Someone who does that should be locked up.’

Subject-less relatives start to occur in late Middle English. The variation discussed in this section concerns how to resolve the CP-TP ‘bottle-neck’. In this bottle-neck area, the phi-features of C are argued to be inherited by T and a number of issues can arise, e.g. indeterminacy may occur because there are too many positions in close proximity to each other through which *wh*-elements must move. In Modern English, a strategy is followed where the complementizer *that* and the CP can delete; in Old English and Dutch, the TP is not present so indeterminacy is avoided that way.

4 Verb-second and TP expletives

In this section, I show that verb-second and the presence of a TP are not compatible from a determinacy point of view, using Dutch as an example. I then examine the history of English where we see a loss of verb-second (V2) at the time of the introduction of a TP, which makes for determinate derivations.

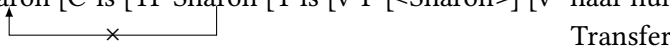
As the name implies, V2 constructions have the finite verb in second position but may or may not have the subject in first position, as (38) and (39) show, respectively, for Dutch.

Elly van Gelderen

- (38) Subject first and V2 Dutch
Sharon is naar huis gegaan.
 Sharon AUX to house gone
 ‘Sharon went home.’
- (39) Adverbial first and V2 Dutch
 Gisteren **is** Sharon laat weggegaan.
 Yesterday AUX Sharon late away.gone
 ‘Yesterday, Sharon left late.’

As is well-known, [den Besten \(1983\)](#) accounts for the complementary distribution between V2 and the presence of a complementizer by suggesting that the V moves to C when the latter isn’t overtly filled with a complementizer. With the expansion of the CP ([Rizzi 1997](#)), strict V2 languages such as Dutch (and German) can be seen as employing the high regions of this split CP, i.e. the ForceP (see [Hinterhölzl 2018](#) and the contributions in [Woods & Wolfe 2020](#)). I will just continue to use the CP for Dutch but, for more complex V2 languages, such as Old English, the split CP will be useful, as there is evidence for both high V2 and low V2.

In an approach such as that of [Chomsky et al. \(2019\)](#), V2 constructions in which the subject moves to pre-verbal position are indeterminate if they involve movement of the subject from the specifier of the vP to that of TP, as schematized in (40) for (38).

- (40) [CP Sharon [C-is [TP Sharon [T is [v*P [<Sharon>] [v* naar huis gegaan]]]]]].


The complement of the phase head v* will transfer to the interfaces and is not relevant for the determinacy violation. What remains is the complement of C and that workspace contains two copies of the subject *Sharon*. That will prevent *Sharon* from moving into the next domain, the CP.

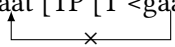
Verbal heads are also subject to the Principle of Determinacy (see [van Gelderen 2022](#)). With a TP present, these would lead to a determinacy violation in (41), as (42) shows.

- (41) Dutch
 Zij gaat weg.
 She goes away
 ‘She is going away.’

3 *Determinacy and the CP-TP Bottleneck*

(42) Transfer

[CP [C gaat [TP [T <gaat> [v*P [v* <gaat> weg <gaat>]]]]]].



If Dutch (and German) lack a TP, as I've argued in Section 3, the possibility of the subject preceding the finite verb in second position, as in (38), is not problematic. If a TP is not present, as in (43), the structure is determinate, both for the subject and for the (auxiliary) verb that moves to C.

(43) CP[Sharon [C is AUXP [AUX <is> v*P[<Sharon> v* naar huis gegaan]]]].
Transfer

If there is no TP, there are no determinacy violations either. Haider (1991, 2010), van Gelderen (1993, 1997), and others have indeed argued that German and Dutch lack a TP and Platzack (1988) combines the T and C in the CP domain. The evidence for such a lack of TP revolves around the optionality of expletives (see van Gelderen 2022), the stranding of German prefixes, the stranding of Dutch subjects in the VP, and the absence of infinitival particles that function independently of the verb, such as English *to*. For reasons of space, I cannot go into these here but, instead, turn to the history of English where a loss of V2 coincides with evidence for the introduction of a TP.

I'll first show that V2 is lost starting in the 14th century and an SV order is introduced instead, with the V moving to T. I then show that, around the same time, expletive subjects become obligatory to meet the SV requirement in cases where an (indefinite) subject fails to move to the subject position. The expletive has been linked to the presence of a TP, with the possibility of the expletive filling its specifier. A link between the loss of V2 and the introduction of expletives hasn't been explored in overview studies of early English syntax (e.g. Fischer et al. 2000, Los 2015, Fischer et al. 2017). Butler (1980: 300) considers them only indirectly related and van Gelderen (1997) links the introduction of T to the appearance of expletives, but not the loss of V2 to obligatory expletives. Why would the TP need an expletive? There could just be the merge of a T and no specifier. This presence only makes sense if, like regular DPs, expletives move to avoid labeling conflicts in the specifier of vP. I therefore explore an approach that considers certain Old English demonstratives and adverbs as CP expletives and others as vP expletives, as in Richards & Biberauer (2005) for German. Once the verb moves to T in Middle English, the vP expletives are reanalyzed as moving to the specifier of the TP.

In the Old and Middle English independent clause, the subject may be first and the finite verb may occur in the second position, as in (44). When there is a non-finite verb, this tends to be at the end (V-last), also shown in (44).

Elly van Gelderen

- (44) V2 and V-last

Wyrd **bið** ful **ared**.

fate AUX fully set

‘Fate is fully determined.’ (DOE, Exeter, Wanderer 5)

Nevalainen (1997) and **Fischer et al. (2000: 132)** put the “sharp decline” of V2 in the mid and late 14th and early 15th centuries. For instance, the beginning passage of Chaucer’s *Astrolabe*, only one out of 12 sentences is V2, namely the initial one, as (45) shows. This sentence may be formulaic since lots of texts start that way.

- (45) **Here beginnid** pe discptnes of the astrolabye. the ring

Thyn astrelabie hap a ring to putten on pe thombe of pi ri3th hōnd in takyng pe heygty of pingys. and take kepe fro from hennes forward **I wil call** pe hepgtij of eny thing take be pe ri3le. pe altitude wtowte mo wordis.

This ring rennyd in a mañ turret fast to pe modir of pe astrelabie yn so rōm a place þt **it distrbit** not the instrume~ t to hange~ aftir his rigtij centre.

The modir of pis astr(o)labie is pe pikkist plate. prcd wt a large hool **pat receiud** yn here woñbe pe pynne plates co~ pownd for dius climatis 7 pe rieth shape~ yn man~ of a nett or of a webbe of a lobbe.

Thys modir is diuidid on pe bakhalf wt a lyne þt **comyd** descending fro pe ring down to the nedirest bordure. **pe whiche lyne fro pe forseide ri~ gvn to the centre of the large hole amydde ys clepid** pe sowth lyne . or ellis pe line m(er)idional. and **pe op deel of pe line dow~ to pe bordure is clepyd** pe norplyne or ell pe lyne of mydnigti.

Ouerdwart pe forseyde loñge lyne **pr crossep** hym a nop~ lyne of pe same length fro pe East to pe West.

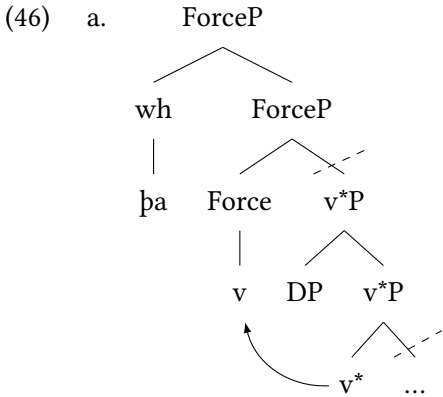
‘Here begins the description of the astrolabe and the ring. Your astrolabe has a ring to put on the thumb of your right hand when taking the height of things. And take care, because from now onwards, I will call the height of anything taken by the rule as the altitude without any more words. This ring goes through a kind of ring fastened to the body of the astrolabe in so large a place that it doesn’t disturb the instrument to hang right down. The body of this astrolabe is the thickest plate and is pierced with a large hole that receives the thin plates made for diverse celestial zones and the reet shaped like a net or border. The which line from the mentioned ring to the center of the large hole in the middle is called the south line or else the line meridional and the other part of the line down to the border is called the north line or else the line of midnight. Across the mentioned

3 Determinacy and the CP-TP Bottleneck

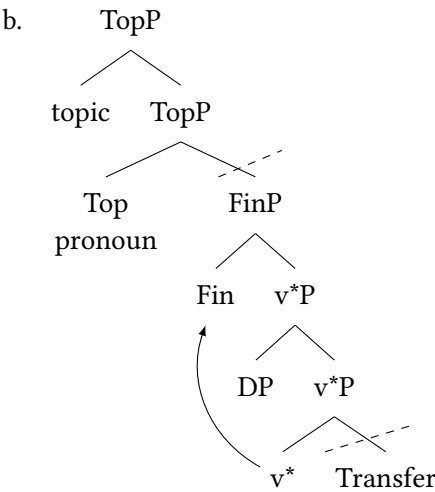
long line there crosses itself another line of the same length from the East to the West.’

Because V2 in Old English is varied (e.g. *wh*-questions are different from topic-initial sentences), two positions are assumed for the subject, a high one and a low one. For [Pintzuk \(1993\)](#), these are the specifier of the CP and that of IP and for [Haeberli 2002b,a](#), these positions are the specifier of CP and of AgrSP (part for the TP-layer), respectively. For others, these positions are the specifiers of TopicP and FinP or the specifiers of ForceP and FinP. I’ll use the latter but not so much hinges on the exact labels. As in later English, *v** is a phase head as well as the highest C, either Force or Top.

An expanded CP-layer, as in (46), contains two positions for the verb to move to, Force and Fin. If the *wh*-element occupies the specifier of ForceP, the verb moves to the Force head and this results in V2, as shown in (46a). The Force head is then the phase head. The intermediate structure of TopP and FinP is not present otherwise there would be a determinacy violation. If there is a pronoun in the head of the TopicP, as in (46b), V-movement to the highest position is blocked and the result will be V3, with the verb moved to the Fin head. Apart from pronouns, the Top head is also the position for some of the deictic markers, as [van Kemenade & Los \(2006\)](#) argue.



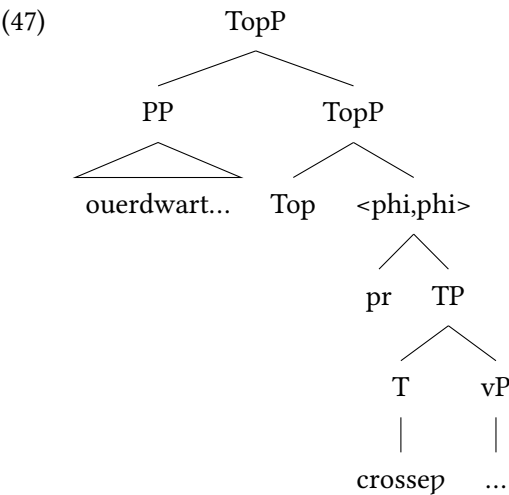
Elly van Gelderen



As before, the two highest v^* P's are labeled after v^* inherits $[\phi]$ -features from the Force and Fin heads, respectively. This is not shown in (46).

(46). provides insight into the possible positions: not all phrases can be present at the same time, only one of the two phase heads can be. For the structure to be determinate, the finite verb can only move to the Force if the lower Fin position is not present or to the lowest Fin position.

From Middle English on, the structure becomes less concentrated on the CP-domain and more on the TP. The Middle English CP includes a TopP, as in (47) for (48), (and a FocP) but its TP can be expanded with many auxiliaries, as shown in (49).



3 Determinacy and the CP-TP Bottleneck

- (48) Ouerdwart pe forseide loñge lyne **pr** **crossep** hym a nop~
 Across the mentioned long line there crosses him an other
 lyne of pe same length.
 line of the same length.
 ‘Across the long line, mentioned before, there crosses itself another line
 of the same length.’ (Chaucer, *Astrolabe*, from Table 5.1)
- (49) If I so ofte **myghte have** ywedded **bee**.
 ‘If I might so often have been married.’
 (Canterbury Tales, Wife of Bath Prologue, 7)

Old English has quite a varied word order, also where V2 is concerned, and that’s why an expanded CP, as in e.g. (46), is appropriate. If the highest C is a phase head, the various choices for verb movement, either to Force or to Fin, are accounted for. By the late 14th century, V2 is mostly lost. The introduction of expletives and the loss of V2 can be linked through the introduction of a TP and a simplification of the CP domain, with C as the phase head. Structurally, it means the verb stops moving to C, except in cases of *wh*-movement, and just reaches T.

Semantically unambiguous existential constructions with ‘there’ are rare in Old and Early Middle English. Most instances of ‘there’ are locative. The criterion I use to determine expletivehood is their position. A demonstrative is a TP expletive if it follows a sentence-initial adverb(ial) and is followed by a verb and indefinite subject, as in (50a). In contrast, if the language is without a TP, an initial, full adverbial will always trigger V to C. An expletive following a complementizer and preceding an indefinite, as in (50b), is also seen as a TP-expletive.

- (50) a. XP, there is/was a/no/some person
 Locative Adverbial EXPL BE INDEFINITE N
- b. C there ... is/was a/no/some person
 C EXPL BE INDEFINITE N

Typical Old English V2 existentials are given in (51) and (52) and these occur without an expletive. As a result, these can be V1, as in (53).

- (51) On þæm dagum wæs an hirde on Ispanium, se wæs Ueriatus
 In those days was a shepherd in Spain, who was Viriathus
 haten.
 called
 ‘In those days, **there** was a shepherd in Spain, who was called Viriathus.’
 (DOE, Orosius, 114.12)

Elly van Gelderen

- (52) On þas kinges dæi Offa, wæs an abbot on Medeshamstede Beonne
In the king's day Offa was an abbott in Medeshamstede Beonna
gehaten.

called

'In the days of King Offa, **there** was an abbot Medeshamstede called Beonna.' (DOE, Peterborough, 777.3)

- (53) Wæs micel fyr onæled on middum þam huse.
was large fire started in middle that house
'**There** was a large fire started in the middle of the house.' (DOE, Bede 180.26)

By Early Middle English, e.g. Layamon's *Brut*, TP expletives start to appear regularly but are not required. Layamon's text remains in two manuscripts, the earlier Caligula and the later Otho. In (54), there is an expletive in Caligula but not in Otho and, in (55) and (56), the reverse is true. Breivik (1983: 262) takes the optionality as an indication that these are expletives and not locatives.

- (54) a. Ah nes hit buten ane while. þat þer com an oðer time.
but NEG.was it but one while that EXPL came an other time
'It was only a short time that there came another style.' (Caligula 5489)
b. þat com an oþer time. (Otho 5489)
that came an other style
- (55) a. Nes næuere na ma. þat don þer mihte Cristin dom.
NEG.was never no man REL do there could Christian ceremony
'There was no man who could perform the Christian ceremony there.'
(Caligula 6618)
b. Nas þar neuere no man. 'There was no man.' (Otho 6619)
NEG.was never no man
- (56) a. For nis nan kinelond. na swa brad næ swa long.
for NEG.is no kingdom not so broad nor so long
'Because there is no kingdom however broad or long.' (Caligula 6660)
b. For nis þar no kinelond.
for NEG.is EXPL no kingdom
'Because there is no kingdom.' (Otho 6660)

3 Determinacy and the CP-TP Bottleneck

There expletives in Layamon are ambiguous, according to Breivik (1983) and van Gelderen (1997: 92–95), and certainly not obligatory. Layamon's word order is quite varied, which the sentences above show. As is common for the instances of surface SV, it is not clear whether they are V2 with the subject in the CP domain or SV with the subject in a specifier of TP. However, after adverbs, like 'thus' and 'now', there is still V2 in this text, so this stage is still V2.

The word order in Layamon's (later) Otho is not remarkably different. There is some indication that the infinitival 'to' is more independent of the verb in Otho (57a) than in Caligula (57b). In the latter, it is still a verbal prefix but, in Otho, it appears together with the complementizer in C. (57a) provides no evidence of *to* as T, however, because *for* and *to* could be one unit.

- (57) a. *Pe king ... wide his men sende. for to hine finde.*
 the king ... wide his men sent for to him find
 'The king sent his men all over to find him.' (Otho 8488–8490)
- b. *Heo wenden hine to finden.*
 they went him to find
 'They went to find him.' (Caligula 8874)

Layamon's V2 stage is compatible with the optional expletives filling the specifier of TP; the lack of a designated T element provides more evidence for the optionality.

Wyclif's works remain in 170 manuscripts and are usually divided in earlier and later versions. The later text is solidly SV and displays expletives, as shown in (58) with S and V are bolded.

- (58) *IN the bigynnyng **God made** of nougt heuene and erthe. Forsothe **the** erthe was idel and voide, and derknessis weren on the face of depthe; and **the** Spiryte of the Lord was borun on the watris. And **God seide**, **Liȝt** be maad, and **liȝt** was maad.*

'In the beginning, God made heaven and earth out of nothing. And, the earth was useless and empty and darkness was on the face of the depth; and the spirit of God was born on the waters. And God said, Let there be light and light was made.' (Wyclif, Genesis 1-3, later version)

Since this stage is SV, we expect TP expletives and there is indeed good evidence for TP expletives in the later edition, as in (59a) and (60a), but fewer in the earlier version, as (59b) and (60b) show, because that translation stays close to the Latin original, where expletives do not appear. Because the Latin text does not include a locative, as in (59c) and (60c), it is likely that the *there* occurring in the later English edition is an expletive.

Elly van Gelderen

- (59) a. **Ther** cam a womman of Samarye. (*John*, IV, 7, later version)
 there came a woman from Samaria
- b. A womman came of Samarie. (*John*, IV, 7, early version)
 a woman came from Samaria
- c. Uenit mulier de samaria.
 came woman of Samaria
 ‘There came a woman of Samaria.’
 (*John*, IV, 7 *Vulgate*)
- (60) a. And **ther** was a man there. (*John*, V, 5, later version)
 and there was a man there
- b. Forsothe sum man was there. (*John*, V, 5, early version)
 truly some man was there
- c. Erat autem quidam homo ibi.
 was indeed some man there
 ‘There was indeed some man there.’
 (*John*, V, 5 *Vulgate*)

There are some *ther* expletives in the early version, but these are fewer compared to the later version: 4 in the early version as opposed to 18 in the later version. In the later version, the verb ‘to be’ followed by an indefinite doesn’t occur without the expletive.

Locative *there* appears in both the early and later Middle English renderings, as in (61a) and (61b), translating *ibi* ‘there’ in the Latin version in (61c).

- (61) a. Forsoth the welle of Jacob was **there**. (*John*, IV, 6, early version)
 truly the well of Jacob was there
- b. And **ther** was the welle of Jacob. (*John*, IV, 6, late version)
 and there was the well of Jacob
- c. Erat autem **ibi** fons iacob. (*John* IV, 6 *Vulgate*)
 was moreover there well Jacob
 ‘Moreover, the well of Jacob was there.’

The insertion of *ther* where the Latin original lacks *ibi* or where another locative *there* occurs, as in (59a), provides evidence that TP expletives are used in Wyclif. If the introduction of expletive *ther(e)* is due to the introduction of a specifier of TP, one expects evidence for a T position as well. Changes in the behavior of modals and of infinitival *to* provide evidence for the occurrence of a special

3 Determinacy and the CP-TP Bottleneck

head position T. In Wyclif, such evidence can be found: modals no longer double and infinitival *to* can be separated from its verbal part as in (62) and (63). This points to a special position (T) being present with expletives in the specifier.

- (62) The prestis ben forfended to eny more **takyn** monee of the puple.
 ‘The priests are forbidden to take any more money from the people.’
 (4 Kings, XII, 8, Visser 1963-1973: 1041)
- (63) Betere to ben stille þan to fewe thingis **writen**.
 ‘Better to be silent than to write too few things.’
 (Prefatory Epistles of St Jerome, VII, 224-5)

Accusative-with-infinitives also occur, as in (64) and (65), and this is evidence of the independence of *to* in a T (Massam 1985; van Gelderen 1993). Lindberg (1978: 42–43) mentions 16 occurrences in the early edition he compiled of *The Prefatory Epistles of St Jerome*. Although Warner (1982: 141ff) attributes the more frequent use of Accusatives-with-Infinitives in the early version to Latin influence, I assume the author would not use an ungrammatical translation.

- (64) Forsothe thei gessinge **him to be in the felowshipe**.
 ‘truly, they guess him to be in the fellowship.’
 (Luke II, 44).
- (65) We knowen **the writer of hem, luke, to ben a physician**.
 ‘We know the writer of them, Luke, to be a physician.’
 (Prefatory Epistles of Jerome, VII, 160, Visser 1963-1973: 1041)

As mentioned, the word order is SV but verbs continue to move to C in questions, of course, as (66) shows.

- (66) **Saw Y thee not** in the 3erd with him?
 ‘Didn’t I see you in the garden with him?’
 (John, XVIII, 26, later version)

In conclusion to Wyclif, there are TP expletive *thers*. This can be related to the appearance of a T-position, evident from the introduction of accusative-with-infinitives and split infinitives.

Chaucer is another author identified as having SV, rather than V2, in (45) above. In the same text, i.e. the *Astrolabe*, there are expletives as well. They occur after topics, as in (67) to (70), and in subordinate clauses, as in (71) to (73), indications that they occupy the specifier of TP.

Elly van Gelderen

- (67) of the astrelabie þat I haue seyn, **there** ben some conclusions þat ...
 ‘of the astrolabe that I have seen, there are some goals that...’ (Astrolabe, Preface 21–22)
- (68) Ouer-thwart this for-seide longe lyne, **ther** crosseth hym a-nother lyne
 ‘Across this already mentioned long line, there crosses itself another line.’
 (Astrolabe, 1.5)
- (69) thorw wich pyn **ther** goth a litel wegge
 ‘through which pin, there goes a little wedge.’ (Astrolabe, 1.14)
- (70) From this senyth, as it semeth, **ther** come a maner krokeke strikes
 ‘from this zenyth, as it seems, there comes a type of curved marks.’
 (Astrolabe, 1.19)
- (71) so general ... þat **ther** nedith no more declaracion
 ‘so general that no more explanation is needed.’ (Astrolabe, 2.2)
- (72) after thin Almenak vp-on wych table **ther** folwith a canon
 ‘according to your almanac, upon which plate there follows a set of rules.’
 (Astrolabe, Preface 94–95)
- (73) ouer the wiche degrees **ther** ben nowmbres of augrym
 ‘over which degrees there are numbers of algorithms.’ (Astrolabe, 1.7)

Searching for a ‘be’ verb with an indefinite without ‘there’ provides no results in the *Astrolabe* so that suggests TP-expletives are as necessary as in Modern English. In Chaucer’s *Canterbury Tales*, *ther* is also used as an expletive, e.g. after topics in (74) to (76).

- (74) And over his heed **ther shynen** two figures.
 ‘And above his head, there shine two images.’ (CT, *Knight’s Tale* 2043)
- (75) With hym **ther wenten** knyghtes many on.
 ‘With him, there went many a knight.’ (CT, *Knight’s Tale*, 2118)
- (76) that in hir coppe **ther was** no ferthyng sene.
 ‘that in her cup, there was no small remnant to be seen.’ (CT, *Knight’s Tale*, 134)

In addition to the frequent use of expletives, is there evidence for a T? Modals and *do* are used in their Modern English meanings, as in (77) and (78).

3 Determinacy and the CP-TP Bottleneck

(77) It **shal** be doon.

‘It shall be done.’ (Chaucer, *The Reeve’s Tale* I, 4034)

(78) how now, what **do** ye heer?

‘How now, what do you hear?’ (Chaucer, *The Reeve’s Tale*, I, 4025)

To is often together with *for* in C, but there is a split infinitive, as in (79), and an Accusative-with-Infinitive in (80) and these are seen as an indication of the independence of *to*.

(79) to seye to the, that art the man that I best triste; And peril non was it **to** the **bywreye**.

‘to say to you, who is the man I most trust; and it was no peril to reveal to you.’

(*Troilus and Criseyde*, III, 365–367, Visser 1963-1973: 1039)

(80) It behoveth by necessite that every thing be ryght as science comprehendeth it **to be**.

‘It is appropriate by necessity that everything is as science understands it to be.’ (Boece, Visser 1963-1973: 2309).

In conclusion to Chaucer, SV word order is prevalent, as are TP expletives, and there is some independent evidence for a T-position as well. Auxiliaries continue to move to C in questions and here I assume that the tense and agreement features can be bundled on C.

Thus, as the word order changes from V2 to SV and evidence for a T head starts to appear in late Middle English, the use of TP expletives solidifies as well. Wycliff (the later version) and Chaucer are from the last part of the 14th century and show TP expletives as well as evidence for a T. For determinacy, this state of affairs means that, as the TP is introduced, movement of V to C will not be a problem because this movement is lost. The question I haven’t been able to answer is: why was the TP introduced when V2 was being lost? Is it the information structure that changes (cf Los 2009) or does the role of C become less prominent? The C has tense and agreement features and doesn’t need T. The reason for the rise of auxiliaries and T must have been an independent change.

When the TP is present in the spine, determinacy requires the expletive in the TP if the subject is moving into the CP domain. Languages show a fluidity between C and T: in some, both C and T are present but T inherits features from C; in some, T can be skipped; and, in some, T is not present. The options are summarized in Table 1.

Table 1: Options for C and T

	V to C	presence of T	Feature Inheritance
Dutch	yes: V2	optional	C to v*/v or C to T
Old English	yes: V2 and V3	no	C to v*/v
later Middle English	no	yes	C to T

5 Conclusions and limitations

There are many ways to achieve optimal computation. For instance, the syntax ‘doesn’t care’ if the subject moves to the specifier of TP before valuation or after, as shown through the work of Obata et al. (2015). They 2015: 15 have argued that “[o]ptimal computation is executable in more than one way.” If computational efficiency is underspecified, the timing of subject raising to TP and C-agreement (and Feature Inheritance) can vary, as two varieties that use a complementizer *ki*, Cape Verdian Creole and Haitian Creole, show. The difference between the two varieties has the effect of skipping T.

Likewise, the C-T complex can be simplified in a number of ways: (a) C transfers phasehood to T and the CP deletes or is not present or (b) TP is not present. Table 2 provides a typology of choices and how the difference is evident linguistically.

Table 2: A typology of choices for the C-T complex

	Optional CP	Optional TP
<i>That</i> -trace sequence	no; English (6)	yes; Old English (23), Dutch (22)
Subject-less RCs	some; English (16)	no
C-less complements	yes; English (13)	no

This is a programmatic paper: “let’s see what a determinacy approach contributes to the CP-TP complex”. Therefore, the picture I have given of e.g. *that*-trace and subject-less relatives is somewhat simplified. More nuanced data on *that*-trace can be found in Chaves & Putnam (2020) and Kiziak (2010), for instance. My discussion has also been limited to Old, Middle, and Modern English, and to Dutch. It has been assumed that the structure of German is similar to that of Dutch but this has to be investigated more.

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Elly van Gelderen

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3 Determinacy and the CP-TP Bottleneck

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Elly van Gelderen

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Chapter 4

A comparative investigation of verbal mood in Old High German, Old Saxon and Old English adverbial clauses

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In this paper, we provide new insights into mood alternations in adverbial clauses in Old High German (OHG), Old Saxon (OS) and Old English (OE). We will show that (non-)veridicality plays a role, but the empirical investigation conducted highlights another relevant factor involved in the choice of mood, namely the type of complementizer. We link this piece of evidence to the presence of dedicated positions in the C-layer, a factor that is corroborated both by cross-linguistic evidence and the corpus data. The analysis we present postulates three different heads in the C-domain that activate respectively the propositional, the illocutive and discourse layer, and that are targeted by the different complementizers and are responsible for the choice of mood. The investigation sheds light on the interaction between verbal mood, clause type and complementizer type, and opens up new avenues for future research.

1 Introduction

It is a well-known fact that Old High German (OHG), Old Saxon (OS) and Old English (OE) display alternation between the indicative and the subjunctive in adverbial clauses, as exemplified by the following examples for OHG, in which

M. Coniglio, Ch. De Bastiani, R. Hinterhölzl & S. Petrova

the indicative and the subjunctive are used in a temporal clause of the same type and introduced by the same adverbial subordinator, respectively.¹

- (1) unz thiu fluot **quam**
until the flood came.IND
'until the flood came'
(DDD-AD-Tatian_1.1 > T_Tat147 (edition 40 - 56))
- (2) unz ir sit giuuatite mit megine fon hóhi
until you are.SUBJ vested with power from above
'until you are given heavenly power'
(DDD-AD-Tatian_1.1 > T_Tat244 (edition 2 19))

Previous accounts of this alternation in West Germanic are non-exhaustive, focusing on individual varieties only (cf. Schrodtt 1983, Petrova 2008, 2013, Coniglio 2017, Coniglio, Hinterhölzl & Petrova 2018 for OHG; Mitchell 1985, Vezzosi 1998 and Van Gelderen 2019 for OE). A unified account comparing the distribution of verbal mood in the early Germanic varieties and the factors determining it is still missing. But there is reason to assume that the selection of mood in these varieties is governed by the same, or very similar, principles. One important observation is that according to traditional descriptions, the functional domains of the subjunctive are very similar across the Early Germanic varieties under investigation: these are, very generally, nonfactual or counterfactual utterances like wishes, commands or hypothetical clauses, often involving a not easily explainable overlapping with the indicative (see Behaghel 1924: 219–246 for OHG, Rauch 2000: 1266 for OS, Faraci 1980, Mitchell 1985: §2562, Mitchell & Robinson 2001, Traugott 1972, Vezzosi 1998 for OE).

Looking at the ratio of the indicative and the subjunctive in corpora of the three varieties, we observe a similar quantitative distribution of verbal moods in these language.² Consider the figures in Table 1 which provide the quantitative

¹Verbal moods are glossed as follows: the indicative is glossed as IND, the subjunctive is glossed as SUBJ. Similarly to Modern German, there are two classes of subjunctives in Early Germanic, depending on the stem form of the finite verb, namely subjunctive I, derived from the present stem, and subjunctive II, derived from the plural of the preterit stem. In the data of the present paper, which comprises adverbial clauses, both classes of the subjunctive occur, but they are semantically equivalent, being governed by *consecutio temporum*, in contrast to Modern German. For simplicity, instead of providing the full morphological specifications (e.g. for tense, person and number), we use the inflected forms of the verbs in the glosses.

²The OHG and OS data presented in this paper are extracted from the *Referenzkorpus Altdeutsch* (750–1050), ReA (Donhauser, Gippert & Lühr 2018), queried through ANNIS3 (Krause & Zeldes

4 Verbal mood in OHG, Old Saxon and Old English adverbial clauses

distribution of verbal moods in adverbial clauses in OHG, OS and OE:³

Table 1: Distribution of mood in West Germanic

	OHG			OS		OE	
IND	3,130	65.0%	828	69.0%	4,242	70.7%	
SUBJ	1,682	35.0%	372	31.0%	1,761	29.3%	
Total	4,812	100.0%	1,200	100.0%	6,003	100.0%	

All three varieties share a ratio of approximately 65–70% of indicative forms of the verb and 30–35% of subjunctive forms in adverbial clauses; this similarity is striking and requires an explanation. As mentioned above, a comparative investigation of the factors determining mood alternation in Old Germanic is still a desideratum (cf. Coniglio, Hinterhölzl & Petrova 2018, Coniglio, De Bastiani, Hinterhölzl & Weskott 2021). In this study, we present the results of a pilot study as part of a larger comparative project aiming at filling this void. For the three languages considered, we will first discuss the two main factors determining mood alternation in adverbial clauses, namely the semantic type of the adverbial clause and the type of adverbial subordinator involved (Section 2). We will then address the question of the existence of specialized complementizer positions and provide cross-linguistic evidence for that, in general, as well as for multiple complementizer positions in OHG, OS and OE, specifically (Section 3). Based on the data comparatively observed, we will offer a comparative syntactic analysis of the phenomena discussed by accounting both for the semantic classes of adverbial clauses and for the type of subordinating elements (Section 4).

2 Factors determining mood alternations in adverbial clauses

In complement clauses, the choice of verbal moods is assumed to depend on the semantic properties of the selecting category (for a recent cross-linguistic

2016), whereas the OE data (850–1150) are extracted from the *York Corpus of Old English Prose*, YCOE (Taylor et al. 2003), queried through the *Corpus Studio Web Application* (Komen 2017) and processed through *Cesax* (Komen 2011). The Middle High German data presented in Section 3 are extracted from the *Referenzkorpus Mittelhochdeutsch (1050–1350)*, ReM (Klein et al. 2016).
³The OHG and the OS dataset comprises all adverbial clauses that are annotated as such in the corpora used, while the OE data refer to the types of adverbial clauses discussed in this paper: causal, temporal, concessive, conditional and consecutive clause, as extracted from the YCOE corpus.

M. Coniglio, Ch. De Bastiani, R. Hinterhölzl & S. Petrova

overview, see Baunaz & Puskás 2022). But differently from complement clauses, mood selection in adverbial clauses is assumed to be completely independent of the properties of the matrix clause (cf. Coniglio, Hinterhölzl & Petrova 2018 for an analysis of mood alternations in different clause types in OHG). It is to be traced back to the semantics of the adverbial phrase itself, and the type of adverbial subordinator. In what follows, these two factors will be taken into consideration while explaining the choice of verbal moods in adverbial clauses in the three language varieties under investigation.

2.1 Factor 1: Semantic type of the adverbial clause

One factor responsible for the mood alternations observable in the data is the semantic class of the adverbial clause. Figure 1 and Figure 2 display the distribution of the indicative and the subjunctive in different semantic classes of adverbial clauses in OHG and OS. The figures show that with the exception of consecutive clauses, the ratio of the indicative and the subjunctive is similar in each of the clause types investigated in the two varieties. More precisely, while the proportion of the subjunctive is low in causal, temporal, local and modal clauses in OHG and OS, it is especially high in concessive and purpose clauses. In conditional clauses, the indicative dominates but the difference between the indicative and the subjunctive is not as big as in the remaining clause types.

Let us compare this picture with the situation in OE shown in Figure 3. The ratio of moods in the individual clause types shows a similar distribution to their OHG and OS counterparts, in some cases patterning with OS and in others with OHG, as commented in the following.⁴ While the indicative dominates in causal and temporal clauses, it is a minority option in concessive clauses. Conditional clauses in OE pattern with those in OHG and OS in that the amount of clauses in the indicative is slightly higher than the respective amount in the subjunctive. Finally, consecutive clauses are similar to those in OS rather than to those in OHG in that the ratio of the indicative is higher than the one of the subjunctive.

In conclusion, comparing the three Germanic varieties under consideration, it is evident that the subjunctive appears to be particularly frequent in concessive and purpose clauses, displaying very similar ratios across all varieties involved. What is the theoretical explanation of the distributional properties of the indicative and the subjunctive arising from the outcomes of the corpus analysis? Given the distribution of verbal mood in adverbial clauses in OHG, Coniglio, Hinterhölzl & Petrova (2018) discover a parallel between the conditions favoring the

⁴These data include only the texts that are indicated in the YCOE as non-translated.

4 Verbal mood in OHG, Old Saxon and Old English adverbial clauses

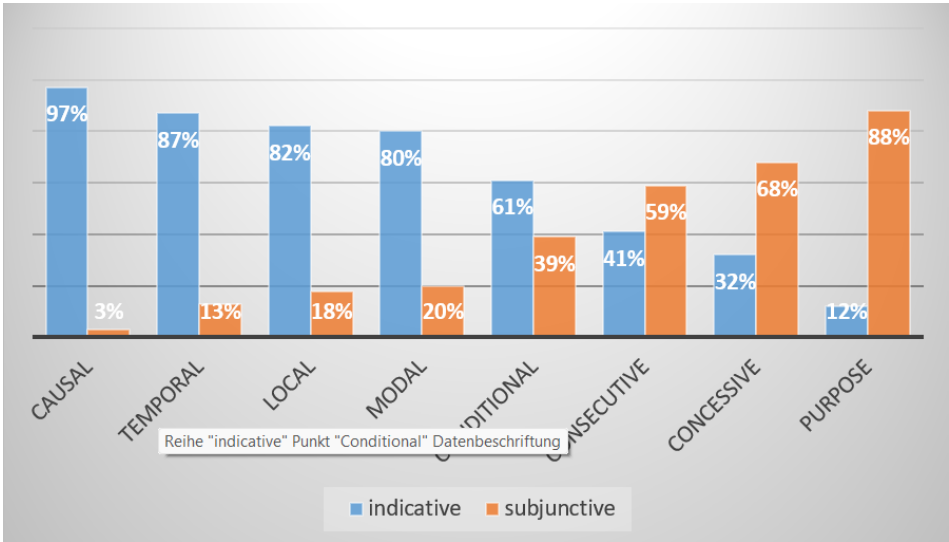


Figure 1: Mood alternations and type of adverbial clause in OHG (Coniglio et al. 2018: 24)

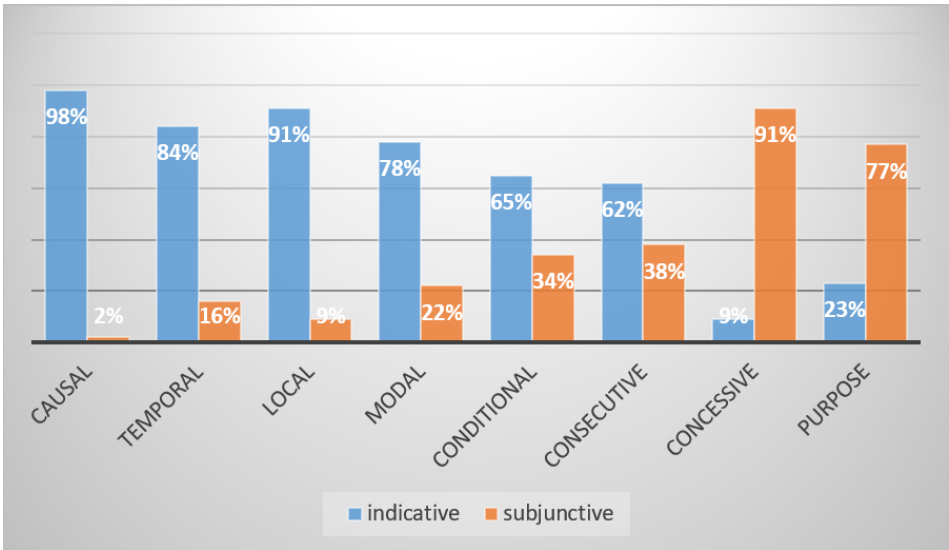


Figure 2: Mood alternations and type of adverbial clause in OS

M. Coniglio, Ch. De Bastiani, R. Hinterhölzl & S. Petrova

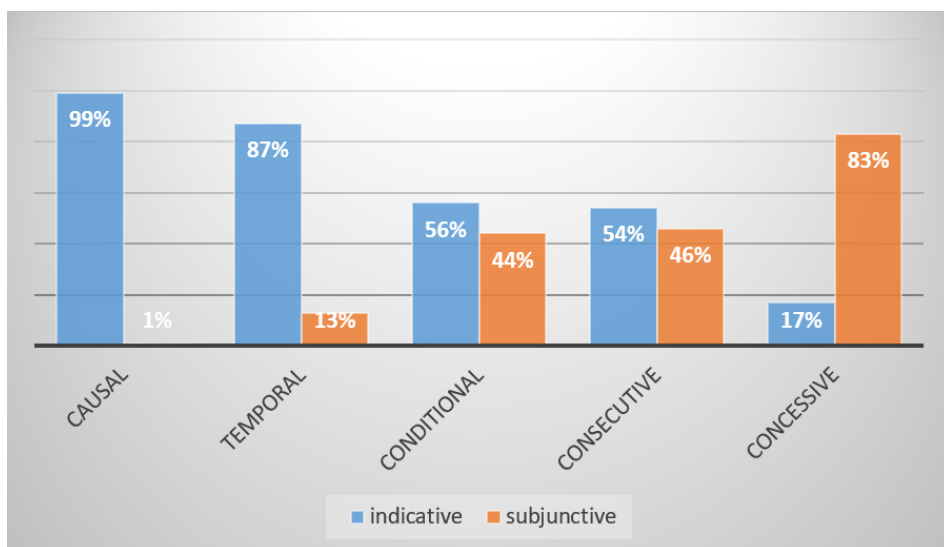


Figure 3: Mood alternations and type of adverbial clause in OE

subjunctives in early Germanic and those triggering the subjunctive in modern Greek and Romance (see Giannakidou 1998 and later, Farkas 1992). In Greek, the subjunctive complementizer *na* occurs in complements of volitional and directive predicates, but importantly, also in combination with other conjunctions in the following types of adverbial clauses: *ja na* ‘for’ in purpose clauses, *xoris na* ‘without’, *prin na* ‘before’, *oste na* ‘with the result that’ (Giannakidou 1994, 1998; Giannakidou & Zwarts 1999). Similarly, in Romance, the subjunctive is present in complements of the same two types of predicates discussed in Giannakidou (1998) (cf. also Farkas 1992) as well as in concessive clauses (Quer 2001), although there is variation here which is subject of an ongoing debate.

Giannakidou models the distribution of verbal mood in Greek and Romance along the lines of the notion of (non-)veridicality (Giannakidou 1998 and later), i.e., she argues that the factors influencing the choice of the indicative and the subjunctive cross-linguistically are related to truth and existence. According to this view, contexts which enforce reference and imply the truth of the proposition are defined as *veridical* ones, while those “captur[ing] the state of an unknown truth value” (Giannakidou 2009: 1889) are classified as *non-veridical*. The subjunctive is defined as a kind of a negative polarity item which is triggered by conditions related to the non-veridicality of the proposition.

Based on the parallels in the selection of the subjunctive in OHG and the cross-linguistic evidence discussed in the literature, Coniglio, Hinterhölzl & Petrova

4 Verbal mood in OHG, Old Saxon and Old English adverbial clauses

(2018) argue that the distribution of the indicative and the subjunctive in OHG correlates with (non-)veridicality, similarly to the distribution known from Greek and Romance (Giannakidou 1998 and later). In the face of the evidence presented for OS and OE, it can be concluded that the relation between the choice of the subjunctive and the inventory of adverbial clauses which trigger the subjunctive cross-linguistically applies to early West Germanic as a whole, as shown in Figure 1-Figure 3 above.

To demonstrate that the decisive factor in the selection of the subjunctive is related to conditions of non-veridicality, we compare the selection of the indicative and the subjunctive in two types of temporal clauses which differ in their selectional properties in Greek and Romance, namely temporal clauses expressing posterior events and introduced by *after* and those expressing anteriority and introduced by *before* (but see Coniglio, Hinterhölzl & Petrova 2018 for a more comprehensive investigation of other clause types, including complement and relative clauses). In Greek and Romance, only the latter select the subjunctive. Giannakidou explains this difference by employing the notion of entailment (Giannakidou 2012: sec. 3.1). She argues that temporal clauses introduced by *before* display conditions of non-veridicality in that the situation in the matrix clause does not entail the event of the *na*-clause, although it is completely in the past. Remarkably, the respective subjunctive complementizer does not occur in *after*-clauses.

We queried the corpora to determine the selection of mood in *as/after*-clauses and *before*-clauses in OHG, OS and OE. The results, given in Table 2, show that in OHG, OS and OE, the subjunctive is practically non-existent in *as/after*-clauses but well-represented in *before*-clauses, which are among the non-veridical contexts described by Giannakidou (1998) for Greek.

Table 2: Percentage of SUBJ in *as/after* vs *before*-clauses in OHG, OS and OE

	OHG	OS	OE
<i>do/tho/ða</i>	0.0%	3.0%	0.6%
<i>er/er/ær</i>	36.8%	41.7%	65.7%
<i>er danne/er than/ær ðan ðe</i>	58.6%	72.7%	77.7%

But at the same time, we see that the subjunctive does not appear categorically in *before*-clauses, although this type of adverbial clauses is one of those displaying conditions of non-veridicality and selecting the subjunctive cross-linguistically. In addition, we can observe different ratios for different adverbial

M. Coniglio, Ch. De Bastiani, R. Hinterhölzl & S. Petrova

subordinators introducing *before*-clauses. It must be noticed that the percentage of subjunctive in *before*-clauses in OE is slightly higher than in its counterparts in OHG and OS.

This observation suggests that mood choice in early Germanic not only depends on the semantic type of the adverbial clause, but also on the lexical type of the adverbial subordinator.

In the next section, we will investigate the role of the lexical subordinator as an additional factor influencing the distribution of verbal mood in early Germanic. As a testing ground, we will use adverbial clauses which represent the non-veridical type of contexts, and we will investigate those of them in which we find different lexical subordinators introducing such clauses in the Germanic varieties under investigation.

2.2 Factor 2: Type of adverbial subordinator

Let us look at concessive clauses first. The canonical subordinators introducing concessive clauses in the varieties under investigation are OHG *doh* (3), OS *thoh* (4) and OE *ðeah* (ðe) (5), which are etymologically related and which require the subjunctive, according to the literature (Mitchell 1985, Valentin 1997, Lühr 1998, Mitchell & Robinson 2001). But next to these subordinators, we find additional lexical items introducing concessive clauses in OHG and in OS, namely OHG *mit diu* (6) and OS *that* or *so* (7).

- (3) Thaz nezzi [...] tho ni brást, **thoh** iro wári sulih
the fishnet though NEG tore although of.them was.SUBJ such
lást
weight
'The fishnet [...] did not tear, although it [the fish within] was so heavy.'
(DDD-AD-Otfrid_1.1 > O_Otfr.Ev.5.13 (edition 222 - 232))
- (4) **thoh** thu nu an sulicoro pînu sîs
although you now in such sorrow are.SUBJ
'although you are suffering now'
(DDD-AD-Heliand_1.1 > Hel_66 (edition 724 - 734))
- (5) On ðære feorðan ylde geceas Israhela folc him sylfum SAUL to
on the fourth age chose Israelites.GEN people them selves Saul to
cyninge ongean Godes willan. **þeah** ðe he hit geðyldelice
king against God.GEN will though that he it patiently

4 Verbal mood in OHG, Old Saxon and Old English adverbial clauses

forbære.

endured.SUBJ

‘In the fourth age the people of Israel chose for themselves Saul as king against the will of God, even though he endured it patiently.’
(cocathom2.o3_ÆCHom_II,_4:35.179.795)

- (6) uuο theser buohstaba uueiz, **mittiu** er sie ni **lerneta**
why DEM letters knows although he them NEG learned.IND
‘Why does he know the letters although he did not learn them.’
(DDD-AD-Tatian_1.1 > T_Tat104 (edition 237 - 247))
- (7) endi ôc seokan man te gehêleanne, **sô** ik ina hrīnan ni
and also sick.ACC man to heal although I him touch NEG
tharf
need.IND
‘and also to heal a sick man although I do not need to touch him’
(DDD-AD-Heliand_1.1 > Hel_28 (edition 427 - 437))

Table 3 provides the distribution of verbal moods in concessive clauses introduced by the different adverbial subordinators attested in OHG and OS, compared to the data in OE.

Table 3: IND vs SUBJ in concessive clauses in OHG, OS and OE

	OHG			OS		OE	
	<i>doh</i>	<i>mit diu</i>	<i>thoh</i>	<i>that</i>	<i>so</i>	<i>ðeah</i>	<i>ðeah ðe</i>
IND	11.6%	100.0%	2.4%	50.0%	100.0%	17.0%	17.5%
SUBJ	88.4%	0.0%	97.6%	50.0%	0.0%	83.0%	82.5%
Total	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%

The figures in Table 3 suggest that the various lexical subordinators differ regarding their selectional properties. In OHG, it is only the subordinator *doh* that predominantly takes the subjunctive. At the same time, concessive clauses headed by *mit diu* are only attested in the indicative. OS provides a similar picture: the subordinator *so* takes only the indicative, while *thoh* predominantly selects the subjunctive. Let us compare the situation described for OHG and OS with that in OE. Since

M. Coniglio, Ch. De Bastiani, R. Hinterhölzl & S. Petrova

concessive clauses in OE, unlike those in OHG and OS, fail to provide the required lexical variation regarding the type of subordinator, we look at conditional clauses which can be introduced by both *gif* or *butan*.⁵ As will be shown, the two complementizers display different selectional properties as far as mood is concerned: whereas *butan* clearly favors the subjunctive, displaying it in 89.10% of the sentences examined in the dataset, *gif* shows a strong preference for the indicative, displaying it in 62.80% of the cases examined.

- (8) **Gif** ðu **lufast** God. þonne fortretst ðu þa worulldican styrunga;
If you love.IND God then tread you the worldly troubles
'If you love God, then you tread down worldly troubles.'
(cocathom2.o3, ÆCHom_II, 28:228.217.5076)
- (9) Ac wite gehwa to gewisson **buton** he his flæsclican lustas. &
but know everyone to certainty unless he his fleshly lusts and
galnysse **gewanige**: þæt he ne hylt his cristendom mid rihtum
wantoness diminish.SUBJ that he NEG holds his Christianity with right
biggenge.
observance
'But everyone should know for sure that, unless he diminishes his fleshly
lusts and wantonness, he does not hold his Christianity with right
observance.'
(cocathom1.o3, ÆCHom_I, 6:227.86.1108)

As shown for concessive clauses in OHG and OS, the lexical variation witnessed in the domain of conditional complementizers in OE correlates with a different choice of verbal moods in the respective clauses.

2.3 Interim conclusions

We have shown that OHG, OS and OE not only share the same ratio of the indicative and the subjunctive in adverbial clauses but that the selection of verbal moods is determined by the same factors in these three varieties. The first factor is the semantic class of the subordinate clause. In Early Germanic, the subjunctive is selected in the same types of adverbial clauses which trigger the subjunctive complementizer *na* in Greek or the subjunctive mood in Romance. These include:

⁵Concession in OE was furthermore expressed in disjunctive clauses introduced by *sam ... sam* or by verb first clauses with no conjunction (Mitchell 1985 and Mitchell & Robinson 2001). Clearly, these strategies to express concession in OE are syntactically different from the adverbial clauses we are considering and a comparison with OS and OHG on the basis of the complementizer and mood alternations in concessive clauses is not possible.

4 *Verbal mood in OHG, Old Saxon and Old English adverbial clauses*

- 1) ‘posterior’ temporal clauses;
- 2) concessive clauses;
- 3) purpose clauses;
- 4) comparative clauses;
- 5) conditional clauses;
- 6) ‘exceptive’ clauses.

(cf. Coniglio, Hinterhölzl & Petrova 2018: 17–18)

Given these cross-linguistic parallels, we argued that non-veridicality as the semantic property unifying the respective types of adverbial clauses is the trigger of the subjunctive in the early Germanic data as well (cf. Coniglio 2017, Coniglio, Hinterhölzl & Petrova 2018).

At the same time, we observed that the subjunctive is not obligatory in the adverbial clauses displaying conditions of non-veridicality in the early Germanic varieties under investigation. We discovered that there is a second factor influencing the choice of verbal mood, namely the lexical type of the adverbial subordinator. The follow-up question is: why do these complementizers select different mood forms in one and the same type of adverbial clause? We believe that there is cross-linguistic evidence suggesting that the different selectional properties of the lexical subordinators is related to differences regarding their positional realization in the complex left periphery of the clause in early Germanic. This evidence will be presented in the next section.

3 Specialized complementizer positions

In the following, we want to pursue the idea that the alternation of verbal moods in adverbial clauses of the same class results from differences regarding the syntactic distribution and the selectional properties of the complementizers introducing these clauses. More precisely, we assume that OHG and OE display a system of multiple complementizers, and that the individual lexical items used to introduce adverbial clauses are localized in different structural positions in the C-domain, also selecting different verbal moods in the respective clause. This idea will be complemented by the syntactic analysis proposed in Section 4, according to which the selectional properties of the complementizers correlate with different syntactic complementizer positions, which in turn correlate with different

M. Coniglio, Ch. De Bastiani, R. Hinterhölzl & S. Petrova

modes of anchoring the proposition expressed by the adverbial clause: with respect to the situation denoted by the matrix verb or with respect to the utterance situation.

In order to show that this idea is on the right track, we want to outline the notion of complementizer doubling, discussing the constitutive properties of this phenomenon on the basis of cross-linguistic data presented in the literature. In a second step, we want to present evidence suggesting that a system of multiple complementizers is present in the historical stages of (High and Low) German and English discussed.

3.1 Cross-linguistic evidence

There is literature (mainly focusing on complement clauses) that provides evidence for complementizer doubling in varieties of Italian, including historical ones (Paoli 2003, Ledgeway 2005, Munaro 2016), and also in French (Dagnac 2012), Spanish (Villa-García 2012, González i Planas 2014), Portuguese (Mascarenhas 2007), Slovenian (Plesničar 2020).

From a Germanic perspective, recent studies have shown that in Cimbrian there are specialized complementizer positions for the declarative complementizers *az* and *ke*, which respectively select the subjunctive and the indicative (cf. for instance Bidese, Padovan & Tomaselli 2012, 2013, Bidese 2017).

Evidence for complementizer doubling in Old Italian and Modern Portuguese is provided in examples (10) and (11) respectively:

- (10) ... però vi priegho in lealtade e fede **che**, se ttue vuoi del mio avere,
 but you pray in loyalty and faith that if you want of my have
 che ttu ne tolghi.
 that you CL take
 ‘... therefore I ask you in loyalty and faith that, if you want my
 belongings, that you take some’
 (Libro della distruzione di Troia, p.155, ll. 26–27, from Munaro 2016: 215)
- (11) Acho **que se** lhe ligasses **que** tudo se resolveria.
 think that if him/her called that all itself solve
 ‘I think that if you called him/her everything would turn out fine.’
 (Mascarenhas 2007, in Munaro 2016: 219)

Examples (12) and (13) show the different selectional properties of the Cimbrian declarative complementizers *az*, which is of Germanic origin, and *ke*, which

4 Verbal mood in OHG, Old Saxon and Old English adverbial clauses

is borrowed from Italian (cf. [Panieri et al. 2006](#)). As commented above, the first selects the subjunctive, whereas the second selects the indicative. This piece of evidence patterns nicely with the data observed for Early Germanic: it is not only the type of clause which plays a role in the selection of mood, but the choice of mood is also determined by the complementizer (cf. [Coniglio, Hinterhölzl & Petrova 2018](#)):

- (12) I bill/sperar, *azz*=ar net au=höar zo spila.
 I want/hope that=he not PART=stops.SUBJ to play
 ‘I want /hope that he will not stop playing.’
 (adapted from [Bidese 2017](#): 134)

- (13) si khött, *ke* dar höart nèt au zo spila.
 she says that he stops.IND not part to play
 ‘She says that he will not stop playing.’
 (adapted from [Bidese 2017](#): 135)

The complementizer (*az*) is selected by a verb of volition (*want*) and a fictional verb (*hope*), respectively, while *ke* is licensed in the complement of a factive verb (*say*). The distribution of the subjunctive and the indicative correlates with the (non-)veridicality of the embedded proposition. Furthermore, as is evident from examples (12) and (13), the finite verb targets a higher position in *ke*-clauses. In example (13), in fact, it precedes the negation, whereas it follows it in the *az*-clause. This evidence is suggestive of a syntactic analysis in which *ke* occupies a higher position in the CP-layer than *az*; we will elaborate on this point in Section 4.

Having seen that there is cross-linguistic evidence within Germanic and Romance for both complementizer doubling and for specialized complementizer positions, the question arises as to whether we can find comparable evidence for the Early Germanic languages investigated in this paper. This question will be explored in the next subsection.

3.2 Evidence for multiple complementizer positions in OHG and OE

In OHG and OS, we find examples like (14) and (15), which display properties constitutive of complementizer doubling in the cross-linguistic evidence shown in (10) to (13). E.g., in OHG (14), the complementizer *thaz* ‘that’ appears both above and below a relative clause integrated into the adverbial clause. Similarly, in OS (15), the question adverb *huô* ‘how’ introducing a dependent interrogative clause is repeated below the intervening two adverbial clauses:

M. Coniglio, Ch. De Bastiani, R. Hinterhölzl & S. Petrova

- (14) **thaz** [thar ih bin] **thaz** ir sit
 that where I am that you are.SUBJ
 ‘that you may also be where I am’
 (DDD-AD-Tatian_1.1 > T_Tat162 (edition 60 - 70))
- (15) Hôriad nu **huô** thie blindun, [sîður im gibôtid uuarð, that sie
 listen now how the blind after them healed was that they
 sunnun liocht gesehen môstun], **huô** si thô **dâdun**.
 sun.GEN.SG light see.INF were.able how they then did.IND
 ‘Listen how the blind people behaved after they were healed such that
 they could see the light of the sun.’
 (DDD-AD-Heliand_1.1 > Hel_44 (edition 632 - 659))

The example in (16), taken from OHG, also displays complementizer doubling, and in addition, it shows a difference in the graphical realization of the higher and the lower complementizer:

- (16) in duom quam ih in thesa uueralt **thaz** [thie dar ni
 in judgement came I in this world in.order.that those PART NEG
 gisehent] **gisehen**, in [thie dar gisehent] **daz** sie **sin** blinte.
 see.3PL.IND see.SUBJ and those PART see.IND that they are.SUBJ blind
 ‘I came to judge, in order that those who do not see become seeing and
 those who see become blind.’
 (DDD-AD-Tatian_1.1 > T_Tat133 (edition 79 - 89))

Evidence of this type is infrequent in OHG and OS. However, Petrova (2020: 271) discusses data from the Middle High German (MHG) period, retrieved from the *Referenzkorpus Mittelhochdeutsch*, in which the complementizer occurs both above and below a sentential constituent that is used as a modifier of an XP within the complement clause (17):

- (17) Unf faget ðch div heilige schrift. **daz** zeder wile [do adam **daz**
 us tells also the holy scripture that at-the time when Adam the
 uerboten obiz eze]. **daz** zû der felben wile der heilige krift. an
 forbidden fruit eat.SUBJ that at the.DAT same time the holy Christ on
 dem heiligen krivce mit gallin unt mit ezziche gitrenkit würde.
 the holy cross with gall and with vinegar drunk was
 ‘The Holy Scripture also tells us that at the time when Adam ate the
 forbidden fruit, the Holy Christ was given to drink gall and vinegar on
 the Holy Cross.’
 (13_1-bairalem-PV-G > M165-G1)

4 Verbal mood in OHG, Old Saxon and Old English adverbial clauses

In contrast to the OHG data in (14) and (16), the complementizer *daz* ‘that’ introduces an argument clause, but the formal properties constitutive of complementizer doubling are still in place. This data shows that complementizer doubling is attested throughout the early history of High German.

The presence of complementizer doubling is corroborated by examples involving left dislocation and adverbial resumption in *that*-clauses found in MHG, see (18). Petrova (2020: 272), who presents this data, interprets the word order in the *that*-clause in (14) by assuming that the verb is fronted to the empty position of the lower complementizer, while the higher complementizer is overtly present.

- (18) Unf faget ðch div heilige fchrift. **daz** zeden felben ftunden. do der
us tells also the holy scripture that at-the same hours when the
erfte man gifchaffen wart [...] zeder felben wile **wart** der gotif fûn
first man created was at-the same time was the God’s son
enphangen uon miner frwen.
conceived by my Lady
‘The Holy Scripture also tells us that at the same time when the First Man
was created [in Paradise], the Son of God was conceived by my Lady.’
(13_1-bairalem-PV-G > M165-G1)

We assume that, in the data discussed so far, the positions occupied by these complementizers are Force and Fin as two functional projections in the C-layer of the clause, as proposed by the seminal approach of Rizzi (1997) on the structure of the clausal left periphery. But we have reason to assume that there is an even higher position above Force (cf. also Section 4). See (19), in which the causal complementizer *wanta* ‘because’ is above the complementizer of the conditional clauses *oba* ‘if’. The same applies to the use of *wanta* as a declarative complementizer ‘that’, see (20). The data in (21) suggests that *wanta* ‘because’ also allows for verb fronting in the respective clause.⁶

- (19) **Uanta oba** in Sodomu uuarin gitanu megin thi u thar gitanu
because if in Sodom were.SUBJ done miracles which PART done
sint in thi, odouuan uuonetin unzan thesan tag
are in you perhaps last.SUBJ until this day
‘because if these miracles that were done in you were also done in Sodom,
they would perhaps still last until today’
(DDD-AD-Tatian_1.1 > T_Tat65 (edition 79 - 94))

⁶Similar serialization patterns involving different types of complementizers in the left periphery of the same clause have been observed for Italian (cf. Rizzi 2001 and Benincà 2001, 2006, *inter alia*).

M. Coniglio, Ch. De Bastiani, R. Hinterhölzl & S. Petrova

- (20) Abur quidu ih iu, **uanta oba** zuuene fon íu gizuftigont sih
again tell I to.you that if two of you unite themselves
obar erdu fon eogilicheru rachu, so uuelichu so sie pittent,
above earth of any thing whatever they ask
uuesent in fon minemo fater ther dar in himile ist.
are.IND to.them by my father who PART in heaven is
‘I am telling you again that if two of you unite above the earth because of
anything, whatever they ask, it will be given them by my father.’
(DDD-AD-Tatian_1.1 > T_Tat98 (edition 126 - 146))
- (21) ther der in mir uuonet inti ih in imo, ther birit mihilan uuahsmon,
DEM which in me live and I in him DEM brings much growth
uanta uzzan mih ni mugut ir niouuiht duon.
because without me NEG can you nothing do
‘whoever lives in me, and I in him, will bring an abundant harvest,
because without me you cannot do anything.’
(DDD-AD-Tatian_1.1 > T_Tat167 (edition 103 - 118))

Similar data is found in OE as well, where the causal complementizer *for ðan* precedes the conditional complementizer *þeah*:

- (22) For nahte he tealde ænig þincg to biddenne buton gesihþe.
for naught he accounted any thing to bid except sight
for ðan þeah se blinda sum þinc **hæbbe**: he ne mæg buton
because though the blind some thing has.SUBJ he not may without
leohte geseon þæt he hæfð.
light see what he had
‘For naught he accounted to ask for anything except for eyesight.
Because, though the blind man had something, he could not see what he
had without light.’
(cocathom1.o3, ÆCHom_I_10:262.107.1900)

We also searched the data for diagnostic evidence suggesting that the higher C-domain is activated in the respective clauses. A set of data displaying such evidence involves embedded clauses in which there is contrastive information realized above nominative subjects. The methodological consideration is that there must be a reason why the higher C-domain is activated in such cases. In fact, we find such data, see (23) and (24):

4 Verbal mood in OHG, Old Saxon and Old English adverbial clauses

- (23) Únde iz náhtê. êr an hîmele stêrnen skînen
 and it becomes.night before on sky stars shine.SUBJ
 ‘and it becomes night before the stars are shining in the sky’
 Lat. *Ac nox funditur desuper in terram. nondum uenientibus astris cêlo*
 (N_DeCon_I_13-15, ReA)
- (24) Pæt fîfte wite wæs cwealm on heora orfe. swa þæt on ðam lande
 the fifth plague was slaughter on their cattle so that on the land
 fornean nan orf ne belaf. buton Israheles þe ansund
 almost no cattle not remained.IND except Israel.GEN that sound
 gestod;
 stood
 ‘The fifth punishment was death on their cattle, so that in that land there
 remained almost no cattle, except for that of Israel, which remained
 untouched.’
 (cocathom2.o3, ÆCHom_II_12.1:112.64.2418)

In (23), the constituent preceding the subject, the adverbial *an hîmele* ‘in the sky’ forms a contrastive pair with the adverbial *in terram* ‘on earth’, which is contained in the Latin original of the example but is left unexpressed in the OHG translation. In the OE example, the PP *on ðam lande*, located before the subject, is contrastive to the PP *buton Israheles*, which follows the finite verb. Concluding this section, we have seen that, although not frequent, there is evidence for complementizer doubling in OHG, OS, MHG and OE; the findings pattern with cross-linguistic evidence for complementizer doubling and different complementizer positions in Romance and other Germanic varieties. It was moreover shown that there is reason to assume that complementizers occupy different positions in the C-layer. Finally, we argued on the basis of the evidence provided by examples such as (19)-(21) that a higher position above Force can be argued for and we illustrated diagnostic evidence that shows that the higher C-domain is activated in embedded clauses.

4 Towards an explanatory analysis: mood and temporal anchoring

In this section, we will combine the cross-linguistic evidence for various complementizer positions with semantic theories of mood alternations by developing

M. Coniglio, Ch. De Bastiani, R. Hinterhölzl & S. Petrova

a structural account of mood licensing that links the definite and indefinite nature of indicative and subjunctive marked clauses to specific positions in the C-domain.

4.1 Previous analyses and their shortcomings

Coniglio, Hinterhölzl & Petrova (2018: 30) developed a purely syntactically oriented account, claiming that an Agree operation affecting different heads in the clausal spine (a C-head and a functional projection devoted to mood) is responsible for the licensing of mood alternation in adverbial clauses (cf. Ledgeway 2012 for a similar approach). They follow Giannakidou's view in arguing that (non-)veridicality plays a crucial role in the licensing of mood alternations in adverbial clauses (cf. Giannakidou 2009 et seq.). This approach is illustrated in (25) and (26), where it is assumed that the selection of IND/SUBJ is due to a lexical semantic property of the embedding conjunction occupying a C-head (cf. Zeijlstra 2012 for the notion of Upward Agree adopted here).

- (25) [CP₁ ... [CP₂-Adv C[iVerid] ... Mood [iInd] [uVerid] ... V[uInd]] *Agree*
 [CP₁ ... [CP₂-Adv C[iVerid] ... Mood [iInd] [uVerid] ... V[uInd]] → IND
- (26) [CP₁ ... [CP₂-Adv C[i¬Verid] ... Mood [iSubj] [u¬Verid] ... V[uSubj]] *Agree*
 [CP₁ ... [CP₂-Adv C[i¬Verid] ... Mood [iSubj] [u¬Verid] ... V[uSubj]] → SUBJ

This analysis is in need of refinement. While it captures very well the fact that different semantic types of adverbial clauses license mood alternations and that the types of the subordinating conjunctions alone are sometimes able to determine the verbal mood (cf. Rivero 1988, Calabrese 1993, Damonte 2010, Ledgeway 2005, 2012, Giannakidou 2009, Padovan 2011), it does not take into consideration that sometimes the same adverbial conjunction is compatible with the indicative and the subjunctive and it also does not consider cross-linguistic facts showing that subordinating elements occupy different positions in the C-layer. Moreover, recent theories proposed for other Germanic and non-Germanic languages (which mainly concentrate on complement clauses) assume that the subordinating elements of adverbial clauses target different head positions in the C-layer (cf. Bhatt & Yoon 1991, Roussou 2000, Grewendorf & Poletto 2011, Bidese, Padovan & Tomaselli 2012 a.o.).

Since we have seen evidence for different complementizer positions in OE, OS and OHG, we propose that different lexical types of complementizers target/are merged in different complementizer slots/projections which are endowed with different features and linked to different selectional properties. As we have seen

4 Verbal mood in OHG, Old Saxon and Old English adverbial clauses

in Section 3, there is evidence for two possible clause-internal complementizer positions and for a clause external position, as is illustrated in (27).

(27) $[_{\pi P} COMP_1 \dots [_{HT(?) } \dots [_{ForceP} COMP_2 \dots [_{ContrP} \dots [_{Mood/FinP} COMP_3 \dots [_{VP} \dots]]]]]$

The highest complementizer is a pure marker of dependency and has no selectional properties within the adverbial clause. For the internal ones, we will explore the possibility that the indicative and the subjunctive are licensed in different C-positions: namely that the intermediate complementizer licenses the indicative, while the lowest complementizer licenses subjunctive responding to a non-veridical or non-factual interpretation. Table 4 illustrates some examples of complementizers occupying different syntactic positions in selected languages.

Table 4: Types of complementizers in selected languages

	COMP ₁	COMP ₂	COMP ₃
OHG/OS/OE	<i>wanta/for ðan/...</i>	<i>do/ða</i>	<i>doh/ðeah</i>
Cimbrian	<i>ke</i>		<i>az (ke)</i>
Italian	<i>(per)ché</i>	<i>che</i>	<i>(che)</i>
etc.			

In the following subsection, we will motivate the relevance of these two positions for mood licensing by relating them to differences in the temporal anchoring of the subjunctive and of the indicative.

4.2 Temporal anchoring

It is well known that main clauses are temporally anchored via the utterance time. We follow Reichenbach (1947) and Klein (1994) in distinguishing two types of temporal relations, one between the reference time, also called the topic time, and the utterance time (tense), and one between the topic time and the time for which the verbal event holds (aspect). In the standard account, tense is analyzed as a predicate on points in time or intervals (cf. Stowell 1995 and much subsequent work). Our proposal rests on the assumption that tense relates situations (cf. Hinterhölzl 2023). In situation semantics (cf. Kratzer 2019), tense expresses a relation between the utterance situation and the Austinian topic situation (cf. Austin 1961, Barwise & Perry 1983). This topic situation, which we identify with the reference situation – we argue – is introduced as an argument of the tensed predicate. Thereby, tense specifies that the reference situation, i.e. the situation

M. Coniglio, Ch. De Bastiani, R. Hinterhölzl & S. Petrova

which the sentence is about, precedes (past tense), follows (future tense) or overlaps (present tense) with the utterance situation. Aspect then has the role to situate the verbal event with respect to the reference situation.

An embedded clause on the other hand is primarily anchored with respect to the matrix event, as is illustrated in (28). Here, the *leaving*-event is predicated to have occurred before (= past tense) the *saying*-event.

(28) *John said that Peter (already) left.*

However, embedded clauses are also anchored with respect to the utterance situation, at least if the embedded predicate is marked with the indicative mood, as the contrast between (29a) and (29b) indicates.

- (29) a. *John said (ten days ago) that Peter will come *yesterday / tomorrow.*
 b. *John said (ten days ago) that Peter would come yesterday / tomorrow.*

While the future tense with the indicative mood indicates that the *coming*-event is to follow both the *saying*-event and the utterance situation, the future tense with the subjunctive mood only indicates that the *coming*-event follows the *saying*-event. The former is also called the double-access reading (cf. Giorgi & Pianesi 1997, Stowell 2007) since the embedded event is both anchored with respect to the matrix event and to the utterance situation, while a tensed embedded verb marked with the subjunctive mood is only anchored to the matrix event.

Let us assume that the same distinction applies to adverbial clauses. In the scope of this paper, we can only do that by way of discussing a limited number of cases. For the sake of the argument, we will thus concentrate on the difference between *before*-clauses and *after*-clauses. Remember that *before*-clauses show variation between the indicative and the subjunctive mood in our three languages of comparison, while *after*-clauses only appear with the indicative mood in OE, OS and OHG.

Let us first discuss the interpretation of a *before*-clause and take as an example a host clause with a past tense predicate and a *before*-clause that is also marked with past tense. If the *before*-clause is marked with the subjunctive mood, its reference situation will only be anchored with respect to the reference situation of the host clause. Its past tense will be interpreted as an agreeing form (with its antecedent) that is semantically vacuous, as is typical of tense in subjunctive clauses (cf. Giorgi & Pianesi 1997). Thus, the reference situation of the adverbial clause will be predicated to follow the reference situation of the predicate in the host clause leaving open whether the event denoted by the *before*-clause occurs before or after the utterance situation. Hence, it is completely unspecified

4 Verbal mood in OHG, Old Saxon and Old English adverbial clauses

whether the adverbial event is factual, that is to say, has occurred at the time of the utterance, or is non-factual because it may occur (or not) after the time of the utterance. In any case, the event denoted by the *before*-clause is unrealized at the point of time that corresponds to the reference situation of the host clause. Both meaning effects constitute typical conditions that have been reported to require/license the subjunctive mood.

If the *before*-clause is marked with the indicative mood, it will be predicated to follow the event of the host clause, but it will also be predicated via its past tense morphology to occur before the utterance situation contributing to a factual interpretation of the adverbial event.

Things are crucially different with *after*-clauses. Note, in particular, that even if the *after*-clause were only temporally anchored to the host clause that refers to an event in the past, it will always have a factual interpretation since it is predicated to precede that past event of the host clause. Thus, a distinction between subjunctive and indicative marking in *after*-clauses in combination with a host clause in the past does not give rise to any difference in meaning. We may therefore assume that the morphological distinction is rendered obsolete in this case by the principle of economy, allowing only for the more unmarked indicative mood to occur in these cases.

4.3 Formal properties of the C-layer

In this section, we discuss the implications of the above considerations for a structural analysis of adverbial clauses marked with the indicative and the subjunctive mood. From the above characterization of the role of indicative morphology, it follows that a verb marked with the indicative mood must have access to the utterance situation, while this is not necessary or excluded for verbs marked with the subjunctive mood. We propose that access to the utterance situation involves the presence of an additional head in the C-domain that is present in indicative clauses, but lacking or not activated in subjunctive clauses.

In other words, certain semantic classes of adverbial clauses will only select for the subjunctive mood, since their meaning contribution does not require access to the utterance situation, for instance because they introduce an alternative situation as is typical for purpose clauses and, by the principle of economy, they will only appear with the minimal structure possible.

Other semantic classes of adverbial clauses, like *after*-clauses, only select for the indicative mood, since they require access to the utterance situation to license their intrinsic factual interpretation. A third semantic class of adverbial clause, as *before*-clauses, allows for the selection for both moods depending on their

M. Coniglio, Ch. De Bastiani, R. Hinterhölzl & S. Petrova

factual/non-factual interpretation which – as we argued above – is connected to the presence or absence of an additional functional head in the C-domain. It is thus plausible that a language distinguishes between different tokens of the same type of conjunction to lexicalize the higher or the lower functional head in a distinct way. This constitutes a case of division of labor in which a particular lexical subordinator within a semantic class of adverbial clauses specializes for the use of one type of mood exclusively. The case of *before* in this scenario then constitutes an alternative solution of inserting the same subordinator in different C-positions.

In particular, we propose that a *before*-complementizer relates two situations, as is given in (30). s_1 will then be identified with the reference situation of the host clause, but the attachment site of the subordinator *before* will crucially depend on the nature of its internal argument s_2 . In an indicative-marked adverbial clause, the internal argument corresponds to a definite situation that has been located with respect to the utterance situation, while in a subjunctive marked adverbial clause, the internal argument corresponds to a set of situations.

(30) *before* (s_1, s_2) := s_1 precedes s_2

In other words, we propose that an indicative-marked adverbial clause involves a definite reference situation, while a subjunctive marked adverbial clause involves an indefinite reference situation. We know that in general definite NPs move to higher positions in the clause with respect to indefinite ones, since the former need to be assigned/get access to contextual values. We propose that contextual values are assigned in FinP, hence a definite reference situation argument has to move to FinP, while an indefinite reference situation argument can be licensed *in situ*, that is, in [Spec, TP].

This take constitutes a first step towards a combined semantic and syntactic analysis of the representation of mood in language and we will leave the exact formal implementation to future research. In conclusion, *before* selecting the subjunctive mood can be inserted in the lower complementizer position, that is, in Fin^0 , and the structure above it will remain inert or be absent and will only be activated for other reasons, like licensing different information structural categories, as we have seen in (23) above. In contrast, *before* selecting the indicative cannot be inserted in the lower complementizer since a definite situation argument will undergo licensing movement to [Spec, FinP] and the lexical subordinator that takes it as argument must thus be inserted in a higher position, that is, Force^0 . To summarize, the use of the indicative always indicates the presence of the illocutionary layer, while the use of the subjunctive typically goes together with the

4 Verbal mood in OHG, Old Saxon and Old English adverbial clauses

presence of the propositional layer only. However, if the illocutionary layer is activated by information-structurally motivated movement types, the subjunctive subordinator is moved to the higher complementizer position. The most important insights and structural claims of the paper are summarized in (31).

- (31) [πP COMP₁ [ForceP COMP₂ [ContrP [FinP COMP₃ [IP...]]]]]
 (because) before.IND before.SUBJ
 discourse layer illocutionary layer propositional layer

5 Conclusions

Starting from the consideration that a comparative investigation of mood alternations in the Germanic languages is a desideratum, we provided first insights by means of a corpus-based investigation of mood in adverbial clauses in OHG, OS and OE.

As already argued in previous literature, there is a correlation between the type of adverbial clauses and (non-)veridicality, but the corpus investigation has also highlighted another factor that plays a role in the choice of mood: the type of lexical subordinator. In fact, within the same type of adverbial clause, notable differences in choice of mood can be observed and related to the type of lexical complementizer.

The question then arose as to what triggers the variation observed and it was hypothesized that different complementizers target different positions in the C-layer, thereby activating the propositional, the illocutive and discourse layers respectively. Support for this hypothesis is given by cross-linguistic evidence involving complementizer doubling and different types of complementizers in Romance and Germanic languages. It was shown that the languages examined display both evidence for complementizer doubling and for the activation of the higher C-layer, when for instance a contrastive element follows the complementizer.

The empirical data is therefore interpreted by assuming three different heads in the C-layer that are targeted by the complementizers and which in turn correlate with the choice of mood, based on whether an adverbial clause is anchored to the utterance situation or to the situation denoted by the matrix verb which in turn largely correlates with (non-)veridicality. As an illustration of this approach, we discussed the process of temporal anchoring relevant for the subordinating conjunctions *after* and *before*. We argued that indicative mood indicates that the adverbial clause is anchored with respect to a definite situation, namely the utterance situation, while the subjunctive mood indicates that the adverbial clause

M. Coniglio, Ch. De Bastiani, R. Hinterhölzl & S. Petrova

is anchored with respect to a set of situations located with respect to the event denoted by the matrix verb.

As a first comparative investigation on mood in older West Germanic, this paper provides new insights into mood alternation in general, and opens up new avenues for future research. In fact, one question to ask is related to developments in mood licensing in the history of West Germanic. As is well known, verbal morphology is quickly eroded in the languages under investigation, and periphrastic forms with modal verbs arose and took up some of the functions expressed by the subjunctive mood. In addition, there were also substantial changes regarding the lexical inventory of adverbial subordinators in the later stages of the languages. It should therefore be asked, how these different diachronic changes affected the licensing of verbal mood illustrated in Section 4, facing a multi-factorial scenario. These questions have to be left for future research.

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Chapter 5

OCP effects in Germanic possession

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Since its original proposal as a phonological constraint, the Obligatory Contour Principle (OCP) has been extended under various guises to morphosyntactic phenomena (e.g. Menn & MacWhinney 1984, Mohanan 1994, Yip 1998, van Riemsdijk 2008, Richards 2010, Nevins 2012). In this paper, we show how a morphosyntactic OCP in the nominal domain can account for data from Germanic possessive constructions, in particular the Possessor Linking Construction (e.g. *dem Mann sein Haus*, ‘the man his house’). Based on ordering restrictions which are exhibited in this construction across present-day Germanic varieties, we argue that the possessive element in the PLC (*sein*) serves as a linker (Franco et al. 2015) which occurs between possessor and possessum in order to prevent an OCP violation. On the basis of data from Middle Low German (c. 1250–1600) and Modern Alemannic, we show how this account can also be extended to possession more generally, including the adnominal genitive and prepositional possessive. We also address the way in which these multifunctional linkers establish the possessive relation, drawing on the notion of anchoring as per Wiltschko (2014).

1 Introduction

The Possessor Linking Construction (PLC) in (1) is widespread across Germanic varieties past and present and has attracted a good deal of attention in the syntactic literature on Germanic possession (Georgi & Salzmann 2011, Haegeman 2004, Kasper 2014, Roehrs 2019, Weiß 2008, 2012).¹

¹The phenomenon we dub the “Possessor Linking Construction” is generally known as “possessor doubling”, due to the realisation of a possessor-DP in addition to the possessive element

- (1) a. Alemannic (SynAlm)²
 des isch [dem Vater **sin** Platz]
 this is the.MASC.SG.DAT father his seat
 ‘this is father’s seat’
- b. Middle Low German (CHLG: Lauremberg)³
 men wat in [disses Mans **syn** Hovet] is begrepen
 but what in this.GEN man.GEN his head is understood
 ‘but whatever in this man’s head is understood’

Despite the extensive literature on the PLC, there are two outstanding issues which emerge from previous accounts. Firstly, there is no consensus on the categorial status and function of the bold-faced element in the examples in (1) (*sin/syn*, henceforth “possessive element”). In most previous analyses, the possessive element is assumed to be a possessive article in D (e.g. [Schoorlemmer 1998](#), [Demske 2001](#), [Weiß 2012](#); [Roehrs 2019](#): 102), but possessive elements in German have also been analysed as adjectives ([Löbel 1996](#), [Leu 2008](#)) and as pronouns ([Olsen 1996](#), [Georgi & Salzmann 2011](#)).⁴

The second unresolved issue concerning the PLC relates to an ordering restriction. Specifically, none of the previous analyses explicitly address the ordering exhibited by the PLC, whereby the possessor must precede the possessum, e.g. (2).⁵

(2) German

sein (‘his’) (e.g. [Haegeman 2004](#), [Norde 2012](#)). Another common label is “possessive dative” (e.g. [Kasper 2016](#)), as the possessor-DP is dative-marked in most German varieties and the diachronic origin is assumed to be an external possessive construction (cf. the discussion in [Fleischer & Schallert 2011](#), as well as [Weiß 2012](#) who explicitly rejects this origin).

²References to SynAlm throughout refer to Alemannic data which was collected as part of the Syntax des Alemannischen project (SynAlm, [Brandner 2015](#)). The data are available at: <https://ilg-server.ling.uni-stuttgart.de/synalm/html/de/index.html>.

³Middle Low German examples throughout either stem from the Corpus of Historical Low German (CHLG, [Booth et al. 2020](#)), or the Referenzkorpus Mittelniederdeutsch/Niederrheinisch (1200–1650) (ReN, [ReN-Team 2021](#)).

⁴A reviewer asks about the impact of the diachronic change of the possessive element *sein*, which developed from a pronominal element via a possessive adjective to a possessive determiner (cf. [Weiß 2012](#), following [Demske 2001](#)). While it is true that in Middle High German possessive elements could co-occur with determiners (indicating adjectival status), there are – at least to our knowledge – no PLCs in which this is attested, pointing towards the determiner status of *sein* in this construction already at this early stage.

⁵Norwegian appears to be an exception here, allowing for two variants of the PLC, with different orderings of possessor and possessum:

5 OCP effects in Germanic possession

- a. dem Mann sein Haus
the.DAT Mann his house
'the man's house'
- b. * sein Haus dem Mann
his house the.DAT Mann
Intended: 'the man's house'

The possessor-possessum ordering restriction on the PLC is particularly striking given that this restriction does not hold in the same way for other adnominal possessive constructions, e.g. adnominal genitive, see (3), and the prepositional possessive construction, see (4).

(3) German

- a. das Haus des Mannes
the house the.GEN man.GEN
'the man's house'
- b. des Mannes Haus
the.GEN man.GEN house
'the man's house' (archaic)

(4) German

- a. das Haus von dem Mann
the house of the.DAT man
'the man's house'
- b. von dem Mann das Haus
of the.DAT man the house
'the man's house'

In this paper, we present a novel account for the Germanic PLC which accounts for the positional restriction in (2) and which also sheds light on the categorial status of the possessive element. In this account, the possessive element

-
- | | |
|---|---|
| (i) a. mannen sitt hus
man.DEF REFL house
'the man's house' | b. huset hans Per
house.DEF his Per
'Per's house' |
|---|---|

However these different orderings occur with different possessive elements which are available in this language (reflexive *sitt* and non-reflexive *hans*), a distinction which is not relevant for German varieties.

Alexandra Rehn & Hannah Booth

in the PLC serves as a linker (e.g. [Franco et al. 2015](#)), i.e. a functional element inserted between nouns and their dependents (i.e. modifiers or complements) in order to prevent a morphosyntactic OCP (Obligatory Contour Principle) violation (e.g. [Yip 1998](#), [Richards 2010](#), [Nevins 2012](#), [Neeleman & Van de Koot 2017](#)).

More broadly, we show how these observations can inform a more general account of adnominal possession (including the adnominal genitive and the prepositional possessive). This is supported by novel diachronic data from Middle Low German (MLG, c. 1250–1600) as well as synchronic data from Modern Alemannic and other present-day German varieties. Both Middle Low German and Modern Alemannic offer interesting studies in relation to possession. MLG represents a language stage during which the gradual erosion of case is underway, accompanied by shifting preferences between a range of possessive constructions. Moreover, no detailed research on the PLC in MLG has been conducted to date, despite the fact that it looms large in the literature on language contact as the generally acknowledged source of the Mainland Scandinavian PLC (e.g. [Norde 2012](#)). Crucially, the recent release of two corpora of MLG, the Corpus of Historical German (CHLG, [Booth et al. 2020](#)) and the Referenzkorpus Mittelniederdeutsch/Niederrheinisch (1200–1650) (ReN, [ReN-Team 2021](#)) make data for the language readily available for syntactic studies of this kind. Modern Alemannic similarly offers an interesting case study in this context, since it no longer has productive genitive case-marking and also offers a range of non-standard possessive constructions, as we discuss below. Relevant Alemannic data is available via the SynAlm project ([Brandner 2015](#)), which collected data on various morphosyntactic phenomena via questionnaires across the Alemannic region.

The paper proceeds as follows. In Section 2, we outline the types of adnominal possessive construction which are attested in Germanic, as well as the critical properties of the Possessor Linking Construction, including its ordering restrictions. Section 3 discusses aspects of the OCP in relation to morphosyntactic phenomena, as well as the crosslinguistic properties and status of linkers, which is essential background for our account of Germanic possession. In Section 4, we show how data from a range of possessive constructions in present-day German varieties, as well as diachronic data from Middle Low German, can be accounted for in terms of a morphosyntactic OCP constraint, and in Section 5 we explore how this proposal can be formalised in the context of previous accounts of Germanic possession. Section 6 concludes the paper.

2 Adnominal possession across Germanic

2.1 Types of adnominal possession construction

“Adnominal possession” is a cover term for possessive constructions in which both the possessor and the possessum are realised in a complex DP. The Germanic languages have various adnominal constructions available to express possession. The adnominal genitive, e.g. (5a), the prepositional possessive, e.g. (5b), and the PLC, e.g. (5c), are three productive constructions in German, though with differences regarding the varieties in which they are attested. While the adnominal genitive and the prepositional possessive both belong to the standard varieties, the PLC is only productive in non-standard German varieties.⁶

- (5) a. standard German
 das Haus des Mannes
 the house the.GEN man.GEN
 ‘the man’s house’
- b. standard German
 der Geschmack *(von) Äpfeln
 the taste of apples.DAT
 ‘the taste of apples’
- c. non-standard German
 dem Mann sein Haus
 the.DAT man his house
 ‘the man’s house’

Across the full range of present-day Germanic, an even wider variety of adnominal possessive constructions can be identified, though again with variation in terms of the precise varieties they are attested in. Table 1 provides an overview, based mainly on Delsing 1998; for a more detailed overview see Dahl 2015, particularly with respect to North Germanic varieties. Note that the specific focus of this paper, the PLC (shown in bold in Table 1), is widely attested across both West and North Germanic.

⁶The prepositional possessive in (5b) is in standard German restricted to contexts where an adnominal genitive construction is ruled out. Thus, in standard German the example in (i) is ruled out, whereas in non-standard varieties which have lost the genitive it is grammatical.

- (i) non-standard German
 das Haus von dem Mann
 the house of the.DAT man
 ‘the man’s house’

2.2 Properties of the Possessor Linking Construction

As already noted, the PLC is widely attested across Germanic varieties and the case-marking and agreement properties of the possessive element vary depending on the Germanic variety. With respect to case-marking, a relevant factor is the general Germanic trend towards a decline of (adnominal) genitive, which is observed to different degrees across the various varieties (Ackermann et al. 2018). Standard German has resisted this trend and adnominal genitive is still productive (Weinrich 2005), while most German dialects have virtually lost genitive and exhibit a three-case system (Kiefer 1910, Fleischer & Schallert 2011, Kasper 2014), with the exception of certain Highest Alemannic varieties spoken in Switzerland which have preserved genitive (Kasper 2014). Case erosion has proceeded even further in Low German and some Bavarian varieties, resulting in a two-case system. This backdrop is relevant for the PLC, as the possessor is generally marked for the most oblique case available in a given variety (Weiß 2008). As such, in many German dialects, i.e. those which have a three case-system, the possessor in the PLC is marked for dative, e.g. (6a). In Alemannic, genitive-marked possessors can be observed, e.g. (6b)⁷, and in two-case systems such as Low German, the possessor is marked for non-nominative case, e.g. (6c).

- (6) a. Bavarian (Weiß 2008: 381)
 am **Sepp** sei Haus
 the.DAT Joseph his house
 ‘Joseph’s house’
- b. Alemannic (Weiß 2008: 385)
 s **Nochbors** ihre Katz
 the.GEN neighbours.GEN their cat
 ‘the neighbours’ cat’
- c. Low German (Berg 2013: 36)
 den **Pasturn** sien ole(n) Hoot
 the.NONNOM priest his old(.WK) hat
 ‘the priest’s old hat’

In terms of agreement, the possessive element generally agrees in gender and number with the possessor. When a MASC.SG possessor is present, the possessive

⁷PLCs with genitive-marked possessors are however somewhat exceptional in Alemannic, and can be considered a remnant of an earlier stage (Middle High German) when the construction became productive, as shown in Weiß (2012) (see also Booth & Rehn 2023).

5 OCP effects in Germanic possession

Table 1: Adnominal possessive constructions across Germanic

construction	example	distribution
POSSESSIVE PRONOUN ^a	<i>hus-et mitt</i> house-DEF my	Norw., Ice., Northern Swe.
	<i>mitt hus</i> my house	all Gmc. languages (North Scand.,
ADNOMINAL GENITIVE	<i>das Haus des Mannes</i> the house the.GEN man.GEN	emphatic) Standard Ger., Ice.
	<i>mannen-s hus</i> man-GEN house	Eng., Standard Dan., (WJut.,
s-POSSESSIVE ^b	<i>Mutter-s Haus</i> mother-POSS house	Dutch/Ger. dialects) Ice. proper names, Ger./Far. also
	<i>dem Mann sein Haus</i> the.DAT man his house	kinship terms Norw. dialects, Dan. (WJut.),
PLC	<i>hus-et hans Per</i> house-DEF his Per	Dutch/Ger. dialects North Swe., Norw., Ice. (proper
	<i>hus-et till Per</i> house-DEF to Per	names/kinship all Gmc. languages terms) except Dan., Swe., Ice.

^a“Possessive pronoun” is a commonly used term for possessive elements like “my/your/his”, although there are crosslinguistic differences with respect to their categorial status. [Schoorlemmer \(1998\)](#) provides a crosslinguistic overview of possessive elements which show article-like behaviour (e.g. Germanic) and those which are adjective-like (e.g. Romance).

^bThe s-possessive features a grammaticalised possessive marker derived from the MASC/NEUT.GEN suffix, which can occur on prenominal feminine possessors.

Alexandra Rehn & Hannah Booth

element is realised as *sein* (MASC.SG), e.g. (6a). With a plural possessor, the possessive element is realised as *ihr* (PL), e.g. (6b), and with a FEM.SG possessor the possessive element is realised as *ihr* (FEM.SG), e.g. (7).

- (7) non-standard German
 der Frau **ihr** Haus
 the.DAT woman her house
 ‘the woman’s house’

In some varieties, however, e.g. Alemannic, it is also possible for some speakers to use *sein* (MASC.SG) with a FEM.SG possessor, e.g. (8a).⁸ This type of variation is also found in Dutch (Piepers & Redl 2018), where masculine and feminine possessors can both occur with the possessive element *zijn* (MASC.SG), e.g. (8b).⁹

- (8) a. Alemannic (SynAlm)
 de Oma **sin** Schurz
 the.DAT granny his apron
 ‘the granny’s apron’
 b. Dutch (Piepers & Redl 2018: 99)
 Piet/Marie **zijn** auto
 Piet/Marie his car
 ‘Piet’s/Marie’s car’

2.3 Ordering restrictions

As already noted, the PLC is subject to an ordering restriction; unlike in the adnominal genitive and the prepositional possessive, in the PLC the possessor must precede the possessum, see (2) above. We outline this for German more fully in Table 2 for a MASC.DAT possessor (*dem Mann*), for a MASC.GEN possessor (*des Mannes*) and for a feminine possessor (*der Frau*) which is ambiguously marked for DAT/GEN due to syncretism in the paradigm. All three types of possessor must precede the possessum in the PLC. Another way to view the restrictions shown in Table 2 is in terms of the behaviour of the possessive element (*sein/ihr*): the possessive element must be placed between the possessor and the possessum, which

⁸A reviewer points out that in some varieties (e.g. Bavarian), this is the only option with a FEM.SG possessor.

⁹The PLC is standardly assumed to be restricted to the third person, cf. **mir mein/sein Haus* (‘me my/his house’)/**dir dein/sein Haus* (‘you your/his house’); since there are diverging accounts for this (e.g. Zifonun 2005, Weiß 2008) and the finer details are not relevant for our claims, we do not address this point further.

5 OCP effects in Germanic possession

is achieved via possessor-possessum ordering. In other words, it is not sufficient for the possessive relation to be marked overtly by *sein/ihr*; rather the possessive element must occur in a particular structural position. This observation will be crucial to our proposal for the categorial status and function of *sein/ihr* below.

Table 2: Ordering restrictions in the German PLC

possessor	possessum	possessor
dem Mann	sein Haus	
	sein Haus	*dem Mann
des Mannes	sein Haus	
	sein Haus	*des Mannes
der Frau	ihr Haus	
	ihr Haus	*der Frau

Another relevant observation concerns ordering restrictions in the adnominal genitive, shown here in Table 3. Although, as we have already noted, there is no ordering restriction in adnominal genitive constructions with a masculine possessor (e.g. *des Mannes*), where both possessum-possessor and possessor-possessum order is possible, with a feminine possessor (*der Frau*), the possessor-possessum order is ruled out. This is striking as, with a possessor that is a feminine proper name combined with a possessive marker *-s* (e.g. *Lenas*), both *das Haus Lenas* and *Lenas Haus* are possible.¹⁰ On the basis of data like that in Table 3, we will later extend our account of the PLC to adnominal possession more generally.

3 OCP effects and linkers in morphosyntax

In this section, we discuss the OCP specifically in relation to morphosyntax which is necessary background for our account of Germanic possession outlined in Section 4 and Section 5.

3.1 The Obligatory Contour Principle (OCP)

The Obligatory Contour Principle (OCP) was originally proposed as a phonological constraint by *Leben (1973)* to account for the observation that, “at the melodic

¹⁰The configuration *das Haus Lenas* is marked “if not ungrammatical” in *Lindauer 1998*: 125 and may only appear when certain pragmatic, semantic and syntactic conditions are met, e.g. syntactic complexity and animacy, as also pointed out in *Ackermann 2021*.

Table 3: Ordering restrictions in the German adnominal genitive

possessor	possessum	possessor
	das Haus	des Mannes
des Mannes	Haus	
	das Haus	der Frau
*der Frau	Haus	
	?das Haus	Lenas
Lenas	Haus	

level, any two adjacent tonemes must be distinct” (as formulated by Goldsmith 1976: 403, following Leben 1973). As a consequence, in a sequence of two identical tones, one is deleted in order to meet the requirement of distinctness, see e.g. (9).

- (9) (Goldsmith 1976: 403)
- a. *HHL
 - b. HL

Since this original formulation, the OCP has come to be characterised by many as a general principle of linguistic organisation which operates across different linguistic dimensions, albeit in language-specific and even construction-specific ways (Mohanan 1994, Plag 1998, Yip 1998, Mohanan & Mohanan 2003, Nevins 2012). In this context, it has been extended to account for various morphosyntactic phenomena (see e.g. Perlmutter 1971, Menn & MacWhinney 1984, Mohanan 1994, Yip 1998, Ackema 2001, van Riemsdijk 2008, Richards 2010, Nevins 2012, Neeleman & Van de Koot 2017). For instance, Menn & MacWhinney (1984) extend the OCP to morphological phenomena in terms of their “Repeated Morph Constraint”, which rules out adjacent identical morphs, in line with the observation that languages often show strategies to avoid such sequences. A relevant example is English where, for instance, both genitive and plural are marked with identical sibilant morphs (/z/), e.g. (10a) and (10b), but the GEN.PL of *girl* bears only a single case/number marker, thus avoiding repetition of /z/ which is ruled out, e.g. (10c).

- (10) English (Menn & MacWhinney 1984: 523)
- a. girl/z/
girl.GEN.SG

5 OCP effects in Germanic possession

- b. girl/z/
girl.PL
- c. girl/z/ (*girl/zəz/)
girl.GEN.PL

The example in (10) shows that **deletion** can be a relevant strategy in order to avoid an OCP violation (cf. “hapology”, Nevins 2012, Neeleman & Van de Koot 2017), but languages frequently make use of other strategies to the same end (see e.g. Zwicky 1969, Menn & MacWhinney 1984, Richards 2010, Neeleman & Van de Koot 2017 for discussion). One such strategy which is commonly employed in this context – albeit not necessarily systematically – is for instance **suppletion**, i.e. realisation of one of the relevant units as an alternative extraparadigmatic form (see e.g. Menn & MacWhinney 1984, Neeleman & Van de Koot 2017 for discussion). The suppletion strategy is commonly employed in the context of clitic clusters, as in e.g. Italian, where both an impersonal and a reflexive clitic are expected to be realised as *si*, cf. (11a); the sequence *si si* being an OCP violation, it is instead realised as *ci si*, e.g. (11b), where *ci* is a form that normally functions as the 1PL pronoun.

- (11) Italian (Bonet 1995: 609)
- a. * Si si lava.
IMPRS REFL washes
Intended: ‘One washes oneself.’
 - b. Ci si lava.
1PL REFL washes
‘One washes oneself.’

Another relevant avoidance strategy in the context of the OCP is **insertion**, e.g. of an element between the relevant units (Neeleman & Van de Koot 2017). For instance, in present-day English which lacks case morphology, insertion of the preposition *of* and the associated structure prevents an otherwise OCP violation between the two nominals, e.g. (12)

- (12) English (adapted from Richards 2010: 38–40)
- a. * [the destruction [a city]]
 - b. [the destruction [of a city]]

Alexandra Rehn & Hannah Booth

In standard German which does have rich case morphology and in particular genitive, the parallel example is grammatical without the preposition, as overt genitive case morphology is present on the determiner, e.g. (13).¹¹

- (13) standard German (adapted from Richards 2010: 133–134)
 [die Zerstörung [ein-er Stadt]]
 the destruction a-FEM.GEN city
 ‘the destruction of a city’

We discuss this insertion as an OCP-related strategy in detail in the context of “linkers” in Section 3.4,

A fourth OCP avoidance strategy is **word order**, i.e. opting for an alternative word order so that identical units are not adjacent. An example of this comes from Serbo-Croatian, which generally permits multiple *wh*-fronting, e.g. (14a), but blocks multiple *wh*-fronting in contexts where two *wh*-phrases within the same domain are marked for the same case, allowing instead only one of the *wh*-phrases to be fronted e.g. (14b).

- (14) Serbo-Croatian (Nevins 2012: 89)
- a. **Ko koga** vidi?
 who whom sees
 ‘Who sees whom?’
 - b. **Šta** uslovljava **šta**?
 what conditions what
 ‘What conditions what?’

At this point, it is important to acknowledge that the OCP relates specifically to what is often referred to as “*accidental* repetition”, i.e. where units occur together which happen to be in some way identical, as distinct from *intentional* repetition, such as reduplication, which is freely permitted across languages, e.g. (15)

¹¹One may wonder why constructions with genitive marking on the noun alone are not grammatical:

- (i) standard German
 *die Verkostung Weines
 the tasting wine.GEN
 Intended: ‘the tasting of wine’

The ungrammaticality of such examples is however explained by the Genitive Rule (Gallmann 2018: 158), which states that a DP/NP can only bear genitive case if another element within the same DP/NP also bears genitive case morphology.

5 OCP effects in Germanic possession

(Radford 1977, Menn & MacWhinney 1984, Nevins 2012, Neeleman & Van de Koot 2017).¹²

(15) Japanese (Hiraiwa 2010)

- a. ie-ie
house-house
'houses'
- b. tabe-te tabe-te tabe-makut-ta.
eat-CONT eat-CONT eat-repeat-PST
'(I) ate and ate and ate.'

Furthermore, even within “accidental repetition”, the OCP is not a hard and fast constraint but rather has language- and construction-specific manifestations, as evidenced by examples such as (16) in Dutch, where the adjacency of identical prepositions is not ruled out (“tolerance”, cf. Neeleman & Van de Koot 2017).

(16) Dutch (Neeleman & Van de Koot 2017: 4380)

Dit is een goede wijn [PP voor [PP voor de televisie]].
this is a good wine for for the television
'This is a good wine for in front of the television.'

In fact, the broad crosslinguistic and intralinguistic variation which is manifest in morphosyntactic OCP effects has been a key focus in the literature in recent decades (e.g. Mohanan 1994, Plag 1998, Yip 1998, Mohanan & Mohanan 2003, Nevins 2012). There are two main aspects to this variation: (i) what type of identity is relevant for the OCP, and (ii) the domain in which the OCP holds. Next, we discuss these two aspects in turn.

3.2 Types of identity

In the original formulation of the OCP as described above, cf (9), the relevant type of identity was assumed to be identity in form, and this is typically the standard context in which the OCP is invoked. However, some authors have suggested that the OCP may operate on the basis of other types of linguistic identity, in particular in relation to morphosyntactic phenomena. Nevins (2012), for

¹²Close apposition seems to be an exception to the OCP in complex DPs because the construction is grammatical even though no linking element is inserted between the two DPs. However, this does not fall under the version of the OCP in Richards (2010) whereby two identical elements cannot be linearised; in close apposition, linearisation is not relevant, as the ordering of the two DPs is variable (Lekakou & Szendrői 2007, Rehn 2019).

Alexandra Rehn & Hannah Booth

instance, states that the (morphological) OCP may operate in relation to both the form and content of morphemes, while Neeleman & Van de Koot (2017) propose that the (syntactic) OCP can also refer to identity in feature content between two linguistic units. On such views, the OCP can operate, at least in relation to morphosyntax, on both a formal and functional level and, as Neeleman & Van de Koot (2017) suggest, perhaps even on both levels at once.

In this context, Nevins (2012) discusses OCP effects that are independent of phonological form, i.e. where an OCP violation arises irrespective of the choice of element, cf. (17), and those that are dependent on phonological form. In the latter, an OCP violation only arises when phonologically identical elements are adjacent, but can be avoided when a different form is chosen for one of the elements. This is the case in Pashto which has two forms for the 1/2.PL subject clitic, *am* and *mo*, which are usually in free variation. The preferred form in Western/literary Pashto is *mo*, e.g. (18a), but when two instances of *mo* would occur adjacently, the first one is realised as *am*, e.g. (18b). Since *mo* and *am* are functionally equivalent, here it is phonological rather than functional identity which triggers the OCP violation.

(17) English (Nevins 2012: 92)

- a. *John is **more more** intelligent than Bill than you are.
- b. *John is **more taller** than Bill than you are.

(18) Western/literary Pashto (Nevins 2012: 104)

- a. motar mo/*am rAwostə.
car PART.PL bought
'We/y'all brought the car.'
- b. motar *mo/am mo rAwostə.
car PART.PL PART.PL bought
'We/y'all brought our/y'alls car.'

Neeleman & Van de Koot (2017) explore identity specifically in terms of syntactic features, providing an example from Dutch where this specific type of identity appears to trigger an OCP effect. Agreement in Dutch is generally insensitive to word order, but 2.SG verb forms in *-t* as in (19a) surface as 1.SG ($-\emptyset$) whenever verb-second gives rise to subject-verb inversion and thus direct adjacency of the verbal affix and the subject, as in (19b) and (19c).

(19) Dutch (Neeleman & Van de Koot 2017: 4398)

5 OCP effects in Germanic possession

- a. Jij loopt dagelijks met een hondje over straat.
you walk.2SG daily with a doggy in-the street
- b. *Dagelijks loopt jij met een hondje over straat.
daily walk.2SG you with a doggy in-the street
- c. Dagelijks loop jij met een hondje over straat.
daily walk.1SG you with a doggy in-the street
'You walk daily with a doggy in the street.'

Neeleman & Van de Koot (2017) interpret this as a syntactic OCP violation in terms of the adjacency of the 2.SG verbal affix and the agreeing subject (see also Ackema & Neeleman 2003), thus being an instance where OCP operates on identity in terms of a syntactic feature (here agreement). This is interesting, as agreement is not normally affected by adjacency in Dutch; identity in this context is normally tolerated. Thus, even in relation to functional identity, languages can exhibit construction-specific and perhaps even lexically-specific OCP effects.

3.3 Relevant domains

In the context of the original OCP, linear adjacency was the relevant domain within which it was assumed to hold, and linear adjacency does seem to be relevant for some specific morphosyntactic OCP effects. So, while linearly adjacent nouns with identical case affixes are strongly degraded in Hindi, cf. (20a), configurations where the two nouns are separated by a modifier of one of the nouns or by a pause are fine, e.g. (20b) and (20c) respectively.

(20) Hindi (Mohan 1994: 186–187)

- a. ?? ramm-ko baccō-ko samhaalnaa paḍaa.
Ram-DAT children-ACC take.care.of.NF fall.PERF
Intended: 'Ram had to take care of the children.'
- b. ramm-ko apnii bahin-ke baccō-ko samhaalnaa paḍaa.
Ram-DAT self's sister-GEN children-ACC take.care.of.NF fall.PERF
'Ram had to take care of his sister's children.'
- c. ramm-ko [pause] baccō-ko samhaalnaa paḍaa.
Ram-DAT children-ACC take.care.of.NF fall.PERF
'Ram had to take care of the children.'

However, in relation to other morphosyntactic OCP effects, some have raised the possibility that the relevant domain can go beyond linear adjacency (Anttila

Alexandra Rehn & Hannah Booth

& Fong 2000, Richards 2010, Nevins 2012, Neeleman & Van de Koot 2017). In particular, the fact that syntactic constraints generally extend beyond linear adjacency suggests that syntactic OCP effects could too. In fact, there is evidence that the case OCP effect in Hindi, while sensitive to linear adjacency, is not completely blind to syntactic structure, since adjacent nouns with *-ko* are permitted when they cross a clause boundary, e.g. (21).

- (21) Hindi (Mohanani 1994: 200)
 ilaa aramm-**ko** baccō-**ko** bulaane b^hejegii.
 Ila-NOM Ram-DAT children-ACC call.NF send.FUT
 ‘Ila will send Ram to call the child.’

Another example of the OCP being sensitive to syntactic structure comes from Japanese (as discussed by Nevins 2012), which bans multiple accusative-marked NPs within the same phrase (vP) in causative constructions, even when they are not linearly adjacent. Japanese causative verbs allow either an accusative or a dative causee, but when the embedded verb is transitive and takes an accusative object, only the dative option is allowed, e.g. (22a). Crucially, this is not improved when the two nominals are separated by an adverb, e.g. (22b), indicating that the domain which is relevant is the broader phrase, irrespective of linear order.

- (22) Japanese (Nevins 2012: 91)
- a. Taro-wa Hanako-ni/*Hanako-o kusuri-o
 Taro-TOPIC Hanako-DAT/Hanako-ACC medicine-ACC
 nomseta.
 caused-to-drink
 ‘Taro had/made/let Hanako drink the medicine.’
 - b. *Taro-wa Hanako-o tika^razukude kusuri-o
 Taro-TOPIC Hanako-DAT forcibly medicine-ACC
 nomaset^a.
 caused-to-drink
 Intended: ‘Taro forcibly made Hanako drink the medicine.’

Elsewhere, Anttila & Fong (2000) propose a Case-OCP for Finnish which prohibits the assignment of the same case to both the head and its sister, i.e. an OCP restriction which operates on the basis of a syntactic structural relation rather than straightforward linear adjacency. Within a phrase-based approach to syntax, Richards (2010) has cast the OCP in terms of a ban on the linearisation of two

identical nodes which are part of the same phase. Thus, it is clear that while linear adjacency can be relevant for the OCP at the morphosyntactic level, the OCP can also operate on the basis of other types of syntactic relations and domains.

3.4 Linkers

In relation to the OCP or the related notion of “identity avoidance”, some authors discuss “linkers”, which are widely attested crosslinguistically (e.g. Ghomeshi 1997, Richards 2010). In the context of this paper, linkers can be broadly defined as functional elements which occur between a modified element and its modifier (e.g. Franco et al. 2015). The Iranian *ezafe* is in many ways what one might consider a prototypical linker, and has been discussed by many on these terms (e.g. Ghomeshi 1997, Franco et al. 2015).¹³ Specifically, when a noun is accompanied by a modifier, the modifier is followed by an enclitic unstressed vowel (see e.g. Lazard 1992 for a traditional description). In Persian, for instance, the *ezafe* is an unstressed vowel that is realised on nominal modifiers including adjectives, possessors, locations and paths (Larson & Yamakido 2008, Windfuhr & Perry 2013), e.g. (23). As Franco et al. (2015) describe, the *ezafe* does not seem to indicate anything about the precise semantic or syntactic nature of the relation between the modifier and modified noun.

(23) Persian (Windfuhr & Perry 2013: 473)

- a. āsmān-e ābi
blue-EZ sky
‘blue sky’
- b. ketāb-e Hasan
book-EZ Hasan
‘Hasan’s book’
- c. rāh-e Tehrān
road-EZ Tehran
‘the road to/of Tehran’

The label of “linker” has also been applied to elements in other languages with diverse properties and, beyond the core case of the *ezafe*, there is little consensus on what counts as a linker, nor on their formal analysis. “Determiner spreading” as in, for instance, Greek, is analysed as a linker by Larson & Yamakido (2008),

¹³See also earlier work which assumes a relative clause construction as the origin of the Iranian *ezafe* (Haider & Zwanziger 1984).

Alexandra Rehn & Hannah Booth

but not by others (Den Dikken & Singhapreecha 2004; see also the discussion in Franco et al. 2015). This involves duplication of the definite determiner between a noun and its modifier(s), which licenses postnominal adjectival modification which is otherwise ruled out, e.g. (24). Like a prototypical determiner, the linker agrees in indefiniteness, number, nominal class and case with the head noun.

(24) Greek (Larson & Yamakido 2008: 15)

- a. to spiti to meghalo to petrino
the house the big the of-stone
'the big stone house'
- b. *to spiti meghalo petrino
the house big of-stone
Intended: 'the big stone house'

Japanese, meanwhile, exhibits a linker, *no*, traditionally classified as a genitive case-marker, which links a noun with another nominal in a genitive or modifying relation, e.g. (25).

(25) Japanese (Larson & Yamakido 2008: 18)

- a. Taroo-no hon
Taroo-GEN book
'Taroo's book'
- b. Nihon-no gakusei
Japanese-GEN student
'Japanese student' ('student who is Japanese')

It is important to stress, however, that many do not necessarily view linkers as OCP-motivated elements. In line with data regarding linkers in diverse languages, some take them to be copulars (Den Dikken & Singhapreecha 2004, cf. also Karimi 2007 on Kurdish), others as case assigners (Samiian 1994 and Larson & Yamakido 2008 on Indo-Iranian), agreement phenomena (Franco et al. 2015 on Albanian and Iranian), or semantic licensors of possession (Koontz-Garboden & Francez 2010 on Ulwa). As such, the term "linker" is rather imprecise, with its assumed categorial status and function varying according to the specific language, phenomenon and theoretical assumptions. Indeed, the only common aspect of the accounts discussed here is that linkers are functional elements which occur between a head and its dependent (see for discussion Samvelian 2007, Toosarvandani & Van Urk 2014). Another property which many linkers share is being identical in form to other elements in the grammar, e.g. articles (as in

5 OCP effects in Germanic possession

the Greek example in (24)). This observation may be extended to the *ezafe*, which e.g. shows agreement with the head noun in the Indo-Iranian language Zazaki, e.g. (26) (cf. Larson & Yamakido 2008, Toosarvandani & Van Urk 2014, Franco et al. 2015), lending further support to the observation that forms which serve as linkers are often multifunctional.

(26) Zazaki (Toosarvandani & Van Urk 2014: ex. (3))

- a. kutk-o girs
dog-EZ.MASC.SG.NOM big
'the big dog'
- b. ju biz-a girs-e
one goat-EZ.FEM.SG big-FEM.SG
'a big goat'

4 The OCP and ordering restrictions in Germanic possession

In this section, we expand on the notion of a syntactic OCP, showing how it can account for variation and change in Germanic possessive constructions, drawing a connection with Richards (2010) who demonstrates OCP effects within complex DPs. Although Richards (2010) does not explicitly discuss possession, some of his examples concern possessive relations, cf. (12) and (13) above. Specifically, we propose that the possessive element in the PLC (*sein/ihr*) fulfils a double function: (i) marking the relation between two DPs as possessive; (ii) serving as a linking element preventing an OCP violation. As we show in this section, this is supported by data from present-day German dialectal varieties, as well as from Middle Low German, and can account for the ordering restriction concerning the possessor and possessum outlined in Section 2.3. We also show how this proposal can be extended to Germanic possessive constructions more broadly, in particular the adnominal genitive, the *s*-possessive and the prepositional possessive.

4.1 Present-day German varieties

As already outlined in Section 2.3, in present-day German varieties, it is not sufficient for the possessive relation in the PLC to be marked overtly by *sein/ihr*; rather, the possessive element must occur specifically between the possessor and the possessum (*dem Mann sein Haus*/**sein Haus dem Mann*), cf. Table 2. At the same time, it is not possible to omit the possessive element altogether (**dem*

Mann Haus). Given the fact that the possessive element must occur specifically between the two nominals, and that its omission leads to ungrammaticality, we propose that the possessive element in the PLC serves as a linker which prevents an OCP violation which would otherwise arise from two adjacent DPs within a single constituent, cf. (27).

- (27) a. * Possessor_{DP} Possessum_{DP}
 b. Possessor_{DP} **sein** Possessum_{DP}

In this sense, the possessive element in the PLC serves a similar function to genitive case morphology in the adnominal genitive construction. When a genitive case suffix occurs on the noun belonging to the possessor-DP, as in the MASC/NEUT.SG, e.g. (28a)-(28b), a prenominal possessor is grammatical. However, in the FEM.SG, where the genitive ending on nominals has been eroded and thus no genitive morphology intervenes between the two nouns, a prenominal possessor is ruled out, e.g. (28c).¹⁴

- (28) a. des Mann-es Haus
 the.MASC.GEN man-MASC.GEN house
 ‘the man’s house’
 b. des Mädchen-s Haus
 the.NT.GEN girl-NT.GEN house
 ‘the girl’s house’
 c. * der Frau-Ø Haus
 the.FEM.GEN woman house
 Intended: ‘the woman’s house’

¹⁴A reviewer points out the contrast in (i) which demonstrates that only genitive case morphology or a possessive element like *sein* can establish the possessive relation between the two DPs. Hence with a dative possessor with case morphology in the relevant position, the *sein* is still required. It may of course be debated as to whether the establishment of the possessive relation and the linking function are realised together or separately, but this is hard to tease out empirically.

- (i) a. dem Komponisten sein Haus
 the.DAT composer.NONNOM his house
 ‘the composer’s house’
 b. * dem Komponisten Haus
 the.DAT composer.NONNOM house
 Intended: ‘the composer’s house’

5 OCP effects in Germanic possession

The ungrammaticality of (28c) is particularly informative, as an alternative with possessum-possessor ordering is fine, as is the case for all of the examples in (28), cf. (29).

- (29) a. das Haus **des** Manne-s
 the house the.MASC.GEN man-MASC.GEN
 ‘the man’s house’
 b. das Haus **des** Mädchen-s
 the house the.NT.GEN girl-NT.GEN
 ‘the girl’s house’
 c. das Haus **der** Frau-Ø
 the house the.FEM.GEN woman
 ‘the woman’s house’

The data in (28) and (29) thus further support the existence of an OCP effect in possessive constructions in present-day German varieties and show that, like *sein/ihr* in the PLC, in the adnominal genitive construction, genitive case morphology is sufficient to prevent an OCP violation between the possessor and possessum, either on the noun within the possessor-DP in the case of possessor-possessum ordering, cf. (30b), or on a(n) (in)definite element belonging to the possessor in the case of possessum-possessor ordering, cf. (30c).

- (30) a. * Possessor_{DP} Possessum_{DP}
 b. Possessor_{DP} GEN Possessum_{DP}
 c. Possessum_{DP} GEN Possessor_{DP}

Furthermore, the genitive case appears to occupy a special position in this context; by contrast, dative case-marking cannot satisfy the OCP-constraint, as evidenced by the fact that a PLC with a dative-marked possessor, together with possessum-possessor ordering, and thus a dative-marked definite element between the two nominals, is not grammatical, e.g. (31).

- (31) a. * sein Haus **dem** Mann
 his house the.DAT man
 Intended: ‘the man’s house’
 b. * Possessum_{DP} DAT Possessor_{DP}

The fact that (31) is ruled out is particularly striking, since dative case morphology can otherwise establish a possessive relation between two nominals in external possessive constructions, e.g. (32).

Alexandra Rehn & Hannah Booth

- (32) a. standard German
 der Peter wäscht **dem** Opa die Haare.
 the Peter washes the.DAT grandfather the hair
 ‘Peter washes his grandfather’s hair.’
 b. Alemannic
 des isch **dem** Leo.
 this is the.DAT Leo
 ‘this belongs to Leo.’

This special status of genitive within the case system is already known from the literature, as e.g. in [Weinrich \(2005: 695\)](#) who observes that genitive is the only one of the four cases that is able to link two nominals, which he thus gives the label “Junktor”.

At the same time, many present-day German varieties exhibit a grammaticalised possessive marker, *-s*, which is derived from the MASC/NEUT.GEN suffix ([Fuß 2011](#), [Rauth 2014](#)) but can occur on feminine and complex prenominal possessors which otherwise lack possessor-final genitive morphology (‘*s*-possessive’, cf. Table 1). So while prenominal feminine possessors in the adnominal genitive construction are ruled out due to the lack of genitive marking on the noun (**der Frau-Ø Haus*, cf. (28c) above), prenominal feminine possessors which bear the grammaticalised *-s* marker are fine, e.g. (33). Similar examples with FEM.SG and PL possessors are exhibited in Low German, e.g. (34).

- (33) Alemmanic (SynAlm)
 Mit der Lena-s nuja Stiefel han ih net laufa kenne.
 with the.DAT/GEN Lena-POSS new boots have I NEG walk can
 ‘With Lena’s new boots I was unable to walk.’
 (34) Low German ([Strunk 2005: ex. \(59\)–\(60\)](#))
 a. Antje-s Bröögam
 Antje-POSS bridegroom
 ‘Antje’s bridegroom’
 b. höör ollen-s hus
 her parent.PL-POSS house
 ‘her parent’s house’

Moreover, the *-s* marker always occurs between the possessor and the possessum, i.e. in the same position as the possessive element in the PLC, and the

5 OCP effects in Germanic possession

(non-grammaticalised) genitive suffix in the adnominal genitive. This is evidenced by the fact that, in *s*-possessive constructions with a possessor involving coordination, the *-s* marker must occur on the second conjunct, i.e. directly adjacent to the possessum, e.g. (35). Likewise, when the possessor involves a complex name, the *-s* marker must attach to the right edge of it, e.g. (36).

(35) Low German (Strunk 2004: 109)

- a. Hinnerk un Anna-s Huus
Hinnerk and Anna-poss house
'Hinnerk and Anna's house'
- b. *Hinnerk-s un Anna Huus
Hinnerk-poss and Anna house
Intended: 'Hinnerk and Anna's house'

(36) Low German (Strunk 2004: 110)

- a. Franz vun Assisi-s "Sünnensang"
Francis of Assisi-poss song-of-the-sun
'Francis of Assisi's song of the sun'
- b. *Franz' vun Assisi "Sünnensang"
Francis.POSS of Assisi song-of-the-sun
Intended: 'Francis of Assisi's song of the sun'

Still further evidence for OCP effects in Germanic possession come from the prepositional possessive construction; as already noted in Section 1, here both possessor-possessum and possessum-possessor ordering is possible, cf. (4), repeated here in (37).

(37) German

- a. das Haus von dem Mann
the house of the.DAT man
'the man's house'
- b. von dem Mann das Haus ~~von dem Mann~~
of the.DAT man the house
'the man's house'
- c. Possessum_{DP} von Possessor_{DP}

The variant in (37b) is possible when the possessor is focussed, and Roehrs (2019) shows there is good reason to assume that in such cases the PP moves from its base position to a higher functional projection. On this assumption, the grammaticality of the preposed variant can be accounted for; underlyingly, the relevant element *von* is still between the possesum and possessor, preventing an OCP violation, cf. (37c).

As an interim summary, we have so far argued on the basis of data from present-day German varieties that various restrictions within possessive constructions can be accounted for by assuming that there is an active morphosyntactic OCP-constraint in this context, cf. (38).

- (38) a. * Possessor_{DP} Possesum_{DP}
 b. Possessor_{DP} **sein** Possesum_{DP}
 c. Possessor_{DP} **GEN** Possesum_{DP}
 d. Possesum_{DP} **GEN** Possessor_{DP}
 e. Possesum_{DP} **von** Possessor_{DP}

Further insights into the status of the OCP in present-day German(ic) possession can be gained from examining constructions where multiple possessive attributes are stacked. For instance, Weinrich (2005: 711–713) observes that it is possible to have several stacked adnominal genitives within one constituent, but only when the inflectional markers are phonologically distinct, cf. (39).¹⁵ By contrast, when several attributes which have the same inflectional genitive marker are stacked, only the first is realised as an adnominal genitive; the other(s) is/are realised as (a) prepositional possessor(s), whereupon the order genitive>PP is obligatory, e.g. (40).

- (39) standard German (Habermas, as cited in Weinrich 2005: 712)
 der Versuch **einer** Rekonstruktion **der** Vorgeschichte **des**
 the attempt a.GEN reconstruction the.GEN record the.GEN
 neueren Postivismus
 newer.wk positivism
 ‘the attempt at a reconstruction of the record of the newer positivism’
- (40) standard German

¹⁵Sequences of more than one genitival attribute are overall rare, however, and are mostly confined to technical language (Weinrich 2005: 712).

5 OCP effects in Germanic possession

- a. * Peters Onkels Buch
 Peter.GEN uncle.GEN book
 Intended: ‘Peter’s Uncle’s book’
- b. * Das Buch **des** Onkels Peters
 the book the.GEN uncle.GEN peter.GEN
 Intended: ‘the book of Peter’s Uncle’
- c. Das Buch des Onkels von Peter
 the book the.GEN uncle.GEN of peter
 ‘the book of Peter’s Uncle’
- d. * Das Buch von Peter des Onkels
 the book of Peter the.GEN uncle.GEN
 Intended: ‘the book of Peter’s Uncle’

Similarly, in the literature on the Germanic PLC, examples which are provided for multiple stacked PLCs involve alternative forms of possessive marking. [Strunk \(2004\)](#), for instance, marks the pair of Low German sentences in (41) as a grammatical/ungrammatical contrast.

(41) Low German ([Strunk 2004](#): 68–69)

- a. Jehann dee uk Markus jenant woat **siene** Mutta Marie **aea** Hus
 John who also Mark called was his mother Mary her house
 ‘John – who was also called Mark – his mother Mary’s house’
- b. * Jehann **sien** Broder **sien** Süster **ehr** Hus
 John his brother his sister her house
 Intended: ‘John’s brother’s sister’s house’

Thus the data from possessor-stacking suggests that phonological form is relevant for the OCP in the context of German(ic) possessive constructions. In the German dialects where the PLC and the prepositional possessive are the dominant strategies for expressing adnominal possession, such as Alemannic, the question arises as to whether these varieties exhibit similar restrictions in sequences of possessive attributes, i.e. whether phonological form is also relevant for the OCP in Alemannic. According to our Swabian (one of the five Alemannic varieties, cf. [Wiesinger 1983](#)) informants, such restrictions are indeed manifest. The version with two stacked prepositional possessors in (42a) is considered highly marked or even ungrammatical, while [Zifonun \(2003: 100\)](#) only considers it “inelegant”. The versions with two stacked PLCs, as in (42b) and (42c), are highly marked if not ungrammatical, but are rated slightly better than the stacked prepositional possessors in (42a) by our informants.

Alexandra Rehn & Hannah Booth

(42) Swabian (Alemannic)

- a. ??/* ds Buach vom Onkl vom Nochbr
the book of-the uncle of-the neighbour
Intended: ‘the book of my uncle’s neighbour’
- b. ??/* em Nochbr seim Onkl sei Buach
the.DAT neighbour his.DAT uncle his book
Intended: ‘the book of the neighbour’s uncle’
- c. ??/* dr Nochbre ihra Dande ihr Buach
the neighbour.FEM her.DAT aunt her book
‘the book of the neighbours’s aunt’

Zifonun (2003) also provides examples with sequences of two PLCs, although in her examples a masculine and a feminine possessor are combined, resulting in a change in phonological form; the inherent features of the possessive elements are not identical, which may be a relevant factor. Our Swabian informants accept these examples as grammatical and unmarked, thus further confirming that phonological identity is relevant in Alemannic.¹⁶ The examples are given in (43a) and (43b).

(43) Swabian (Alemannic) (as based on *Zifonun 2003*: 100)

- a. dr Doris ihram Mo sei Partei
the.DAT Doris her.DAT husband his party
‘Doris’ husband’s party’
- b. in meim Nochbr seiner Schweschdr ihram Garda
in my.DAT neighbour his.DAT sister her.DAT garden
‘in my neighbour’s sister’s garden’

By way of further confirmation of this, when two different types of possessive construction are stacked, i.e. when the PLC is combined with a prepositional possessive, the examples are perfectly grammatical, e.g. (44).

(44) Swabian (Alemannic)

- a. ds Buach vom Nochbr seim Onkl
the book of-the.DAT neighbour his.DAT uncle
‘the book of the the neighbour’s uncle’

¹⁶Possessor stacking was not included in the SynAlm project, so in this context we interviewed three native Swabian speakers. The judgements provided here are based on their responses.

5 OCP effects in Germanic possession

- b. ds Buach von dr Nochbre ihra Dande
the book of the.DAT neighbour her.DAT aunt
‘the book of the neighbour’s aunt’

This is also the case for Low German, where similar examples are grammatical according to [Strunk \(2004\)](#), e.g. (45).

- (45) a. Low German ([Strunk 2004](#): 25)
se sungen dat Leed von Herrn Pastor sien Koh
they sang the song of mister pastor his cow
‘They sang the song of the pastor’s cow.’
b. Low German ([Strunk 2004](#): 59)
de Fru van usen Wärt sienen Sühne
the wife of our inkeeper his son
‘the wife of our inkeeper’s son’

4.2 Middle Low German

We now examine possessive constructions in Middle Low German to see to what extent OCP effects are exhibited in this particular language stage. Middle Low German (MLG) is an umbrella term for several Continental West Germanic scribal dialects which were in use across northern German and the north-eastern Netherlands in c. 1250–1600. Crucially in the context of possessive constructions, the erosion of morphological case is underway in the period and beyond ([Lasch 1914](#), [Härd 2000](#), [Askedal 2005](#)), resulting in the modern Low German two-case system where only a distinction between nominative and non-nominative remains in the MASC.SG, cf. Table 4. By contrast, MLG texts generally attest a much richer case morphology, albeit with some case-levelling already evident, in particular between accusative and dative. Crucially, genitive morphology remains present on the majority of MASC/NEUT nouns, and on many FEM nouns, although here the genitive ending is already optional, cf. Table 5.

Table 4: The case system in modern Low German ([Askedal 2005](#): 6)

	MASC.SG	NEUT.SG	FEM.SG
NOM	de dicke Kopp	dat oole Book	de oole Kann
NON-NOM	den dicken Kopp	dat oole Book	de oole Kann

Table 5: The case system in Middle Low German, based on Härd (2000: 1431)

	MASC.SG	NEUT.SG	FEM.SG
NOM	dach	wort	wîse
ACC	dach	wort	wîse
DAT	dage	worde	wîse(n)
GEN	dag-es	word-es	wîs-e(n)

As pointed out already, in present-day German varieties which lack genitive morphology on FEM.SG nouns, adnominal genitive possessive constructions with a prenominal FEM.SG possessor are ruled out, as we have argued, due to the OCP (cf. **der Frau-Ø Haus*). By contrast, in MLG, where genitive morphology can still be present on FEM.SG nouns, one expects such constructions with a prenominal FEM.SG possessor to be attested, as the genitive case-marker will prevent an OCP violation between the two nouns. This is confirmed by an investigation of adnominal genitive constructions in the CHLG, e.g. (46).

(46) Middle Low German

- a. (CHLG: Arznei)

dar aff eyn plaster geleget vppe der vrowe-n hemelike stad
 thereof a plaster placed on the.GEN woman-GEN secret place
 ‘thereof a plaster placed on the woman’s secret place’
- b. (CHLG: Bremen)

des ersten Sondaghes na vser vrowe-n daghe
 the.GEN first.GEN Sunday.GEN after our.GEN lady-GEN day.DAT
 ‘the first Sunday after our lady’s day’
- c. (CHLG: Engelhus)

de was eyner arme-n ghevange-n vrowe-n sone.
 he was a.GEN poor-GEN captive-GEN woman-GEN son
 ‘He was the son of a poor captive woman.’
- d. (CHLG: Flos)

dorch clarisse-n willen mochte se des nicht laten.
 through Clarisse.GEN will wanted she so NEG leave
 ‘On account of Clarisse’s will she didn’t want to leave so.’

5 OCP effects in Germanic possession

As expected, prenominal genitive-marked MASC./NEUT.SG possessors are also widely attested in MLG, e.g. (47).

(47) Middle Low German

a. (CHLG: Engelhus)

He dodede na syns wieu-s rade Achapp-es dochter.
 he killed after his.GEN wife-GEN advice Achapp-GEN daughter
 ‘On his wife’s advice he killed Achappe’s daughter.’

b. (CHLG: Griseldis)

Ick bin naket vth mynes vader-s hus gegan.
 I am naked out my.GEN father-GEN house gone
 ‘I went out of my father’s house naked.’

c. (CHLG: Braunschweig)

Do was Europ-s konningk Siaoniorum.
 then was Europe-GEN king Siaoniorum
 ‘Then Europe’s king was Siaoniorum.’

In the examples in (47), where the prenominal possessor consists of multiple items, each item is marked for genitive, as is the case in modern standard German (*de-s Mann-es Haus*). Similar examples are provided for “Older Low German” by Strunk (2004), alongside their modern Low German equivalents, where crucially only the right-most element in the prenominal possessor bears genitive morphology, cf. the pairs in (48) and (49).

(48) a. Older Low German (Strunk 2004: 106)

myn-s Grotvader-s older Möme sprack
 my-GEN grandfather-GEN old.GEN grandmother language
 ‘the language of my grandfather’s old grandmother’

b. Modern Low German (Strunk 2004: 107)

mien-Ø Grotvader-s olle Möme-s spraak
 my grandfather-POSS old grandmother-POSS language
 ‘the language of my grandfather’s old grandmother’

(49) a. Older Low German (Strunk 2004: 107)

de-s Künige-s Tochter
 the-GEN king-GEN daughter
 ‘the king’s daughter’

- b. Modern Low German (Strunk 2004: 107)
 de(n) König-s Tochter
 the.NOM.(ACC) king-POSS daughter
 ‘the king’s daughter’

The fact that the genitive marker becomes confined to the right-most element in the prenominal possessor in later stages of Low German, as the pairs in (48) and (49) show, lends further support to our claim that possession constructions in Germanic are conditioned by the OCP.

With respect to the PLC, this is robustly attested in the CHLG/ReN data. As expected given the ongoing erosion of case morphology in MLG and the fact that the possessor in the Germanic PLC is typically marked for the most oblique case available, the possessor is marked for a range of different cases in the data, i.e. NOM/ACC, e.g. (50a), DAT, e.g. (50b) and GEN, e.g. (50c) (see Booth & Rehn 2023 for discussion and further examples).

(50) Middle Low German

- a. (ReN: Chr._Wass._Duisberg)
 Ende was hartoch Jan van cleue **syn** suster.
 and was duke.NOM/ACC Jan van Cleve his sister
 ‘And (she) was Duke Jan van Cleve’s sister.’
- b. (ReN: Hildesh._StR_1300)
 Sterft emme mann-**e** **sin** wif.
 dies a.DAT man-DAT his wife
 ‘If a man’s wife dies.’
- c. (ReN: Lauremberg_1652)
 Wen idt wehre des Her-**n** **sin** Wille.
 if it were the.GEN sir-GEN his will
 ‘If it were the sir’s will.’

In examples like (50c), where a PLC occurs with a possessor which has genitive-marking on the right-most element, the possessive element is arguably redundant from the perspective of the OCP (cf. also Weiß 2012), since the genitive-marking on the possessor should be sufficient to prevent an OCP violation, as evidenced by adnominal genitive constructions which lack the possessive element, which are widely attested in MLG, e.g. (51a) (see also (46) and (47) above). Indeed, parallel constructions where two possessive markers co-occur on the PLC are ungrammatical in modern German varieties, as noted by Roehrs (2019), cf. (51b).

5 OCP effects in Germanic possession

- (51) a. (CHLG: Braunschweig)
 unses alde-n herre-n tiden
 our.GEN old-GEN sir-GEN time
 ‘our old sir’s time’
- b. modern German (Roehrs 2019: 109)
 * Peter-s sein Wagen
 Peter-GEN his car
 Intended: ‘Peter’s car’

By contrast, with the gradual loss of genitive in the history of Low German, when the possessor is marked for NOM/ACC/DAT, the possessive element in the PLC is itself necessary in preventing an otherwise OCP violation. In this light, examples like (50c) can be viewed as witness of a transitional stage in the diachronic development of the expression of possession in Low German. Moreover, on this view, the erosion of (in particular genitive) case in Low German and the associated OCP-effects is expected to be a crucial factor in the overall consolidation of the PLC. The connection between the loss of genitive and the PLC is further supported by the complementary distribution of productive adnominal genitive to express possession and the PLC more widely in Germanic; for instance, modern Icelandic and modern standard German both have productive genitive but lack the PLC. This also suggests that while the OCP is standardly invoked in synchronic settings, it can also provide useful insights in the context of diachrony.

5 The multifunctionality of possessive elements

So far we have shown that possessive constructions in Germanic are conditioned by a syntactic OCP, and that certain elements serve as linkers. However, all of these constructions also involve a possessive relation between the two nominals which must also be established in the syntax. In this section, we claim that the possessive element in the PLC is multifunctional in the sense that it serves not only as an OCP-related linker but also marks the possessive relation, albeit in an abstract way, in line with the fact that linking elements crosslinguistically are typically multifunctional (see Section 3.4). To formalise this idea, we draw on the the idea of “anchoring” as proposed within the Universal Spine Hypothesis of Wiltschko (2014), as we discuss further below, extending this to possession as one type of complex DP.

5.1 Previous accounts

The idea that the possessive element in the PLC is required to mark the possessive relation between the two nominals is central to previous accounts by [Strunk \(2004, 2005\)](#) and [Charters \(2014\)](#) within the constraint-based parallel architecture of Lexical Functional Grammar (LFG, [Bresnan & Kaplan 1982](#), [Bresnan et al. 2016](#), [Dalrymple et al. 2019](#)). [Strunk \(2004, 2005\)](#) and [Charters \(2014\)](#) start from the question of why the possessive element cannot be dropped in the PLC, cf. the contrast in (52), although they do not go beyond this or address the ordering restriction discussed in Section 2.3 specifically.

(52) Low German ([Strunk 2005](#))

- a. (d'en Jung) sien Vadder
the boy his father
'the boy's father'
- b. *d'en Jung Vadder
the boy father
Intended: 'the boy's father'

Rather than invoke the OCP as a motivation for the obligatoriness of the possessive element, both [Strunk \(2005\)](#) and [Charters \(2014\)](#) assume that the possessive element (*sien*) in (52a) is obligatory because the possessor (*den Jung*) does not itself contribute the grammatical function poss at LFG's f(unctional)-structure, i.e. establish the possessive relation; thus the possessive element is needed to contribute the poss feature, i.e. to establish the possessive relation between the possessor and the possessum.

We argue that both things can be true of the possessive element at once; *sein* marks the possessive relation and also serves as a linker in order to prevent an OCP violation. This can also be extended to the other elements (genitive inflection and the preposition); they also serve both as linkers and establish the possessive relation. Beyond LFG, this idea is also reflected in [Roehrs \(2019\)](#), who observes similar behaviour across the PLC, prenominal genitive and the prepositional possessive in German, and addresses the fact that the elements which we have claimed to be linkers cannot be left out, cf. (53).

(53) German ([Roehrs 2019](#): 109)

- a. Peter *(sein) Wagen
Peter his car
'Peter's car'

5 OCP effects in Germanic possession

- b. Peter*(s) Wagen
Peter(GEN) car
'Peters car'
- c. *(von) Peter der Wagen
of Peter the car
'Peter's car'

Roehrs (2019), similar to Weiß (2008), aims to provide a unified structure from which all German(ic) possessive constructions can be derived.¹⁷ In such accounts, either: (i) the possessor is in complement position, or (ii) the possessor is in some specifier position exhibiting subject-like properties. However, such analyses face problems given the data which we have outlined here. Firstly, the assumption that the possessor is in complement position is challenged by the ordering restriction in the PLC (see Section 2.3), namely the fact that the possessor in the PLC is never grammatical in postnominal (i.e. complement) position. Secondly, another challenge for such accounts is the fact that the prenominal genitive and the PLC – which both involve prenominal possessors – show an animacy restriction, whereby only animate possessors are permitted, as we discuss in detail next. If all possessors are assumed to originate in complement position, it is not automatically clear how movement of only an animate subset of possessors can be accounted for.

5.2 Narrow and broad possession

Although we have so far argued that the linkers in German(ic) possession (*sein*, *von* and adnominal genitive) are functionally equivalent in the sense that they each prevent an OCP violation, it is not yet clear how they establish the possessive relation, in particular because in this aspect some subtle interpretational differences between the various possessive constructions can be observed. Possession as a term has both a narrow and broad sense. Narrowly construed, possession expresses ownership; in the broad sense, however, possession goes beyond ownership and spans multiple semantic patterns, including also kinship, property, body part, part-whole and agent/theme (e.g. Lindauer 1998, Alexiadou

¹⁷Since the PLC is a recurring topic within research on Germanic possession, many other theoretical accounts have been put forward (Corver 1990, Olsen 1996, Gavrusseva 2000, Haegeman 2004, Weiß 2008, Georgi & Salzmann 2011, Weiß 2012). However, the accounts focus on other aspects of the construction, including the ban on possessor extraction (Olsen 1996, Gavrusseva 2000, Haegeman 2004), agreement and case assignment (Georgi & Salzmann 2011) and diachrony (Weiß 2012), so we do not discuss these further here.

Alexandra Rehn & Hannah Booth

et al. 2007, Bjorkman & Cowper 2016). In terms of the adnominal genitive in present-day German, the construction can in principle express all these semantic patterns, both with the prenominal and postnominal genitive, e.g. (54) and (55) respectively. However, there is a subtle animacy restriction on the prenominal genitive, whereby the genitive-marked possessor can only be a proper name or a particular kinship term (see also Table 1), which is why the example in (54e) with an inanimate possessor is archaic and does not represent productive genitive.

(54) present-day German: adnominal genitive (prenominal)

a. OWNERSHIP

Peter-s Katze

Peter-GEN cat

‘Peter’s cat’

b. KINSHIP

Peter-s Onkel

Peter-GEN uncle

‘Peter’s uncle’

c. PROPERTY

Peter-s gute Singstimme

Peter-GEN good sing-voice

‘Peter’s good singing voice’

d. BODY PART

Peter-s Nase

Peter-GEN nose

‘Peter’s nose’

e. PART-WHOLE (archaic)

des Meer-es Wellen

the.GEN sea-GEN waves

‘the sea’s waves’

f. AGENT/THEME

Peter-s Operation

Peter-GEN operation

‘Peter’s operation’

(55) present-day German: adnominal genitive (postnominal)

a. OWNERSHIP

5 OCP effects in Germanic possession

- die Katze des Nachbarn
the cat the.GEN neighbour.GEN
'the neighbour's cat'
- b. KINSHIP
der Onkel des Nachbarn
the uncle the.GEN neighbour.GEN
'the neighbour's uncle'
- c. PROPERTY
die Höhe des Baumes
the height the.GEN tree.GEN
'the tree's height'
- d. BODY PART
das glänzende Fell des Hundes
the shining pelt the.GEN dog.GEN
'the dog's shining pelt'
- e. PART-WHOLE
die Zweige des Baumes
the branches the.GEN tree.GEN
'the tree's branches'
- f. THEME
die Operation des Mannes
the operation the.GEN man.GEN
'the man's operation'
- g. AGENT
die Operation des Arztes
the operation the.GEN doctor.GEN
'the doctor's operation'

The prepositional possessive is a true semantic equivalent of the postnominal genitive; it can express all the same relations and also does not exhibit the specific animacy restriction of the prenominal genitive, e.g. (56).

(56) present-day German: prepositional possessive

- a. OWNERSHIP
die Katze vom Nachbarn
the cat of-the neighbour
'the cat of the neighbour'

Alexandra Rehn & Hannah Booth

- b. KINSHIP
 der Onkel vom Nachbarn
 the uncle of-the neighbour
 ‘the uncle of the neighbour’
- c. PROPERTY
 die Höhe von dem Baum
 the height of the tree
 ‘the height of the tree’
- d. BODY PART
 das glänzende Fell von dem Hund
 the shining pelt of the dog
 ‘the shining pelt of the dog’
- e. PART-WHOLE
 die Zweige von dem Baum
 the branches of the tree
 ‘the branches of the tree’
- f. THEME
 die Operation von dem Mann
 the operation of the man
 ‘the operation of the man’
- g. AGENT
 die Operation von dem Arzt
 the operation of the doctor
 ‘the operation of the doctor’

With respect to the PLC, this shows similar behaviour to the prenominal genitive in terms of the semantic patterns which can be expressed and by having an animacy restriction on the possessor; however the animacy restriction is broader here, all animates being possible possessors, not just proper names and certain kinship terms, e.g. (57).¹⁸ As a result of this animacy restriction, part-whole relations cannot be expressed by the PLC, e.g. (57e), with the exception of body part-whole relations which coincide with animacy, e.g. (57d).

(57) present-day German: PLC

¹⁸In the examples in (57), we only provide a proper name as the possessor for ease of exposition, but such examples are equally good with animate possessors in general.

5 OCP effects in Germanic possession

- a. OWNERSHIP
dem Peter seine Katze
the.DAT Peter his cat
'Peter's cat'
- b. KINSHIP
dem Peter sein Onkel
the.DAT Peter his uncle
'Peter's uncle'
- c. PROPERTY
dem Peter seine gute Singstimme
the.DAT Peter his good sing-voice
'Peter's good singing voice'
- d. BODY PART
dem Peter seine Nase
the.DAT Peter his nose
'Peter's nose'
- e. ?? PART-WHOLE
dem Meer seine Wellen
the.DAT sea his waves
Intended: 'the sea's waves'
- f. AGENT/THEME
dem Peter seine Operation
the.DAT Peter his operation
'Peter's operation'

5.3 Anchoring and the Universal Spine Hypothesis (Wiltschko 2014)

As mentioned, we draw on ideas from Wiltschko (2014) as part of her Universal Spine Hypothesis in order to formalise the possessive relation involved in the constructions discussed in this paper. In addition, the way in which we make use of Wiltschko's ideas for complex DPs reflects adding structure as one way of avoiding a syntactic OCP violation, as discussed in Richards (2010) (see also discussion in Section 3.1).

Wiltschko (2014) extensively discusses the status of categories in morphosyntax and aims to reconcile a widespread theoretical position, namely the assumption that morphosyntactic categories are universal, with the apparently contradictory empirical observation that languages differ in the particular morphosyntactic categories which they employ. Her solution is to propose that there is a

Alexandra Rehn & Hannah Booth

universal set of functional elements, “*k*”, which are devoid of substantive content and from which functional categories are constructed on a language-specific basis. The *k* elements acquire substantive content when combined with language-particular elements (“Units of Language”, UoL), e.g. words/morphemes.

Each member of the set of universal categories (*k*) is defined by a unique function, and here Wiltschko (2014) proposes four (linking, anchoring, point-of-view, classification). Crucially, *k* is assumed to constrain the derivation of language-specific categories, and so it follows that categorial patterns, if not the categories themselves, are expected to be universal. The various members of *k* are assumed to be hierarchically organised, following the blueprint in (58) in the clausal domain and likely also the nominal domain (“Universal Spine”).

(58) (Wiltschko 2014: 87)

$C_{UG} = k:discourse\ linking > k:anchoring > k:point\ of\ view > k:classificat$

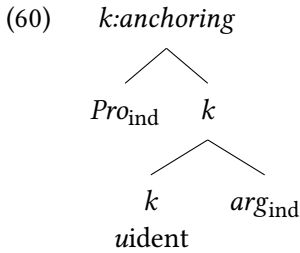
Units of Language are assumed to associate with the spine in (58) and to acquire the function of the area they associate with. In this sense, the relation between a Unit of Language and its interpretation is mediated by its categorial identity.

For Wiltschko (2014), anchoring is the function whereby an event or individual is anchored to the utterance. At the clause-level, anchoring concerns the coincidence of event time and utterance time, which is prototypically marked via tense. In this view, present tense is +coin(cidence) and past tense is –coin(cidence), cf. the contrast in (59).

- (59) a. Peter is cooking soup.
b. Peter cooked soup.

Wiltschko (2014) also discusses anchoring in the nominal domain, not in the context of complex DPs but rather as applied to pronouns, demonstratives and case within simple DPs. In this specific context, anchoring concerns the identity of the referent of the NP and the utterance individual (i.e. the speaker). Thus the coincidence feature in the nominal domain is an identity feature. This means that 1st person pronouns are +ident(ity), as the referent of the noun phrase and the utterance individual (speaker) coincide, i.e. are identical. By contrast, in the 3rd person, the referent of the noun phrase and the utterance individual do not coincide. Thus, nominal anchoring is defined by an identity feature (relating to the speaker-addressee relation) and the nominal anchoring category is D (as the nominal counterpart to I), cf. (60).

5 OCP effects in Germanic possession



5.4 Germanic possession and anchoring

As briefly summarised in Section 5.1, in previous approaches which start from the parallel make-up of the nominal and clausal domain, possessors are often considered to have subject-like properties (i.e. they do not enter into a specific thematic relation). The functional projection that establishes the possessive relation is therefore assumed to be similar to IP (see e.g. discussion in [Alexiadou et al. 2007](#)). If this is indeed the case, then in Wiltschko's framework the relevant part of the structure for establishing the possessive relation in adnominal constructions is $k:anchoring$, see (58). Anchoring in Wiltschko's framework means that the event is anchored to the utterance via the feature $[\pm coincidence]$. In languages with tense, this means that anchoring is based on whether utterance and event time coincide (present tense) or not (past tense). Alternatively, anchoring may be realised as something else, as in Blackfoot, which distinguishes between local and non-local participants and where the event is thus anchored to the utterance via $[\pm coincidence]$ of event and utterance participant (i.e. participancy). In other words, anchoring categories have the same formal and functional properties but differ in their substantive content. What these categories also have in common is the fact that they are transitive and thus involve heads that mediate between two arguments of which one is in Spec and the other in complement position. As already noted, the anchoring relation is established via coincidence, which expresses whether or not two arguments coincide in time, space, participancy or reality, as discussed at length in [Wiltschko \(2014: 140\)](#). As possessive constructions can also be considered transitive, the basic idea of anchoring may be applicable to possession – if it is possible to identify a relation based on coincidence for complex DPs.

For the constructions discussed in this paper, different functional elements seem to have the same or at least similar properties: all possessive elements (genitive, *sein*, *von*) establish a dependency relation between the two DPs involved. In the context of [Wiltschko \(2014\)](#), the dependency relation in possessive constructions can be argued to be based on a nominal version of $[\pm coincidence]$,

specifically coincidence in space (i.e. location). As has been suggested in previous work on possessive constructions, possession involves the identification of the location of the possessor relative to the possessum, i.e. coincidence in space (cf. Freeze 1992, Kayne 1993, Stassen 2009, Kasper 2014). Thus, we argue that all possessive constructions – as broadly defined in Section 5.2 – involve *k:anchoring* based on coincidence in space and the possessive elements thus express [+coincidence_{space}] in order to establish the possessive relation. In this view, possession is merely an abstract location relation, which distinguishes possessive constructions from other types of complex DP, such as close apposition which does not involve anchoring at all but rather complete identity (see Section 3.1), and coordination, which is [–coincidence_{space}].

At the same time, possession is a semantically broad label and allows for a range of semantic patterns which distribute in particular ways across the three possessive constructions we have discussed (adnominal genitive, PLC, prepositional possessive). As we showed in Section 5.2, each of the constructions can express different semantic patterns which fall under broad possession (e.g. ownership, kinship, property, part-whole, agent/theme), albeit not in a fully equivalent way. This has been well recognised in the literature, as in for instance Pittner (2014: 5–6), who identifies twelve different semantic patterns that can be expressed via the German adnominal genitive alone. As a result, the linking elements in each of the possessive constructions merely serve to value the coincidence feature as [+coincidence_{space}]; they do not encode the specific semantic interpretation.¹⁹ This underlying semantic contribution which we cast in terms of [+coincidence_{space}] is also expressed in earlier work on the adnominal genitive, such as by Kiefer (1910: 19) and Pittner (2014: 5).²⁰

We propose that the underlying universal anchoring category which is relevant for complex DPs is [$\pm$ coincidence_{space}]. This differs from the original proposal in Wiltschko (2014). Under our proposal, the anchoring category is not

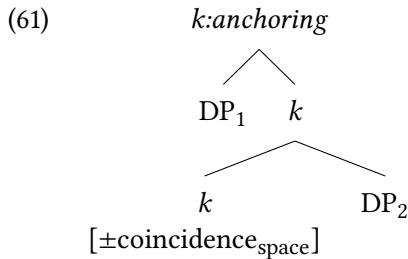
¹⁹We assume that the specific semantic interpretation is tied to the lexical elements involved in the possessive relation, in particular their status wrt. animacy, as this seems to restrict the range of semantic interpretations available, see Section 5.2.

²⁰Kiefer (1910: 19), for instance, provides the following definition in his work on the decline of adnominal genitive focusing on possessive constructions: “Tritt ein Genitiv als nähere Bestimmung zu einem Substantiv, so wird durch diese Verbindung nichts weiter ausgedrückt, als daß zwischen beiden Worten ein Verhältnis der Abhängigkeit besteht.” [When a genitive occurs as a closer definition to a noun, nothing more is expressed by this connection than that there is a relationship of dependence between the two words.]

A similar definition can be found in Pittner (2014: 5): “Der attributive Genitiv drückt eine Zugehörigkeitsrelations aus, die inhaltlich sehr unterschiedlich gefüllt sein kann.” [The attributive genitive expresses a relation of affiliation, which can be filled very differently in terms of content.]

5 OCP effects in Germanic possession

located somewhere between N and D, cf. (60), in other words it is not a general part of a potential universal nominal spine. The nominal anchoring category is rather a general part of complex DPs, see the schema in (61), at least for complex DPs involving possession and coordination.²¹ As such, we apply Wiltschko's idea of anchoring in a way that clearly distinguishes it from the clausal anchoring category.



In sum, in this section we have motivated how the possessive relation is established. However, what is not yet clear from the schematic structure in (61), is how the observed ordering restrictions arising from the OCP can be accounted for. As discussed in detail in Section 4, the possessive element must be placed between possessor and possessum because it functions as a linking element whilst at the same time establishing the possessive relation. All adnominal possessive constructions have in common that the possessive element that is responsible for valuing $[+coincidence_{space}]$ is either part of the possessum DP or the possessor DP. In other words, the possessive element forms a constituent with one of the DPs involved and this in turn plays a role in determining the ordering of possessor and possessum. By contrast, in a complex DP involving coordination, the coordinating conjunction does not form a constituent with either of the two DPs, which explains why there is no ordering restriction in these cases; nevertheless, the OCP still applies, as evidenced by the fact that leaving out the coordinating conjunction leads to ungrammaticality, cf. the contrast in (62).

(62) German

- a. der Peter und die Lena gehen heim.
 the Peter and the Lena go home
 ‘Peter and Lena go home.’

²¹It may be possible that anchoring is involved in further types of complex DPs, but we leave this for future research.

Alexandra Rehn & Hannah Booth

- b. * der Peter die Lena gehen heim.
 the Peter the Lena go home
 Intended: ‘Peter and Lena go home.’

6 Conclusion

In sum, we have discussed three types of possessive construction in Germanic (the PLC, adnominal genitive and prepositional possessive), specifically as attested in present-day German dialects, as well as in Middle Low German. We have argued that the positional restrictions which are exhibited in this context can be attributed to a morphosyntactic OCP effect within complex DPs, and moreover that this can also account for subtle differences between diachronic stages of a language, e.g. Middle versus Modern Low German. In line with the general view of linking elements, the possessive elements are multifunctional in the sense that they not only prevent the OCP violation but also establish a possessive relation, albeit in an abstract way. In order to bring these two aspects of Germanic possessive constructions together, we drew on ideas from within the Universal Spine Hypothesis of [Wiltschko \(2014\)](#), namely that complex DPs involve an anchoring function which is based on a specific version of coincidence, namely coincidence in space. Specifically, we proposed that it is the possessive elements (i.e. the linkers) which contribute the $[+\text{coincidence}_{\text{space}}]$ feature, which is characteristic for possession overall. This $[+\text{coincidence}_{\text{space}}]$ feature distinguishes possession from other types of complex DP, for instance, coordination, which involves the anchoring feature valued as $[-\text{coincidence}_{\text{space}}]$.

As such, our account for Germanic possession does not conform to the specific type of unified account previously suggested (e.g. [Haegeman 2004](#), [Weiß 2008](#), [Roehrs 2019](#)), whereby all possessors start out in either complement or specifier position, or within a small clause. However, our approach is unified in the sense that all possessive constructions are “mediated” by a nominal anchoring category in the sense of [Wiltschko \(2014\)](#). Under this account, the possessor DP is either based in Spec or complement position, depending on whether the element which expresses the relevant features for anchoring is part of the possessor or the possessee. We view this “variation” in positioning as being tied to the OCP, as it is neither the structure nor the semantic pattern that requires the specific ordering but the fact that the element must be placed between DP_1 and DP_2 . If on the right track, this connection between anchoring and restrictions on possessive constructions in Germanic opens up the possibility that further morphosyntactic OCP effects could be related to the interpretation of relations between morphosyntactic entities.

Abbreviations

CHLG	Corpus of Historical Low German
MLG	Middle Low German
OCP	Obligatory Contour Principle
PLC	Possessor Linking Construction
ReN	Referenzkorpus Mittelniederdeutsch/Niederrheinisch (1200–1650)
SynAlm	Syntax of Alemannic
UoL	Unit of Language

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Will be added later

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Alexandra Rehn & Hannah Booth

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Alexandra Rehn & Hannah Booth

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Chapter 6

Some remarks on the properties and analysis of expletive *es* in German

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1 Introduction

All present-day Germanic V2 languages make use of expletive elements as an apparent Last Resort to fill the clause-initial position in root clauses in case no other material undergoes fronting. Some relevant examples are given in (1) (cf. e.g. [Vikner 1995](#) for an overview).

- | | | | |
|-----|----|---------------------------------------|---|
| (1) | a. | Es ist ein Junge gekommen. | (German, Vikner 1995 : 70) |
| | | EXPL is a boy come | |
| | b. | Es iz gekumen a yingl. | (Yiddish, Vikner 1995 : 70) |
| | | EXPL is come a boy | |
| | c. | Það hefur komið strákur. | (Icelandic, Vikner 1995 : 70) |
| | | EXPL has come (a) boy | |
| | d. | Tað búðu nógv fólk í bynum. | (Faroese, Vikner 1995 : 204) |
| | | EXPL lived many people in town-the | |

Eric Fuß

- e. Det er løver i Afrika. (Norwegian, Breivik 1989: 57)
EXPL are lions in Africa
- f. Der boede mange folk i byen (Danish, Vikner 1995: 204)
EXPL.LOC lived many people in town-the
- g. Er is hier veel sneeuw. (Dutch, Breivik 1989: 57)
EXPL.LOC is here much snow

(2) *Generalization: CP expletives and V2 in Germanic*
rigid V2 syntax $\nleftrightarrow$ CP expletives

Hence, there seems to be a correlation between a rigid V2 syntax and the existence of expletive elements related to the CP domain of the clause, quite similar to the connection between an obligatory subject position and subject (or, TP-related) expletives, cf. (2). However, while there is a bulk of literature on the theoretical analysis of subject expletives (cf. e.g. the – already dated – overview in Svenonius 2002), CP expletives have received considerably less attention. Based on a review of the properties of CP expletives (with a focus on German *es* ‘it’), this paper presents a new analysis that attributes the insertion of a CP expletive to a repair operation (cf. Brandt & Fuß 2013 on the notion of grammatical repair) that applies to satisfy an EPP feature linked to C. More precisely, it is argued that a CP expletive is externally merged in SpecCP to rescue a derivation where C contains an unvalued discourse-related δ -feature. It will be shown that the relevant mechanism not only captures the syntactic properties and functions of the expletive *es*, but also sheds new light on the information structural properties of relevant constructions.

The paper is organized as follows. Section 2 presents a set of basic data each theoretical account of CP expletives must account for. Section 3 summarizes the descriptive facts and highlights a number of open research questions. Section 4 briefly reviews previous analyses before it lays out an alternative approach that assumes that the insertion of a CP expletives is the result of a repair operation that is required to patch up a syntactic derivation in which the EPP feature on C cannot be satisfied. Section 5 discusses how this approach accounts for the core properties of expletive *es* in German. Section 6 wraps up and presents a brief concluding summary.

6 *Some remarks on the properties and analysis of expletive es in German*

2 Properties of CP expletives

2.1 Syntactic distribution

This section briefly reviews some basic characteristics of CP expletives, with a focus on German *es* ‘it’ (including some side-remarks on other languages). As is well-known, pure CP expletives are confined to the clause-initial position, immediately to the left of the finite verb (henceforth prefinite position) as in the impersonal passive in (3a). The CP expletive may not occur in postfinite position (i.e., the TP domain), neither in embedded clauses as in (3c) nor in cases where some other element undergoes fronting to prefinite position as in (3b).

- (3) a. *Es* wird getanzt. (CP expletive *es*)
 EXPL is danced
 b. Heute wird (**es*) getanzt.
 today is EXPL danced
 c. dass (**es*) getanzt wird
 that EXPL danced is

As shown in (4) and (5) no such restriction holds for other non-referential variants of *es* such as the quasi-argument occurring e.g. with weather verbs and the cataphoric pronoun pointing to an extraposed clausal argument.¹

- (4) a. *Es* regnet heute. (quasi-argument *es*)
 it rains today
 b. Heute regnet *(*es*).
 today rains it
 ‘Today, it is raining.’
 c. dass *(*es*) heute regnet
 that it today rains
 ‘that it is raining today’
- (5) a. *Es* hat mich überrascht, [dass *es* regnet]. (cataphoric *es*)
 it has me surprised that it rains

¹The other non-referential instances of *es* do not show a uniform behavior. Quasi-arguments are obligatory, while the insertion of a cataphoric pronoun is often merely optional, although it is close to obligatory in some contexts depending on factors like choice of matrix predicate and (non-)finiteness of the clausal complement (cf. e. g. Pütz 1986², Sudhoff 2003, Sternefeld 2008: 347f.). As shown in (4a) and (5a) quasi-argumental and cataphoric instances of *es* can also occupy SpecCP, sometimes leading to ambiguous expressions where the status of the clause-initial element (cataphoric pronoun or prefinite expletive) is not entirely clear.

Eric Fuß

- b. Mich hat ?(es) überrascht, [dass es regnet].
me hat it surprised that it rains
'It surprised me that it rained.'
- c. dass ?(es) mich überrascht hat, [dass es regnet]
that it me surprised has that it rains
'that it surprised me that it rained'

The same basic facts hold for *það*, the corresponding CP expletive in Icelandic. Note that in contrast to other Germanic languages, Icelandic lacks other non-referential pro-forms, even with weather verbs:²

- (6) a. Það rigndi í gær.
EXPL rained yesterday
- b. *Í gær rigndi það.
yesterday rained expl
'It rained yesterday.'
(Booth 0218: 57)

Similar to the other Germanic V2 languages, Mainland Scandinavian requires that the prefinite position must be filled by an expletive if no other element undergoes fronting. However, in contrast to German and Icelandic, the expletive is not restricted to clause-initial position; rather, it can also show up (and is even required) in postfinite (subject) position, as in (7b) (similar to English *there/it*). This is usually taken to suggest that the expletive is inserted in a lower position from which it undergoes fronting to prefinite position (cf. e.g. Vikner 1995).

- (7) a. Det dansades i går.
EXPL dance.PST.PASS yesterday
- b. I går dansades det. (Swedish, Booth 0218: 143)
yesterday dance.PST.PASS EXPL

²But note that in contrast to German *es*, *það* can occur in embedded clauses to the right of a complementizer as in (i). This difference between German and Icelandic has to do with the fact that Icelandic is a symmetric V2 language, which exhibits generalized V2 in embedded clauses, while in German, V2 is confined to root clauses (so-called asymmetric V2). (i) *Prófessorinn sagði að það hefði einhver lokið ritgerðinni í gær.* professor-the said that EXPL had someone finished thesis-the yesterday (ii) *Prófessorinn sagði að í gær hefði einhver lokið ritgerðinni.* professor-the said that yesterday had someone finished thesis-the (Vangsnes 2002: 47)

6 Some remarks on the properties and analysis of expletive *es* in German

2.2 Lack of A-properties

In contrast to subject expletives, CP expletives do not take part in the valuation of case and agreement. This is illustrated in (8) – while *det* ‘it’ triggers singular number agreement on the participle in Swedish, verbal agreement is solely controlled by the associate DP (if present) in Icelandic and German:³

- (8) a. Det blev skjutet älgar.
 it was shot.NEUT.SG moose
 ‘There were some moose shot.’
 (Swedish)
- b. Það voru skotnir hvalir.
 it were.3PL shot.NOM.MASC.PL whale
 ‘There were some whales shot.’
 (Icelandic)
 (Sigurðsson 2010: 171)
- c. Es wurden viele Fragen beantwortet.
 it were.PL many questions answered
 ‘Many questions were answered.’
 (German)
 (NKU02/MAR.09224 Nordkurier, 27.03.2002)

Likewise, it is usually assumed that CP expletives do not enter into a dependency with their associate (which may be absent) or partake in case checking. This has inspired the wide-spread idea that they are directly base-generated in SpecCP, an A-bar-position (see Richards & Biberauer 2005 for conceptual arguments for the assumption that CP expletives are not part of the inflectional/A-domain of the clause).

2.3 Definiteness effects

As is well-known, the insertion of expletive pronouns typically leads to a definiteness effect, that is, a cooccurring thematic subject (traditionally called the associate) must usually be indefinite (Milsark 1974).⁴ (i) There were the same

³The bulk of the German examples is taken from the German Reference Corpus (DeReKo), as indicated by the relevant text sigla as in (8c).

⁴The exact nature of the definiteness effect is still an open issue. There is a long tradition, at least going back to Milsark (1977), to explain the non-availability of a definite associate DP in syntactic terms. However, it has repeatedly been pointed out that the acceptability of examples like (i) is at odds with a purely syntactic account, perhaps suggesting a pragmatic explanation (cf. e.g. Ward & Betty 1995).

Eric Fuß

people at both conferences. (Newmeyer 1998: 65) However, languages differ with respect to the class of indefinites that are compatible with the use of an expletive. As shown in (9), subject expletives such as English *there* tolerate only weakly quantified associate DPs.

- (9) a. There are (**some/three/many/few/no**) **children** in the garden. (weak quantifiers)
 b. * There is (are) **every/each/all/most child(ren)** in the garden. (strong quantifiers)
 c. * There are **the children** in the garden.
 (Hinterhölzl 2019: 199)

It has been observed that the definiteness effect is somewhat less strict in languages that exhibit pure CP expletives such as Icelandic. While definite DPs are generally ruled out in connection with the expletive *það* (cf. the examples in (10)), strongly quantified DPs may occur in this context, but only in a position to the left of the non-finite verb as in (11a). No such restriction holds for weakly quantified DPs. This is shown in (11b).

- (10) a. **það hefur verið flugan í súpunni.*
 there has been fly-the in soup-the
 b. **það hefur flugan verið í súpunni.* (Icelandic, Thráinsson 2007: 323)
 there has fly-the been in soup-the
 (11) a. *það hafa nokkrir kettir/ allir kettirnir verið í eldhúsinu.*
 there have some cats/ all cats-the been in kitchen-the
 b. *það hafa verið nokkrir kettir/* allir kettirnir í eldhúsinu.*
 there have been some cats/ all cats-the in kitchen-the
 (Icelandic, Thráinsson 2007: 324)

German *es* is even more permissive. As shown in (12) and (13), strongly quantified subjects and even definite DPs are generally available:

- (12) *Es haben alle sehr konzentriert gearbeitet.*
 EXPL have all very concentrated worked
 ‘All worked very hard.’
 (A01/FEB.09687 St. Galler Tagblatt, 23.02.2001)

6 *Some remarks on the properties and analysis of expletive es in German*

- (13) a. Es **haben** die besten Mannschaften des Landkreises Gifhorn
 EXPL have the best teams of-the district Gifhorn
 zugesagt.
 assented
 ‘The best teams of the district Gifhorn have accepted the invitation.’
 (BRZ12/JUN.00554 Braunschweiger Zeitung, 01.06.2012)
- b. Es **hat** der Mut gefehlt.
 EXPL has the courage lacked
 ‘There was a lack of courage.’
 (A12/OKT.07167 St. Galler Tagblatt, 18.10.2012)

However, German still exhibits some form of the definiteness effect in that unstressed subject pronouns may not cooccur with the prefinite expletive *es* (cf. Erdmann 1886, Curme 1922: 460).⁵ As shown in (14), a somewhat weaker effect can be observed in connection with proper names:^{6,7}

⁵The Finnish expletive *sitä* seems to exhibit a similar behavior in that it may cooccur with definite DPs while unstressed personal pronouns are ruled out (cf. Holmberg & Urpo 2002).

⁶There are some contexts where the use of proper names does not seem to create problems, for instance in event notices that announce a speaker (arguably the most important information, i.e., an information focus) as in (i) or if the subject (or subject pronoun) is contrastively focused as in (ii).

- (i) Am Donnerstag, 12. Januar, findet um 19 Uhr im Gasthaus Krämer in Argenthal eine Veranstaltung der Züchtervereinigung Hunsrück statt. Es spricht **Prof. Dr. Georg Dusel** von der FH Bingen zu aktuellen Fragen der Schweinefütterung.
 ‘On Thursday, January 12th, at 7 p.m., the Hunsrück Breeders’ Association will hold an event in the Gasthaus Krämer in Argenthal. Prof. Dr. Georg Dusel from the FH Bingen speaks on current issues of pig feeding.’
 (RHZ06/JAN.09077 Rhein-Zeitung, 12.01.2006; Infos rund um Schweinefutter)

- (ii) Es könnte ja auch mal **PEter/ER** den Abwasch machen.
 expl could prt also once Peter/he the washing make
 ‘Peter/he could also do the dishes for once.’

⁷While subject pronouns referring to persons such as *ich* ‘I’, *du* ‘you’, *er* ‘he’, *sie* ‘she’ etc. are ruled out in connection with a prefield expletive, instances with an additional subject pronoun *es* ‘it’ do show up in corpora, compare the following examples taken from the German Reference Corpus:

- (i) a. Es **hat es** mir sehr gefallen.
 EXPL has it me very pleased
 ‘I really enjoyed it.’
 (M08/JUN.49737 Mannheimer Morgen, 28.06.2008, S. 24)

Eric Fuß

- (14) Es hat *er/?Ludwig niemanden gesehen.
 EXPL has he/Ludwig nobody.ACC seen
 ‘He/Ludwig saw nobody.’

No such effect is observed in connection with pronominal objects:

- (15) Es hat ihn niemand gesehen.
 expl has him nobody seen
 ‘Nobody saw him.’

A related (arguably more subtle) contrast emerges with full DPs. While expletive *es* does not seem to be compatible with a topical subject DP (compare (16)), examples with a topical object DP seem to be more acceptable as shown in (17).

- (16) a. Context: Ich erzähle dir etwas über den Hausmeister.
 ‘I tell you something about the janitor.’
 b. ?? Es hat [der Hausmeister] heute den Baum im Vorgarten
 EXPL has the janitor today the tree in-the front
 gefällt.
 yard felled
 ‘The janitor felled the tree in the front yard.’
- (17) a. Context: Ich erzähle dir etwas über den Baum in unserem Vorgarten.
 ‘I tell you something about the tree in our front yard.’
 b. Es hat ?[den Baum] heute ein Windstoß umgeblasen.
 EXPL has the tree today a gust of wind down-blown
 ‘A gust of wind blew the tree down today.’

This contrast between subjects and objects is captured by the generalization in (18) (cf. e.g. Rögnvaldsson 1984, Sigurðsson 1989 on Icelandic, Speyer 2008, 2009 on German).⁸

-
- b. Es hat es gekracht, Tränen flossen, Ehefrau Renate (45) warf ihn raus.
 EXPL has it crashed tears flowed wife Renate threw him out
 ‘There was a row, tears flowed, his wife Renate (45) threw him out.’
 (HMP06/JUN.01949 Hamburger Morgenpost, 19.06.2006, S. 1–34–35)

⁸The contrast between subjects and objects is perhaps linked to a structural difference. It is well-known that given material is typically placed at the left edge of the TP domain in German (cf. e.g. Lenerz 1977, Frey 2006a). Note however, that it is usually assumed that objects and subjects reach this position as a result of different movement operations: While topical/given subjects undergo A-movement to SpecTP, given objects are usually assumed to undergo scrambling to an A-bar-position (cf. Grewendorf & Sabel 1999). So, descriptively speaking, we might say that the insertion of expletive *es* is blocked by topical elements that reach the left edge of TP via A-movement.

6 Some remarks on the properties and analysis of expletive *es* in German

(18) Context for the insertion of CP expletives

A CP expletive may be inserted if the clause lacks a topical subject.

2.4 Position of the associate subject

It is a well-known fact that (nominal) subjects do not need to move to SpecTP in German (cf. e.g. Haider 2010). This holds in particular for indefinite DPs, which tend to stay in their vP-internal base position. This leads us to expect that the indefinite subject of clauses introduced by expletive *es* should also preferably occupy a lower position to the right of high sentence adverbs such as *wahrscheinlich* ‘probably’ and *leider* ‘unfortunately’. This expectation seems to be borne out by the facts:

- (19) a. Es haben **leider** in den vergangenen Jahren [viele
EXPL have unfortunately in the last years many
Wandervereine] aus Altersgründen oder Personalmangel
hiking clubs out-of reasons-of-age or staff-lack
schließen müssen.
close must
‘Unfortunately, many hiking clubs have had to close in recent years
for reasons of age or lack of staff.’
(RHZ06/AUG.25174 Rhein-Zeitung, 28.08.2006; Neue Wanderroute
feierlich eröffnet)
- b. Es haben sich **wahrscheinlich** während der Pandemie [so viele
EXPL have self probably during the pandemic so many
Menschen] gegen Grippe impfen lassen wie nie zuvor.
people against flu vaccinated let as never before
‘More people have probably gotten the flu vaccine during the
pandemic than ever before.’
(U21/FEB.01604 Süddeutsche Zeitung, 15.02.2021, S. 14; Corona
bekämpft, Grippe besiegt)

In a pilot study carried out in the German Reference Corpus (DeReKo), the vast majority of examples follows the pattern in (19). There seem to be only very few cases where an indefinite subject appears to the left of a sentence adverb as in (20).

Eric Fuß

- (20) Es sind [nur zwei Lieferanten] **leider** etwas stärker
 EXPL are only two suppliers unfortunately somewhat stronger
 betroffen.
 affected
 ‘There are to suppliers who are unfortunately a little more affected.’
 (NON12/JUN.12811 Niederösterreichische Nachrichten, 21.06.2012)

Note that the subject in (20) represents a contrastive focus, which is known to license exceptional patterns and outweigh all other factors that govern the word order in the German middle field (cf. e.g. [Lenerz 1977](#)).

What about definite subjects? Initial results from a corpus study in the German Reference Corpus suggest that even with definite subjects there is a tendency for the subject to occur to the right of a sentence adverb when the clause is introduced by an expletive, compare the examples in (21).⁹

- (21) a. Es haben sich **leider** [die Mehrheiten] nicht geändert.
 EXPL have self unfortunately the majorities not changed
 ‘Unfortunately, the majorities have not changed.’
 (PSN/W03.00072 Protokoll der Sitzung des Parlaments Sächsischer Landtag am 10.12.2002)
- b. Es hat ja **leider** [die ÖVP] Ende des Jahres 2008 nicht
 EXPL has PRT unfortunately the ÖVP end of-the year 2008 not
 nachgeben wollen.
 give-in want
 ‘Unfortunately, the ÖVP didn’t want to give in at the end of 2008.’
 (PNO/W17.00019 Sitzungsbericht der 19. Sitzung des Landtages von Niederösterreich)

To summarize, it appears that the subject prefers to take a lower, vP-internal position in clauses introduced by expletive *es*. This is expected if the subject is part of the focus domain in these constructions. However, let me stress that the

⁹A first search in the archive “W-öffentlich” of the German Reference Corpus has not produced any examples in which a definite subject occurs to the left of *wahrscheinlich/leider*. Note that the search was restricted to finite instances of the frequent verbs ‘have/be/go/come/drive’ (e.g. ‘(Es hat (die oder der oder das)) /+w2:4 leider’). This result is somewhat surprising since the placement option in (i) seems to be generally available (at least according to my own intuitions). For some reason, this variant seems to be less favored in the written language. (i) Es haben sich [die Mehrheiten] **leider** nicht geändert. EXPL have self the majorities unfortunately not changed ‘Unfortunately, the majorities have not changed.’

6 Some remarks on the properties and analysis of expletive *es* in German

above observations should be taken with a grain of salt – as is well known, the word order in the German middle field is very free and influenced by a large number of different factors (cf. recently [Sternefeld 2023](#)). In any case, the preliminary results reported here should be checked against additional, ideally experimental data.

2.5 Contexts of use and pragmatic functions

2.5.1 Lack of a topical subject

As already mentioned, CP expletives typically appear in clauses that lack a topical subject and therefore a topic-comment structure, giving rise to a broad focus reading (in which the subject is part of the focus domain). Relevant contexts include subjectless constructions (e.g., impersonal passives as in (3) above), existential sentences, which often specify the location of a discourse-new entity as in (22) and presentational constructions, which describe a specific event in which a discourse-new entity participates as illustrated in (23). In both cases the subject is typically an indefinite element.

- (22) Es liegt sehr viel Neuschnee auf dem Berg.
 EXPL lies very much new-snow on the mountain
 ‘There is very much new snow on the mountain.’
 (NUN94/FEB.00475 Nürnberger Nachrichten, 05.02.1994, S. 2; Wo Afrika am höchsten ist - Um den Sonnenaufgang auf dem Kilimandscharo zu erleben, braucht man eine gute Kondition)
- (23) Es kam ein Elefant daher, er begab sich das Etschtal hinauf.
 EXPL came an elephant along he took himself the Etsch valley up
 There came an elephant along. He went up the Etsch valley.
 (A00/OKT.69309 St. Galler Tagblatt, 09.10.2000)

It is a long-standing idea that the insertion of a clause-initial expletive serves to put emphasis on the subject, which occurs in a low postverbal (i.e., vP-internal) position where it receives the main accent/sentence stress (which is typically assigned to the rightmost element in the middle field), cf. e.g. [Curme \(1922: 461\)](#), [Erben \(1966: 195\)](#), and [Jung \(1967: 337\)](#). The connection between prosody and pragmatics is discussed in some more detail by [Pütz \(1986²: 47ff\)](#), who argues that the use of a prefield expletive makes available a sentence with a standard/unmarked intonation contour where the postverbal subject receives the main sentence stress. As a result, the subject is interpreted as part of the

Eric Fuß

comment (i.e., the focus domain of the clause). Moreover, Pütz distinguishes two types of sentences with the prefield *es*: On the one hand, there are cases where the insertion of *es* seems to be obligatory. These may include examples like (24) where the use of *es* makes available a certain reading that cannot be achieved otherwise (since it requires that the main accent is placed on the subject), or cases like (25) where in prefinite *es* is required to form a grammatical sentence (since the clause contains no frontable material):

- (24) a. Es lebe der König.
EXPL live-OPT the king
'Long live the king.'
- b. Der König LEbe.
the king live-OPT
'The king lives.'
(Pütz 1986²: 51)
- (25) a. Es ist nicht durchzukommen.
EXPL is not through-to-come
'It's impossible to get through.'
- b. ?? Nicht ist durchzukommen.
- c. ?? Nicht durchzukommen ist. (Pütz 1986²: 51)

On the other hand, there are contexts where the insertion of the expletive seems to be optional. For example, Pütz (1986²) argues that there is no obvious prosodic or interpretational difference between the variant with (26a) and without the expletive (26b), see Section 2.6 for further discussion:

- (26) a. Es wurde gestern Abend viel getanzt.
EXPL was yesterday evening much danced
'There was a lot of dancing last night.'
- b. Gestern Abend wurde viel getanzt.
yesterday evening was much danced
'There was a lot of dancing last night.'
(Pütz 1986²: 52)

2.5.2 Introducing a new situation

In addition to introducing new discourse referents, sentences that begin with a CP expletive also serve functions at the level of the discourse/text in that they

6 Some remarks on the properties and analysis of expletive *es* in German

typically introduce a new situation and/or tense interval (cf. [Fuß & Hinterhölzl 2023](#)). This has already been noted by [Brugmann \(1917: 36f\)](#), who observes that sentences introduced by a CP expletive are typically found at the beginning of texts/text sections:

“Mit dem syntaktischen *es* werden seit mhd. Zeit gerne Sätze gebaut, die die Einleitung zu Erzählungen bilden [...]; in der ahd. Zeit bedurfte man in diesem Fall noch keiner Deckung der Anfangsstellung des Verbuns. Doch hat sich unser *es* auch vielfach da eingestellt, wo man im Ahd. *thō*, im Mhd. *dō* (nhd. *da*) bei Anfangsstellung der Verbform verwendete.”

‘Since Middle High German (MHG) times, the syntactic *es* is used to build sentences which form the introduction to narratives [...]; in Old High German (OHG), it was still not necessary to cover the clause-initial position of the verb in these cases [by placing another element in front of the verb]. However, our *es* has often developed in contexts where OHG used *thō* and MHG *dō* (present-day German *da*) when the verb would otherwise occur in clause-initial position.’

In [Fuß & Hinterhölzl \(2023\)](#) it is argued that this discourse function can be explained by the assumption that *es* is a weak deictic element which can be used to anchor an event to the utterance situation/the speaker. This general idea receives further support from a number of additional facts that appear to involve some form of contextual anchoring.

2.5.3 Contextual anchoring

In minimal pairs like (27a, b), the use of expletive *es* seems to imply that there is only single situation/event. Thus, while (27a) suggests that there was a specific situation where three wizards came in (possibly at the same time), no such interpretation is forced by (27b).¹⁰

- (27) a. Es kamen drei Zauberer herein.
EXPL came three wizards inside
‘Three wizards came in.’

¹⁰Note that the contrast between (27a) and (27b) is quite subtle. For instance, no all speakers agree that (27a) implies that the wizards entered the room together while others report that the very same interpretation is available with (27b) (even if it less prominent than in (27a)). Certainly, more empirical work is necessary to settle this matter.

Eric Fuß

- b. Drei Zauberer kamen herein.
 three wizards came inside
 ‘Three wizards came in.’

The subtle difference in interpretation between (27a) and (27b) is expected under the assumption that expletive *es* serves to link the event to a new and/or specific situation, as has been proposed by Fuß & Hinterhölzl (2023).

Another subtle interpretative effect that might be attributed to the idea that expletive *es* anchors an event to the utterance situation/the speaker is illustrated with the examples in (28). Here it appears that the use of expletive *es* contextualizes the event expressed by the clause; in particular, clauses introduced by *es* seem to imply that the speaker has a specific event/situation in mind (for related ideas on the discourse function of expletives cf. Haegeman et al. 2017, Sluckin 2021):

- (28) a. Es wurde getanzt.
 EXPL was danced
 ‘There was dancing.’ ⇒ There was a specific situation in which people danced.
 b. Es ist ein Haus abgebrannt.
 EXPL is a house burned down
 ‘A house burned down.’ ⇒ There was a specific situation where a house burned down.

In other words, the utterances in (28) appear to convey some form of evidentiality, in the sense that the speaker has either witnessed the event or has sufficient background information on it (see also Sluckin 2021 on English *there*).

In addition, expletive constructions can be used to highlight significant and/or important events/actions that are part of a larger situation or chain of events, as in (29)-(32). Note that in this usage, the subject is often a definite DP, which may also correspond to given material as in (32).¹¹

¹¹In addition, a definite subject may occur in idioms/formulaic expressions as (i) or in cases where the thematic subject represents a contrastive focus as in (ii).

- (i) a. Es kam [die Idee] auf, aus den Früchten selber Apfelsaft zu produzieren und zu verkaufen. Die Klassenfirma «Apfelkraft» wurde geboren.
 ‘The idea came up to produce and sell apple juice from the fruit itself. The class company ‘Apfelkraft’ was born.’
 (A11/DEZ.04109 St. Galler Tagblatt, 09.12.2011, S. 37; Mit Initiative und Apfelsaft)

6 *Some remarks on the properties and analysis of expletive es in German*

- (29) Es hat [die Regierung] dem Hause eine Reihe von Vorlagen unterbreitet mit dem Ersuchen, dieselben einer möglichst raschen Beratung und Verabschiedung zuzuführen.
 ‘The government submitted a series of bills to the House with the request that they be discussed and passed as quickly as possible.’
 (Stenographisches Protokoll, 101. Sitzung des Nationalrates der Republik Österreich, 26. Mai 1925)
- (30) Mit Mehrheit angenommen! Es haben [die Abgeordneten der FPÖ, der ÖVP und des Liberalen Forums] zugestimmt. (Zustimmung ÖVP, FPÖ, LiF; Ablehnung SPÖ.)
 ‘Accepted with a majority! The MPs of the FPÖ, the ÖVP and the Liberal Forum agreed. (Approval ÖVP, FPÖ, LiF; rejection SPÖ.)’
 (PNO/W14.00030 Sitzungsbericht der 30. Sitzung der Tagung 1995/96 der XIV. Gesetzgebungsperiode des Landtages von Niederösterreich. Donnerstag, den 9. November 1995. Plenarprotokoll, Sankt Pölten (AT), 1995)
- (31) Der durchs Programm führende Präsident Rudi Hieronimus sagte die nächste Lachnummer an: Es kam [die allen bekannte Edith Pauly], die dieses Mal als ”Blonde Blondine” auf die Bühne trat.
 ‘President Rudi Hieronimus, who was leading the program, announced the next laugh: Edith Pauly, known to everyone, came along, this time as ‘blonde blonde’ entered the stage.’
 (RHZ99/FEB.01508 Rhein-Zeitung, 03.02.1999; ”Heulemänner” ließen die Post abgehen)
- (32) Aufatmen beim FC Tuggen: Nach zwei Niederlagen in Folge gewann das Team von Trainer Adrian Allenspach am Vorwochenende gegen Eschen/Mauren mit 2:0. Damit sollte eine positive Grundlage geschaffen

b. Es hat [der letzte Biss] gefehlt.
 it has the last bite lacked

‘The last bite was missing.

(BRZ06/DEZ.04744 Braunschweiger Zeitung, 11.12.2006; Broistedt spielt zu kompliziert)

(ii) Es haben [die] gewonnen, die größeren Hunger hatten.
 it have those won who bigger hunger had

‘Those who were more hungry won.’

(HMP06/JUL.01095 Hamburger Morgenpost, 11.07.2006, S. 3; PRESSE-ECHO)

Eric Fuß

sein, vom Uto Grund zumindest nicht als Verlierer, besser noch als Sieger vom Feld gehen zu können. Es hat [der Erfolg gegen Eschen] in allen Belangen gut getan.

‘A sigh of relief at FC Tuggen: After two defeats in a row, coach Adrian Allenspach’s team won 2-0 against Eschen/Mauren last weekend. This should create a positive basis for Uto Grund to be able to leave the field at least not as the loser, better still as a winner. The win against Eschen did us good in all respects.’

(SOZ08/SEP.01146 Die Südostschweiz, 06.09.2008; FC Tuggen gastiert heute in Zürich)

This rather special usage of the expletive *es* seems to be confined to certain registers and text types of the written language (such as newspaper reports and minutes), possibly suggesting that we deal here with a secondary function that developed from the earlier usage as a general text-structuring device.¹² Note that in all these usages, the definite subject is typically focused, either contrastively or as part of the information focus (which may consist of several XPs, cf. e.g. Höhle 1982), that is, it represents the most important information conveyed by the sentence (or is at least part of it).

2.6 Apparent optionality

In Section 2.5.1, it was already briefly mentioned that there are some cases like (26) in which the insertion of expletive *es* seems to be truly optional. This apparent optionality seems to run counter the traditional idea that the use of expletives is some kind of last resort operation. In what follows, I want to take a closer look at relevant cases involving expletive *es* in German, focusing on clauses that contain material that usually undergoes fronting to SpecCP. In addition to cases with

¹²First results of a corpus study in the German Reference Corpus (DeReKo) suggest that this usage is most popular in Austria and – to some extent – Switzerland. Independent support for this conclusion comes from the fact that northern speakers do not readily accept examples like (27)–(30) (Daniel Gutzmann, p.c.). Interestingly, it turns out that this rather exceptional use in connection with definite subjects is already found in the earliest attestations of expletive *es*. (i) represents of the earliest clear examples of expletive *es* (dating from the first half of the 12th century); all in all, 8/37 (21.6%) of early cases (12th century) exhibit a definite thematic subject similar to (i) in the data extracted from the Reference Corpus of Middle High German reported on in Fuß & Hinterhölzl (2023).

- (i) Ez ist geborn [der heilant] der die werlt alle erlosen sol
 it is born the Saviour who the world all deliver should
 ‘The Saviour was born, who should deliver the world...’
 (12:1; *Londoner Predigt(fragment)*, 39va,9–10)

6 Some remarks on the properties and analysis of expletive *es* in German

definite subjects (cf. the examples in (27)-(30)), relevant problematic data include clauses that contain temporal frame adverbials such as *gestern* ‘yesterday’ which are usually taken to be unmarked prefield elements, similar to definite subject DPs (cf. Frey 2006a,b, Speyer 2008, 2009):

- (33) a. Es ist **gestern** eine Chance verpasst worden.
 EXPL is yesterday an opportunity missed been
 ‘Yesterday, an opportunity was missed.’
 (HAZ16/NOV.03599 Hannoversche Allgemeine, 26.11.2016, S. 1)
- b. Es ist **gestern** das aufgegangen, was wir schon die ganze
 EXPL is yesterday that worked-out what we already the whole
 Zeit trainiert haben.
 time trained have
 ‘Yesterday, what we had been training for the whole time worked out.’
 (NON12/NOV.05320 Niederösterreichische Nachrichten, 08.11.2012)

Moreover, note that frame adverbials can even cooccur with a definite subject in clauses introduced by *es* (but note that the definite occupies a very low position in (34), which clearly shows that it is part of the focus domain):

- (34) Es ist **gestern** auf die Kleine Anfrage meiner Kollegin Martina
 EXPL is yesterday on the small inquiry of-my colleague Martina
 Renner hierzu [die entsprechende Information] mitgeteilt worden.
 Renner here-to the relevant information given been
 ‘The relevant information was given to my colleague Martina Renner
 yesterday in response to a small inquiry.’
 (PTH/W05.00049 Protokoll der Sitzung des Parlaments Thüringer
 Landtag am 24.03.2011. 49. Sitzung der 5. Wahlperiode 2009-
 Plenarprotokoll, Erfurt, 2011)

Another class of elements that tend to be placed in clause-initial position are demonstratives (Speyer 2008, 2009). Since demonstratives typically mark topical material, we expect that subjects modified by a demonstrative determiner (or, pure nominative demonstrative pronouns) are not readily compatible with expletive *es*. And indeed, it appears that relevant examples are quite marginal, as illustrated in (35).¹³

¹³ A relevant search in the German Reference corpus (public subcorpus W1, search strings „Es hat dieser“, „Es haben diese“) produced only a couple of relevant examples, which seem to be

Eric Fuß

- (35) a. Context: Ich erzähle dir etwas über den Hausmeister.
 ‘I tell you something about the janitor.’
 b. ?? Es hat **dieser Hausmeister** heute den Baum im Vorgarten
 EXPL has the janitor today the tree in-the front yard
 gefällt.
 felled
 ‘This janitor felled the tree in the front yard.’

A curious exception is a peculiar predicative construction that makes use of a postfinite demonstrative that is placed in a high position to the left of sentence adverbials (and seems to be primarily used by Southern German speakers, cf. [Fricke & Swantje 2018](#), [Fricke 2020](#)), as shown in (36). In this context, the definite element regularly appears to the left of the sentence adverb:

- (36) a. Es sind [dies] **wahrscheinlich** Leute, die kaum je eine Kirche
 EXPL are this probably people who hardly ever a church
 aus religiösen Gründen betreten haben.
 out-of religious reasons entered have
 ‘These are probably people who have hardly ever entered a church
 for religious reasons.’
 (A10/JUL.07207 St. Galler Tagblatt, 29.07.2010, S. 25; Kloster St. Gallen
 schwimmt im Geld)

linked to special registers/stylistic levels of the written language, compare (i) and (ii). The vast majority of hits are adverbial expressions (e.g. *dieser Tage* ‘these-DAT days’) or cases where the demonstrative modifies a dative/genitive object (see above for the observation that topical objects combine more readily with expletive *es*):

- (i) Nun, dem Schreiber dieser Zeilen ist zufällig bekannt, daß es bei Hamburgs Behörden seit kurzem sogenannte Dienstliche Beurteilungen gibt. Sie erklären vieles. **Es** haben [diese Beurteilungen] die Form eines vierseitigen Fragebogens, in dessen Rubriken die Vorgesetzten ihre Beamten einzuordnen hatten. ‘Well, the writer of these lines happens to know that Hamburg’s authorities have recently introduced so-called official appraisals. They explain a lot. These appraisals take the form of a four-page questionnaire into whose categories the superiors had to classify their civil servants.’ (Z56/NOV.00447 Die Zeit, 22.11.1956, S. 11)
- (ii) **Es** hat dieser Gestus etwas Kindlich-Naives, aber auch Liebenswertes: [...] ‘There is something childishly naïve about this attitude, but also endearing: [...]’ (Z83/NOV.00241 Die Zeit, 11.11.1983, S. 82)

6 Some remarks on the properties and analysis of expletive *es* in German

- b. Es sind [dies] **nur leider** die Worte von Joachim Gauck,
 EXPL are this only unfortunately the words of Joachim Gauck
 dem Vielbeklatschten.
 the much-applauded
 ‘These are unfortunately only the words of Joachim Gauck, the much
 applauded.’
 (NZZ10/SEP.00635 Neue Zürcher Zeitung, 04.09.2010, S. 53;
 Biedermann als Brandstifter)

At first sight, it seems that the demonstrative assumes topic character in cases like (36); however, note that *dies* ‘this’ does not qualify as the subject of this construction (*pace* Fricke 2020: 42), since verbal agreement is controlled by the low (plural) subject. So while the exact status of *dies* is unclear (perhaps a predicative element, which fits in with the observation that it is a ‘pure’ uninflected demonstrative), the acceptability of examples like (36) seems to be consistent with the observation (see above) that expletive *es* is compatible with topical/backgrounded elements as long as the clause lacks a topical subject. Still, the observation that *es*-insertion appears to be optional in contexts where frontable material is available is a challenge for any theoretical analysis of expletives.

3 Preliminary summary of the empirical findings

By now, it should be clear that the difference between CP expletives and TP expletives does not reduce to their location – CP expletives are not merely positional variants of subject expletives. Table 1 summarizes the contrasting properties of these two types of elements.

The next section develops a new theoretical account that focuses on the connection between *es*-insertion and the absence of a topical subject (see (16) above, repeated for convenience):

(37) Context for the insertion of CP expletives

A CP expletive may be inserted if the clause lacks a topical subject.

Below, I will argue that a successful explanation of (37) also provides an answer to a number of additional questions concerning e. g. the low structural position of subjects and the incompatibility of expletive *es* and subject pronouns. In addition, I will address further syntactic properties of expletive *es* in German such as the restriction to the first position of finite root clauses (which is perhaps the

Table 1: Differences between TP expletives and CP expletives

	TP expletives	CP expletives
structural position	SpecTP	SpecCP
distribution	finite and non-finite clauses	initial pos./finite root clauses
A-properties (case/Agr)	yes	no
definiteness effect	strong	weak (Icelandic)/absent (German; incompatibility with subject pron./topics)
position of the thematic subject	vP-internal	vP-internal
optional insertion	unclear (e.g. obligat. in existentials: <i>there are some problems</i>)	yes (non-topical subject)
information structure	broad focus (including the subject)	broad focus (including the subject)
discourse function	introducing a new situation (?)	introducing a new situation

most conspicuous property of CP expletives, cf. e. g. [Sigurðsson 2010: 172](#)),¹⁴ the absence of A-properties and definiteness effects (but see section 4.3.6 for some remarks on pragmatic effects of *es*-insertion), and the observation that *es*-insertion seems to be optional in certain contexts.¹⁵

¹⁴[Sigurðsson \(2010\)](#) assumes that *það* tucks in directly to the right of (phonetically empty) Fin and that it blocks movement of the finite Verb (to Fin), which remains in the TP domain, thereby ruling out patterns in which *það* appears in post-verbal position. However, it is fairly clear that this analysis does not carry over to German, where T is usually assumed to be head-final. Moreover, Sigurðsson’s approach to the definiteness effect (cf. fn. 20) does not capture the fact that this restriction only applies to personal pronouns (and, to some extent, proper names) in German.

¹⁵The observation that CP expletives do not show an identical behavior in Icelandic and German suggests that we presumably should not strive for a fully uniform analysis of *það* and *es* (there might be some basic structural similarities, though). In any case, an in-depth comparative analysis would go well beyond the scope of this paper. I therefore leave that issue for future research.

4 Towards a theoretical account

Similar to their TP-related counterparts, CP expletives are traditionally analyzed as syntactic placeholders (marking a position that must be obligatorily filled), which apparently do not contribute much to the (truth-conditional) meaning of the clause (as is reflected by labels such as “Vorfeld/prefield-expletives”, Bayer & Peter 1997; “positional expletives”, Booth 2018; or “expletive topics”, Faarlund 1990).¹⁶ From that perspective, expletives seem to be a prototypical case of a form-meaning mismatch – elements without semantic content that are obligatory in certain syntactic contexts (cf. Ross 1967, McCawley 1970, Chomsky 1981, 1995, 2001).¹⁷ This mismatch seems to be even more profound in the case of CP expletives. In contrast to subject expletives, CP expletives do not realize a syntactic function and do not play a role in case and agreement checking/valuation. Rather, their presence seems to be mainly due to a requirement linked to linear order, namely that the clause-initial position be filled (i.e., the V2 constraint). This condition is usually attributed to the presence of a generalized EPP/edge-feature in C (Chomsky 2000, Fanselow & Mahajan 2000, Fanselow 2004, Roberts & Anna 2002, Holmberg 2015 for an overview). However, relevant theoretical approaches differ with regard to the question of whether the presence of an expletive in SpecCP (or some other specifier in the left clausal periphery) is the result of external merge or movement/internal merge.

4.1 Previous analyses

4.1.1 CP expletives are externally merged in SpecCP

According to what might be called the standard view, CP expletives are dummy prefield fillers, which are directly inserted in SpecCP to circumvent a violation of the V2 constraint (cf. e.g. Thráinsson 1979, Rögnvaldsson 1984, Platzack 1987, Sigurðsson 1989 on Icelandic *það*; Breckenridge 1975, Lenerz 1985, Grewendorf 1990, Tomaselli 1990, Mohr 2005, Heck & Müller 2007 on German *es*). As Wurmbrand (2006: 205) puts it:¹⁸

¹⁶However, note that the use of a prefinite expletive usually signals declarative mood.

¹⁷The question whether expletives bear some (abstract) semantic content is actually highly disputed. In fact, there is a long tradition of assigning some semantic content or at least pragmatic function to elements such as English *there* (cf. Allan 1971, Bolinger 1977, Jenkins 1975, Lakoff 1987, Langacker 1991, 2009, Williams 1994, 2006, Moro 1997, Hazout 2004, Hartmann 2008, Tsiakmakis & M. Teresa 2022; for the latter position Sells 2005, Hinzlin 2009, Corr 2017, Haegeman et al. 2017; Kaiser 2019).

¹⁸A variant of the insertion account is proposed by Craenenbroeck (2022) who assumes that at least some types of CP expletives are main clause complementizers.

Eric Fuß

“A straightforward account for these expletives is to assume (..) that CP expletives in German and Icelandic are simply phonological fillers of the initial position (SpecCP) when no overt XP has moved there.”¹⁹

So, in contrast to subject expletives (cf. e.g. [Richards & Biberauer 2005](#)) it is commonly assumed that CP expletives do not undergo movement to their target position (but see Section 4.1.2 for a different view). Rather, they are externally merged in SpecCP to satisfy an EPP/edge feature on C. Note that relevant accounts differ with regard to the question of whether the insertion of an expletive is linked to checking/valuation of additional features and the identity of these feature(s) ([Heck & Müller 2007](#): [Person], [Sigurðsson 2010](#): [$-\lambda$, $-\text{Fin}$],²⁰ [Vangsnes 2002](#): [deixis], [Mohr 2005](#): [subject of predication], ...). Purely positional approaches assuming that expletives satisfy an EPP-feature only include [Fanselow \(2002\)](#) and [Frey \(2006a\)](#) (see also [Holmberg & Urpo 2002](#) on Finnish *sitä*).

The idea that CP expletives are inserted into a specifier position in the left periphery of the clause is often taken to be the null hypothesis since it can account without further ado for a number of basic characteristics of CP expletives. First, it is strongly supported by the fact that the expletive is restricted to clause-initial position and may never appear in the TP domain. Put differently, any account that assumes that the expletive reaches SpecCP via movement must address the question why it cannot show up in a lower position. Second, assuming that the expletive is base-generated in an A-bar-position directly explains why it lacks A-properties and does not seem to be involved in case and agreement checking. Third, the weakening of the definiteness effect seems to indicate that there is a fundamental structural difference between CP expletives and TP expletives (even if the reasons for this contrast are not entirely clear).

Still, there are a number of questions for which the base-generation approach does not seem to offer an obvious answer. For example, it does not immediately explain why definiteness effects are weaker (in Icelandic) or confined to unstressed subject pronouns (in German). Moreover, since – in contrast to TP

¹⁹But note that the exact nature of CP expletives is linked to the nature of the EPP on C, which – similar to the traditional subject requirement in TP – might be reconstructed as a syntactic or a phonological condition (cf. [Heck n.d.](#) for discussion).

²⁰[Sigurðsson \(2010\)](#) assumes that *það* matches negative values for finiteness and speaker/hearer features. The latter are taken to be heads in the left clausal periphery that partake in the licensing of person features (and which he refers to as the *logophoric agent/patient*, λ_A/λ_P). His explanation of the definiteness effect rests on the assumption that definite DPs positively match λ and that positive matching takes precedence over negative matching.

6 Some remarks on the properties and analysis of expletive *es* in German

expletives – there is arguably no structural relation between the thematic subject and the expletive, it is quite mysterious why the subject usually remains in a lower (vP-internal) position.

4.1.2 CP expletives are internally merged in SpecCP

As already briefly mentioned above, there is an alternative line of thinking in the literature that is based on the assumption that the expletive is base generated in a lower position before it undergoes obligatory movement to SpecCP (cf. e.g. Vikner 1995, Jónsson 1996, Vangsnes 2002, Hinterhölzl 2019, Fuß & Hinterhölzl 2023). For example, Vangsnes (2002) assumes that in Icelandic, the expletive *það* is inserted in SpecAgrsP to supply a [deixis] feature (which he takes to be necessary for subject anchoring) in cases where the associate subject DP stays in a lower position. Subsequently, the expletive – which is the hierarchically highest element in the inflectional domain – undergoes obligatory movement to SpecCP to satisfy the V2 condition.²¹

It is immediately clear that this line of thinking is challenged by the very same characteristics that lend support to the base-generation approach. Under the assumption that the expletive is inserted into the TP-domain (or even lower) before it moves into the left periphery, it comes as a surprise that (i) the expletive is confined to the initial position of finite clauses, (ii) does not play a role in case and agreement phenomena, and (iii) does not give rise to the same kind of definiteness effects as TP expletives (see Richards & Biberauer 2005 for additional conceptional arguments against a movement analysis of CP expletives).

²¹To support his idea that *það* is inserted in a lower position before it undergoes movement to SpecCP, Vangsnes cites examples where the expletive occurs after an interrogative complementizer in an embedded question as in (i). The ungrammaticality of (ii) is supposed to show that the position to the right of the complementizer may not be targeted by topicalization and is therefore not to be identified with SpecCP.

- (i) Prófessorinn langaði að vita hvort það hefði einhver lokið ritgerðinni
 professor-the wanted to know if EXPL had someone finished thesis-the
 í gær.
 yesterday
- (ii) *Prófessorinn langaði að vita hvort í gær hefði einhver lokið ritgerðinni.
 professor-the wanted to know if yesterday had someone finished thesis-the
 (Vangsnes 2002: 47)

However, note that the German expletive *es* may never appear to the right of a complementizer. This suggests that the possibility of (i) is presumably linked to the fact that Icelandic (but not German) allows generalized embedded V2 and that the analysis put forward by Vangsnes does not extend easily to German.

Eric Fuß

However, there are also a number of properties for which a movement approach seems to offer a better or at least more obvious explanation. For instance, assuming that the expletive is inserted into the specifier of TP offers an account for the observation that the thematic subject typically occupies a vP-internal position. The incompatibility of *es* and unstressed subject pronouns (cf. (14) above) may then be attributed to either a requirement that unstressed subject pronouns must move to SpecTP (either for licensing purposes or to escape the focus domain), or a locality condition such as Relativized Minimality that bans movement of the (unstressed) expletive across an unstressed subject pronoun that occupies the so-called Wackernagel-position directly to the right of C.²² In what follows, I will line out an alternative theoretical approach that aims at combining the strengths of the movement and base-generation models of CP expletives. In particular, I will argue that the connection between expletive insertion and a low subject position can be captured by assuming that T carries a feature that may be licensed either (i) by movement of topical subjects to SpecTP, or (ii), in case the derivation lacks a topical subject, by insertion of a CP expletive after the unchecked feature has been carried up to C via T-to-C movement.

4.2 A new proposal

The analysis proposed in this section adopts the traditional idea that expletives are semantically and phonologically light elements that are inserted as a last resort to realize a certain structural position (Chomsky 1995). If this is generally true of expletives, they can therefore be taken to signal the presence of obligatory EPP-features that require a structural position to be (overtly) filled: SpecTP in English and Mainland Scandinavian (and, to some extent, Dutch), SpecCP in all Germanic V2 languages (cf. e.g. Haider 2010). Moreover, as we have seen above, CP expletives are only licensed in contexts that lack a topical subject. The information-structural impact of this property is highlighted by the rephrased version of (37) in (38):

(38) CP expletives signal the lack of a topical subject.

²²In addition, this type of analysis would immediately account for the grammaticality of examples like (i), which show that the pronoun does not have to move to SpecCP: Due to the structural/featural difference between unstressed subject pronouns and frame adverbials (or DP objects), the latter can cross the former without inducing a locality violation:

- (i) a. Gestern hat er den Hausmeister getroffen.
 yesterday has he the janitor met
 ‘Yesterday, he met the janitor.’
 b. Den Hausmeister hat er gestern getroffen.

6 Some remarks on the properties and analysis of expletive *es* in German

What I would like to propose is that this effect can be derived by assuming that an expletive is inserted in SpecCP when C contains an unvalued discourse-related δ -feature that has not been valued/licensed by a topical/given subject at a previous stage of the syntactic derivation (i.e., in the TP domain). More precisely, I take it that German T inherits an unvalued δ -feature from C in addition to ϕ -features (Miyagawa 2009, 2017). The unvalued δ -feature seeks a subject DP that carries a matching interpretable δ -feature. Crucially, I assume that in German, only [Topic] (and [Contrast], see below) is a viable value for an interpretable δ -feature. In other words, [Topic] is a syntactically active feature, while information focus corresponds to the absence of a (valued) δ -feature, see 4.2.2 below for discussion. Therefore, only subjects that represent given information and carry an interpretable [Topic] feature can value the δ -feature on T. In contrast, subject DPs that are part of the information focus cannot match the δ -feature on T. Since German is a V2 language, the unvalued δ -feature is then carried up to C by verb movement, where it triggers insertion of expletive *es* as a last resort to satisfy the EPP on C. In this way, it is possible to account for the connection between *es*-insertion and a low subject position without invoking a movement analysis – due to the V2 property, the syntax keeps track on what happened (or rather didn't happen) at a previous stage of the derivation. Before I turn to the specifics of that proposal, let me first introduce some general assumptions about the structure of the clause and the workings of the syntactic computation.

4.2.1 Background assumptions

I adopt a fairly standard minimalist approach to the structure of the clause, assuming a tripartite architecture as in (39) (possibly a stand in for more elaborate systems), where lexical properties linked to the verb's argument structure are mapped to the syntax in the vP domain, while TP encodes inflectional features such as tense, verbal mood, case, agreement, negation (and aspect in languages with a grammaticalized aspect system). CP hosts features linked to the expression of clause type/mood, operator scope and the main/embedded distinction. In addition, it provides the interface to the extra-clausal discourse context and is in many languages used to encode information-structural categories such as topic and focus.

(39) CP – TP – vP

Following Chomsky (2000 et seq.) I assume that there is only a single structure building operation Merge, which builds larger structures by combining two syntactic objects at each step of the syntactic derivation. Merge comes in two vari-

ants depending on whether the merged category is taken from the lexicon (external Merge) or from the structure that has been assembled so far (internal Merge = Move). The syntactic derivation is driven by the need to value/license uninterpretable formal syntactic features (so-called *probes*) that are part of the functional layer of the clause (i.e., CP/TP/vP) and seek a matching *goal* with interpretable features in their c-command domain as part of the operation Agree. I adopt the ‘traditional’ minimalist idea (Chomsky 1995, 2000, 2001) that there is a close connection between Agree and movement/internal Merge, assuming that syntactic movement (triggered by EPP/edge features) is parasitic on Agree/feature matching or even an integral part of it, which is needed to establish a local relation between a probe and its goal (cf. e.g. Miyagawa 2009, Sigurðsson 2010).²³

I follow current minimalist theorizing since Chomsky (2000 et seq.) in assuming that the syntactic derivation proceeds in cycles (so-called phases). At the end of each phase, the complement of the phase head is transferred to the post-syntactic components of grammar (morpho-phonology and semantics). The phase heads are C and v. All syntactic operations are triggered by phase heads. As a result, the set of ϕ -features traditionally associated with T are taken to be actually a property of the phase head C, which passes down (a subset of) its features to T via a process of *feature inheritance* (cf. Chomsky 2004, 2008).

Furthermore, I adopt the view that (prefield) expletives are semantically light deictic elements (cf. Fuß & Hinterhölzl 2023) that primarily serve a syntactic function, namely filling a specifier position the projection of which is required by an EPP/edge-feature (on C). I will try to show that the apparent semantic/pragmatic impact of expletives is by and large the result of other properties of the syntactic derivation that lead to the insertion of an expletive (in combination with its deictic character, see Section 5 below for details).

Finally, let me add out some background assumptions on the analysis of the V2 phenomenon. Pursuing a standard approach, I take it that XP-fronting in V2-languages is triggered by a generalized EPP-/edge-feature on C, cf. e.g. Haegeman (1996), Grewendorf (2002), Roberts (2004), Holmberg (2015).²⁴ Since movement is constrained by locality restrictions such as the Minimal Link Condition

²³Note that in more current minimalist theorizing (cf. e.g. Chomsky 2007) it is often assumed that movement is independent of Agree, being triggered by the free insertion of edge features. Note that this approach seems to resurrect the old idea of Government and Binding Theory that movement always comes for free (move- α), with unwanted effects being ruled out by independent constraints and filters.

²⁴To maintain the above assumption that movement (internal merge) is parasitic on a previous Agree relation (Chomsky 2001), i.e. the presence of an unvalued/uninterpretable formal feature [uF] that must be valued in the course of the syntactic derivation, we might simply assume a generalized feature [uF] that can be matched by any featural property of the attracted element.

6 Some remarks on the properties and analysis of expletive *es* in German

(MLC, Chomsky 1995), the EPP-feature triggers movement of the closest matching XP to SpecCP (i.e., the highest element in the TP-domain, often corresponding to a topical subject). To account for obligatory fronting of *wh*-operators, I adopt the view that, in addition to the EPP/edge-feature, C may host an operator feature (perhaps linked to [Contrast], [+*wh*], or a combination of these) that acts as a probe and selects the closest matching *wh*-phrase (or contrastive focus) as a goal, which is then attracted to SpecCP (cf. e.g. Fanselow 2002, 2004, Grewendorf 2002, and in particular Frey 2006a; Coniglio 2012 on Middle High German).

4.2.2 On δ -features and subject positions in German

According to the proposal advanced in this paper, expletive *es* is inserted in a derivation where a discourse-related δ -feature fails to be licensed by a topical subject in the TP domain. In this section, I will therefore take a closer look at the relevant stage of the derivation before the unvalued δ -feature is raised to the left periphery via T-to-C movement. I am going to argue that there are (at least) two subject positions in the German middle field. Subjects representing given information may move to SpecTP, while subjects that are part of the presentational focus/information focus stay in-situ (in their vP-internal base position). German has thus preserved a typical trait of earlier Germanic languages (cf. e.g. van Kemenade & Bettelou 2006, van Kemenade & Tanja 2012, Biberauer & van Kemenade 2011 on OE). However, before I turn to the relevant properties of German, let me first lay out some basic assumptions about the syntactic representation of information-structural categories. Following previous work by Horvath (1995), Holmberg & Urpo (2002) and Miyagawa (2009, 2017), I take it that discourse-related categories such as topic and focus do not head separate projections in the syntax. Rather, relevant discourse-related δ -features may be added to other (phasal) heads of the clausal spine. I assume that, similar to ϕ -features, δ -features enter the derivation unvalued, acting as a probe that seeks a matching goal (i.e., an interpretable/valued δ -feature) in its c-command domain as part of the operation Agree. The goal may then undergo movement if the probe is linked to a (optional) movement-inducing EPP-feature (which is parasitic on a previous Agree relation). However, not all languages make use of syntactic movement to express one and the same information-structural category (cf. e.g. the differences between English and Italian with regard to focus-marking). In other words, the syntactic impact of individual information-structural categories seems to be subject to language-specific parametric choices. To account for these differences, I assume that languages may differ in as to whether a given information-structural category is syntactically active (i.e., corresponds to a grammaticalized syntactic

Eric Fuß

feature that is visible to syntactic operations such as movement). In what follows, I will briefly address how this approach can be put to use to account for the different impact of the categories topic and focus on the word order facts in German. As is well-known, there is a strong tendency to move given information to the left of the German middle field, while information focus is assigned to the constituent immediately to the left of the right sentence bracket (i.e., to the preverbal position in embedded clauses and the position immediately to the left of the non-finite verb in non-embedded clauses), cf. [Lenerz \(1977\)](#), [Höhle \(1982\)](#), [Krifka \(1998\)](#) and many others. Compare the following pairs of question and answers:

(40) Question:

Was hat er dem Studenten gegeben?
 what.ACC has he the.DAT student given?
 ‘What did he give to the student?’

Answer:

Er hat [dem Studenten] [ein BUCH] gegeben.
 he has the.DAT student a.ACC book given
 ‘He gave a book to the student.’

(41) Question:

Wem hat er das Buch gegeben?
 who.DAT has he the.ACC book given
 ‘Who did he give the book to?’

Answer:

Er hat [das Buch]_i [einem STUDENTEN]_{FOCUS} t_i gegeben.
 he has the.ACC book a.DAT student given
 ‘He gave the book to a student.’

In each case, there is a strong tendency to place the object DP that represents the information focus (i.e., which corresponds to the *wh*-pronoun in the preceding question) directly to the left of the non-finite verb. In (40), the relative order of indirect and direct object corresponds to the base order in the German middle field (subject > indirect object > direct object), while in (41), the accusative object (representing given information) has undergone leftward movement (scrambling), leading to inversion of indirect and direct object.

Furthermore, note that so called focus projection, where the focus assigned to the rightmost element in the middle field extends to other material further to the left (giving rise to broad focus) is only available as long as the constituents of the clause appear in the unmarked basic word order (cf. [Höhle 1982](#)):

6 Some remarks on the properties and analysis of expletive *es* in German

(42) Question:

Was ist geschehen?

‘What happened?’

Answer:

- a. Anna hat [heute Morgen] [mit ihrer SCHWester] telefoniert.
 Anna has today morning with her sister telephoned
 ‘Anna talked with her sister on the phone this morning.’
- b. # [Anna] hat [mit ihrer Schwester]_i [heute MORgen] t_i telefoniert.

While (42a) is a possible answer to a question like *What happened?*, deviations from the basic/unmarked word order lead to a restriction of the focus potential: In (42b), the PP *mit ihrer Schwester* has been scrambled to the left. As a result, it is no longer possible to extend the information focus to the entire sentence: the sentence is merely a potential answer to a question such as *When did Anna phone her sister?*

These facts are commonly taken to suggest that in German, new material that represents information/presentational focus stays in-situ (in its vP-internal base position), while given material undergoes displacement to the left, evacuating the domain of presentational focus, which corresponds to the vP (cf. e.g. Heim 1982, Diesing 1992, Drubig 1992, Meinunger 1996).²⁵ Drawing on wider range of typological data, Drubig (2003: 5) highlights the fact that this asymmetry between the encoding of given information and presentational focus follows directly from Diesing’s (1992) Mapping Hypothesis.²⁶

“An important aspect of MH, however, is that it predicts a markedness asymmetry between presentationally focused and given portions of the sentence, which is supported by substantial cross-linguistic evidence: presentationally focused material occurs in situ and is incorporated into the wider sentential domain of presentational focus, whereas given or presupposed material is predicted to require additional coding. Cross-linguistic comparison of givenness marking has revealed that it is represented by a wide range of coding devices: [...]”

Accordingly, Drubig assumes that presentational focus is freely assigned to an XP that resides in the focus domain, while the coding of given information

²⁵Note that EPP-driven movement of the subject to SpecCP (*Anna* in (42)) does not interfere with a broad focus reading.

²⁶Mapping Hypothesis (Diesing 1992: 10): “Material from VP is mapped into the nuclear scope. Material from IP is mapped into a restrictive clause.”

Eric Fuß

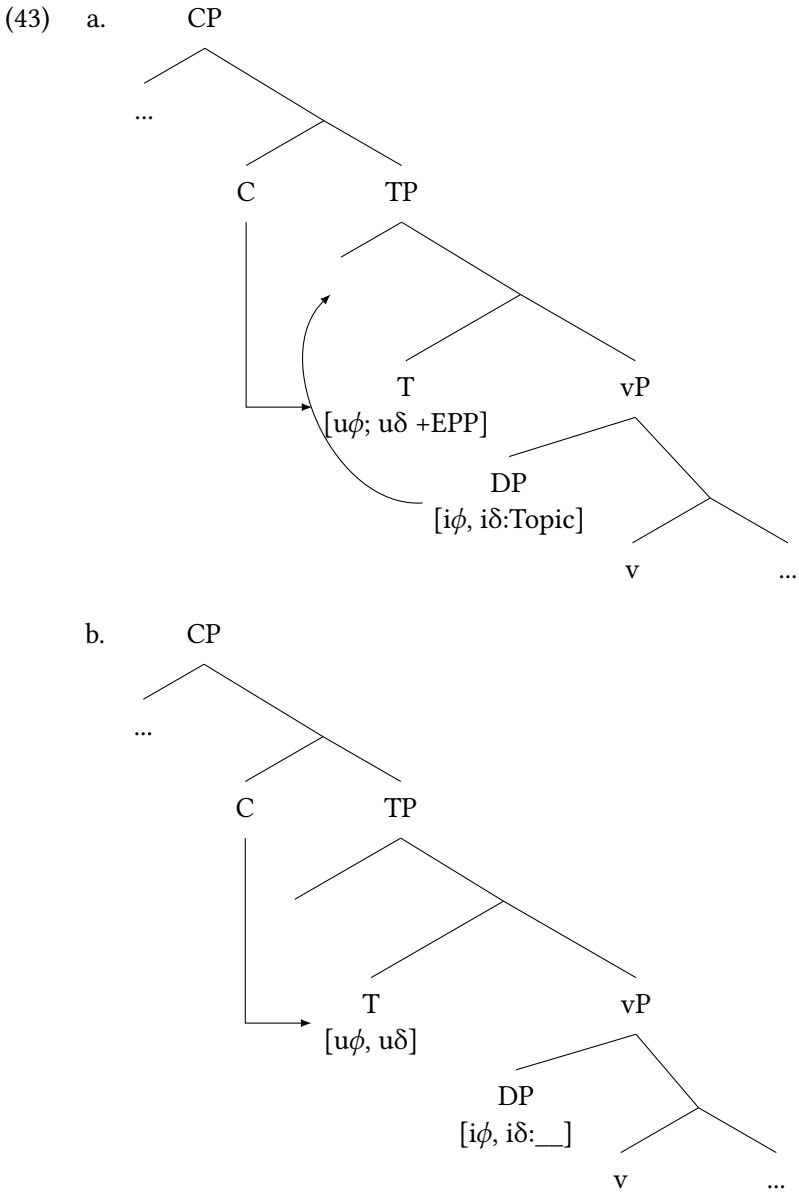
requires additional devices such as syntactic movement. In a system making use of δ -features this basic insight can be captured if we assume that (interpretable) features relating to topicality/givenness (and perhaps contrast) may be optionally added to lexical items that are part of the numeration/lexical array from which syntactic structures are built, while presentational focus represents a default value (see Zeller 2008 and Biberauer & van Kemenade 2011 for the idea that given XPs contain more features/structure than focused elements). More precisely, I propose that – at least in languages such as German where presentational focus is not syntactically encoded – XPs that represent new information carry an unvalued δ -feature that is assigned the default value [Focus] at the interfaces (similar to cases where 3sg is assigned as a default value to a set of agreement features; for the general idea that unvalued probes can be repaired by the insertion of default values cf. e.g. Preminger 2014; see also Noyer 1998, Harley & Ritter 2002, and Harbour 2003). As a result, uninterpretable δ -features added to functional heads of the clausal spine can only be valued by XPs carrying an interpretable δ -feature linked to given information (and perhaps contrast, see below). For the sake of concreteness, I assume that the syntactic correlate of givenness is a δ -feature with the value [Topic],²⁷ which may induce IS-driven leftward movement (scrambling, see e.g. Meinunger 1996, Frey 2004a).

Let us now take a look at how this approach accounts for the word order facts in the German middle field, focusing on the position of the subject (see fn. 30 on scrambling of non-subjects). There are two basic possibilities, depending on the featural content of the external argument. These are illustrated in (43).²⁸

²⁷Note that [Topic] is used here as a general term for all kinds of given, deaccented material.

²⁸I adopt the traditional assumption that TP and vP are head-final in German; they are represented as head-initial projections in (43) for expository reasons only.

6 Some remarks on the properties and analysis of expletive *es* in German



In both cases, T inherits an unvalued δ -feature ($[u\delta: _]$) from C, together with a set of ϕ -features (at least number, person, and features linked to the assignment of nominative case).²⁹ $[u\delta]$ is a probe that seeks a matching goal with a

²⁹Note that in the early Germanic languages, the prefinite position in declarative root clauses

valued δ -feature. Due to the co-occurrence with ϕ -features, $[\text{u}\delta]$ on T can only be licensed/valued by the DP that controls agreement and is assigned nominative case.³⁰ In (43a), the external argument (merged in SpecvP) contains an interpretable δ -feature with the value $[\text{Topic}]$ and therefore matches and values both $[\text{u}\phi]$ and $[\text{u}\delta]$ on T (via Agree). Since $[\text{u}\delta]$ is bundled with an EPP feature,³¹ the topical DP subsequently undergoes movement to SpecTP. In (43b), however, the external argument merely carries an unvalued interpretable δ -feature (interpreted as information focus at the interfaces by default).³² As a result, the DP can only value T's ϕ -set, but leaves $[\text{u}\delta]$ unvalued. Under the assumption that in German, the EPP-/edge-feature on T is parasitic on the valuation of $[\text{u}\delta]$ (in contrast to English, where it is linked to $[\text{u}\phi]$), non-topical subjects cannot raise to SpecTP even if an optional EPP feature were added to T (or, rather $[\text{u}\delta]$).³³ As a result, they occupy a low vP-internal position (recall that German marks information focus by a combination of prosodic and syntactic means – focused material typically occurs at the right edge of the middle field in a position where sentence stress is assigned).

However, the absence of a specified interpretable δ -feature on the subject does not automatically entail that $[\text{u}\delta]$ leads to a crashing derivation. What I would like to propose is that the V2 property opens the possibility that $[\text{u}\delta]$ on T can be matched at a later point in the same derivational cycle/phase (see Heck & Müller 2000 for a related proposal that ensures that for every probe there is a matching goal in the current phase (Phase Balance)). More precisely, I will argue

was confined to topical material (Axel 2007). This suggests that C originally did not donate a $\text{u}\delta$ -feature to T, limiting SpecCP to topical XPs (and wh-operators in cases where C hosted a corresponding $[\text{+wh}]$ /operator feature such as $[\text{Contrast}]$). The change of the relevant parameter from KEEP to DONATE (Ouali 2008) seems to be part of a general diachronic tendency towards encoding information structural distinctions in the middle field (cf. e.g. Speyer 2015 on German, Struik & Gert-Jan 2023 on the history of Dutch).

³⁰To facilitate scrambling of non-subjects to the left of a topical subject, we can either assume that T may carry multiple δ -features, giving rise to a configuration with multiple specifiers (where the first-merged specifier must host the subject), or we can follow Miyagawa (2009), who proposes that C may pass down additional δ -features to a separate head in between CP and TP, the specifier of which hosts scrambled non-subjects.

³¹See Pesetsky & Esther (2001) for the idea that the EPP is a property of a feature of a head and not a property of the head itself.

³²See Pesetsky & Esther (2007) for the idea that interpretability and valuation of features are independent concepts.

³³I take it that the EPP-/edge-feature bundled with $[\text{u}\delta]$ becomes active only upon valuation of $[\text{u}\delta]$. If $[\text{u}\delta]$ does not find a matching $[\text{iTop}]$, its EPP-/edge feature remains inert and is ignored at the interfaces (similar to other uninterpretable features that fail to undergo valuation, along lines laid out in Preminger 2014). Note that in this way, a potential lookahead problem can be avoided. I am indebted to Fabian Heck (p.c.) for clarifying this issue.

6 Some remarks on the properties and analysis of expletive *es* in German

that after T-to-C movement, the unmatched δ -feature can either be valued (i) by a topical non-subject XP or (ii) as presentational focus by default. The latter option is accompanied by the insertion of expletive *es* in order to satisfy the EPP feature on C. In the following two sections I will discuss these two options in more detail.³⁴

4.2.3 Fronting of topical non-subjects

The fact that fronting of an arbitrary XP leads to subject-verb inversion is the defining characteristic of the Germanic V2-property. In this section, I will discuss how fronting of topical non-subjects is linked to the presence of $[u\delta]$ on C (after it has reached the left clausal periphery as a side-effect of verb movement).

In German, word order variation in the so-called middle field plays an important role in signaling information structural distinctions. In particular, given material regularly undergoes movement to a left-peripheral (pre-subject) position in the TP-domain (cf. [Lenerz 1977](#), [Höhle 1982](#), [Frey 2004a](#), [Grewendorf 2005](#), and many others). What I want to propose is that this typological trait of German opens up the possibility that after T-to-C movement, $[u\delta]$ can be satisfied by a non-subject phrase that is marked [Topic]. More precisely, I assume that scrambling to pre-subject position targets an outer specifier of T (an A-bar-position not linked to case and agreement).³⁵ A non-subject scrambled to this position may then value $[u\delta]$ on C, since $[u\delta]$ is not any longer connected to ϕ -features (after valuation the latter are marked for deletion and are therefore syntactically inert).³⁶ The [Topic] phrase in the outer SpecTP is then attracted by C's EPP-feature to SpecCP, since it is the highest accessible goal in the TP domain (see also [Frey 2006a,b](#)):

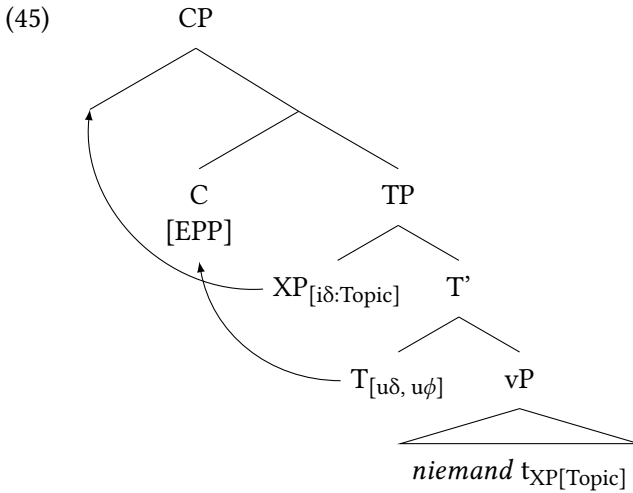
- (44) Den Fehler hat niemand bemerkt.
 the.ACC mistake has nobody.NOM noticed
 ‘Nobody noticed the mistake.’

³⁴The same mechanism can be assumed to take place in embedded clauses, albeit with the following modifications: (i) C obviously lacks an EPP feature that may trigger expletive insertion; (ii) the complex head resulting from T-to-C movement is spelled out by the complementizer ([Pesetsky & Esther 2001](#)), while the finite verb stays behind in vP/VP ([Haider 2010](#)).

³⁵Alternatively, we may assume that pre-subject scrambling targets the specifier of a separate projection that is linked to ground/givenness and is located between C and TP, see [Grewendorf & Sabel \(1999\)](#), [Frey \(2004a\)](#) and [Miyagawa \(2009\)](#) for discussion.

³⁶Note that this approach is somewhat similar to the analysis proposed in [Rizzi \(1996\)](#) for so-called residual V2 effects in English.

Eric Fuß



4.2.4 The insertion of expletive *es*

Let us now take a look at cases in which the derivation does not contain topical material as in (46), where SpecCP is occupied by expletive *es*.

- (46) Es ist niemand gekommen.
it is nobody.NOM come
‘Nobody came.’

Again, C hosts $[u\delta]$ as a result of T-to-C movement, but in (46) there is no way $[u\delta]$ can be valued during the syntactic derivation. Moreover, it is at first sight unclear how the EPP on C can be fulfilled if movement/internal merge is parasitic on Agree – if *niemand* fails to value $[u\delta]$ on C, it shouldn’t be able to check the EPP, either. In what follows, I will argue that both problems can be overcome by a repair procedure in which an expletive is inserted as a last resort to satisfy C’s EPP feature, while $[u\delta]$ is interpreted as [Focus] by default. In this way, the presence of an expletive can actually be taken to signal that the clause lacks a topic-comment structure, inducing a broad focus reading.

First, recall that the presence of an unvalued δ -feature is not necessarily a problem if we adopt the idea that unvalued features are identified with default values at the interfaces (cf. e.g. Preminger 2014). More precisely, what I would like to propose for cases like (46) is that the presence of an unvalued δ -feature (interpreted as focus) in C demarcates the size of the focus domain, giving rise to a broad focus reading and the characteristic ‘all-new’ reading of existential/presentational sentences (broad/wide focus, including ‘focus projection’ effects).

6 Some remarks on the properties and analysis of expletive *es* in German

Turning now to the question of how the EPP can be satisfied (arguably the more important condition), I would like to pursue the traditional idea that CP-expletives are externally merged in SpecCP as a last resort/repair to rescue a derivation where the EPP on C cannot be satisfied otherwise (cf. Blümel 2017 for a general repair perspective on V2). While this is a fairly standard account, it still raises a couple of questions. First of all, it should capture the descriptive generalization that expletive *es* is primarily inserted in clauses that lack a topical subject. Moreover, more has to be said about the feature content of the expletive, in particular if it is assumed that merge is dependent on matching/valuation of another feature in addition to the EPP. As we will see, these two issues are intimately connected.

One way to proceed is to assume that expletive *es* satisfies EPP only without matching [uδ] on C. This approach highlights the exceptional character of this operation: external merge of an expletive is a last resort to rescue an otherwise doomed derivation, which is exempt from the requirement of a previous/additional Agree operation. Moreover, this repair is not an optimal solution and therefore not freely available: if both [uδ] and the EPP can be satisfied by another XP, this XP must undergo movement to SpecCP as e.g. in (44) (Maximize matching effects, Chomsky 2001: 15). As a result, expletive insertion is strictly confined to contexts in which C hosts an unvalued δ-feature. However, this account seems to be challenged by the apparent optionality of expletive insertion, compare (46) with (47).³⁷

- (47) Niemand ist gekommen.
 nobody.NOM is come
 ‘Nobody came.’

Alternatively, we may assume that the insertion of the expletive is accompanied by an Agree relation in which [uδ] is matched by a corresponding feature of the expletive. For the sake of concreteness, let us say that expletive *es* carries an unspecified δ-feature that matches [uδ] on C, but fails to value it.³⁸ We would

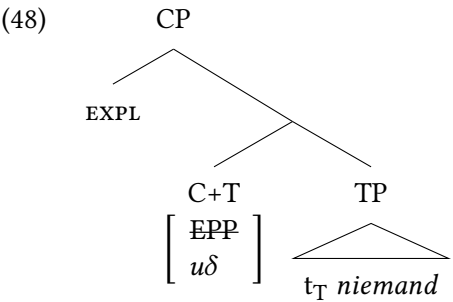
³⁷Note that even in cases like (44), the insertion of an expletive seems to be a (somewhat marked) option:

- (i) Es hat den Fehler niemand bemerkt.
 it has the mistake nobody noticed
 ‘Nobody noticed the mistake.’

³⁸Further support for the idea that non-referential *es* might be linked to non-given information comes from a subtle contrast in the interpretation of cataphoric pronouns in German (cf. Pütz 1986², Sudhoff 2003). As shown in (i), either *es* ‘it’ or *das* ‘that’ may be used to refer cataphori-

Eric Fuß

therefore take a weaker stance on the idea that merge is parasitic on Agree, in the sense that feature matching is a sufficient precondition on merge. Again, this assumption would capture the observation that expletive insertion is confined to the presence of $[u\delta]$ on C. The optionality of (47) could then be attributed to a derivation in which the unvalued δ -feature on *niemand* matches $[u\delta]$ on C, followed by EPP-driven fronting of the subject. However, this approach fails to account for the fact that indefinite subjects trend to occupy a low vP-internal position (see above) – if an indefinite can raise to SpecCP to match $[u\delta]$, this should also allow for movement of indefinites to SpecTP at an earlier stage of the derivation. Moreover, it perhaps predicts that expletive insertion should be more freely available, since its application is not necessarily less optimal than other choices involving a combination of feature matching and movement. So, I tentatively opt for the first approach (expletive *es* satisfies the EPP only, as illustrated in (48)), which seems to be more restrictive (see Section 5 below for additional discussion of the apparent optionality of expletive insertion). Crucially, however, both cases seem to successfully capture the link between expletive insertion and a broad focus reading (due to the presence of $[u\delta]$ on C).



5 Deriving the properties of CP expletives

This section addresses the question if and how the proposed analysis is able to account for the set of properties highlighted in Section 2 above:

- restriction to clause-initial position of root-clauses

cally to an extraposed clause. However, *es* differs from *das* in that the use of the latter clearly signals that the extraposed clause represents given information (Fuß & Hinterhölzl 2023: 44f.).

- (i) Es/Das ist eine Schande, dass Peter zu Hause blieb.
it/that is a shame that Peter at home remained
'It is a shame that Peter remained at home.'

6 Some remarks on the properties and analysis of expletive *es* in German

- lack of A-properties
- incompatibility with topical subjects
- almost complete absence of definiteness effects
- low, vP-internal position of the associate DP
- variation between insertion of an expletive and XP-fronting
- discourse functions

5.1 Restriction to clause-initial position and lack of A-properties

The last resort character of expletive insertion leads us to expect that this operation only becomes available after all other options have been exhausted (typically at the end of a derivational cycle). As a result, expletives may only be inserted into a specifier of a phase head, as has been argued for on independent grounds by [Richards & Biberauer \(2005\)](#). Moreover, the generalization that expletives are linked to the presence of obligatory EPP-features that require a structural position to be (overtly) filled (cf. e.g. [Haider 2010](#)) suggests that ‘pure’ CP expletives are inserted into SpecCP in languages such as German or Icelandic. Expletive insertion should not be available for positions that may only host optional EPP-features (such as SpecTP in German). It is thus correctly predicted that CP expletives are confined to clause-initial position. Since SpecCP is the prototypical A-bar position, it is also expected that CP expletives lack A-properties.

5.2 Incompatibility with topical subjects

The incompatibility with topical subjects is a direct result of the circumstances under which expletive *es* is inserted. Since the use of prefinite expletives is confined to contexts in which C hosts an unvalued δ -feature, a CP expletive is generally incompatible with topical subjects. The latter always value $[\text{u}\delta]$ at an earlier stage of the derivation (in the TP domain) and therefore prevent $[\text{u}\delta]$ from reaching C. More generally, as has already been pointed out in connection with (44)/(45), topical non-subjects may also value $[\text{u}\delta]$ on C before they undergo further movement to SpecCP. But note that definite objects may also stay in-situ as part of the information focus as in (49), which clearly shows that the [Topic] feature is only optionally added to the feature content of the object *den Fehler*.

Eric Fuß

- (49) Es hat niemand den Fehler bemerkt.
 it has nobody.NOM the.ACC mistake noticed
 ‘Nobody noticed the mistake.’

Thus, it is only when an unvalued [uδ] feature is moved to C that expletive *es* is inserted as a last resort to satisfy C’s EPP feature. In all other cases, C must attract topical material sitting at the left edge of TP. This effect is particularly strong with subject pronouns, since these are inherently deaccented topical elements, while the [Topic] feature is merely optional with phrasal elements:³⁹

- (50) * Es hat er niemanden gesehen.
 it has he nobody.ACC seen
 ‘He saw nobody.’

However, this account seems to be contradicted by the fact that in contrast to subject pronouns, object pronouns are readily compatible with expletive insertion (cf. (15) above, repeated here for convenience):

- (51) Es hat ihn niemand gesehen.
 it has him nobody.NOM seen
 ‘Nobody saw him.’

I suspect that the acceptability of (51) is related to the fact that in German, fronting of unstressed object pronouns is generally quite marginal and only possible under certain circumstances (Frey 2004b, 2006b, Duden 2022: 70; for further discussion and experimental results cf. Bader & Yvonne 2019, 2021). The relevant effect is particularly strong with simple personal pronouns referring to non-animate entities (cf. Frey 2004b: 40):

³⁹At first sight, this analysis seems to predict that subject pronouns should also be incompatible with the fronting of non-topical material. As shown in (i), this expectation is not borne out by the facts.

- (i) Niemanden hat er getroffen.
 nobody-ACC has he met
 ‘He met nobody.’

However, note that there is strong tendency to interpret the fronted element in examples like (i) as a contrastive focus, which also usually requires a special intonation pattern. This suggests that in examples like (i), XP-fronting is not triggered by a generalized EPP-feature, but rather by a feature linked to contrast/focus which is also at work in *wh*-questions, cf. Frey (2006a).

6 Some remarks on the properties and analysis of expletive *es* in German

- (52) *Ihn/es hat Maria repariert.
 him/it.ACC has Maria repaired.
 ‘Maria repaired it.’

Seen in this light, the acceptability of (51) might be attributed to the fact that unstressed object pronouns (in contrast to subject pronouns) generally resist fronting to clause initial position, which again calls for repair by insertion of an expletive to satisfy the EPP on C.

5.3 Almost complete absence of definiteness effects

I would like to suggest that the absence of definiteness effects in German is ultimately a consequence of the idea that the insertion of an expletive is linked to the lack of topical material in a clause. Of course, topics are often definite DPs, but there is no one-to-one correlation between topicality and definiteness. In particular, as has been shown in Section 2, there are many cases where a definite subject seems to be part of the information focus, that is the most important information conveyed by a sentence. This is captured by the proposed analysis. The assumption that the insertion of an expletive is linked to the presence of an unvalued feature [u δ] on C neither posits any dependency with the associate nor does it impose any restrictions on the definiteness of the associate. It is merely required that the subject is not marked [Topic]. We thus do not expect any strong definiteness effects in German.⁴⁰

5.4 Low, vP-internal position of the associate DP

The observation that non-topical subjects typically occupy a low, probably vP-internal position also follows immediately from the analysis proposed above: Due to the fact that SpecTP is taken to be reserved for [Topic] elements in German, focused subjects must remain in a lower position. It is only in this context that the insertion of a CP expletive is possible.

5.5 On the apparent optionality of expletive insertion

In Section 4.2.4 I opted for a restrictive approach to XP-fronting where merge as SpecCP is confined to either (i) movement of topical material (satisfying both [u δ] and EPP on) or (ii) insertion of an expletive (satisfying EPP only). In what follows,

⁴⁰Of course, this raises the question of how the present analysis can account for the different behavior of the expletive *það* in Icelandic. I leave that question for future research.

I explore how this approach might account for the impression that expletive insertion is sometimes seemingly more freely available. I am going to argue that the apparent variation can be traced to two sources: (i) the fact that phrases can assume different information-structural functions (i.e., the feature [Topic] is merely optionally added to the feature content of any given phrase) and (ii) the independently motivated idea that movement to clause-initial position might be triggered by another feature, namely [Contrast] (cf. e.g. Frey 2006a).

Consider the following pair of clauses (see (26) above, repeated here for convenience):

- (53) a. Es wurde gestern Abend viel getanzt.
 EXPL was yesterday evening much danced
 ‘There was a lot of dancing last night.’
 b. Gestern Abend wurde viel getanzt.
 yesterday evening was much danced
 ‘There was a lot of dancing last night.’
 (Pütz 1986²: 52)

At first sight, it appears that expletive insertion is merely optional in (53). However, upon closer inspection, it turns out that (53a) and (53b) are produced by two completely different syntactic derivations, depending on the information structural status of the frame adverbial *gestern Abend* ‘yesterday evening’. In (53a), the frame adverbial is arguably part of the information focus. Accordingly, it enters the derivation without a [Topic] feature. Consequently, it cannot value [uδ], which calls for the expletive insertion to satisfy the EPP on C, inducing a broad focus reading (possibly, this is also the case in the *es ist dies*-construction discussed in Section 2.6). In contrast, (53b) can receive a reading where *gestern Abend* is a topical element.⁴¹ If this is the case, [uδ] can be satisfied by moving the frame adverbial to an outer specifier of TP, from where it is attracted to SpecCP. In other words, the apparent optionality of expletive insertion in (53) is not real, but rather the result of different derivations/numerations, where the frame adverbial constitutes a focused XP in (53a), but a topical element in (53b).

In addition, the impression of syntactic optionality might arise due to the fact that SpecCP may also be filled “by means of true $\bar{A}$ -movement to the prefield, which may move (cyclically) any constituent in the middle field, and which always has the effect of inducing stress on the moved item and its contrastive interpretation” (Frey 2006b: 167). For the sake of concreteness, I assume that “true

⁴¹Note that frame-setting elements such as adverbials of time and location are commonly taken to represent potential topics, cf. Chafe (1976), Jacobs (2001).

6 Some remarks on the properties and analysis of expletive *es* in German

A-bar-movement” is triggered by a feature [Contrast] on C (see already footnote 39). [Contrast] is a feature of C proper and not donated to T. It forces *wh*-fronting in interrogatives and might also attract contrastively focused non-*wh* material (and contrastive topics). Now, consider the pair in (54).

- (54) a. Es ist niemand gekommen.
 it is nobody.NOM come
 b. Niemand ist gekommen.
 nobody is come
 ‘Nobody came.’

Again, what I would like to propose is that (54a) and (54b) are only superficially similar. In fact, they involve two different derivations. In (54a), *niemand* ‘nobody’ is part of the information focus, and the presence of an unvalued δ -feature on C calls for expletive insertion (see the discussion of (47) above). In contrast, fronting of *niemand* in (54b) seems to be linked to a contrastive interpretation of the indefinite, which suggests that fronting is linked to the presence of [Contrast] on C.⁴²

5.6 Discourse functions

The description in Section 2 has shown that sentences introduced by expletive *es* can serve a variety of different pragmatic functions, an exhaustive discussion of which would go well beyond the scope of this paper. In what follows, I will therefore add only some brief and admittedly rather speculative comments on a selection of relevant properties.

Lack of a topic-comment/focus-background structure

Since expletive *es* can only be inserted when the clause lacks topical material that can satisfy [$u\delta$] on T/C, it is predicted that the outcome of the syntactic derivation gives rise to the characteristic ‘all-new’ reading of existential/presentational sentences (broad/wide focus). Moreover, as suggested in Section 4.2.4, [$u\delta$] may be taken to mark the upper bound of the extended focus domain.

Contextual anchoring/introduction of a new (specific) situation

⁴² Admittedly, this contrast is quite subtle, but my own intuitions have been confirmed by a number of native speakers I consulted.

Eric Fuß

The CP expletive *es* has somewhat peculiar properties in that it seems to be a personal pronoun that is inserted into an A-bar position. It does not fill an argument slot and therefore seems to violate the Theta Criterion (and the Principle of Full Interpretation). As a result, a form of *es* that is externally merged in SpecCP cannot be taken to refer to an individual. Of course, this corresponds to the traditional characterization of expletive elements, which are taken to lack semantic content and must therefore be deleted (at LF) since they cannot be interpreted at the interface to the semantic component. Still, it seems that *es* does contribute to the interpretation and potential use contexts of the sentence as argued for in Section 2.5. In particular, we have seen that clauses introduced by expletive *es* are taken to introduce a new situation. What I would like to suggest is that this effect results from a semantic repair that facilitates a re-interpretation of the pronominal element at the interface (perhaps some form of (conventional) implicature). As a result, expletive *es* is taken to relate to a specific situation linked to the speaker/utterance context (Hinterhölzl 2019, Fuß & Hinterhölzl 2023, Hinterhölzl 2024: a situational argument of tense; see Brandt 2019 for a general repair perspective on related interpretative effects). The link to the speaker/utterance context might also provide the basis for the impression that relevant constructions appear to convey some form of evidentiality (cf. Sluckin 2021 on English *there*). Perhaps, the capability to anchor an event to the utterance context is linked to the fact that *es* can be characterized as a weak deictic element (Hinterhölzl 2019, Fuß & Hinterhölzl 2023, Hinterhölzl 2024). This characterization is supported by historical and dialectal evidence. First, traditional comparative historical linguistics has disclosed that all Germanic expletives go back to demonstrative/deictic elements (*d/th*-Forms < PIE demonstrative root **so/*tod* ‘this, that’; *it*-forms < PIE **he-*, neut. **hid* ‘that, the (just named)’; cf. Beekes 2011: 226). Second, note that in many present-day dialects of German, CP-*es* is realized as proclitic /s=/, as shown in (55).

- (55) a. S=isch es Haus abpränt.
 it=is the house burned down
 ‘The house burned down.’ (Zurich German, Weber 1964: 240)
- b. S=wird geschafft.
 it=is worked
 ‘People are working.’
 (Swabian, Fabian Heck p.c.)

Interestingly, the weak proclitic /s=/ does not alternate with the full form /es/ ‘it’, but rather with a corresponding strong d-form (e.g., /des/), cf. e.g. Mayerthaler

6 Some remarks on the properties and analysis of expletive *es* in German

& Mayerthaler (1990: 403) on Bavarian:⁴³ “[The] pronoun *es* does not exist in Bavarian; its function is divided between a stressed form *des*, which is identical with the demonstrative pronoun, and a clitic form *s* ([əs], [sə]).” This suggests that *es* (or, rather, /s=/) is a weak deictic element that is part of the demonstrative paradigm, which, as a CP expletive, cannot point to entities, but can be used to anchor a sentence to the utterance context (cf. Hinterhölzl 2019, 2024 for details).⁴⁴

6 Concluding summary

Based on a review of selected properties of the prefinite expletive *es* in German, this paper developed a new analysis of CP expletives that pays special attention to a set of observations that at least in part have not received much consideration in the literature. In addition to the well-known fact that CP expletives are typically used in clauses that lack a topical subject, these include (i) the low, vP-internal position of the subject, (ii) the almost complete absence of the definiteness effect in German and (iii) the fact that in many contexts there seems to be variation between XP-fronting and insertion of an expletive. To account for these properties, I have proposed an analysis based on the traditional assumption that expletives are semantically light elements that primarily serve a syntactic function, namely filling a specifier position that is demanded by an EPP/edge-feature linked to C. However, expletive insertion is only available when C hosts an unvalued discourse-related δ -feature that has not been matched by a topical phrase in the course of the syntactic derivation. More precisely, I have argued that C donates an unvalued δ -feature ([u δ]) to T, which may attract a topical subject (marked with the feature value [Topic]) to SpecTP (before the subject undergoes fronting to SpecCP). The observation that non-topical subjects cannot undergo movement to SpecTP has been accounted for by assuming that they contain an unspecified interpretable δ -feature, which cannot value [u δ] on T, but is identified with the default value [Focus] at the interfaces. When the clause does not contain a topical subject, [u δ] is carried up to the left clausal periphery via T-to-C movement. Upon reaching C, [u δ] may be valued by topical non-subjects (which subsequently move to SpecCP). If the clause does not contain any topical material, the EPP on C is satisfied by expletive insertion, while [u δ] is interpreted as

⁴³See also Weiß (1998: 101) on Bavarian; Rothmund (1932: 35) on Alemannic; Baumgärtner (1959: 39) on East Central German; Schönhoff (1908: 181) and Appel (2007: 290f.) on Low German.

⁴⁴This seems to be in line with earlier proposals that treat the prefield *es* as a declarative operator (cf. e.g. Tomaselli 1990).

Eric Fuß

[Focus] by default, demarcating the upper bound of the focus domain (giving rise to a broad focus reading). I have then explored a set of consequences of this analysis, arguing that the apparent optionality of expletive insertion is not real but rather the result of different syntactic derivations starting out from non-identical numerations. Finally, I have added some speculative remarks on the discourse functions of expletive *es* in German, arguing that these can be attributed to the weak deictic character of *es*, which makes available an (pragmatically induced) interpretation in which *es* is taken to point to the utterance situation.

I am aware of the fact that the paper leaves a number of questions unaddressed, concerning in particular the comparative and the historical perspectives and the array of pragmatic functions linked to the insertion of CP expletives (which were only briefly commented on). Still, I hope that some of the ideas developed here contribute to our understanding of syntactic expletives and that this research avenue can be further advanced in future work, particularly focusing on the historical development of CP expletives.

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6 *Some remarks on the properties and analysis of expletive es in German*

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Chapter 7

An extraction restriction with complement-less prepositions in British English but not dialectal German

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This paper explores a potential parallel between two Germanic dialects regarding complement-less prepositions. British English permits “Prepositional Object Gaps”, e.g. *This box has papers in (it)*; while Northern German permits “*da*-drop”, e.g. *Heute habe ich die Zeit nicht (da)zu gehabt*, ‘Today I have not had the time for (it)’. Novel experiments show that the sentence object (viz. *papers*) cannot be A-bar extracted in British English, whereas there is no such restriction in Northern German. We suggest that the British English restriction follows from ungrammatical crossing A-bar dependencies, which do not arise in German due to A-scrambling. This syntactic analysis forms part of a wider comparison of the two constructions from semantic, typological and diachronic perspectives.

1 Introduction

This paper explores a potential parallel between two Germanic dialects regarding complement-less prepositions. For our purposes, “complement-less prepositions” refers to two very specific constructions: British English “Prepositional Object Gaps” and Northern German “*da*-drop”.

In many varieties of British English (BrEng), inanimate pronominal complements of certain locative prepositions can be omitted under certain circumstances, as illustrated in (1):

Richard Stockwell, Anke Himmelreich & Carson T. Schütze

- (1) a. The box has papers in (it). BrEng
b. Your tie has a stain on (it).

In other varieties, e.g. North American English (NAEng), the pronoun is obligatory, as indicated in (2):

- (2) a. The box has papers in *(it). NAEng
b. Your tie has a stain on *(it).

We will see that BrEng has not simply reassigned *in* and *on* to the class of optionally transitive prepositions. With prepositions of that class, such as *inside* (3), pronoun omission is possible for all speakers of English (AllEng):

- (3) The box has papers inside (it). AllEng

Rather, the omission of the pronouns in (1) is restricted in ways that the omission in (3) is not. Prepositional Object Gaps thus cannot simply be a case of optional phonological omission.

Similarly, in Low- and Middle-German Dialects (for simplicity, Northern German, NGer), inanimate pronominal complements of certain prepositions can sometimes be omitted; namely the *da(r)*- proclitic of “Pronominaladverbien”, as in (4):

- (4) a. Heute habe ich die Zeit nicht (da)zu gehabt.
today have I the time not (DA).for had
‘Today I have not had the time for it.’
b. Im Sommer sollte man auch gelegentlich einen Wassernapf
in summer should one also occasionally a water.bowl
(da)neben stellen.
(DA).next.to put
‘In summer, one should occasionally put a bowl of water next to it.’

As discussed further in Section 3, *da* is homophonous with the locative ‘there’, but in this context means ‘it’. We will pursue an analysis under which, as with Prepositional Object Gaps, *da*-drop is not simply a case of phonologically dropping *da*, but is syntactically controlled.

In outline, the next two sections review the basic patterning of Prepositional Object Gaps and *da*-drop, before Section 4 draws a direct comparison. Sections 5 and 6 present novel experimental evidence designed to further explore their syntactic structures, particularly with respect to A-bar movement. Section 7 then

7 Complement-less prepositions and extraction

discusses the implications of our results for the question of whether these two complement-less P phenomena call for similar syntactic analyses. Section 8 concludes with some directions for further study.

2 Background on British English Prepositional Object Gaps

The British English (BrEng) complement-less locative prepositions we are concerned with were first analyzed¹ by Griffiths & Sailor (G&S) (Griffiths & Sailor 2015a,b, 2017; Sailor & Griffiths 2017) under the moniker Prepositional Object Gaps (POGs). As indicated in (5), the pronoun (a) or gap (b) obligatorily corefers with an overt DP. Only in BrEng is (b) acceptable, where it is synonymous with (a):

- | | | |
|-----|---|--------|
| (5) | a. This film _i has monsters in it _i /*j. | AllEng |
| | b. This film _i has monsters in ___ _i /*j. | BrEng |

This section introduces four essential features of POGs: (i) POGs generally require locative *have* or *with*; (ii) the most broadly accepted prepositions are *in* and *on*; (iii) the “missing” pronoun must be inanimate; and (iv) POGs are importantly different from other constructions involving complement-less prepositions.

First, POGs are best licensed in the context of locative *have*, as in (5) above, or *with*, as in (6) (Swan 1995: 433):

- (6) the film_i with monsters in (it_i)

Beyond these particular predicates, POGs are degraded. In (7), for example, the existential (a) is bad, despite the apparent synonymy of (b) and (c) (G&S):²

¹This phenomenon has been observed in the descriptive literature (Swan 1995, Algeo 2006:197) and (foot)noticed by syntacticians (Belvin & den Dikken 1997:168, fn. 17, McIntyre 2005:5).

²Although the judgement on sentences like (7a) has been uncontroversial among BrEng linguists, including the first author, the naïve BrEng participants in the experiment discussed in Section 5 did not uniformly share it. For the sentence in (i), their ratings ranged from the lowest of 1 to the highest of 7 with a mean of 5.2 and a median of 5, and were unimodally distributed:

- (i) [TV shows]_i are more exciting when there are monsters in ____i.

The item in (i) differs from (7a) in potentially relevant ways. For one, the antecedent for the gap is in the same sentence. Moreover, like in all of the preceding examples, the antecedent c-commands the gap. To assess the importance of c-command we tested (ii), where the antecedent does not c-command the gap. The ratings for (ii) were not much lower than for (i):

Richard Stockwell, Anke Himmelreich & Carson T. Schütze

- (7) a. Don't watch that film_i – there's a monster in *(it_i)!
- b. There's a monster in that film.
- c. That film_i has a monster in (it_i).

Even verbs that can be roughly synonymous with *have* do not license POGs, as shown in (8) (Stockwell & Schütze 2019). Neither do non-locative uses of *have* (Ritter & Rosen 1997, Harley 1998, Myler 2016), as shown in (9):³

- (8) a. This lift_i can **have** up to 14 people in (it_i).
- b. This lift_i can **hold/accommodate** up to 14 people in *(it_i).
- (9) a. For a film_i to be successful, monsters have (got) to be in *(it_i)!
[modal *have (got) to*]
- b. The boiler_i had its_i tank collecting water in *(it_i). [experienter *have*]
- c. The film_i's director had there be lots of monsters in *(it_i).
[causative *have*]

Second, regarding prepositions, POGs are possible with *in*, as above, and *on*, as in (10), throughout BrEng:

- (10) a. This box_i has spots on (it_i).
- b. a utility bill_i with your address on (it_i)

the mean was 4.9, the median 5. Thus, having the antecedent in the same sentence may be more important:

- (ii) I agree with you about [that film]_i, even though there are superheroes in ___i.
- ³Additional licensing verbs are plausibly built from *have* (cf. Sailor & Griffiths 2017:10): (i) illustrates for *need*, which has been argued to contain possessive *have* (Harves & Kayne 2012); and (ii) for *get*, which has been claimed to be the inchoative of *have* (Kimball 1973, Emonds 1994:164, i.a.). We note that, as in (8b), roughly synonymous verbs fail to license POGs:

- (i) This film_i {needs/??requires/??demands} more monsters in ___i.
- (ii) The guestbook_i {got/*received/*acquired} so many rude entries in ___i last year that it had to be thrown away.

At a greater stretch from *have*, we find that the same is true for *put* (iii) (and lack of c-command does not seem to matter; cf. note 2):

- (iii) a. (At Wimbledon:) If the court_i starts getting too hard, we'll
{put/??sprinkle/??spray/*hose} more water on ___i.
- b. If the box_i weighs more than 50lbs, you've {put/??placed/??loaded} too many papers in ___i.

7 Complement-less prepositions and extraction

The availability of POGs with other locative prepositions is subject to inter-speaker variation (G&S).⁴ With a view to encompassing the broadest range of BrEng speakers in the experiment in Section 5, we limit ourselves to *in* and *on* here.

Third, the corresponding “missing” pronoun must be inanimate (11). Accordingly, POGs are possible with *it* counterparts, as above, and with inanimate *them*, as in (a). POGs are impossible, however, with first and second person pronouns, as in (b), and third person animate pronouns, as in (c) (G&S):

- (11) a. These boxes_i have papers in (them_i).
 b. I/You have {poison/radioactive chemicals} in *(me/you).
 c. That guy_i looks like he has ten pints of beer in *(him_i).

Finally, POGs differ from other constructions involving complement-less prepositions. There are environments beyond just *have/with*-frames where complement-less prepositions are possible in all Englishes. Three distinct instances of prepositions with no overt complement are surveyed in (12) – “projective” prepositions (a) (Svenonius 2010), directional particles (b), and predicates of wearing (c):

- (12) a. There was a box on the table. Inside ((of) it) was fine Swiss chocolate.
 b. They fell in (the hole).
 c. John had a hat on. [N.B. ≠ John_i had a hat on him_i – i.e. John had a hat on his person, e.g. in his pocket]

The next section turns to complement-less prepositions in another Germanic dialect – Northern German.

3 Background on Northern German *da*-drop

3.1 Pronominaladverbien

In German, the weak neuter pronoun *es* (‘it’) cannot normally be the complement of most prepositions.⁵ As alternatives, all prepositions allow a neuter demonstrative like *das/dieses* (‘this’) or *jenes* (‘that’), and approximately 20 prepositions

⁴G&S (2017) assert that POGs are also licensed – with inter-speaker variation – by *behind*, *inside*, *below*, *above*, *beyond*, *around*, *through*, *across*, *along*, *over*, *under*, *past*, *between*, *up* and *down*.

⁵There are only a few prepositions that can combine with *es*, for example *ohne* (‘without’) – at least in the majority of dialects that lack the form *darohne*.

Richard Stockwell, Anke Himmelreich & Carson T. Schütze

allow an “R-pronoun” *da(r)* ‘it’, which procliticizes to the preposition. These combinations of the R-pronoun with various prepositions are called “Pronominaladverbien” (pronominal adverbs) (van Riemsdijk 1978, Gallmann 1997, Haider 2010, Koopman 2010, Abels 2012, Noonan 2017, i.a.). This phenomenon is illustrated in (13), where (b) could be used to convey the same message as (a):

- (13) a. Fritz hat gestern an sein Auto gedacht.
 F has yesterday about his car thought
 ‘Fritz thought about his car yesterday.’
 b. Fritz hat gestern {daran /*an es} gedacht.
 F has yesterday DA.about about it thought
 ‘Fritz thought about it yesterday.’

Beyond neuter singular, (14) shows that with third person singular masculine (a,b), feminine, and plural (c,d) complements to P the pronominal adverb alternates with the canonical order of preposition–pronoun if the referent is inanimate. The pronominal adverb is excluded if the referent is human (e,f) (Müller 2000, 2002).^{6,7}

- (14) a. Maria musste noch oft an ihren Lieblingsrock denken.
 M had.to still often about her favorite.skirt think
 ‘Maria still had to often think about her favorite skirt.’
 b. Maria musste noch oft {daran /an ihn} denken.
 M had.to still often DA.about about it(MASC) think
 ‘Maria still had to often think about it.’
 c. Maria musste noch oft an ihre Lieblingspuppen denken.
 M had.to still often about her favorite.dolls think
 ‘Maria still had to often think about her favorite dolls.’
 d. Maria musste noch oft {daran /an sie} denken.
 M had.to still often DA.about about them think
 ‘Maria still had to often think about them.’

⁶When pronominal adverbs are used as in (13) and (14) with the *da* portion representing a personal pronoun they are always stressed on the second syllable. The same orthographic form can also be pronounced with stress on *da*, which is then interpreted as a demonstrative pronoun.

⁷For most speakers, *da(r)*- is degraded with nonhuman animate antecedents; though see Thun (1985).

7 Complement-less prepositions and extraction

- e. Maria musste noch oft an ihre Lieblingsschwester denken.
 M had.to still often about her favorite.sister think
 ‘Maria still had to often think about her favorite sister.’
- f. Maria musste noch oft {an sie /*daran} denken.
 M had.to still often about her DA.about think
 ‘Maria still had to often think about her.’

Note that *da* cannot occur in an out-of-the-blue context. Rather, *da* takes the discourse topic as its antecedent, which often occurs in a previous sentence.

While the pronominal adverb construction is available in all German dialects, it has two interesting properties in NGer, where the *da(r)* portion can be displaced leftwards or, similar to BrEng POGs, dropped.

3.2 *Da*-fronting

The R-pronoun *da(r)* can optionally be displaced leftward from the preposition in NGer, a construction we refer to as “*da*-fronting” (cf. “Spaltungskonstruktion” ‘split construction’ in [Fleischer 2002](#); see also [Müller 2000](#)).⁸ In this construction, *da(r)* frequently appears sentence-initially, as in (15), but it can also show up in the middle field, as in (16):

- (15) a. Colloquial Northern German
 Da kommen sie viel billiger bei weg.
 DA come.3PL they much cheaper by away
 ‘They come away from it much cheaper.’
- b. North Saxon ([Lindow et al. 1998:274](#))
 (Dar) kaamt se veel billiger bi weg.
 DA come.3PL they much cheaper by away
 ‘They come away from it much cheaper.’

- (16) a. Colloquial Northern German

⁸ *Da*-fronting is proscribed in standard German: “Heute gilt die Trennung der Pronominaladverbien nicht als hochsprachlich; sie ist umgangssprachlich, besonders norddeutsch:... *Da kann ich nichts für*. Hochsprachlich: *Dafür kann ich nichts*.” ([Berger et al. 1972: 532](#)) [‘Today the separation of the pronominal adverbs is not considered high-level language; it is colloquial, especially northern German: *DA can I nothing for*. High-level: *DA.for can I nothing* (‘It’s not my fault.’)]. “Ein weiterer Fehler, wieder vor allem in der gesprochenen Sprache, ist die Aufsplitterung des Pronominaladverbs” ([Götze & Hess-Lüttich 2002: 301](#)) [‘Another mistake, again especially in the spoken language, is the splitting of the pronominal adverb’].

Richard Stockwell, Anke Himmelreich & Carson T. Schütze

- Sie kommen **da** viel billiger **bei** weg.
 they come.3PL **DA** much cheaper by away
 ‘They come away from it much cheaper.’
- b. East Pomeranian (Stübs 1938: 140)
 Se sünd **doa** sehr besorgt **üm**.
 they are **DA** very worried about
 ‘They are very worried about it.’

3.3 *Da-drop*

It is also possible to drop the otherwise obligatory *da(r)* morpheme in NGer, a construction that Fleischer refers to as “Präposition ohne overte Ergänzung” ‘preposition without overt object’ and which we call “*da-drop*”. Fleischer notes the optionality of *da(r)* in (17). The two examples are drawn from the same page of a dialectal German source. In (a), *da(r)* is phonologically reduced and appears in the pronominal adverb *drin*. In (b), on the other hand, *da(r)* completely disappears, leaving only the preposition *in*:

- (17) North Saxon (Feyer 1939: 27)
- a. Ja, aver Hinnerk, man dröögt sik doch de Han’n nich **drin** af!
 yes but H one dries self yet the hands not **DA.in** off
 ‘Yes, but Hinnerk, one does not dry off one’s hands in it!’
- b. Dat hangt anne Wand un lett witt, un man dröögt sik de
 that hangs on=the wall and looks white and one dries self the
 Han’n **in** af.
 hands in off
 ‘It hangs on the wall and looks white, and one dries off one’s hands in it.’

3.4 Distribution of *da-fronting* and *da-drop*

According to Fleischer (2002), the regions where *da-fronting* occurs are a superset of those where *da-drop* occurs, as can be seen in Figure 1 and Figure 2. *Da-drop* and *da-fronting* are possible only with consonant-initial prepositions in the lighter shaded areas, but with both vowel- and consonant-initial prepositions in the darker shaded areas. (Note the location of Berlin and Potsdam in the lighter shaded areas; this will be relevant for the experiment described in Section 6.)

In modern colloquial German, *da-fronting* is more widespread than Figure 1 suggests. Fleischer’s data are based on dialect atlases and dictionaries that rely on

7 Complement-less prepositions and extraction



Figure 1: Attestation of *da*-fronting with all Ps (darker shading) and with only C-initial Ps (lighter shading). From Fleischer (2002: Appendix).

Richard Stockwell, Anke Himmelreich & Carson T. Schütze



7 Complement-less prepositions and extraction

attested written examples, so the absence of a construction from a given source could be accidental, and speech may be more liberal than writing. In other words, his maps are conservative. By contrast, questionnaire studies, e.g. by [Elspaß & Möller \(2003ff.\)](#), while still showing a predominantly Northern distribution for *da*-fronting, find occasional attestations in even the southernmost states of Germany (Baden-Württemberg, Bavaria). It would thus be misleading to claim this is an exclusively Northern phenomenon.

The *da*-drop construction, on the other hand, is not found in southern dialects. Furthermore, [Oppenrieder \(1991\)](#) states that not all speakers who accept *da*-fronting accept *da*-drop. This is reflected in comparing Figure 1 and Figure 2, where e.g. the Mosel area allows *da*-fronting but not *da*-drop according to Fleischer's data. Thus, there seems to be a one-way implication: all speakers who can *da*-drop can *da*-front, but not vice-versa.

As an initial causal link between *da*-drop and *da*-fronting, [Fleischer \(2002: 408\)](#) points out that if *da* can be fronted to first position, then another drop construction, namely Topic Drop (= dropping of the constituent in Spec,CP), could explain why *da(r)* is absent from that first position in a V2 clause, resulting in a V1 order such as the response in (18):

- (18) Q: Wie ist's mit Bruckner?
 how is.it with B
 'How's it going with Bruckner?'
 A: Ø Kenn ich eigentlich nicht so viel von.
 know I actually not so much about
 'I really don't know much about it.' (Negele 2012: 119)

But there remain many examples of *da*-drop that exclude this analysis because the first position (XP preceding the finite verb) in a matrix clause is filled or because they occur in embedded clauses.⁹ These include the standard German examples in (19) collected by [Oppenrieder \(1991\)](#) to make this point, and the dialect examples in (20) and (21) from Fleischer:

- (19) a. der Otto Flasnöcker kann ein Lied von singen
 the O. F. can a song about sing.INF
 'Otto Flasnöcker can tell you a thing or two about it.'
 [idiom; lit. 'sing a song about it']

⁹Such examples have been independently attested in corpus analyses by several authors: [Breindl \(1989\)](#), [Negele \(2012\)](#), [Jürgens \(2013\)](#), [Otte-Ford \(2016\)](#), [Freywald \(2017\)](#).

Richard Stockwell, Anke Himmelreich & Carson T. Schütze

- b. ...dann sind sie abends oft so müde, daß sie sich überhaupt
 then are they evenings often so tired that they self at.all
 nicht mehr zu aufraffen, dann sich auch noch mal um ihre
 not more to bring then self also again about their
 Kinder zu kümmern
 children to care

‘...then they are often so tired in the evenings that they no longer can
 bring themselves to it at all: to also take care of their children once
 again.’ (Breindl 1989: 146)

- (20) Hamburgish (Saltveit 1983: 323)
 Also büst du wedder nich bi wesen.
 so are you again not at been
 ‘So you weren’t there again.’

- (21) Brandenburgish (Landemann 1956: 338)
 a. Der hät den janßen Noamiddach bei tuejeracht.
 he has the whole afternoon at spent
 ‘He spent the whole afternoon at it.’
 b. Der hät lange föö jespäält.
 he has long for played
 ‘He [an organ grinder] played [music] for a long time for it [a penny].’

In (19a), the subject of the clause occupies first position.¹⁰ Similarly, in (20) and (21), first position is overtly filled. Moreover in (19b), *da*-drop occurs in an embedded clause, where Topic Drop is not possible (Cardinaletti 1990).

Thus, Topic Drop cannot explain the absence of *da(r)* in (19)–(21) and cannot explain any general correlation between a speaker’s allowing *da*-fronting and *da*-drop. Nonetheless, the geographic relationship discussed above suggests there could be such a link, a possibility we return to in Section 7.

4 Comparison

Having introduced BrEng POGs and NGer *da*-drop, this section draws a comparison between them. While the two complement-less preposition constructions

¹⁰(19a) also features the colloquial use of a definite article before the personal name. In colloquial German, it is acceptable for virtually all speakers to have a personal name optionally preceded by a definite determiner that contributes no additional semantics. See Kokkelmans (2018) for discussion of such ‘preproprial’ articles across Germanic, and Stockwell (2017) for a potential case in dialectal English.

7 Complement-less prepositions and extraction

share broad distributional and descriptive similarities, there seem to be deeper syntactic and semantic differences, as detailed below.

Starting with the similarities, we can observe that both constructions are dialectally restricted and that there is considerable regional variation as to which prepositions they occur with. Another similarity is that the omission of the proform is restricted to inanimates. (For the German *da*, this is trivial since it can only refer to inanimates.)

Turning to the differences, the omitted proform in German is homophonous with the locative proform, which is not the case in English (viz. *there*). Further, the BrEng POG construction shows a semantic restriction on the prepositions, whereby only locative/spatial prepositions allow the omission of the proform. In NGer, while most of the prepositions that combine with *da(r)*-, and hence allow *da*-drop, have locative uses, at least one (*für*) does not, and examples above show that the same prepositions also have many non-locative uses (e.g. 13b, 16b, 18). NGer instead has a phonological restriction: for many speakers, *da*-drop is possible only with consonant-initial prepositions.

Another difference between the two constructions concerns the predicate. In BrEng, the predicate must be *have* or *with* to allow pronoun omission; recall (8), repeated as (22):¹¹

- (22) a. This lift_i can **have** up to 14 people in (it_i).
 b. This lift_i can **hold/accommodate** up to 14 people in *(it_i).

NGer *da*-drop, on the other hand, is not restricted to predicates built on *have/with* – see the examples throughout Section 3. In fact, the literal translation of such a sentence comprising locative *have* and a small clause is highly marked. As shown in (23), the preferred pronominal form here would be a reflexive *sich* (a) rather than *da* (b). Dialectal *da*-drop seems impossible in this environment (c), and *da*-fronting is out of the question (d,e). Binding theory could be at issue here; where *da* usually refers to a topic supplied by prior discourse, in (b-e) it may have too local of an antecedent.¹²

- (23) a. Das Hotel hat einen Golfplatz neben sich.
 the hotel has a golf.course next.to itself
 ‘The hotel has a golf course next to it.’
 b. ?? Das Hotel hat einen Golfplatz daneben.
 the hotel has a golf.course DA.next.to

¹¹Or a predicate built on *have/with* – recall fn. 3.

¹²The strings in (23d,e) allow *da* to receive only the interpretation ‘there’, not ‘it’.

Richard Stockwell, Anke Himmelreich & Carson T. Schütze

- c. ?* Das Hotel hat einen Golfplatz **neben**.
the hotel has a golf.course next.to
- d. * Das Hotel hat **da** einen Golfplatz **neben**.
the hotel has DA a golf.course next.to
- e. * **Da** hat das Hotel einen Golfplatz **neben**.
DA has the hotel a golf.course next.to

The final set of differences concerns the interaction of proform omission with movement. For NGer, we observed that *da*-drop occurs only in regions where *da*-fronting also occurs. Our analysis in Section 7 will reify this co-occurrence into having *da*-drop depend on an earlier derivational step of *da*-fronting.

In BrEng, the pronoun cannot be overtly separated from the preposition, as shown in (24):

- (24) a. * Them_j, these boxes have papers in t_j.
b. * These boxes <them_j> have <them_j> papers <them_j> in t_j.

Still, we will argue that silencing of the pronoun depends on fronting in BrEng just as in NGer, with indirect evidence coming from interactions between proform omission and the displacement of other constituents. In particular, what we call the “sentence object” (e.g. *papers* in 24, i.e. the DP following *have*) cannot be extracted if the pronoun is dropped in BrEng. In NGer, on the other hand, *da*-drop does not inhibit movement of the direct object – we will argue due to the availability of scrambling in German. The next two sections present new experimental data to substantiate this difference. Analysis of the interaction of object extraction and proform omission will follow in Section 7.

Table 1 summarises the similarities and differences between BrEng POGs and NGer *da*-drop:

5 Object extraction and POGs: New data

Based on native speaker intuitions, Stockwell & Schütze (2019) suggested that A-bar movement of the sentence object is impossible with POGs, as in (25):

- (25) What_j does this shirt have t_j on *(it)? BrEng

At the same time, they reported that there is no such restriction with other complement-less prepositions (26) like *inside* (a) or the particle use of *on* in *have DP on* meaning ‘be wearing DP’ (b) (cf. section 2.4):

7 Complement-less prepositions and extraction

Table 1: Conditions on omission of P complements

	BrEng	NGer
<u>Variability</u>		
Dialectally restricted	yes	yes
Regional variation in Ps	yes	yes
<u>Properties of proform & antecedent</u>		
Omission restricted to inanimates	yes	yes
Proform homophonous with locative	no	yes
Omission restricted to locative Ps	yes	no
Locative <i>have/with</i>	required	not possible
<u>Interaction with movement</u>		
Option to separate proform from P	no	yes
Omission blocks extraction of object	yes (Section 5)	no (Section 6)

- (26) a. What_j does this box have *t_j* inside?
b. [What colour hat]_j does your guest have *t_j* on?

Here we report a pilot experiment confirming the intuitions in (25) and (26) that A-bar movement of the sentence object is impossible only with POGs.

5.1 Method

The participants were 60 speakers from England, recruited via Amazon Mechanical Turk.¹³ They were paid US\$5 for their participation, which took approximately 10 minutes. Participants were tasked with providing acceptability ratings on a 1–7 Likert scale (7=best) of target sentences containing the configuration *DP_i HAVE...in/on {it_i/them_i vs. Ø_i}*. The experiment employed a 2×3 design: the prepositional complement was either (i) an overt pronoun or (ii) null (Ø); while the sentence structure involved either (a) no extraction, (b) A-bar extraction of the direct object, or (c) A-bar extraction of a non-object – the subject or an adjunct.¹⁴

¹³In addition to meeting the Amazon Mechanical Turk criterion of Location = United Kingdom, participants had to answer Yes to the following two questions: (i) Did you live in England from birth until (at least) age 13? (ii) Did you speak English in the home? We restricted the location to England because we are aware of parts of the British Isles where POGs are unacceptable, e.g. Glasgow, Belfast.

¹⁴The adjuncts tested included *sometimes*, *usually*, and *often*—translation equivalents of three of the frequency adverbials used in the German experiment described in Section 6.

Richard Stockwell, Anke Himmelreich & Carson T. Schütze

Each of the target items thus consisted of a 6-tuple of sentences. Five types of A-bar movement structures were tested across items: *which/what N* interrogatives (27), *how much/many N* interrogatives (28), restrictive relative clauses (29), topicalization structures (30), and *it*-clefts (31). There were four token sets for each of these types, yielding 20 6-tuples of target items. Each participant saw only one member of each 6-tuple, along with 28 filler sentences, for a total of 48 items to be rated:¹⁵

- | | | |
|------|---|--------------|
| (27) | <i>Which/What N</i> questions | [Extraction] |
| | a. Those jeans have stains on them/ $\emptyset$. | [none] |
| | b. [What stains] _i do those jeans have t_i on them/ $\emptyset$? | [object] |
| | c. [Which jeans] _j t_j have stains on them/ $\emptyset$? | [subject] |
| (28) | <i>How much/many</i> questions | [Extraction] |
| | a. These bags have forged banknotes in them/ $\emptyset$. | [none] |
| | b. [How many forged banknotes] _i do these bags have t_i in them/ $\emptyset$? | [object] |
| | c. [How many bags] _j t_j have forged banknotes in them/ $\emptyset$? | [subject] |
| (29) | Relative clauses | [Extraction] |
| | a. I like that the cake has the purple icing on it/ $\emptyset$. | [none] |
| | b. I like [the purple icing] _i that the cake has t_i on it/ $\emptyset$. | [object] |
| | c. I like [the cake] _j that t_j has the purple icing on it/ $\emptyset$. | [subject] |
| (30) | Topicalization | [Extraction] |
| | a. Cinema popcorn often has too much salt on it/ $\emptyset$. | [none] |
| | b. [Too much salt] _i , cinema popcorn often has t_i on it/ $\emptyset$. | [object] |
| | c. Often _j , cinema popcorn t_j has too much salt on it/ $\emptyset$. | [adjunct] |
| (31) | <i>It</i> -clefts | [Extraction] |

¹⁵The fillers included one sentence that is grammatical in BrEng but not in most other Englishes – *Have you any idea how dangerous that is?* – which participants had to rate at least 4 out of 7; sentences (i) and (ii) of fn. 2 and another similar to (ii); eight ungrammatical catch trials, twelve grammatical catch trials, and four miscellaneous trials. The grammatical catch trials included nine with a final preposition (which might be prescriptively disfavored, like the targets with null pronouns). Participants could make no more than two errors on catch trials, where an error was defined as a rating greater than 4 on an ungrammatical item or less than 4 on a grammatical item. Data from an additional seven participants was excluded on this basis.

7 Complement-less prepositions and extraction

- a. This envelope has the final scores in it/ $\emptyset$. [none]
- b. It's [the final scores]_i that this envelope has t_i in it/ $\emptyset$. [object]
- c. It's [this envelope]_j that t_j has the final scores in it/ $\emptyset$. [subject]

5.2 Results

The mean ratings for the 20 target 6-tuples are summarized in Table 2. Numeri-

Table 2: Mean ratings by condition

Extraction	i) Overt pronoun	ii) Null pronoun	Difference
a) None	6.40	5.46	0.94
b) Object	4.46	3.21	1.25
c) Non-object	6.02	5.43	0.59

cally, the null pronoun condition is always rated lower than the corresponding overt pronoun condition, a difference that is largest in object extraction. Consistent with the hypothesis that POGs are ungrammatical just when the object is extracted, only this condition is rated below 4 (the midpoint of the scale) and more than a full point below its counterpart with an overt pronoun.

To what extent are these patterns in the global means representative of the behaviour of the five A-bar movement constructions considered separately? Table 3-Table 7 present the mean ratings by condition for each one. Across all five construction types, POGs are always rated lower than their counterparts with overt pronouns. Moreover, object extraction with a null pronoun is always the lowest-rated of the six conditions, and is below the 4.0 midpoint in all except the *how much/many* questions.

Table 3: Which/what N questions

Extraction	i) Overt pronoun	ii) Null pronoun	Difference
a) None	6.48	5.43	1.05
b) Object	5.20	3.63	1.57
c) Subject	6.28	5.93	0.35

Richard Stockwell, Anke Himmelreich & Carson T. Schütze

Table 4: *How much/many* questions

Extraction	i) Overt pronoun	ii) Null pronoun	Difference
a) None	6.35	5.50	0.85
b) Object	6.00	4.28	1.72
c) Subject	5.08	4.50	0.58

Table 5: Relativization

Extraction	i) Overt pronoun	ii) Null pronoun	Difference
a) None	5.93	5.50	0.43
b) Object	4.78	3.45	1.33
c) Subject	6.48	5.60	0.88

However, the pattern seen in Table 2 whereby the decrement in acceptability from overt to null pronoun is greatest for object extraction is carried only by the first three construction types across Table 3-Table 5 – *which/what* and *how much/many* questions plus restrictive relatives. It is not maintained for topicalization or *It*-clefts in Table 6 and Table 7, so we examine these two constructions in more detail.

Table 6: Topicalization

Extraction	i) Overt pronoun	ii) Null pronoun	Difference
a) None	6.53	5.75	0.78
b) Object	2.25	2.03	0.22
c) Adjunct	6.35	5.83	0.52

For Topicalization, it appears that object extraction with both overt and null pronouns may have been subject to a floor effect, with most item means below 2.5 and some below 2.0. We suspect that, out of context, the lack of anything explicit to contrast with the topicalized object made these sentences very awkward, de-

7 Complement-less prepositions and extraction

Table 7: *It*-clefts

Extraction	i) Overt pronoun	ii) Null pronoun	Difference
a) None	6.70	5.13	1.57
b) Object	4.08	2.68	1.40
c) Subject	5.90	5.30	0.60

spite our best efforts, as in (30b) above. As a result, we exclude the topicalization data from further discussion and analysis.

In the *It*-cleft condition, examination of item means revealed that three of the four items followed the overall patterns observed in Table 2, while one item showed an extreme degradation in the rating for object extraction with the overt pronoun, making it actually worse than with the null pronoun. Obviously we would hope to explore in follow-up work whether this was more than a random glitch. For the moment we assume that it was just that, and exclude that item from further analysis. Table 8 shows the means across the remaining three *it*-cleft 6-tuples, which conform to the pattern of Table 2:

Table 8: Three *it*-cleft items

Extraction	i) Overt pronoun	ii) Null pronoun	Difference
a) None	6.63	5.00	1.63
b) Object	4.33	2.50	1.83
c) Subject	5.63	5.57	0.06

Before looking at the POG data in more detail, let us return briefly to the contrast between object extraction with POGs versus with optionally transitive prepositions like *inside* and particle ‘wearing’ *on*. We tested these using the 6-tuple in (32) and two triplets like (33), respectively:¹⁶

¹⁶The null symbol in (32) is not intended to endorse any particular analysis of the intransitive use of the preposition; we adopt it purely for notational convenience. A reviewer claims that *inside of it* is much more natural than *inside it* for many speakers of British English. The first author doesn’t share this intuition, though it may be something to take into account in future experiments. *Inside of* without *it* is distinctly ungrammatical for the first author.

Richard Stockwell, Anke Himmelreich & Carson T. Schütze

- (32) *Inside* [Extraction]
- a. This drawer has money inside it/Ø. [none]
 - b. What_i does this drawer have *t_i* inside it/Ø? [object]
 - c. [Which drawer]_j *t_j* has money inside it/Ø? [subject]
- (33) ‘Wearing *on*’ [Extraction]
- a. The performer has a costume on. [none]
 - b. [What costume]_i does the performer have *t_i* on? [object]
 - c. [Which performer]_j *t_j* has a costume on? [subject]

A simple examination of the means in Table 3, Table 4, Table 5 and Table 8 vs. Table 9 and Table 10 provides strong *prima facie* evidence that the latter two constructions do not degrade like POGs when their objects are extracted:

Table 9: *Inside* (32)

Extraction	i) Overt pronoun	ii) Null pronoun	Difference
a) None	6.70	6.70	0
b) Object	6.42	6.43	−0.01
c) Subject	6.65	6.60	0.05

Table 10: ‘Wearing *on*’ (33)

Extraction	
a) None	6.77
b) Object	6.56
c) Subject	6.68

5.3 Analysis

Having excluded Topicalization and one *it*-cleft item, the remaining items containing POGs were grouped into interrogatives and declaratives. Table 11 and Table 12 present the mean ratings by condition for interrogatives (8 items: 4 *which/what* N and 4 *how much/many* questions) and declaratives (7 items:

7 Complement-less prepositions and extraction

4 relative clauses and 3 *it*-clefts), respectively. As before, POGs are always rated lower than their counterparts with overt pronouns; object extraction with a null pronoun is the lowest-rated condition; and the decrement in acceptability from overt to null pronoun is greatest for object extraction:

Table 11: Interrogatives

Extraction	i) Overt pronoun	ii) Null pronoun	Difference
a) None	6.41	5.46	0.95
b) Object	5.60	3.95	1.65
c) Subject	5.68	5.21	0.47

Table 12: Declaratives

Extraction	i) Overt pronoun	ii) Null pronoun	Difference
a) None	6.23	5.29	0.94
b) Object	4.59	3.04	1.55
c) Subject	6.11	5.59	0.52

All of this is consistent with the hypothesis that POGs are ungrammatical just when the object is extracted. To be sure of this conclusion, what we would like to know about these data is the following: given that null pronouns are always rated lower than their overt pronoun counterparts, are null pronouns in object extraction sentences rated even worse than one would expect by virtue of that main effect, plus whatever effect object extraction generally has in the relevant type of sentence? This is the essence of how Sprouse (e.g., [Sprouse & Villata 2021](#)) argues that degradation should be identified as the result of a genuine grammatical constraint violation in need of theoretical explanation: a configuration is rated substantially worse than the sum of the components that are shown to independently degrade ratings on sentences of the relevant type.¹⁷

¹⁷Consider *whether*-islands, for example, whose level of unacceptability is often taken to call for positing a grammatical constraint. However, they might instead just be hard to process, since they involve extraction from an embedded clause and the presence of an embedded polar question. To adjudicate between these possibilities, [Sprouse & Villata \(2021\)](#) make the

Richard Stockwell, Anke Himmelreich & Carson T. Schütze

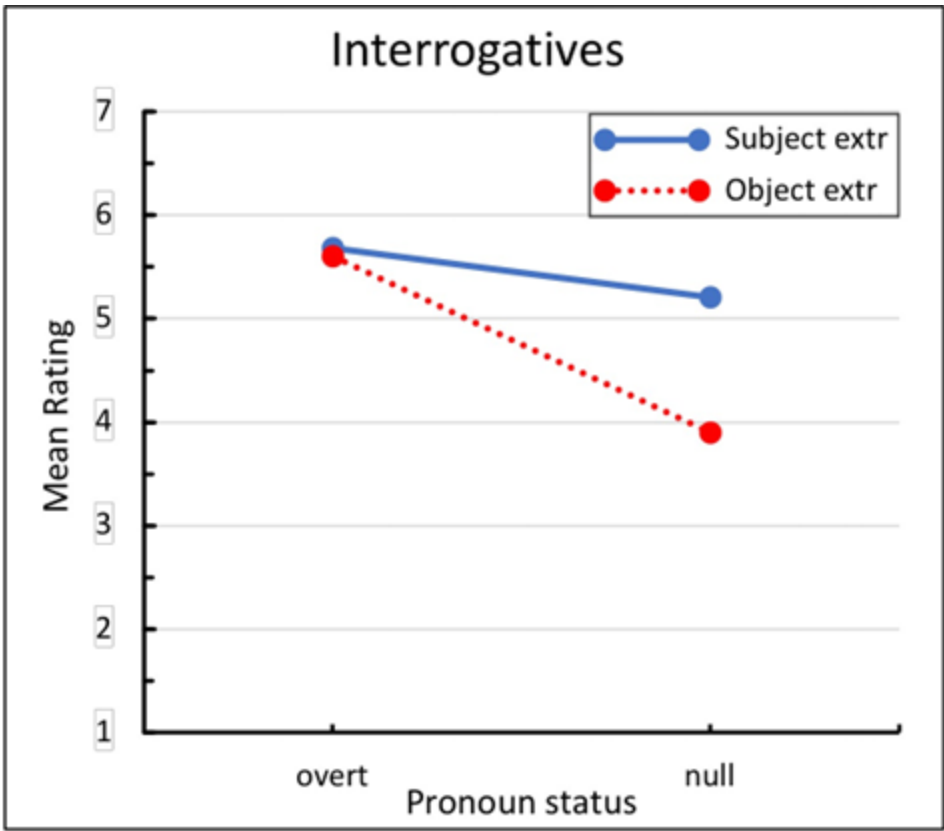


Figure 3: Condition means for BrEng interrogatives

following series of comparisons of ratings among the paradigm in (i). The first two establish baseline effects: (b) vs. (a) measures the cost of long-distance vs. local extraction; while (c) vs. (a) measures the cost of processing an embedded polar question vs. an embedded declarative within a matrix *wh*-question. The third comparison, (d) vs. (a), can then show if the *whether*-island effect is greater than the sum of its parts; that is, whether long-distance extraction from a polar question yields a greater decrement in ratings than the sum of the long extraction effect (b vs. a) plus the presence of polar question effect (c vs. a). If so, there is an interaction, not just two main effects, implicating the workings of a grammatical constraint:

- (i) a. Who *t* thinks that Lisa invented the algorithm?
- b. What do you think that Lisa invented *t*?
- c. Who *t* wonders whether Lisa invented the algorithm?
- d. What do you wonder whether Lisa invented *t*?

Sprouse’s logic generally relies on an assumption that seems problematic in light of our

7 Complement-less prepositions and extraction

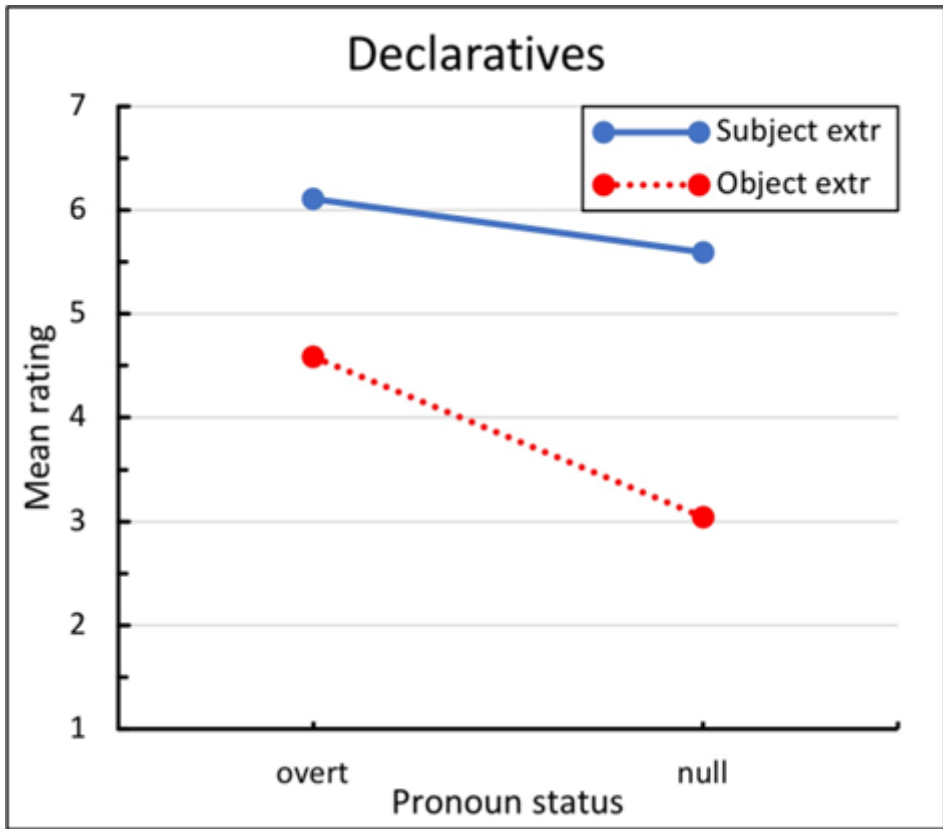


Figure 4: Condition means for BrEng declaratives

data, namely that the degradation contributed by each factor to the target sentence type is independent of all other factors. In Table 3, Table 4 and Table 8 the degradation due to a null pronoun is considerably less when the subject is extracted than in the baseline condition where there is no extraction. Absent an explanation for this difference, it is not obvious which structure – no extraction or subject extraction – provides a better estimate of the expected effect of null pronouns on object extraction. We suggest the comparison of subject vs. object extraction is the most straightforward to interpret, in that the sentences being compared are maximally similar: both involve a *wh*-operator of the same type in Spec,CP (*which/what N*, *how much N*, *how many N*, or OP_{rel}) and a semantic property (interrogation, focus, restrictive modification) not found in the no-extraction condition.

But there could be other, subtler confounds that cut the empirical pie differently. For example, we did not control for the extent to which the sentence would be acceptable (and retain the same meaning) if the entire PP were omitted (in all dialects). Among our stimuli we find a full range of acceptability of such omissions, whose endpoints are illustrated in (ii):

Richard Stockwell, Anke Himmelreich & Carson T. Schütze

In our case, such a scenario manifests itself as an interaction between two independent factors: whether the pronoun complement to the preposition is overt or null and whether the extraction is from object versus subject position. This pattern is clearly visible in Figure 3 and Figure 4, which plot the means from Table 11 and Table 12, respectively.

For interrogatives, ratings for subject extraction fall slightly from overt to null. The far greater fall with object extraction can be attributed to its ungrammaticality with POGs. The same goes for declaratives. Object extraction starts out with lower acceptability than subject extraction even with overt pronouns. But the steeper fall for object than subject extraction with null pronouns can be attributed to the ungrammaticality of object extraction with POGs.

The next section reports a parallel experiment for NGer to investigate whether object extraction is similarly ungrammatical in combination with *da*-drop.

6 Object extraction and *da*-drop: New data

This section reports a second experiment, which finds no evidence for a restriction on object extraction in NGer *da*-drop.

6.1 Method

The participants were 34 speakers from the Berlin/Brandenburg region recruited from the University of Potsdam subject pool,¹⁸ from whom we collected accept-

-
- (ii) a. Expensive dresses usually have fancy embroidery (on them).
b. I saw that the new play has that famous actress *(in it).

When the pronoun complement to the preposition is null, then to the extent that one can deaccent the preposition (which is sentence final in all our stimuli except for some of the Topicalization items), one can “hear” the sentence as if the entire PP were omitted. This might alleviate the degradation associated with POGs for sentences like (iia) but would obviously be of no benefit to sentences like (iib). It is plausible that prosodic differences among the three structures (object extraction, non-object extraction, no extraction) differentially affect deaccenting and thus, indirectly, pronoun drop acceptability. In fact, the authors’ ratings of PP optionality as in (ii) on a 5-point scale correlated significantly or marginally (sometimes positively, sometimes negatively) with experimental acceptability ratings in each of the six conditions for the data in Table 11 and Table 12, with r^2 ranging from 0.15 to 0.31. We hope to explore this in future work.

¹⁸Of the original 39 subjects, five were excluded because they did not self-identify as native speakers of German from the Berlin/Brandenburg region or they gave three or more anomalous scores on catch trials. An anomalous score was defined as either a score on a grammatical catch trial that was lower than some score the participant gave on an ungrammatical catch trial, or a

7 Complement-less prepositions and extraction

ability ratings on a 1–7 Likert scale (7=best) of the final sentence in a multi-sentence two-person dialog. They were paid 10€ for their participation, which took approximately 30 minutes. The target sentence either contained a *da* fronted to the middle field or omitted *da*, and used one of the following eight consonant-initial prepositions: *zu, bei, für, von, gegen, hinter, vor, neben*.¹⁹ The experiment employed a 2×2 design: *da* was either (i) overt or (ii) omitted ($\emptyset$); while the sentence structure involved A-bar extraction of either (a) the direct object or (b) a non-object (subject or AdverbP). Two types of A-bar extraction were tested: matrix topicalization (i.e., fronting to first position in a V2 declarative clause), which applied to direct objects and AdverbPs; and restrictive relativization, which applied to direct objects and subjects. The 16 target dialogs thus involved 4-tuples of final sentences. In addition there were 32 filler dialogs (among which 10 had final sentences designated as grammatical catch items and 7 had final sentences designated as ungrammatical catch items), for a total of 48 items to be rated (each participant saw only one member of each target 4-tuple).

A target dialog exemplifying topicalization is given in (34) and another exemplifying relativization in (35). The question (a) was to be read as uttered by one interlocutor, with another interlocutor responding with (b) followed by either (c) or (d). Participants were tasked with judging the final sentence, (c) or (d):

- (34) *Topicalization* [Extraction]
- a. Was sind die Nebenwirkungen der Tabletten?
 what are the side.effects of.the tablets
 ‘What are the side effects of the tablets?’
 - b. Das ist bei jedem unterschiedlich...
 that is for everyone different
 ‘That is different for everyone...’
 - c. Aber einen Ausschlag_i kann man da/ $\emptyset$ häufig t_i von bekommen.
 but a rash can one DA/ $\emptyset$ often from get
 ‘But one can often get a rash from them.’ [object]

score on an ungrammatical catch trial that was higher than some score the participant gave on a grammatical catch trial. We did not exclude participants who stated that they did not speak Berlin/Brandenburg dialect day-to-day themselves; doing so would have shrunk the subject pool in half, but comparing them to active dialect speakers could prove interesting in future research.

¹⁹In these dialects, vowel-initial prepositions are strongly dispreferred in this construction – viz. the location of Potsdam in Figure 1 and Figure 2 – so we could not test the closest counterparts to BrEng *in/on (in/auf)*. *Mit* was also avoided – see Appendix.

Richard Stockwell, Anke Himmelreich & Carson T. Schütze

- d. Aber häufig_j kann man da/Ø *t_j* einen Ausschlag von bekommen.
 but often can one DA/Ø a rash from get
 ‘But often one can get a rash from them.’ [AdvP]

(35) *Relativization* [Extraction]

- a. Wohin kann ich mich mit Computerproblemen wenden?
 where can I myself with computer.problems turn
 ‘Where can I turn with computer problems?’
- b. Das kommt drauf an. In unserem Betrieb gibt es eine
 that comes there.on on in our company gives it a
 Computerabteilung.
 computer.department
 ‘That depends. In our company, there is a computer department.’
- c. Es gibt einige Fragen, die_i Service-Mitarbeiter da/Ø dort
 it gives a.few questions which service-employees DA/Ø there
 sicher *t_i* zu beantworten können.
 surely to answer can
 ‘There are a few questions that service staff there are sure to be able
 to answer about those [computer problems].’ [object]
- d. Es gibt dort Service-Mitarbeiter, die_j da/Ø sicher *t_j* einige
 it gives there service-employees who DA/Ø surely a.few
 Fragen zu beantworten können.
 questions to answer can
 ‘There are service staff there who are sure to be able to answer a few
 questions about those [computer problems].’ [subject]

The stimuli were presented in Standard German orthography rather than attempting to represent dialectal pronunciation (as in [Henneberg 2017](#)), since people are not accustomed to reading the latter.

6.2 Results

The mean ratings by condition are summarized in Table 13 and Table 14 and plotted in Figure 5 and Figure 6 for Topicalization and Relativization, respectively:

The results show potential interactions in opposite directions for the two clause types. Topicalization appears to fit the pattern from BrEng POGs, whereby *da*-drop degrades object extraction more than non-object (AdvP) extraction. However, NGer Topicalization lacked a direct counterpart in the BrEng data.

7 Complement-less prepositions and extraction

Table 13: Topicalization

Extraction	i) <i>da</i> overt	ii) <i>da</i> dropped	Difference
a) Object	4.56	3.13	1.43
b) AdvP	4.71	3.68	1.03

Table 14: Relativization

Extraction	i) <i>da</i> overt	ii) <i>da</i> dropped	Difference
a) Object	3.56	2.65	0.91
b) Subject	4.28	2.70	1.58

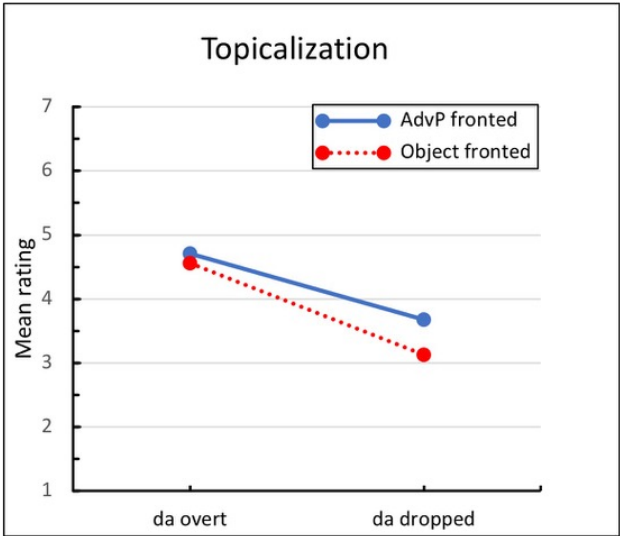


Figure 5: Condition means for NGer topicalization

Richard Stockwell, Anke Himmelreich & Carson T. Schütze

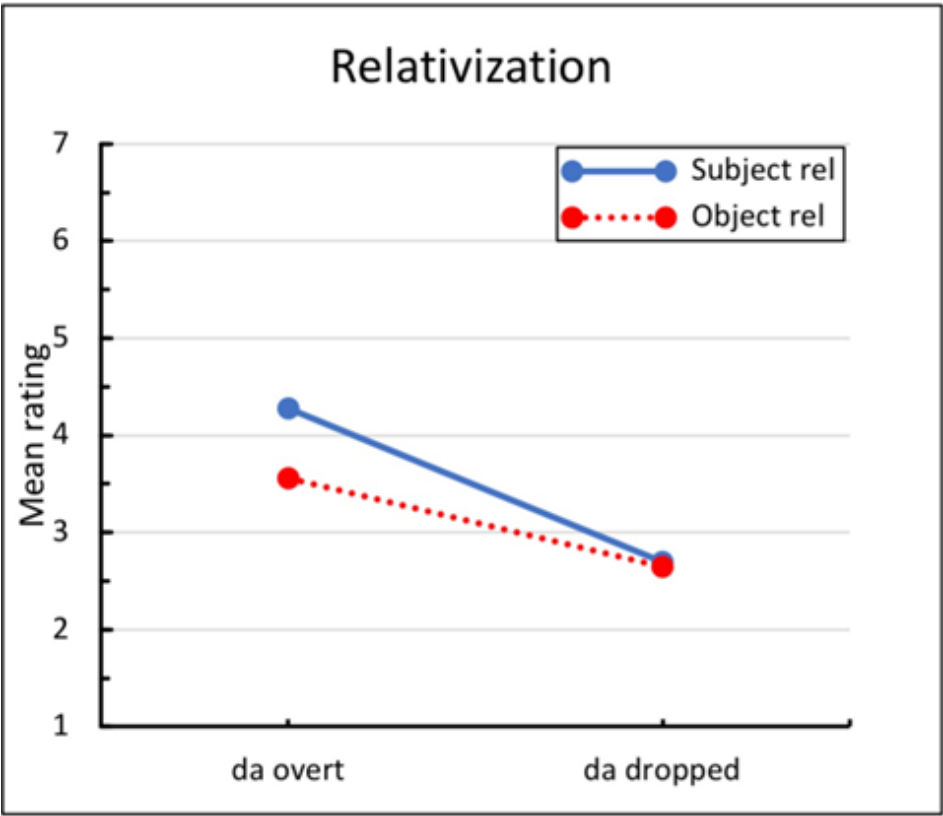


Figure 6: Condition means for NGer relativization

English “Topicalization” is already semantically and syntactically quite different from German Topicalization; and in any case the Topicalization data had to be excluded from the BrEng analysis due to a floor effect.

These caveats urge giving greater weight to NGer Relativization. Here the potential interaction is numerically larger and in the opposite direction compared to NGer Topicalization: subject extraction appears to be more degraded by *da*-drop than object extraction. (This is also in the opposite direction from English Relativization, cf. Table 5.) Before making too much of the apparent greater difficulty of subject relativization combined with *da*-drop, however, it should be noted that the critical stimuli were long sequences of sentences where the judgement hinged on the presence/absence of the same very short word (*da*) in each case. It would be desirable to conduct a follow-up experiment where participants read the sentences out loud, to rule out artifacts that could arise from skipping

7 Complement-less prepositions and extraction

da when it is present, or subconsciously inserting it when it is absent. And perhaps most importantly, both of the NGer numerical interactions (Figure 5 and Figure 6) are roughly half the size of the BrEng interactions (Figures 3 and 4), despite involving almost identical numbers of items.

Taken together, we interpret the results of the NGer experiment as failing to provide any clear evidence for any interaction between A-bar extraction and *da*-drop. The analysis in the next section seeks to reconcile the difference between NGer and BrEng in this regard.

7 Towards an analysis of the extraction facts

This section presents an idea for how to derive the difference between English POGs and German *da*-drop with respect to object extraction. In overview, we claim that the object extraction restriction with BrEng POGs follows from the pronominal complement of the preposition A-bar moving to the left of the sentence object for silencing. The restriction would not follow on G&S's previous A-movement analysis of POGs. The A-properties of German scrambling can, however, explain why there is no such extraction restriction in NGer.

7.1 The object extraction restriction in BrEng POGs

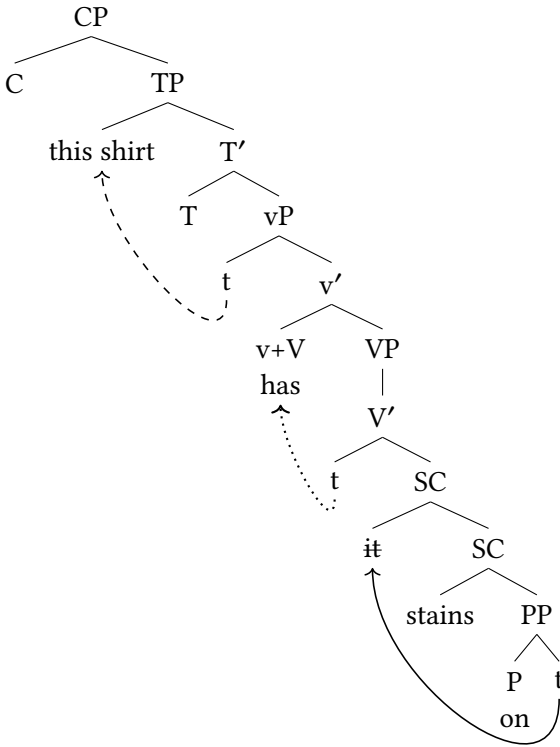
Our analysis overall involves three crucial assumptions, two of which are relevant to BrEng POGs. The first is that pronominal complements of P need to move in order to be silenced. Circumstantial evidence for this comes from the reliance of *da*-drop on *da*-fronting in NGer. We extend this suggestion to BrEng POGs in assuming that the complement of P moves for silencing. The claim that certain elements must move in order to delete draws on several precedents in the literature; for example, Chomsky's (1973, 1977) classic analysis of Comparative Deletion. See also the analyses in Johnson (2001), Fitzpatrick (2006), and Schirer (2008).

More precisely and with respect to POGs, we stipulate that *it/them* must A-bar move to the edge of the small clause (SC) complement of locative *have* in order to be silenced; (36) illustrates.²⁰

(36) This shirt has stains on.

²⁰We use different arrows to indicate different kinds of movement: solid for A-bar movement, dashed for A-movement, and dotted for head movement.

Richard Stockwell, Anke Himmelreich & Carson T. Schütze



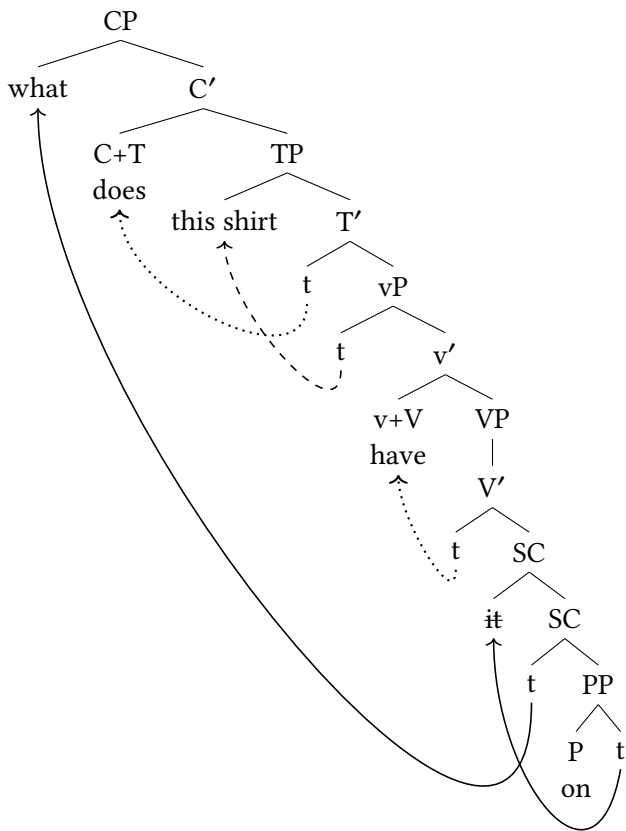
Our second assumption is that crossing A-bar dependencies are ungrammatical. That is, we adopt Pesetsky's (1982) Path Containment Condition. Crossing, in contrast to nested, A-bar dependencies yield ungrammaticality in a range of structures. The crossing (a) vs. nested (b) 'i' and 'j' dependencies in (37) illustrate this for *tough*-movement combined with *wh*-movement. Compare also the relative (un)acceptability of the crossing vs. nested *wh*-island violations in (38):

- (37) a. * [Which sonata]_i is [this violin]_j easy [OP_j PRO to play *t*_i on *t*_j]?
 b. [Which violin]_j is [this sonata]_i easy [OP_i PRO to play *t*_i on *t*_j]?
 (38) a. * Who_i do you know [CP [what subject]_j PRO to talk to *t*_i about *t*_j]?
 b. ? [What subject]_j do you know [CP who_i PRO to talk to *t*_i about *t*_j]?
 (39) * What does this shirt have on?

Applied to POGs, A-bar movement of the sentence object from the specifier of SC combined with A-bar movement of *it* creates crossing A-bar dependencies. Hence (39) is ungrammatical:

- (39) * What does this shirt have on?

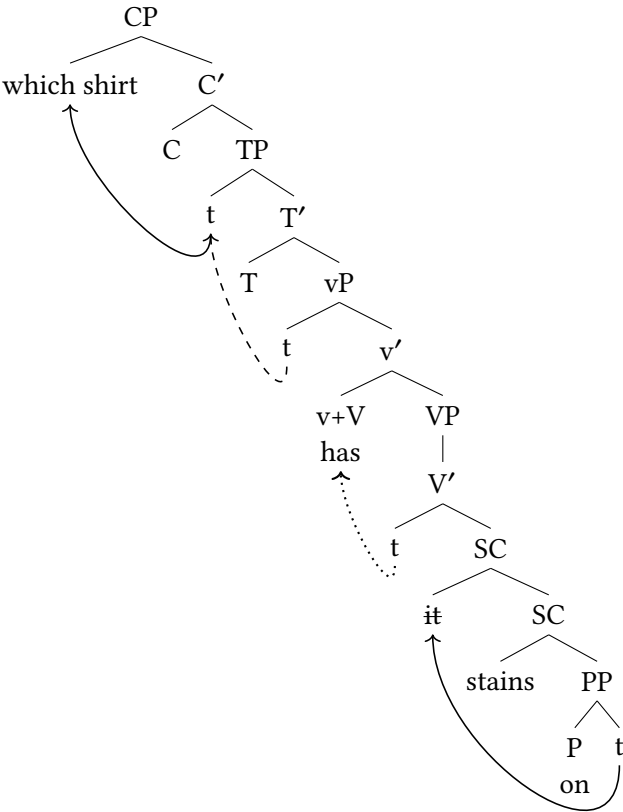
7 Complement-less prepositions and extraction



A-bar movement of subjects or AdvPs from above SC, on the other hand, does not intersect with A-bar movement of *it*. Hence (40) is grammatical:

- (40) Which shirt has stains on?

Richard Stockwell, Anke Himmelreich & Carson T. Schütze



7.2 Comparison with G&S’s analysis of POGs

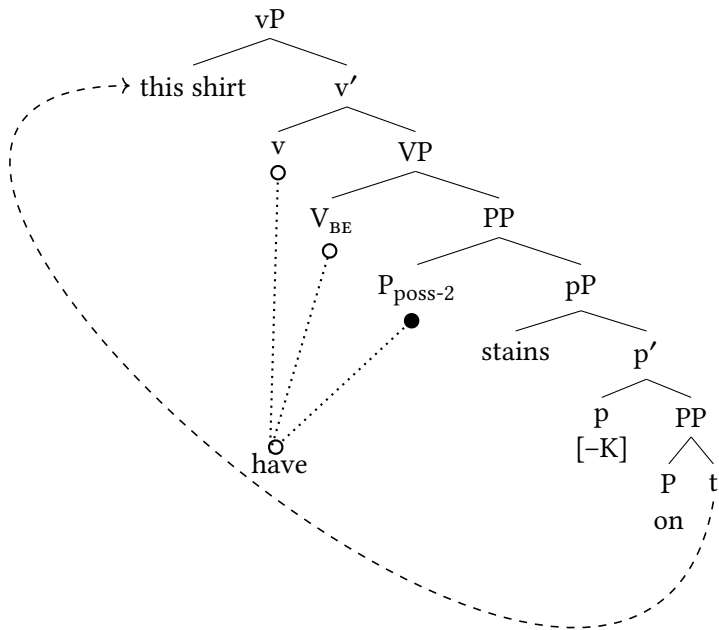
Our analysis in terms of A-bar movement contrasts with G&S’s analysis involving A-movement. G&S propose that POGs are derived by A-movement of the complement of P to subject position via Spec,vP, as in (41).²¹

- (41) This shirt has stains on.
- [G&S analysis]

²¹In other words, for G&S POGs are in a Case alternation with existentials. Whereas p is not a Case assigner [-K] in (41), it would be [+K] in (i):

(i) There are stains on this shirt.

7 Complement-less prepositions and extraction



The object extraction restriction would not be expected on this A-movement analysis. Crossing of an A-chain (j) and an A-bar chain (i) is not excluded, as shown in (42):²²

- (42) Who(m)_i does John_j strike t_i as (being) t_j selfish?
-

7.3 No object extraction restriction with NGer da-drop

Turning to NGer, we maintain the two assumptions presented above for BrEng POGs: crossing A-bar dependencies are ungrammatical and pronominal complements of P must move in order to be silenced. As noted above, *da-drop* is possible only where *da-fronting* is also possible, suggesting that *da-fronting* feeds

²²Stockwell & Schütze (2019) argue further that the structure in (41) makes incorrect predictions regarding variable and anaphor binding. In (i), for example, since Principle A should be satisfiable prior to A-movement, (b) should be able to convey the same (trivial) thing as (a). Instead, (b) is as bad as (c) with an overt pronoun:

- (i) a. Of course [my car]_i is in the picture of itself_i.
b. * Of course [the picture of itself]_j has [my car]_i in t_j.
c. * Of course [the picture of itself]_j has [my car]_i in it_j.

Richard Stockwell, Anke Himmelreich & Carson T. Schütze

da-drop. Adopting Frey's (2004a) assumption of a medial topic position, we take movement of *da* to be to a middle field position, analogous to movement of *it/them* to adjoin to SC in POGs. However, there is no analogous object extraction restriction with NGer *da*-drop, due to the third crucial assumption in our analysis: NGer has local scrambling, which is not A-bar movement.

This assumption is founded on German local scrambling having several A-movement properties. As shown in (43) and (44), for example, scrambling of the direct object feeds binding interactions with the indirect object (Haider 2010):

- (43) a. dass wer den Schülern_i einander_i zeigen wird
 that someone the students(DAT) each.other(ACC) show will
 'that someone will show the students each other'
 [base order, IO > DO]
- b. dass wer die Schüler_i einander_i *t* zeigen wird
 that someone the students(ACC) each.other(DAT) show will
 'that someone will show the students to each other'
 [scrambled order, DO > IO]
- (44) a. *dass man Peter_i Peters_i Vater *t* nicht übergeben hat
 that one Peter(ACC) Peter's father(DAT) not surrendered has
 'that one has not surrendered Peter to Peter's father'
- b. dass man [den Hut des Polizisten_i] dem Polizisten_i *t*
 that one [the hat of.the policeman](ACC) [the policeman](DAT)
 nicht übergeben hat
 not surrendered has
 'that one has not surrendered the policeman's hat to the policeman'

In (43), scrambling from the base order in (a) allows the direct object to bind the reciprocal indirect object *einander* in (b). Meanwhile in (44), scrambling of the direct object over the indirect object feeds evaluation of Condition C, triggering a violation in (a) and voiding one in (b).

In the rest of this section, we show that the A-properties of scrambling and other middle field movements create a different situation than in English. There are no crossing A-bar dependencies involved in *da*-drop, since *da* scrambles to the left edge of the middle field. This contrasts with English, where *it/them* undergoes A-bar movement.

Concretely, we follow the approach put forward in Frey (2003, 2004a,b, 2006). Frey argues that topics can appear at the left edge of the middle field in a projection he calls FP. This projection is generated just above the position of sentence

7 *Complement-less prepositions and extraction*

adverbials (like *unfortunately*, *probably*, etc.). The head F agrees with the (potentially multiple) topics in a feature [Top] and has an optional EPP feature that triggers movement of the topic(s) to Spec,FP. We assume that it is in this Spec,FP position that dropping can occur.

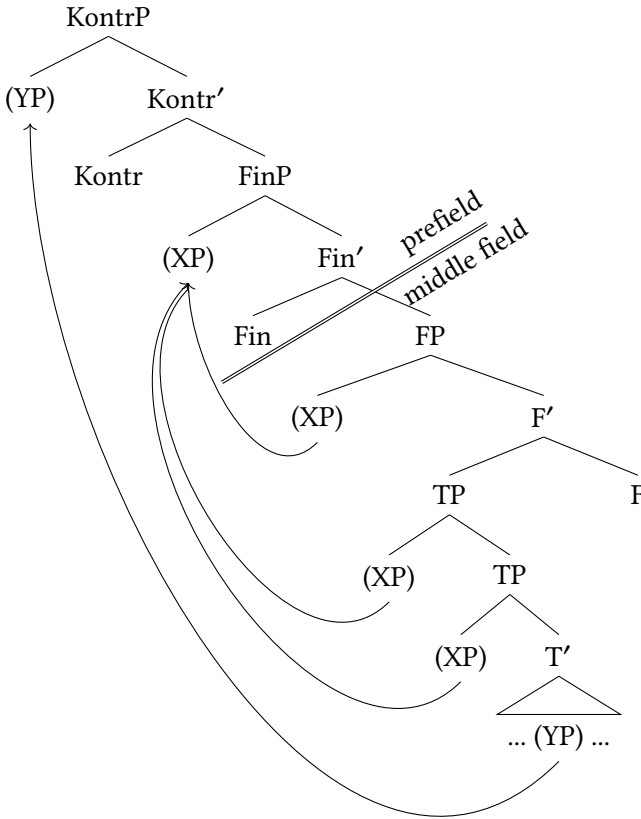
As for the prefield, Frey (2004b) assumes that it is made up of multiple projections – i.a. KontrP, FinP – only one of which can be filled at a time (with the specifier as the prefield element and the finite verb filling the head position). KontrP is filled by contrastive foci or contrastive topics that are A-bar moved to the prefield. FinP is the projection that is filled in most V2 sentences, namely by moving the highest element of the middle field to Spec,FinP.

If a topic has moved to Spec,FP, it will inevitably be the highest element and is henceforth promoted to Spec,FinP. If the topic remains low, then another element (e.g. the subject or a high adverbial) is promoted. The tree in (45) summarizes Frey's assumptions about the structure of the German prefield:²³

- (45) Relevant parts of Frey's (2003 et seq.) structure of the German prefield:
only one projection can be filled at a time

²³Note that German is standardly assumed not to have an obligatory EPP feature on T (Grewendorf 1989: ch.3), with the consequence that movement to TP is not necessary for agreement and case assignment; Agree would suffice for that. Rather, the TP bears tense information and might serve as a high scrambling position for the subject (cf. Heck & Himmelreich 2017).

Richard Stockwell, Anke Himmelreich & Carson T. Schütze



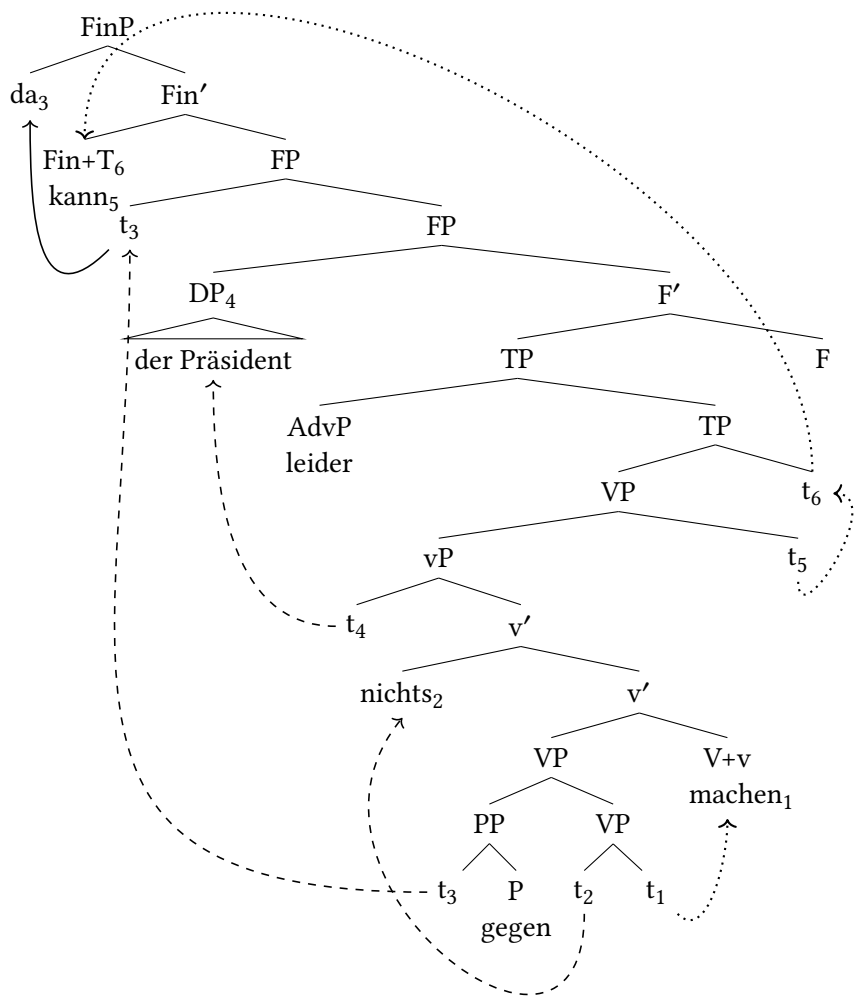
With respect to German *da*, this means that *da* can appear in the prefield if it first goes through Spec,FP. This movement is dialectally restricted to NGer, just as movement of *it/them* to adjoin to SC is restricted to BrEng. The structure in (46) illustrates:²⁴

- (46) Da kann der Präsident leider nichts gegen machen.
 DA can the president unfortunately nothing against do
 ‘The president can unfortunately do nothing against this.’

²⁴Note that we do not assume that Spec,FP is the only scrambling position for *da*. For word orders of embedded clauses like (i) it is necessary to assume a scrambling position below FP. We will not discuss this issue further as it does not impact our analysis of *da*-drop:

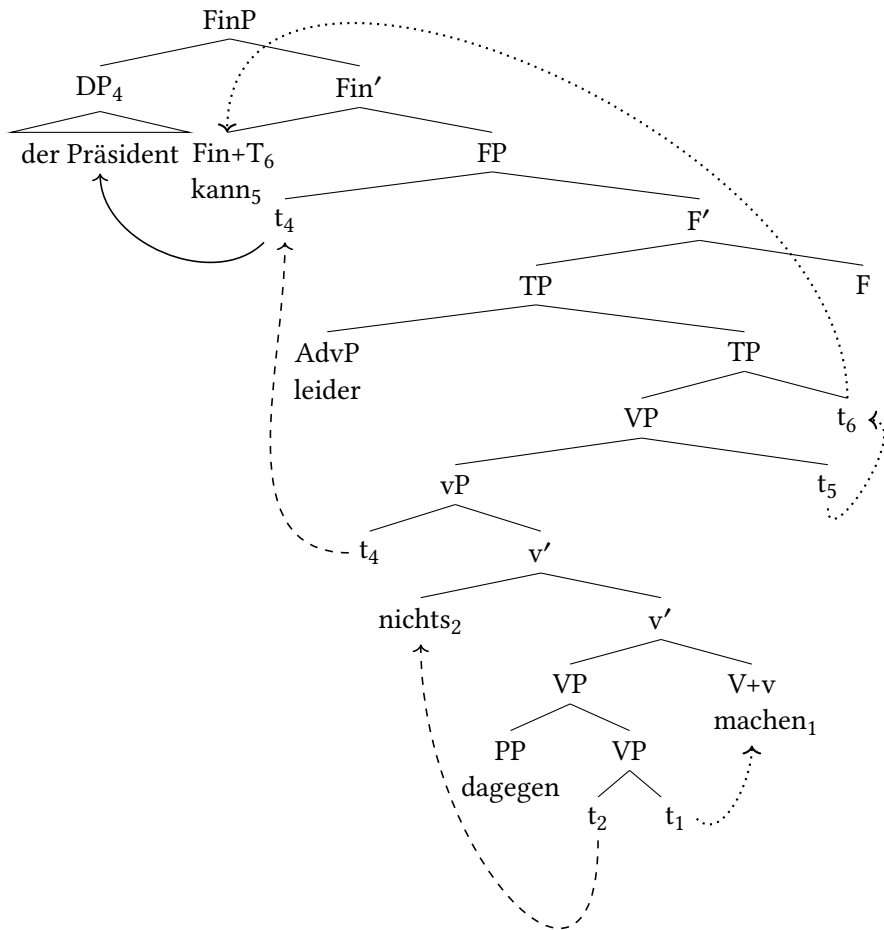
- (i) dass [_{FP} der Präsident [_{TP} leider da nichts gegen machen kann]]
 that the president unfortunately DA nothing against do can
 ‘that the president can unfortunately do nothing against this.’

7 Complement-less prepositions and extraction



If, on the other hand, *da* stays in its base position, the subject can move to Spec,FinP, as in (47):

- (47) Der Präsident kann leider nichts dagegen machen.
 The president can unfortunately nothing DA.against do
 ‘The president can unfortunately do nothing against this.’



Turning to dropped *da*, we assume that dropping occurs in the medial Spec,FP topic position. Deletion of *da* in Spec,FP is dialectally restricted to a subset of the speakers who allow *da* to overtly front to this position.²⁵ If a silenced *da* topic is still promoted to the prefield, the result is a topic drop construction, as in (48) (cf. 18). The representation would be the same as for (46), except with the *da* that reaches Spec,FinP having been silenced in Spec,FP:

²⁵And among these, deletion is further restricted for a subset of speakers to cases where the stranded preposition is consonant-initial.

7 Complement-less prepositions and extraction

- (48) Ø kann der Präsident leider nichts gegen machen.
 (DA) can the president unfortunately nothing against do
 ‘The president can unfortunately do nothing against this.’

To account for dropped *da* in the middle field, we assume that if multiple topics are moved to FP, *da* can sit and be deleted in any Spec,FP. As in (49), the prefield is then filled by moving a lower element to Spec,KontrP (a, c) or the highest topic to Spec,FinP (b):

- (49) a. Häufig kann man Ø einen Ausschlag von bekommen. cf. (34d)
 often can one (DA) a rash from get
 ‘Often one can get a rash from it.’
 b. Man kann Ø häufig einen Ausschlag von bekommen.
 One can (DA) often a rash from get
 ‘One can often get a rash from it.’
 c. Einen Ausschlag kann man Ø häufig von bekommen. cf. (34c)
 a rash can one (DA) often from get
 ‘One can often get a rash from it.’

The crucial question now is whether movement to medial topic Spec,FP behaves like A-bar movement. Frey (2004a: 33f) applies the same tests as in (43) and (44) to movement to Spec,FP and concludes that it creates new binding relationships, just like scrambling to other middle field projections. In (50), movement to Spec,FP allows the direct object to bind the reciprocal indirect object. And in (51), movement to Spec,FP feeds evaluation of Condition C to trigger violations in (a-b) and void them in (c-d):²⁶

- (50) dass die Schüler_i wahrscheinlich der Lehrer einander_i *t*
 that the students(ACC) probably the teacher each.other(DAT)
 gezeigt hat
 shown has
 ‘that the teacher has probably shown the students to each other’

²⁶We admit that this test does not definitively rule out that movement to Spec,FP could be A-bar movement. Since there are many scrambling positions in the German middle field, high scrambling (to TP) could be responsible for changing the binding relations. Still, the data are at least analogous to the behaviour in the lower part of the middle field. We leave this issue open here and simply adopt Frey’s argumentation. If movement to Spec,FP turns out to be A-bar movement, our analysis would have to include more scrambling operations to achieve a structure without crossing A-bar dependencies.

Richard Stockwell, Anke Himmelreich & Carson T. Schütze

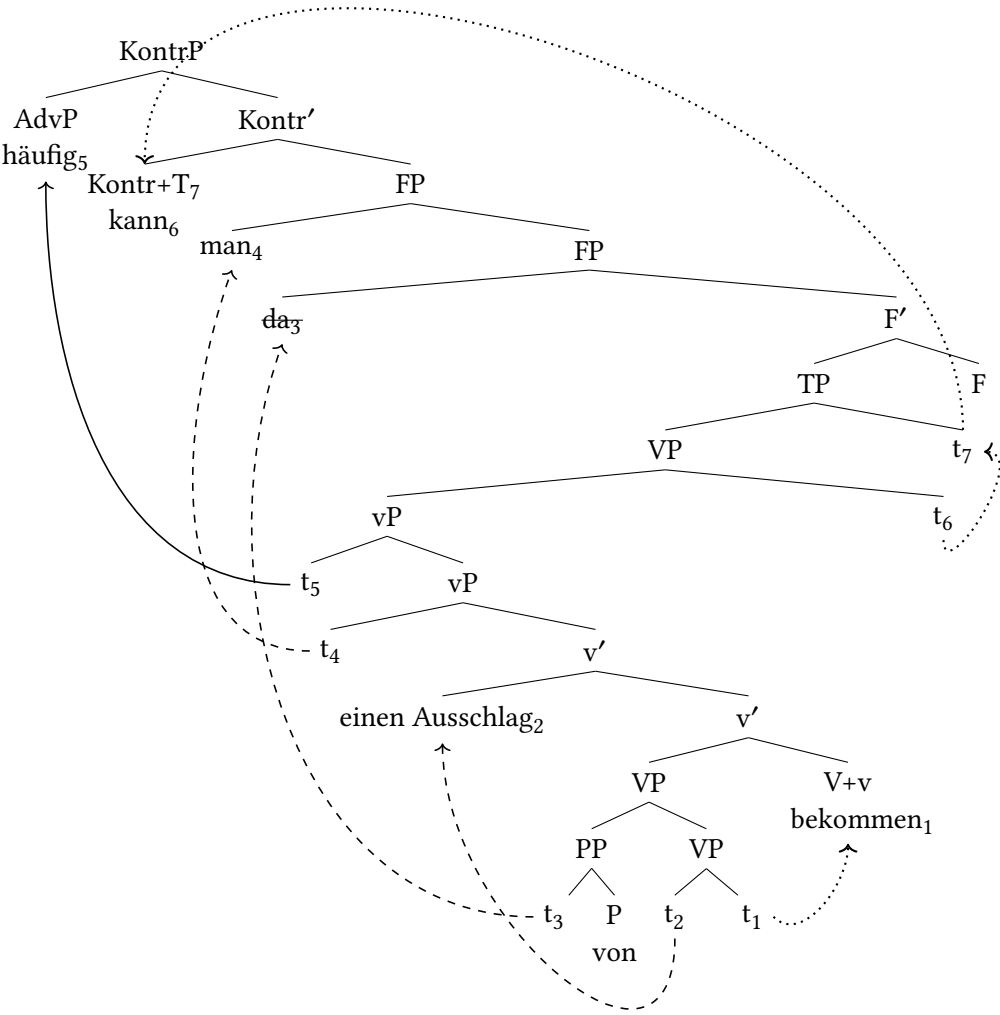
- (51) a. *dass den Peter_i wahrscheinlich die Polizei Peters_i
 that the Peter(ACC) probably the police(NOM) Peter's
 Vater *t* übergeben hat
 father(DAT) surrendered has
 'that the police has probably surrendered Peter to Peter's father'
- b. *dass den Peter_i wahrscheinlich Peters_i Vater der
 that the Peter(ACC) probably Peter's father(NOM) the
 Polizei *t* übergeben hat
 police(DAT) surrendered has
 'that Peters father has probably surrendered Peter to the police'
- c. dass den Hut des Polizisten_i wahrscheinlich der
 that the hat(ACC) of.the policeman probably the
 Polizeidirektor dem Polizisten_i *t* übergeben hat
 police.director(NOM) the policeman(DAT) surrendered has
 'that the police director has probably surrendered the hat of the
 policeman to the policeman'
- d. dass den Hut des Polizisten_i wahrscheinlich der Polizist_i
 that the hat(ACC) the policeman probably the policeman(NOM)
 dem Polizeidirektor *t* übergeben hat
 the police.director(DAT) surrendered has
 'that the policeman has probably surrendered the hat of the
 policeman to the police director'

From this we can conclude that movement to Spec,FP has A-movement properties. Consequently, we predict that A-bar moving an element above the dropped *da* will not result in ungrammaticality, since, trivially, there are no crossing A-bar dependencies. The trees in (52) illustrate for the examples in (49):²⁷

- (52) a. Häufig kann man Ø einen Ausschlag von bekommen.
 often can one (DA) a rash from get
 'Often one can get a rash from it.'

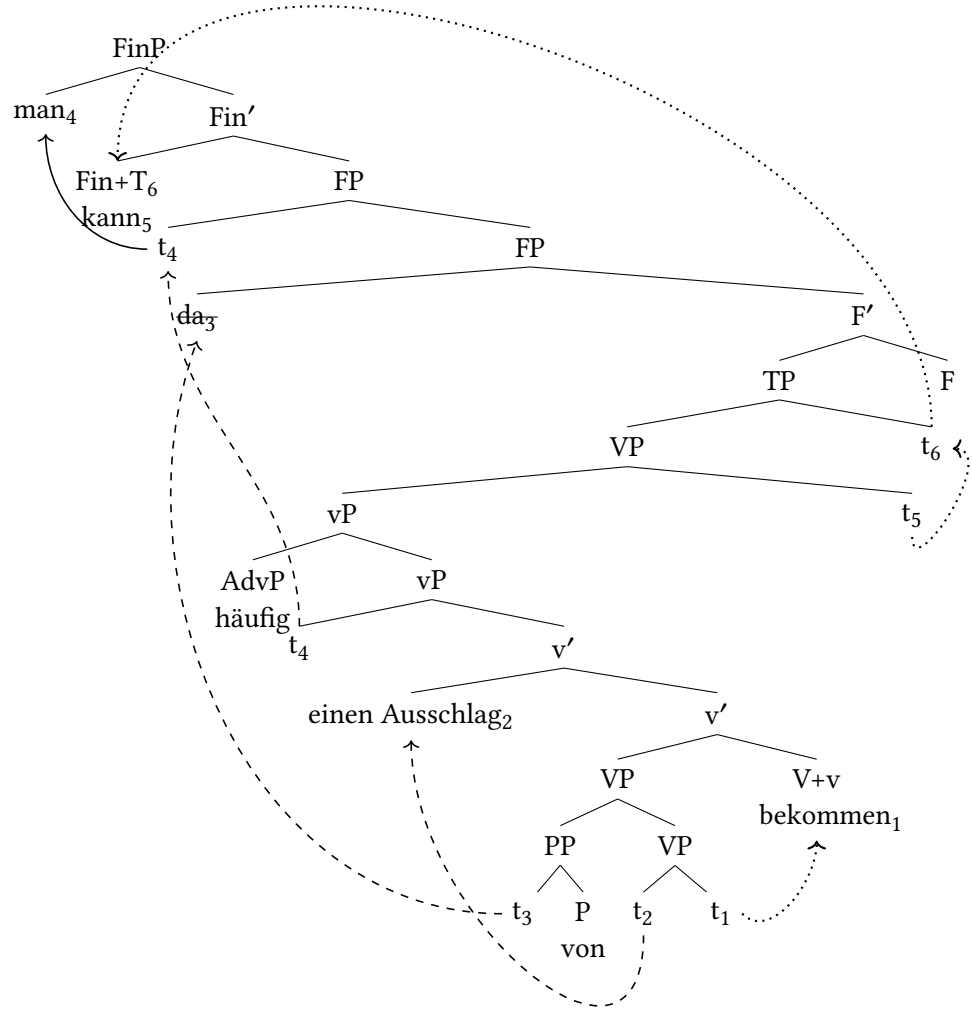
²⁷Note that FinP is absent, on the assumption that only necessary structure is generated (Frey 2006: 251).

7 Complement-less prepositions and extraction



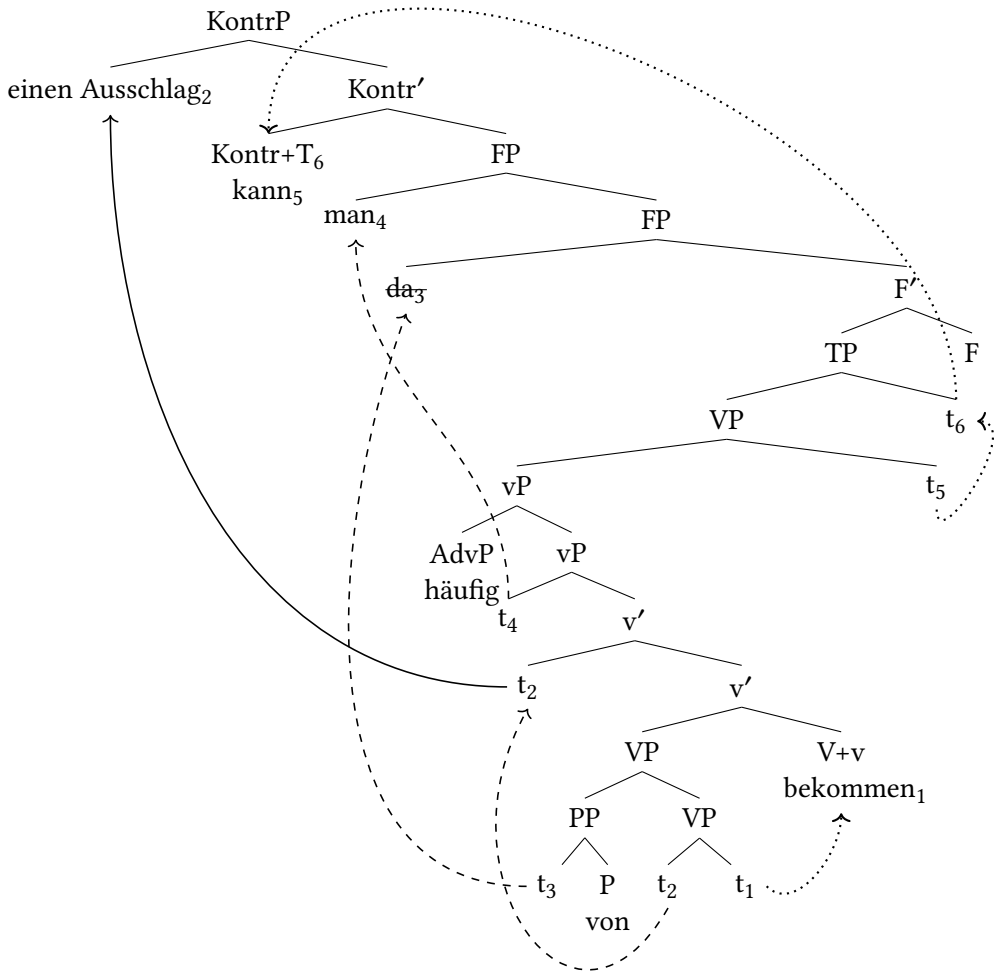
- b. Man kann Ø häufig einen Ausschlag von bekommen.
One can (DA) often a rash from get
'One can often get a rash from it.'

Richard Stockwell, Anke Himmelreich & Carson T. Schütze



- c. Einen Ausschlag kann man Ø häufig von bekommen.
a rash can one (DA) often from get
'One can often get a rash from it.'

7 Complement-less prepositions and extraction



Finally, we show what happens in relative clause constructions. As with topicalization, since movement of *da* to Spec,FP is not A-bar movement, we predict a grammatical outcome regardless of whether the object or subject is relativized. Frey does not discuss relativization, though he mentions that *wh*-movement targets Spec,KontrP (Frey 2006: 253). Thus, we assume that relativization also targets KontrP. Relevant derivations for object and subject relatives are shown in

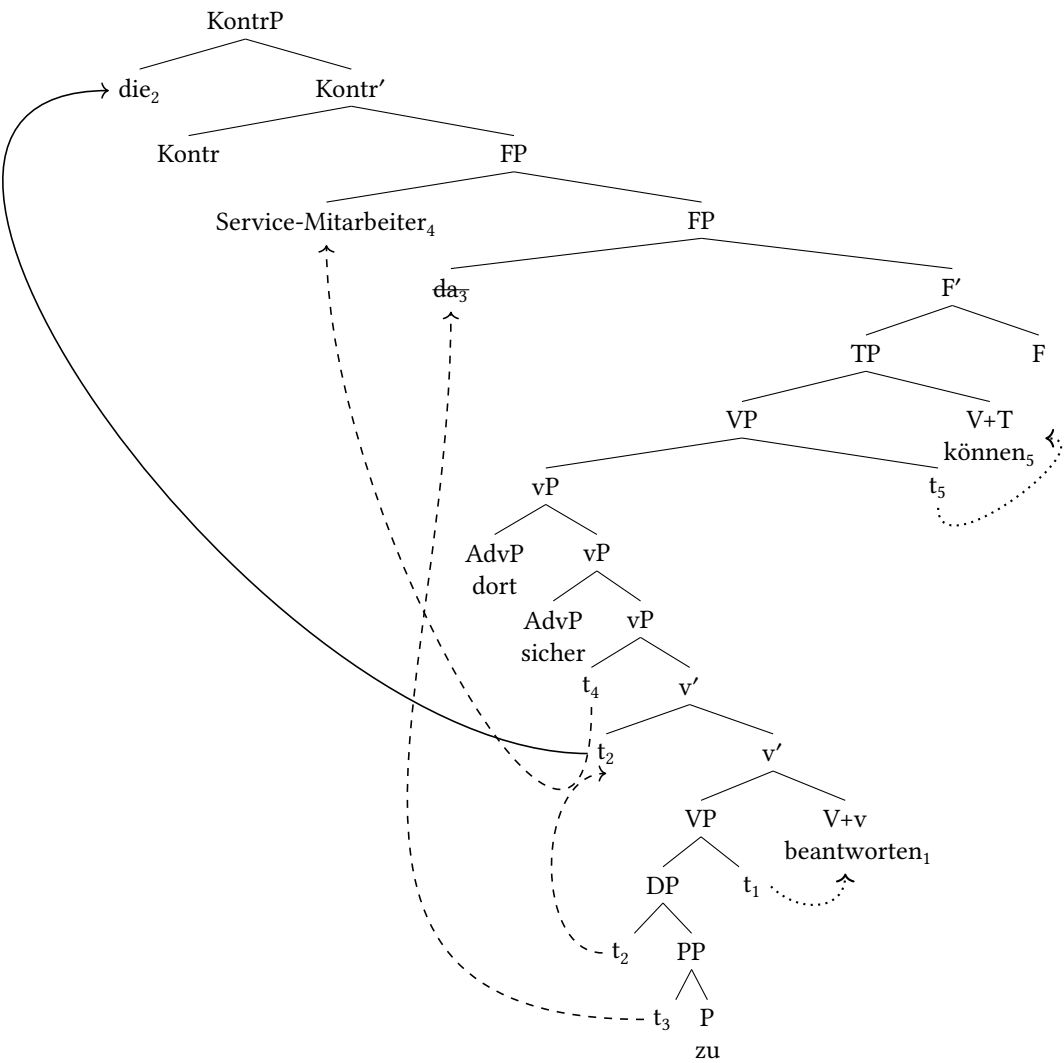
Richard Stockwell, Anke Himmelreich & Carson T. Schütze

(53) and (54), respectively:²⁸

- (53) Es gibt einige Fragen, die_i Service-Mitarbeiter da/Ø dort sicher t_i
it gives a.few questions which service-employees DA/Ø there surely
zu beantworten können. cf. (35c)
to answer can
'There are a few questions that service staff there are sure to be able to
answer about those [computer problems].'

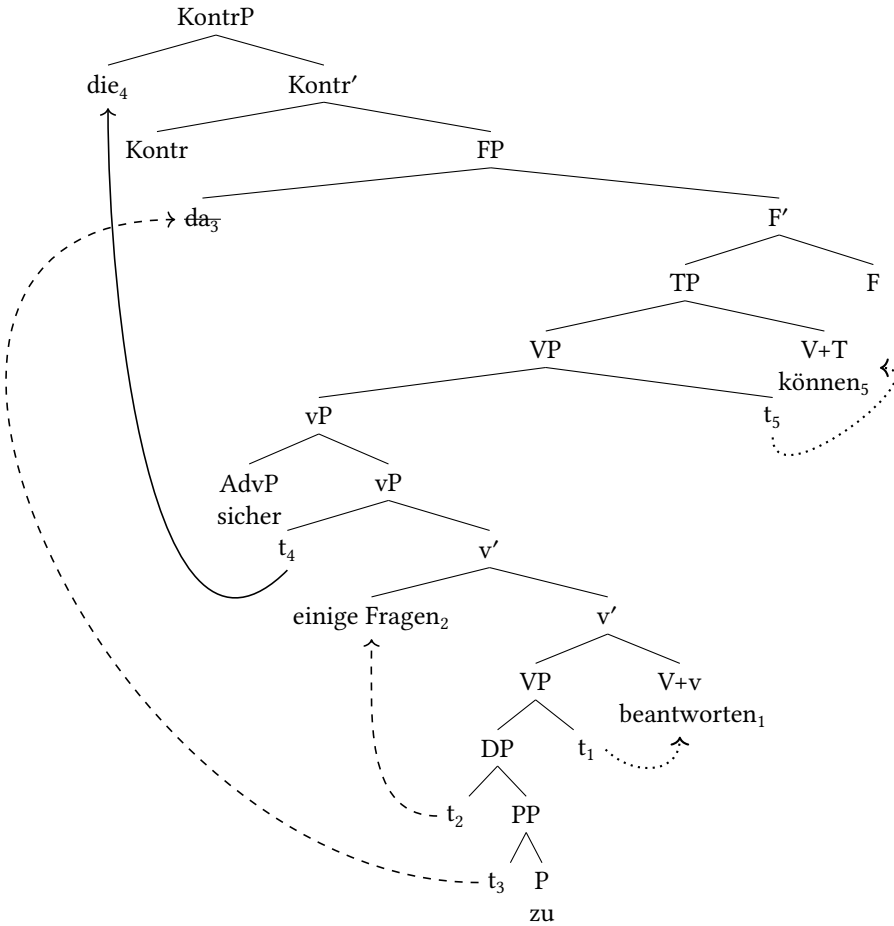
²⁸We assume that the PP is modifying the noun *Fragen* 'questions' here. Even if this assumption is incorrect, as long as scrambling and movement to FP are not A-bar movement, the analysis predicts this sentence to be possible.

7 Complement-less prepositions and extraction



- (54) Es gibt dort Service-Mitarbeiter, die_j da/Ø sicher t_j einige Fragen
it gives there service-employees who DA/Ø surely a.few questions
zu beantworten können. cf. (35d)
to answer can
'There are service staff there who are sure to be able to answer a few
questions about those [computer problems].'

Richard Stockwell, Anke Himmelreich & Carson T. Schütze



In sum, and in contrast to BrEng, movement for dropping of the pronoun in NGer is A-scrambling. Hence no crossing A-bar dependencies arise, with the result that there is no object extraction restriction in NGer *da*-drop.

7.4 Summary

This section has presented an analysis of the difference between BrEng and NGer with respect to object extraction in constructions where a pronoun complement of P is omitted. The analysis is based on three assumptions. First, pronominal complements of P need to move in order to be silenced. In BrEng, the pronoun A-bar moves to the edge of *have*'s small clause complement; while in NGer, *da* moves to a medial topic position, Spec,FP. Second, crossing A-bar dependencies

7 Complement-less prepositions and extraction

cause ungrammaticality (Pesetsky 1982). And third, middle field and other scrambling movements in German, including of *da*, do not proceed by A-bar movement.

With these assumptions, we can account for the differing sensitivity of BrEng POGs and NGer *da*-drop to object extraction. In BrEng, the pronoun has to A-bar move to the edge of the small clause in order to be silenced. This movement traps the sentence object, which is no longer able to A-bar move without creating a crossing A-bar dependency. In NGer, the pronoun still has to move for silencing. However, like scrambling to medial positions throughout German, *da* does not front by A-bar movement. Consequently, A-bar movement of the object trivially does not cross another A-bar movement path.

8 Further directions

Overall, it seems clear that BrEng POGs and NGer *da*-drop require different syntactic analyses. Our proposals above are admittedly preliminary and intended to inspire further research in this area. Although not all of the differences between the languages, nor for that matter the seemingly enticing descriptive similarities (cf. Table 1), have been explained by our accounts, ultimately a comprehensive analysis must of course accomplish as much within the confines of a restrictive theory of universal grammar. Moreover, the fact that these languages are historically related imposes further constraints on possible analyses: the theory must provide an account of the paths of historical change that led from Old English and Old German (which, as we see below, were very similar in relevant respects) to the present-day dialects (and their counterparts that lack the P-complement dropping constructions). For the purpose of enriching future work with this broader empirical context, this concluding section summarizes relevant observations from the literature on diachrony and variation.

8.1 Diachrony

The history of complement-less prepositions could shed further light on their analysis. We know almost nothing about the historical origins of POGs or *da*-drop, and whether they arose independently in (earlier) German and English.²⁹ More is known about *da*-fronting in the history of both languages.

Allen (1980) and van van Kemenade (1987) show that Old English had some properties in common with dialectal German: the locative proform ‘there’ could

²⁹Visser (1963-1973: 535) cites half a page of complement-less prepositions, but offers no discussion of what factors may have been licensing complement omission, or indeed whether the relevant prepositions were simply optionally transitive at the time.

Richard Stockwell, Anke Himmelreich & Carson T. Schütze

also serve as an inanimate 3rd person pronominal proclitic complement to adpositions (55),³⁰ and it could strand those adpositions (56) by moving to the left edge of VP (a)³¹ or to the left periphery (b),³² (c)³³ (Allen refers to this as scrambling; van Kemenade calls it A-bar movement):

- (55) Awyrtwala grædignysse of ðinre heortan, and aplanta þaeron þa
uproot greediness from your heart and plant therein the
soðan lufe.
true love
'Uproot greediness from your heart and therein plant true love.'
- (56) a. Be þæm þu meahst ongietan ðæt þu þær nane myrhðe on næfdest.
by that you might perceive that you there no joy in not.had
'From that, you could understand that you found no joy in it.'
- b. þæt ðær nan cinu on næss gesewen
that there no chink in not was.seen
'that no chink was seen in it'
- c. ac ðær comon munecas to on ðæs mannes forðsiðe
but there came monks to at the man's death
'but monks came to it when the man died'

However – and perhaps more surprisingly – personal pronouns could also move away from the preposition of which they were the complement, to the same two landing zones (57) – the left edge of VP (a)³⁴ and the left periphery (b),³⁵ (c).³⁶ This is different from what we saw with POGs in (24):

³⁰ *The Homilies of the Anglo-Saxon Church*, ed. Benjamin Thorpe, Ælfric Society, London, 1844. Vol. 2, pg. 410.

³¹ *King Alfred's Old English version of Boethius* "De consolazione Philosophiae", (mid-10th century), ed. Walter John Sedgefield, Clarendon Press, Oxford, 1899. Section VII, pg. 15, line 11.

³² *The Homilies of the Anglo-Saxon Church*, ed. Benjamin Thorpe, Ælfric Society, London, 1844. Vol. 2, pg. 154.

³³ *Homilies of Ælfric: a Supplementary Collection*, ed. John Collins Pope, Early English Text Society vols. 259 & 260, Oxford University Press, 1967. Homily XI, line 171. (10th-11th centuries).

³⁴ *King Alfred's Old English version of Boethius* "De consolazione Philosophiae", (mid-10th century), ed. Walter John Sedgefield, Clarendon Press, Oxford, 1899. Section II, pg. 8, line 11.

³⁵ *The Homilies of the Anglo-Saxon Church*, ed. Benjamin Thorpe, Ælfric Society, London, 1844. Vol. 1, pg. 32.

³⁶ *Homilies of Ælfric: a Supplementary Collection*, ed. John Collins Pope, Early English Text Society vols. 259 & 260. Oxford University Press, 1967. Homily X, line 174. (10th-11th centuries).

7 Complement-less prepositions and extraction

- (57) a. þa wendon hi me heora bæc to.
 then turned they me their backs to
 ‘Then they turned their backs to me.’
 b. þæt him eal middangeard to beh
 that him all world to bowed
 ‘that all the world bowed to him’
 c. ac him côm fyr to færllice ehsynes
 but him came fire to suddenly visibly
 ‘but suddenly a light came to him visibly’

We do not know if there was ever a time when some P+pronoun combinations could be expressed only using *there*+P (cf. German, 13), such that prepositional complement drop would have been unambiguously *there*-drop. But if so, and if that is when dropping arose, then whatever analysis was posited for complement-less preposition sentences would have had to change once *there*+P ceased to be productive. It seems plausible that the counterparts of *it/them*, true personal pronoun complements to P, were more restricted in their behaviour than *there*.³⁷ This could have forced a reanalysis of the complement-less preposition construction that induced the additional restrictions found in modern BrEng but not NGer (e.g., the restriction to *have/with* etc.). The reanalysis may simply have failed in NAmEng. Östermann (1997) and Müller (2000) argue that, while pronominal adverbs (*thereat*, *therewith*, *therefore*) were still well-attested in Shakespeare, they ceased to be productive after 1400. Müller proposes a theory on which their disappearance is tied, among other things, to the loss of Wackernagel movement of pronouns. By Middle English, P-stranding by ‘there’ or personal pronouns was no longer possible (Fischer & Wurff 2006).

As for the history of German, Müller (2000) cites Paul (1919: sect. 139) and Lockwood (1968) for the claim that while pronominal adverbs were attested in Old High German, the possibility of separating *da(r)* from the preposition was an innovation that emerged in Middle High German, as illustrated in (58) from the works of Walther von der Vogelweide (c. 1170–c. 1230):

³⁷Indeed, German is suggestive in this regard. In NGer, proform *da* can occupy first position in a V2 clause (cf. 15), but object *es* cannot (other object pronouns may be degraded to varying degrees):

- (i) * Es habe ich gesehen.
 it have I seen
 ‘I have seen it.’

Richard Stockwell, Anke Himmelreich & Carson T. Schütze

- (58) Dâ mugent ir alle schouwen wol ein wunder bî.
 DA could.2PL you all see PRTC a miracle at
 ‘You all could see a miracle in this.’

However, Fleischer (2008) claims *da*-fronting was already attested in Old Low German, as in (59a) from the *Hêliand* (9th century); likewise Russ (1982) and Miller (2004) find examples in Old High German such as (59b) from Notker’s *Martianus Capella* (c. 1000), and note that they are abundant in Middle Low and High German. A reviewer points out that there are many examples to be found in the Corpus of Historical Low German (Booth et al. 2020):

- (59) a. Tho forun thar thie liudi to
 dann gingen da die Leute zu
 then went DA the people to
 ‘Then the people went to it.’
 b. Dâr spráng ínne éin brúnno hímeliskes liehtes ...
 DA sprang in a fountain heavenly light
 ‘In it a fountain of heavenly light sprang...’

Fronting of *da(r)* continued to be well attested everywhere through the 16th century, but after that became geographically restricted to Northern Germany.

Da-drop is less well documented historically. Fleischer (2008) can find clear examples only with *mit* (see the Appendix) in the Old German period. In Middle Low German other clear cases emerge, such as the following (from *Reynke de vos*, 1498):

- (60) De quam ghelopen myt ereme wocken, / Dar se des dages
 die kam gelaufen mit ihrem Spinnrocken da sie des Tages
 she came running with her distaff as she the.GEN day.GEN
 hadde by gheseten
 hatte bei gesessen
 had at sat
 ‘She came running with her distaff, since she had sat by it all day.’

In High German, cases not involving *mit* remain rare throughout all periods; one such is from a letter by Goethe (to Sophie v. La Roche in 1775) (Paul 1919: 159):

- (61) Ich weiß kein Wort von.
 I know no word of
 ‘I don’t know a word about it.’

8.2 Other dialects and languages

Further insights could come from additional comparative analysis. Stranding of prepositions under *da*-fronting is attested elsewhere in Germanic, as the examples in (62)–(64) show:³⁸

- (62) a. Dutch (Zwaarts 1997: 1092)
 Daar staat iemand **achter**.
 DA stands someone behind
 ‘Someone’s standing behind it.’
 b. Dutch (Hoekstra 2001: 781)
 Ik wilde **daar** niet **op** wachten.
 I wanted DA not for wait
 ‘I didn’t want to wait for it.’
- (63) a. Festlandnordfriesisch (North Frisian) (Walker 1990: 23)
 Deer wiitj ik nint **foon**.
 DA know I nothing of
 ‘I know nothing of it.’
 b. Saterfriesisch (East Frisian) (Fort 2001: 418)
 Deer weet iek iks **fon**.
 DA know I nothing of
 I know nothing of it.
- (64) a. Old Danish (c. 1500) (Falk & Torp 1900: 315)
 ther bedher ieg **om**
 DA ask I for
 ‘I asked for it.’
 b. Jutlandic Danish (Jensen 1971: 19)
 en dunk, og **der** drak de **af**
 a pitcher and DA drank they from
 ‘a pitcher, and they drank from it’

However, Fleischer (2002) is hard-pressed to find any Germanic languages besides German that display (the counterpart of) *da*-drop (except as the result of Topic Drop).³⁹ The only candidates he puts forward are Jutlandic Danish (65) and North Frisian (66):

³⁸ According to Longbotham (2010), Delsing (1995) reports that P-stranding with R-pronouns is also found from the earliest stages of Swedish.

³⁹ A reviewer suggests that BrEng-style POGs are attested in Afrikaans, e.g. (i):

Richard Stockwell, Anke Himmelreich & Carson T. Schütze

- (65) Jutlandic Danish (Jensen 1971: 33)
 sådan en stor stykke noget stiv gullig papir med en klat rød lak
 such a big piece of some stiff golden paper with a blood red blob
 på
 on
 ‘such a big piece of stiff golden paper with a blood red blob on (it)’
- (66) North Frisian (Hoekstra et al. 2006)
 Ik hee en Dååler far deen, wen’t ä wân weer.
 I have.SBJV a dollar for given if=it not been was
 ‘I would have given a dollar for it, if it had not been (true).’

Intriguingly, Fleischer reports there was one German dialect (spoken in Cat-tenstedt, Nordharz, Eastphalian) that was described by Damköhler (1927: 37) as dropping *da(r)* just in the presence of the verbs *give*, *have* and *get* (and only before the preposition *of*):

- (67) a. Jif mek wat fon.
 give me some of
 ‘Give me some of it.’
- b. Ek wil wat fon hebn.
 I want.to some of have
 ‘I want to have some of it.’
- c. Dû drist nischt fon.
 you get nothing of
 ‘You get none of it.’

Appendix: *mit* – a preposition like no other

Fleischer (2000, 2002) notes that there are many non-Northern German dialects where *da*-fronting and *da*-drop are possible only with *mit*, including High Ale-

-
- (i) Die boks het papiere in.
 the box has papers in
 ‘The box has papers in (it).’

(We have not independently verified whether all the criteria discussed in Section 2 are met.)
 Afrikaans allows R-pronouns, but they are not compatible with POGs (ii, cf. 23a):

- (ii) * Die boks het papiere daarin.
 the box has papers DA.in
 ‘The box has papers therein.’

7 Complement-less prepositions and extraction

mannic, Low Alemanic, Swabian, East Franconian, Upper Saxon, and Silesian: “In all these dialects the stranding construction and the orphan preposition construction [*da*-drop] are totally unknown with prepositions other than *mit*” (Fleischer 2000: 138); see (68) and (69):

- (68) Zurich Swiss German (High Alemannic) (Fleischer 2002: 152)
 etz mus i die Tabäle usenèè und da han i Müe mit
 now must I this chart take.out and DA have I trouble with
 ‘Now I must take out this chart and I am having trouble with it.’
- (69) Colmarien (Alsatian) (Muller 1983: 260)
 i nimm d’rüet un schlâ-di mit.
 I take the=rod and hit=you with
 ‘I take the rod and hit you with it.’

For Zurich Swiss German, van Riemsdijk (1975: 196f.) already noted that *mit* and its negative *ooni* ‘without’ (70) license omission of inanimate complements, while no other prepositions do. (71) is an example van Riemsdijk considers idiomatic, with the understood complement being context-dependent but lacking a linguistic antecedent; Standard German would not use *damit* here:

- (70) yaʒ nyʏt maye ooni
 you.can nothing do without
 ‘You can’t do anything without it.’
- (71) iʒ daz {mit/ooni} (zervis)?
 is that with/without service
 ‘Is service included/extra?’

As Fleischer notes, Zurich Swiss German, like most varieties of German, lacks an R-pronoun built on *ooni*, casting the first doubt on the idea that complement-less *mit* is really derived from *damit* in this dialect. Moreover, citing some descriptions of other non-Northern dialects listed above, Fleischer suggests that the pronominal adverb *damit*⁴⁰ has virtually ceased to be used.⁴¹ He concludes

⁴⁰ As distinct from the subordinating conjunction *damit* ‘in order that’.

⁴¹ E.g., from Muller’s (1983) description of Colmarian: “Dans plusieurs locutions où l’allemand dit “damit”, l’alsacien se contente d’un simple “mit” [‘In several phrases where German says “damit”, Alsatian is satisfied with a simple “mit”]; “Das Umstandswort *damit* wird 1930 Pforzheim überhaupt abgelehnt” [‘the adverb *damit* is completely rejected in Pforzheim as of 1930’] (Badisches Wörterbuch I: 412) [*Badisches Wörterbuch*. 1925–. Ed. by Ernst Ochs, Karl Friedrich Müller & Gerhard W. Baur. Lahr: Moritz Schauenburg.]; “In moderner Mundart ist

Richard Stockwell, Anke Himmelreich & Carson T. Schütze

on this basis that “*da*-fronting” cannot be literally correct as the analysis for examples like (68), and hence that dropping of a fronted *da* also cannot be the correct analysis for examples like (69). Rather, in these Southern dialects, there is no grammatical source with *da* in situ in the PP, so one must posit a silent pro-form within PPs headed by *mit/ooni*, and base-generate the overt *da* outside the PP in examples like (68). Without further details, this is just an argument that a process that deletes *da* is not necessary. Since *da* is not the complement of P but merely doubles it, it could simply be absent when we do not hear it. But recall that with prepositions other than *mit/ooni*, these dialects show no NGer-style *da*-drop: *da* always surfaces procliticized to P, and is optionally doubled. This would fit with Fleischer’s observation that (only) Southern dialects are where one finds *da*-doubling, as in (72):⁴²

(72) Bernese Swiss German (Greyerz & Bietenhard 1981:87)

Da hani gar nüüt dergäge!
 da hab=ich gar nichts dagegen
 DA have=I at.all nothing DA.against
 ‘I have nothing against it at all.’

Given the likelihood that complement-less *mit* calls for its own analysis separate from *da*-drop, we have avoided any examples involving *mit* in the discussion in the main text.

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das Wort [*damit*] selten; häufiger einfach Adv. *mit*” [‘In modern dialect, the word is rarely *da-mit*; more often simply the adverb *mit*.’] (Schwäbisches Wörterbuch II: 44 [*Schwäbisches Wörterbuch*. 1904–1936. Ed. by Hermann Fischer & Wilhelm Pfeleiderer. Tübingen: H. Laupp.].

⁴²What remains mysterious is why only *mit* (and sometimes *ohne*) requires the resumptive to be silent, thereby allowing doubling to masquerade as fronting.

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Chapter 8

Pseudo-resumptives in Pennsylvania Dutch: A preliminary analysis

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Pennsylvania Dutch (PD) exhibits a peculiar syntactic construction that distinguishes it from most, if not all, other colloquial and diasporic varieties of German. Although PD exhibits *da*-Komposita, it does not license *wo*-Komposita. To compensate for the lack of *wo*-Komposita, PD licenses **pseudo-resumptive** structures, in which both the *w*-element and the *d*-element associated with *da*-Komposita are lexicalized. In this chapter, I adopt a neo-constructivist approach to the syntax-lexicon interface, exploring the challenges that these structures pose for (any) formal analysis. Building on recent proposals put forward by [Fisher et al. \(2022\)](#) and [Natvig et al. \(2025\)](#), my analysis of pseudo-resumptives in PD, based on previous work by [Noonan \(2017\)](#), requires unified sequences of syntactic structure (i.e., *spans*) to occasionally be lexicalized with multiple exponents. I show how this can be achieved in a simple, conceptually-appealing way through minor adjustments to the Insertion Heuristic at the syntax-lexicon interface.

1 The empirical puzzle

A recent trend in generative theorizing that has gained significant traction and support over the last two decades is that ‘small words’, such as functional elements, exhibit detailed underlying structure. For example, [Hachem \(2015\)](#) provides a fine-grained analysis of *d*(eterminer)- and *w*(h)-elements in two closely related Germanic languages (e.g., Dutch and German), demonstrating the descriptive adequacy of such an approach (see also [Leu 2015](#) for related arguments). An outcome of this radical decomposition of ‘small words’ is that lexical/morphological operations are conditioned by syntactic structures and

Michael T. Putnam

operations. In this sense, morphology proper is not afforded a separate and independent status in the architecture of grammar, hence, it is *distributed* across other multiple modular domains of grammar.

It is from this position of the radical decomposition of ‘small words’ that I investigate a puzzling property of Pennsylvania Dutch that has heretofore received no attention in the literature (at least that I am aware of). Pennsylvania Dutch is truly a language that has “outgrown its name” (Keiser 2012: 1). Once the koinè of a few thousand German-speaking immigrants who initially settled in South Eastern Pennsylvania in the late 18th century, Pennsylvania Dutch has now become the first language of over 400,000 people.¹ To illustrate the impact of this analysis on enhancing a more general understanding of Germanic morphosyntax, here I adopt a comparative approach, juxtaposing Pennsylvania Dutch with Standard German and its colloquial forms. The empirical focus of this chapter is the underlying structure of adpositional units that contain so-called *r*-pronouns, that are manifest as *da*-Komposita in declarative contexts and *wo*-Komposita in interrogative ones.² The examples in (1) from Standard German illustrate the distribution of these structures:

(1) **Standard German**

- a. Ich warte **auf** den Bus.
I wait P the.ACC bus
‘I wait on the bus.’
- b. Ich warte **da**rauf.
I wait da-r-P
‘I wait on it.’
- c. **Wo**rauf wartest du?
wo-r-P wait you
‘What are you waiting for/on?’

Example (1a) consists of a prepositional phrase with an inanimate object. In (1b) we observe a *da*-Kompositum, consisting of a *d*-element, an ‘*r*’-element (more on this to follow), and the preposition *auf*. Importantly, *da*-Komposita are only possible when the object of the preposition is inanimate. Finally, we find an example

¹See Louden (2016) for a detailed historical overview of this language.

²Admittedly, this distinction is overly simplistic, since there are specific instances in which *da*-Komposita can appear in interrogative contexts (i.e., *Seit wann wartest du darauf?* ‘Since when are you waiting on this/that?’) as well as *wo*-Komposita appearing under a pure indefinite construal in spoken varieties of German (see Gärtner 2009: 11 for a discussion). It is unclear if either of these contexts license Komposita in PD, which is a topic I leave for future research.

Michael T. Putnam

- c. **Vun was** sin(d) die Schtanne g'macht?
 from what are the stars made
 'What are the stars made out of/from?' [PD: piped-piped]
- d. ***Was** sin(d) die Schtanne g'macht vun?
 what are the stars made from
 'What are the stars made out of/from?' [PD: P-stranding]

The primary focus of this chapter is to provide an account of the underlying structure of PD pseudo-resumptives, thus serving a two-fold purpose: First, the analysis advanced here seeks to build upon ongoing treatments of the detailed underlying syntactic structure associated with adpositions (Noonan 2017, Koopman 2010, Terzi 2010, Pretorius 2017). In particular, the analysis sketched out here takes the work of Noonan (2017) as a fundamental basis. Second, my treatment of pseudo-resumptives enriches our understanding of the typology of the (complex) syntax of pronominal adverbs, and adpositions more generally. The remainder of this chapter adheres to the following structure: In §2 I sketch out the core assumptions of the underlying structure of pronominal adverbs in German(ic). As alluded to above, I adopt an *exo-skeletal* approach to grammar, according to which morphology is conditioned by syntactic structures and operations. Bearing these core assumptions in mind, I return to the distribution of PD pseudo-resumptives in §3, highlighting particular ways in which these structures distinguish themselves from other non-standard forms found in regional and colloquial German. In §4 I develop an analysis of PD pseudo-resumptives, discussing the common attributes these structures share with 'typical' *da*-Komposita, while simultaneously focusing on the unique aspects of the structure and exponency associated with them. This paper concludes in §5.

2 Theoretical preliminaries

In this chapter I adopt a *neo-constructivist* approach to the syntax-lexicon interface. More concretely, research carried out within this family of theoretical approaches³ shares these important architectural traits:

- Elements that function as 'heads' can be morphemic,

³Here I do not engage in a detailed comparison of the differences that exist between frameworks such as *Distributed Morphology* (Embick & Noyer 2007), *Exo-skeletal syntax* (Borer 2003), and *Nanosyntax* (Starke 2009), because for our purposes here, their similarities far outweigh their differences. For a summary and comparison of these models, see Lohndal (in press).

8 Pseudo-resumptives in Pennsylvania Dutch: A preliminary analysis

- The syntactic structure of function, or ‘smaller’, words are more often than not quite complex

The conceptualization of FEATURES advanced here is one in which they function as individualized syntactic projections in a functional sequence (*fseq*), or *spine* (Wiltchko 2014). This hierarchical ordering of FEATURES results in a One Feature One Head (OFOH) Architecture (Putnam 2020, Kayne 2005, Cinque & Rizzi 2008).⁴ This spine-like structure found in the syntax is generative, and provides the blueprint from which ‘lexical items’ can be created. In essence, lexical items, which are nothing more than syntactic objects, and their exponency are derived by parsing and storing smaller treelet structures in the post-syntactic lexicon.

For the purposes of this chapter, I define these aforementioned syntactic objects as instances of SPANS. This term has been defined in various ways in the extant literature⁵, however for the purposes of our treatment of pseudo-resumptives in PD, I adopt the definition of spans in (4):

(4) Span:

An n -tuple of heads $\langle X_n, \dots, X_1 \rangle$ is a span in a syntactic structure S , iff $X_{n-1}P$ is the complement of X_n in S . (Blix 2021b: 7)

According to (4), any contiguous set of heads in a single complementation line is a licit span. The notion of spanning in this system plays in an important role not only in the syntax, but also in the post-syntactic lexical items (and the processes that direct and shape their formation). The role of the (narrow) syntax is to create spans via the successive merging of formal features, generating an **S-span**. S-spans then interface with stored feature constellations (i.e., lexical **L-spans**) to determine which is the ‘best fit’ for the realization of these structures with regards to its exponency. A fundamental assumption that a *spanning*-approach to lexicalization makes is that spans (as defined in (4)) are the locus of lexical insertion (and not terminal nodes). Singleton terminal nodes are not insignificant however. A conceptually-appealing economy principle that guides the lexicalization process in *spanning*-approaches is the **Exhaustive Lexicalization Principle** in (5), which ensures that each and every node in an S-span will be associated with an exponent:

⁴The major difference between neo-constructivist models that embrace this concept of OFOH vs. those that do not is that the latter models permit feature bundling in syntactic structures.

⁵For example, see Blix (2021a,b) and Svenonius (2020) for related, yet competing approaches.

Michael T. Putnam

- (5) **Exhaustive Lexicalization Principle:** Every syntactic feature must be lexicalized. (Fábregas 2007: 167)

One core principle that guides this matching algorithm ensures an ‘optimal match’ when the features realized on a given L-span contain a superset of the features found in the S-span. This state of affairs is conceptualized as the **Superset Principle** in (6):

- (6) **Superset Principle:**
In case a syntactic span does not have an identical match in the lexical repertoire, select a L(exical)-span which contains a superset of the features present in the S(yntactic)-span.
(adapted from Fábregas & Putnam 2020: 40)

An important point to emphasize here is that the ‘optimal’ L-span need not be a perfect (i.e., one-to-one) match for a given S-span, rather, the L-span containing the least amount of superfluous features (when matching a given S-span) represents the ‘best fit’. Throughout the remainder of this section I provide a novel sketch of the syntax-lexicon interface that adheres to these principles as developed in recent collaborative efforts (Fisher et al. 2022, Natvig et al. 2025). Figure 1, taken from Natvig et al. (2025), illustrates how matching at the syntax-lexicon interface plays out in this system.

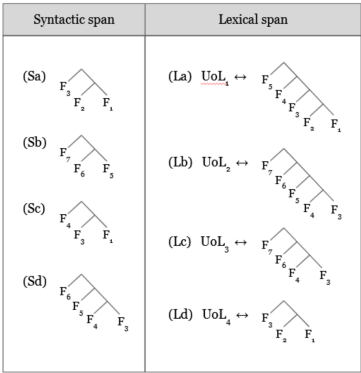


Figure 1: Syntax-Lexicon Mappings

The left column in Figure 1 represents S-spans, whereas the right column contains L-spans (which Natvig et al. 2025 also refer to as *Units of Language* (UoLs),

8 Pseudo-resumptives in Pennsylvania Dutch: A preliminary analysis

which does not impact the theoretical preliminaries introduced here). Turning first to (Sa), the L-spans (La) and (Ld) are eligible candidates to compete with one another since they contain all of the features found on (Sa), i.e., F_1 , F_2 , and F_3 . In contrast, the L-spans (Lb) and (Lc) are inferior candidates since they do not contain all of the features comprising (Sa). Of these two eligible L-spans, (Ld) is the better candidate, since it contains fewer superfluous/extra features than (La). The case of (Sb) in Figure 1, (Lb) is the only L-span that qualifies since it is the only one that contains all of the features found in (Sb), and for this same reason, (La) is the only L-span that fulfills the requirements of the Superset Principle for (Sc). Lastly, (Lb) is the best fit to lexicalize the S-span (Sd) since it is the only L-span candidate that contains all of the features on (Sd).

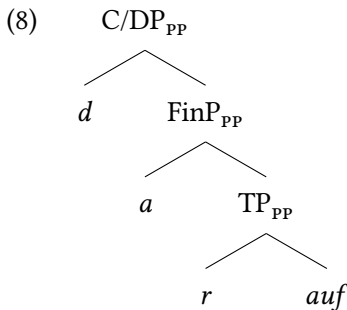
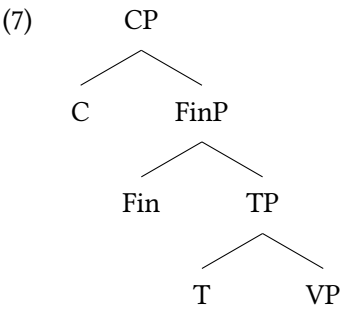
Equipped with these theoretical preliminaries, in the subsequent subsections we turn our attention to the underlying structure of *da*-Komposita and *wo*-Komposita (§2.1). This discussion of *da*/*wo*-Komposita builds upon the proposals advanced by Abels (2012: Ch. 7), Leu (2015), and Noonan (2017), with a focus on Standard Dutch and Standard German. Although I do not want to implicate that Standard German represents some sort of ‘baseline’ for PD⁶, the structure of *da*-Komposita in both Standard German and PD are remarkably similar. In a related sense, even though PD lacks *wo*-Komposita, a brief review of the syntax of these units is necessary in order to understand how pseudo-resumptives in PD somehow stand in as a proxy for them.

2.1 Syntactic structure of *da* & *wo*-Komposita

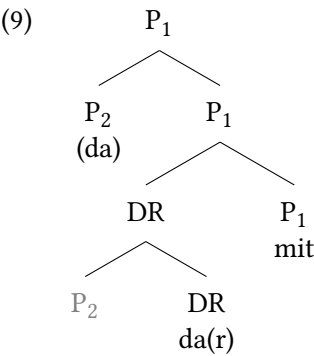
Da-Komposita are perhaps most accurately classified as categorial hybrids, as the *d*-element represents - at least in Standard German - a non-animate argument, which also functions as the object of a preposition. Our proposed underlying structure of these elements sketched out below owes much to previous work by Abels (2012: Ch. 7), Leu (2015), and Noonan (2017), and in line with these analyses, any treatment of these structures must account for: (i) the *d*-element, (ii) the R-pronoun, and (iii) the preposition. Again, in line with other neo-constructivist work on function words, we see a high degree of symmetry between clause-level and lexical-level syntax. To illustrate this point, compare the break down of the clausal-level tree structure in (7) with the one proposed for the lexical-level representation of *dar-* in (8) (NB: *dar-* is to be understood as the *d*-element in a *da*-Komposita such as *dar-auf* ‘on top of that’ (from Noonan 2017: 222):

⁶See Polinsky (2018: §1.4) for the persistent challenges that involve establishing baseline varieties for heritage languages.

Michael T. Putnam



Let's explicate some important points about the tree structure in (8). Starting from the top down, the *d*-element in Spec,CP specifies the deictic-nature of *d*- and *w*-elements and should be understood as a complementizer (see [Hachem 2015](#) for an approach that adopts a similar approach to syntactic structure as the one developed here). The vowel *a*- is a bona fide syntactic head, and functions as a silent *pro*.⁷ This proposal aligns with earlier suggestions by [Postal \(1966\)](#) and [Déchaine & Wiltschko \(2002\)](#) that pronouns are in fact D-elements that contain an unpronounced 'noun'. The *r*-element represents a source of controversy, with [Abels \(2012: Ch. 7.2.1\)](#) presenting arguments in favor of its *non*-pronominal status, nor should they be treated as complements of P (Ch. 7.2.2). In the OFOH-architecture adopted here, the *r*-element should by default occupy a syntactic head-position, however [Abels \(2012: 241\)](#), based on previous work by [Gallmann \(1997\)](#), proposes that the projection DR exists in the syntax as a complex head that contains both P- and *r*-elements as shown in (9):



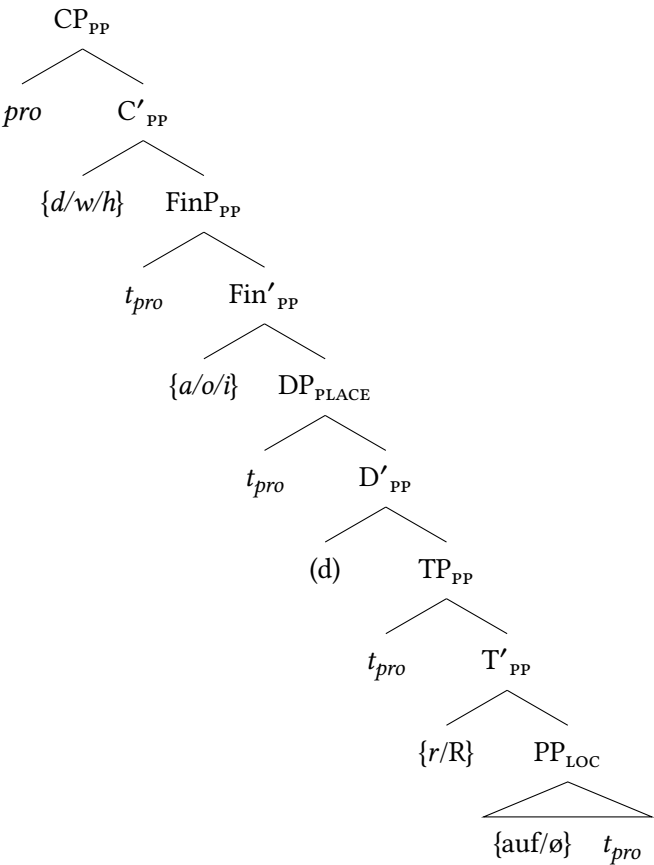
Although the structure in (9) can account for dialectal variants of German that permit doubling of the *d*-element (see e.g. (2b) above for an example), it

⁷This part of the proposed structure of *da/wo*-Komposita aligns with Leu's (2015) generalization of x-to-C movement across Germanic languages.

8 *Pseudo-resumptives in Pennsylvania Dutch: A preliminary analysis*

comes at the expense of proposing the existence of a complex head. There are two additional difficulties that the structure in (9) faces: First, it appears that the complex head DR houses not only P- and r-elements, but also two complex P-elements (in addition to the r-element). Second, Abels’ analysis is also predicated on the need to check case features on P, which commonly is checked on positions other than D (which is arguably an A’-position). I do not contribute further to the debate of the exact nature of r-elements, however, it is worth stating that I agree with Noonan (2017: 221, 234), that these elements should not be considered deictic heads. Moreover, in this treatment of pseudo-resumptives in PD, however, I wish to point out that to address Abels’ concerns, doubled d-elements in variants of German can be dealt with in an extended projection consisting of a DP embedded within a CP. In fact, this is the strategy taken by Noonan (2017: 225), as shown in (10):

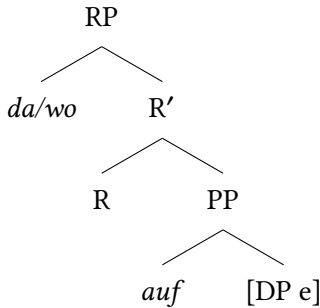
(10) Colloquial German doubled structures (e.g., *da/wodrauf*)



8 Pseudo-resumptives in Pennsylvania Dutch: A preliminary analysis

than resorting solely to lexical idiosyncrasies which offer little predictive or explanatory value ((12) is adopted from Noonan 2017: 216, 222).

(12) R-stranding



Two final points are in order before we move on to our discussion of the characteristics of pseudo-resumptives in PD in the subsequent section. The first concerns the requirement in Standard German that the null DP be inanimate. For ease of exposition, let us simply assume that something like a [H(uman)] feature (projected as a syntactic head) is responsible for indicating whether or not the DP is animate.⁹ As discussed in the following section, this distinction is not relevant in PD. Second, with respect to *wo*-Komposita, I do not postulate any significant structure differences between these and their *da*-Komposita counterparts. Rather, the key distinction rests in the **featural content** of the leftmost projection. Again, for the sake of ease of exposition, in the remainder of this chapter, I claim that the key difference between *wo*- and *da*-Komposita boils down to whether (or not) the topmost project contains a *d*- or *w*-feature/element (Hachem 2015).

3 Distribution of Penn Dutch pseudo-resumptives

We now turn to an overview of the general properties of pseudo-resumptives in PD. The data discussed in this section have been collected through a variety of methods, i.e., anecdotal evidence, translation requests, judgment task results, and the analysis of recorded narratives and free speech over the course of several years. The purpose here is to point out particular ways that these structures

⁹Of course, I fully recognize this is an oversimplification of the treatment of animacy in an OFOH-approach. The interested reader is referred to Wiltchko & Ritter (2015) for a more nuanced and detailed exposition.

Michael T. Putnam

differ from what is found in colloquial and regional German, as well as highlight aspects of their uniqueness within the broader typology of West Germanic languages.

The first notable feature found in PD pseudo-resumptives that distinguishes them from German equivalents is the lack of animacy effects (cf. (1) & (13)). As shown in (13a) and (13b), pseudo-resumptives in PD can be formed regardless of whether or not the object of the preposition is inanimate (i.e., (13b) would be ungrammatical in German and most of its colloquial and regional forms). The pied-piped topicalized form in (13c) is also possible with animate and inanimate objects.

(13) Lack of animacy effects

- a. Was warscht du n warte druff?
what were you PROG wait it-on
'What were you waiting on?'
- b. Wer warscht du n warte druff?
who(m) were you PROG wait it-on
'Who(m) were you waiting on?'
- c. [Uff wer/was] warscht du n warte?
on who(m)/what were you PROG wait
'Who(m) were you waiting on?'

As noted in the introduction (§1), English-style *r*-stranding is generally disfavored in PD; however, there are notable exceptions. As a case in point, compare the behavior of pseudo relatives that appear in combination with the preposition *uff* 'on' (14) and *mit* 'with' (15). The ability to license a pseudo-resumptive containing *uff* (14a) as well as a pied-piped topicalized structure (14b) represents the most common strategy for the overwhelming majority of prepositions. However, although *r*-stranding is not possible with *uff* (14c), it interestingly is preferred in combination with the preposition *mit* (15a).¹⁰ More concretely, pseudo-resumptives are not possible in combination with *mit* (cf. (15a) & (15c)), while pied-piped topicalized structures are still acceptable (15b). Interestingly, *mit* appears to display exceptional behavior in both historical and contemporary non-

¹⁰The dispreference of stranding vowel-initial prepositions in PD (14c) is also found in varieties of German closely related to PD (Fleischer 2002b: 123). I thank an anonymous reviewer for pointing this out.

8 Pseudo-resumptives in Pennsylvania Dutch: A preliminary analysis

standard varieties of German, including High and Low Alemannic dialects (Fleischer 2008, van Kemenade 2020).¹¹

- (14) Pseudo-resumptives with the base preposition *uff* are grammatical
- a. **Wer** hoscht du gewart **druff**?
 who(m) have you waited it-on
 ‘Who did you wait on/for?’
 - b. **Uff wer** hoscht du gewart?
 - c. ***Wer** hoscht du gewart **uff**?
- (15) Pseudo-resumptives with the base preposition *mit* are ungrammatical
- a. **Wer** hot er geschwetzt mit?
 who(m) has he talked with
 ‘Who did he talk with?’
 - b. **Mit wer** hot er geschwetzt?
 with who(m) has he talked
 ‘Who did he talk with?’
 - c. ***Wer** hot er geschwetzt **demit**?

Different types of *w*-elements – such as those that involve ‘complex’ units (16) and D-linked items (17) require pseudo-resumptives in PD. Irrespective of the type of *w*-element that co-occurs with pseudo-resumptives, the same sort of ban on R-stranding for most prepositions such as *wege* ‘about’ (16c) is still in effect, however, in contrast to examples with ‘simple’ *w*-elements, both pseudo-resumptives (17a) and R-stranding (17c) are viable options for D-linked *w*-elements that co-occur with the preposition *mit*.

(16) Complex *w*-elements

¹¹As a brief aside, this peculiar behavior of *mit* across time and geographic space lends support to our neo-constructivist analysis put forward here. Otherwise, we’re forced to assume that this is merely a lexical idiosyncrasy rather than a predicted structural effect. With its comitative semantics, *mit* is distinct from other prepositions that commonly denote directional semantic paths, including features such as PATH, PLACE, DIR(ECTION), & AXIAL PARTS. I do not provide a full analysis of the syntactic structure of *mit* in this chapter, however, it should be clear at this point to the reader that I infer that the difference between *mit* and more directional-oriented prepositions can be analyzed in terms of featural content, and hence, syntactic structure. I leave these finer-grained distinctions for future research.

Michael T. Putnam

- a. **[Welle Schtann]** hot er gedraunt dewege?
 which star has he dreamed it-of/about
 ‘Which star did he dream about?’
 - b. **[Wege welle Schtann]** hot er gedraunt?
 about which star has he dreamed
 ‘Which star did he dream about?’
 - c. ***[Welle Schtann]** ... wege? (P-stranding)
- (17) D-linked w-elements
- a. **[Welli Gael]** hat er geschwetzt zu die Sally dewege?
 which horses has he talked to (the) Sally it-about
 ‘Which horses did he talk to Sally about?’
 - b. **[Weller Mann]** hat er argued mit?
 which man has he argued with
 ‘Which man did he argue with?’
 - c. **[Weller Mann]** war er n argue-a mit?
 which man was he PROG argue with
 ‘Which man was he arguing with?’

We observe similar, expected patterns in relative clauses that appear with prepositional phrases in environments that would necessitate pseudo-relatives. As seen in examples of relative clauses that appear with the preposition *vun* ‘from’ (18) and those that appear in combination with *wege* ‘about’ (19), the pseudo-resumptive is required, while pied-piping and R-stranding are ungrammatical.¹²

(18) Relative clause with *vun*

- a. Des sin(d) die Schtanne, **as** der Himmel gemacht is **devun**.
 these are the stars RP/c the sky made is it-of
 ‘These are the stars that the sky is made out of.’
- b. Des sin(d) die Schtanne, ***vun** die ... (no pied-piping)
- c. Des sin(d) die Schtanne, ***die** ... **vun** (no R-stranding)

(19) Relative clause with *wege*

¹²To be fair, contemporary sectarian PD no longer employs relative pronouns (Van Ness 1994: 436), hence, these examples could be ruled out independently. This point notwithstanding, they remain here for the sake of exposition.

8 Pseudo-resumptives in Pennsylvania Dutch: A preliminary analysis

- a. Es iss die Frau, as mir gedraamt henn **dewege**.
it is the woman RP/C we dreamed have it-about
'She is the woman that we have dreamed about.'
- b. ***wege** as ... (no pied-piping)
- c. *as ... **wege** (no R-stranding)

Staying consistent with previous observations concerning the preposition *mit* 'with', we once again observe its exceptional behavior in relative clause formation, allowing R-stranding as seen in (20).

(20) Relative clause with *mit*

Er iss da Mann, as mir geschwetzt hen **mit** gestern Owet.
he is the man RC/P we talked have with yesterday evening
'He's the guy that we talked with last night.'

Finally, we turn to different types of A'-movement. Turning first to instances of long-distance extraction (21) (i.e., those licensing an intermediate gap), *w*-elements that undergo long-distance movement still require a pseudo-resumptive (21a), while topicalized pied-piped structures are slightly marked (21b) (Mark Loudon, p.c.).¹³ As expected, R-stranding in long-distance structures is ungrammatical (21c).

(21) Long-distance extraction

- a. **Was** denkscht (du) hot er gedraamt **dewege**?
what think (you) has he dreamed it-about
'What do you think he dreamed about?'
- b. ?**Wege/weeich was** denkscht (du) hot er gedraamt?
- c. ***Was** denkscht (du) hot er gedraamt **wege**?

The distinction between adjunct and argument *w*-elements is worth mentioning. Taking a closer look at the examples in (22), we confirm that pseudo-resumptives do not appear with adjunct *w*-elements (cf. (22a) & (22b)), however, as seen throughout this section, they are obligatory in contexts where the *w*-element functions as the object of the preposition (22c).

¹³As pointed out by an anonymous reviewer, an analysis in terms of V1-parentheticals (Reis 2002) would also be viable here. With respect to PD, more evidence involving extraction from *that*-headed embedded clauses is required.

Michael T. Putnam

(22) Adjunct vs. argument *w*-elements¹⁴

- a. **Ferwas** bischt sell n duh?
 why are(you) that PROG do
 ‘Why are you doing that?’
- b. ***Ferwas** bischt sell n duh **de**for?
 why are(you) that PROG do it-for
 Intended: ‘Why are you doing that?’
- c. **Was** bischt sell n duh **de**for?
 what are(you) that PROG do it-for
 ‘What are you doing that for?’

In summary, pseudo-resumptives are overwhelmingly preferred in PD among native and highly-proficient speakers in most environments (with the primary exception being when the preposition *mit* ‘with’ appears in these contexts) as a repair strategy to cope with the absence of *wo*-Komposita in PD in order to avoid English-like R-stranding (see Müller 2000 for an earlier approach to these structures being associated with a repair strategy-analysis). These elements appear in structures exhibiting long-distance *A'*-movement and in instances where the *w*-element is D-linked, and, as expected, they are obligatorily linked with arguments and do not appear in combination with adjunct *w*-elements. In the following section, I develop a neo-constructivist analysis of pseudo-resumptives as a repair strategy.

4 The syntax of Penn Dutch pseudo-resumptives

At this point, let’s take inventory of the key factors that differentiate *da*- and *wo*-Komposita in Standard German and PD summarized below in Table 1. What we can take from these key differences is the following: the implementation of pseudo-resumptives can be interpreted as a repair strategy of sorts. The doubled *d*-elements compensate for the absence of *wo*-Komposita in PD. Their presence in the grammar avoids the appearance of English-like R-stranding in most contexts. This strategy is not uncommon in heritage and contact languages, in which the minority language engages in the hyperextension of certain structures in order to (tacitly) avoid patterns commonly found in the dominant language (for example,

¹⁴ As a point of clarification, the *w*-element is considered to be atomic, hence no longer consisting of separate lexical elements. It can therefore be appealed to for drawing general conclusions about adjunct clauses in PD.

8 *Pseudo-resumptives in Pennsylvania Dutch: A preliminary analysis*

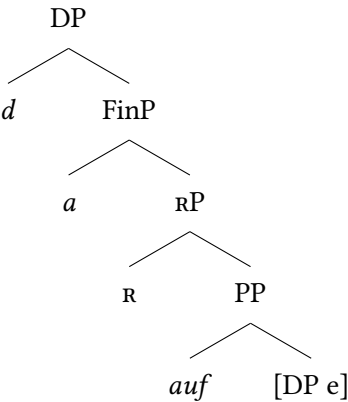
see Putnam & Søfteland’s (2022) treatment of non-finite complementation in American Norwegian). This applies to all elements, both animate and inanimate referents, in PD (as opposed to Standard German).

Table 1: Comparison of German & Penn Dutch

	German	Penn Dutch
<i>da</i> -Komposita	yes	yes
<i>wo</i> -Komposita	yes	no
animacy effect	yes	no
pseudo-resumptives	no	yes
R-stranding	limited	limited

In line with our neo-constructivist analysis that we develop here, the task at hand is to propose L-span structures that represent the lexicalization of segments of the *fseq*, or S-span. Making use of the theoretical machinery and assumptions introduced in §2, I propose the structure in (23) to hold for the majority of cases for *da*-Komposita in PD. The observant reader will note that the structure in (23) is an abbreviated syntactic structure of (10). In line with Noonan’s (2017) treatment of the underlying syntax of *d/w*-elements, in (23), Spec,DP houses the deictic element, functioning as a complementizer in sentential clauses. Spec,FinP is a true syntactic head, and lexically specifies the vocalic elements in these units. The R-head functions as the silent pronominal *pro*.

(23) Syntactic structure proposed for *da*-Komposita in Penn Dutch (most cases)



Michael T. Putnam

This leads us to the formalization of and motivation for the existence of pseudo-resumptives in PD. Simply put, PD lacks the L-spans responsible for lexicalizing *wo-Komposita*, requiring the lexicalization of both (i) the *w*-element (which undergoes *A'*-movement) and (ii) the doubling of the span of syntactic structure containing $\{d + a\}$. An interesting and valid point raised by an anonymous reviewer at this juncture concerns whether or not we are dealing with $\{D + \text{FinP}\}$ or $\{D + \text{TP}\}$ spans. This is due to the lack of an overt vowel immediately following the *d*-exponent (DP). Examples from above with containing *druff* ‘in-on’, i.e., (13) & (14), suggest the latter. This is an instance of *selective opacity*, i.e., it is certainly not the case that PD lacks *w*-elements and the rich array of features these elements contain, rather, *PD lacks L-structures of Komposita that contain w-features in the leftmost periphery*.

The eligibility of an exponent for insertion into a given S-span depends on whether or not the L-span encoded for that exponent is a good match for this particular S-span. Under ideal conditions, the Exhaustive Lexicalization Principle (5) and the Superset Principle (6) would be all that is needed to model the lexicalization process. Fisher et al. (2022: 11–12) introduce two additional principles that guide the lexicalization process in cases where the Superset Principle alone cannot account for all attested patterns of insertion. As we shall see momentarily, these additional principles play a central role in our treatment of pseudo-resumptives. The first of these is known as **Subsect S-span** (24), according to cases in which no exponent in a lexicon is associated with an L-span containing a superset of the span:

- (24) **Subsect S-span:** In case no exponent contains a superset of the features present in the S-span,
- a. select the exponent whose L-span contains as many features present in the S-span as possible, then
 - b. apply (a) until the Exhaustive Lexicalization Principle obtains.

The principles guiding insertion relate to one another by means of a heuristic process, which I outline below in (25).

- (25) **Insertion Heuristic:** When an S-span is spelled out, exponents are inserted according to (a). If (a) cannot obtain, exponents are inserted according to (b). If (b) cannot obtain, then (c) applies:
- a. Superset Principle
 - b. Subsect S-span

8 *Pseudo-resumptives in Pennsylvania Dutch: A preliminary analysis*

c. No insertion

When considered in tandem, the Superset Principle represents the default strategy for lexicalization; however, the Insertion Heuristic introduced in (25) activates the Subject S-span strategy (24), which results in the S-span being lexicalized by multiple exponents. This need and function of the Subject S-span is explicated in Figure 2 (from Fisher et al. 2022: 12):

S-span	L-span (exponents)
(Sa)	(La) ↔
(Sb)	(Lb) ↔
	(Lc) ↔
	(Ld) ↔

Figure 2: Subject S-spans

The scenario in Figure 2 supposes that (Sa) and (Sb) are S-spans at the syntax-lexicon interface that are eligible for lexicalization. The L-spans in the right column represent those that are competing for lexicalization. The situation above reveals that none of the L-spans are a perfect match for either (Sa) or (Sb). Considering the S-span (Sa) first, (La) and (Lb) are both eligible candidates due to the Superset Principle (because they contain all of the features in (Sa)). The L-spans (Lc) and (Ld) fail to meet this requirement. The L-span (Lb) represents the ‘best fit’ for (Sa) since it is comprised of fewer superfluous features than (La) (when compared with (Sa)).

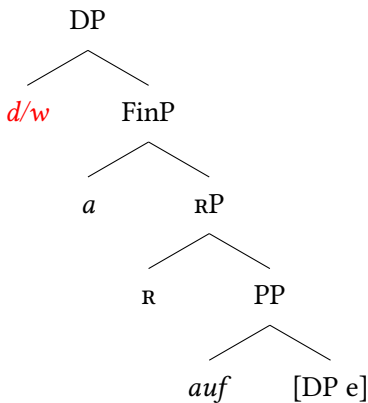
The situation surrounding the lexicalization of the S-span (Sb) is different. Once again, there are no candidate L-spans that contain all of its features, thus preventing a perfect match. At this juncture, the Superset Principle cannot hold, thus forcing the Subject S-span to come into play. To prevent the lack of any insertion, the best case scenario for the lexicalization of the S-span (Sb) is to make use of multiple exponents, namely, (Lc) and (Ld) to lexicalize this S-span.

The critical question we face at this juncture is whether or not the Subject Principle (24) and the Insertion Heuristic (25) provide the necessary tools to model pseudo-resumptives in PD. Rephrasing this question slightly, what is the difference - from the perspective of featural content - between *d*- & *w*-Komposita in a

Michael T. Putnam

language like Standard German that licenses *da*- & *wo*-Komposita? In Hachem's (2015) detailed decompositional account of *d*- and *w*-elements (see in particular §4.3.5 and §7.4.4), she proposes that the leftmost element corresponds to Wiltchko's (2014) *Linking Domain*, in which propositional semantics are situated.¹⁵ Adopting Hachem's (2015) proposed underlying S-span structure for the *Linking Domain*, i.e., left edge of *d*- & *w*-elements, we arrive at the following S-span structure in (26)¹⁶:

(26) *da*-Komposita S-spans in PD (slightly revised)



The PD grammar is presented with a dilemma: (i) although it has L-spans that match and lexicalize both *d*- and *w*-elements, (ii) it does not possess an L-tree that permits the lexicalization of *w*-elements in the context of *da*-Komposita. The solution that the PD grammar opts for is to have its cake and to eat it too; namely, to lexicalize *both* the *w*-element *and* the *d*-element. As a proof of concept, consider (3b), repeated below as (27):

(27) Was sin(d) die Schtanne g'macht devun?
 what are the stars made it-from
 'What are the stars made out of/from?' [PD: default]

The PD grammar selects *both* the L-span for the *w*-element *was* 'what' and the L-span for the *d*-element associated with the *da*-Komposita. In this respect, it

¹⁵It is beyond the scope of book chapter to determine whether or not (*in*)*definiteness* is determined/situated (solely) in this projection (Boef 2013, Gillon 2009, Heim 2013, Wiltchko 2009).

¹⁶Hachem (2015) appeals to differentiating *d*- and *w*-structures by means of animacy features, which is a moot point in the context of PD *da*-Komposita and related pseudo-resumptives as seen in §3.

8 *Pseudo-resumptives in Pennsylvania Dutch: A preliminary analysis*

technically does not violate the Superset Principle¹⁷ or the Insertion Heuristic we have proposed above. The selection of multiple L-spans to suffice syntax-lexicon mapping requirements is not explicitly ruled out by either the Superset Principle or the Insertion Heuristic, however it raises a host of issues if this is possible. PD addresses this potential dilemma rather simply, but ensuring that the *w*-element undergoes A'-movement; however, *if* it were to remain *in situ*, it would open the door for a host of questions centering on the possible existence of complex specifiers and how they should be properly lexicalized (for a similar and related discussion in Nanosyntax, see [Starke 2018](#)).

4.1 A recent development: The rise of R-stranding

Although PD serves as the L1 for many Old Order Amish and Conservative Menonite practitioners, it is also spoken by many who grew up in these congregations yet never officially became members (i.e., were baptized). The conditions of no longer speaking the L1 in the home and in the close-knit church community result in a state of affairs in which some speakers have been *heritage* speakers of PD. This lack of usage, or more aptly stated *activation*, of heritage languages can have both a temporary and potentially lasting impact not only on linguistic production, but also on the underlying representations that constitute linguistic competence ([Putnam & Sánchez 2013](#), [Scontras et al. 2018](#)).

While collecting judgment data, a small number of younger, heritage speakers of PD reported that structures exhibiting English-like R-stranding were a (slightly) degraded, yet possible option in their grammars. It is also worth noting that all of these speakers found structures with pseudo-resumptives to be acceptable, too. Although this state of affairs requires a separate, detailed study to qualify and quantify these incipient observations, they provide evidence of the long-standing position that the grammatical knowledge of bi- and multilinguals represents a complex nexus of shared information ([Hartsuiker & Bernolet 2017](#), [Gámez & Vasilyeva 2020](#), [Melinger et al. 2014](#), [Putnam et al. 2018](#)).

The *spanning*-approach sketched out above, based largely on [Fisher et al. \(2022\)](#) and [Natvig et al. \(2025\)](#), is compatible with other late-insertion approaches to bi/multilingual grammars ([López 2020](#), [Putnam et al. 2019](#), [Perez-Cortes et al. 2019](#), [Alexiadou & Lohndal 2018](#), [Lohndal & Putnam 2024](#)), and provides a relatively straightforward way to capture this apparent change in the PD grammar of certain speakers. Recall that R-stranding is possible for most PD speakers in

¹⁷Interestingly, Boef's (2013: 71) *Closest Match Principle* in connection with her treatment of the morphosyntactic variation of Dutch relative clauses in relation to the selection of *d*- and *w*-exponency bears a strong resemblance to the Superset Principle.

Michael T. Putnam

connection with the preposition *mit* ‘with’. The rise of *r*-stranding represents an instance where a lexicalization strategy present in the L1 (Penn Dutch) is amplified by the continued, frequent activation of the sample - or, very similar - strategy in L2 (English). The increase of English-style *r*-stranding in a heritage language grammar that does not (commonly) license such structures does occur (see, e.g., Pasqual y Cabo & Gómez Solar’s 2015 research on prepositional stranding in Heritage Spanish). Once again, these scenarios are adduced as further evidence for the integrated nature of bi/multilingual grammars, which represents an exciting new frontier of research on the syntax-lexicon interface from a *spanning*-perspective (Fisher et al. 2022, Natvig et al. 2025, Sugimoto & Baptista 2022).

5 Conclusion

Pseudo-resumptives in PD appear to be a morphosyntactic coping mechanism motivated by two related, yet separate phenomena: First, PD lacks *wo*-Komposita, which are common in Standard German and the vast majority of its non-standard variants. Second, PD, at least in the case of highly-proficient, non-heritage speakers of the language, does not license English-style *r*-stranding in most environments (except for in the context of *mit* ‘with’). The lexicalization strategy employed in this *spanning*-approach to syntactic structure operates on a ‘best fit’ approach to associating L-spans to S-span representations. Following Fisher et al. (2022), the Subject S-span (24) and the Insertion Heuristic (25) provides a predictive and explanatory account for how syntax-lexicon mismatches can be addressed when the Superset Principle in isolation fails to deduce a clear solution. My tentative proposal advanced here is that when presented with a fork in the road, the grammar of PD “takes it”, opting to lexicalize the L-span for the (simplex) *w*-element in addition to the *d*-element at the edge of the *da*-Komposita. If this tentative analysis is on the right track, it would provide a path forward in capturing the variable doubling and stranding patterns found in colloquial and diasporic varieties of German in connection with *da/wo*-Komposita under a unified approach, which is an exciting prospect for future research.

Abbreviations

fseq	functional sequence	
OFOH	One Feature One Head	...
PD	Pennsylvania Dutch	...
PROG	Progressive (aspect)	

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Chapter 9

The phenomenon of Complementizer Agreement: New data from South Tyrol

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The aim of this paper is to present new syntactic data on complementizer agreement from three South Bavarian dialects spoken in three different areas of South Tyrol, a German-speaking region in northern Italy. Although data on this phenomenon in Bavarian dialects have been extensively discussed in the literature, the South Bavarian varieties spoken in South Tyrol have rarely been considered. Our data show, first, that the phenomenon is present in these varieties, albeit concentrated in the second person plural. Second, our data shed light on how the complementizer agreement emerged, suggesting two evolving pathways. The basic step is in both cases the emergence of a new inflectional morphology on finite verbs in C°/Fin° via reanalysis of subject clitics that become part of the inflectional system. The necessary condition for the second person plural is the presence of the personal pronoun *es* 'you' encliticizing on the finite verb. After that, the extension of the phenomenon to new contexts can proceed in two different ways: Either the new morphology is transferred first to complementizers and then to final verbs, or first to verbs in the end position and then to complementizers, giving rise in both cases to a parallel agreement.

Keywords: complementizer agreement; double agreement, emergence of; South Tyrol; South Bavarian.

1 Introduction: What is Complementizer Agreement (CA)?

Though complementizers are lexical items usually considered indeclinable, there are several forms of agreement of complementizers with other lexical items.

This phenomenon is called complementizer agreement (CA) (van Koppen 2017). Roughly speaking, depending on the direction of the agreement relationship, two forms of CA can be distinguished: upward and downward agreement.¹

Upward agreement occurs when the complementizer agrees with a word of the superordinate clause on certain features. A common form of upward agreement appears with complementizers introducing relative clauses. These complementizers can agree with the head noun with respect to features like animacy as in the Old Ligurian examples in ((1a), (1b)):²

- (1) Old Ligurian (quoted after Poletto & Sanfelici 2018: 275)
- a. questa femena chi ...
this woman who ...
 - b. questa cità que ...
this town which ...

In the case of agreeing relative clause complementizers, the probe of the AGREE relation is the head noun, which embeds the relative clause. But embedding does not appear to be a necessary requirement, since upward agreement is also attested without a probe directly embedding a goal. This form of upward agreement is very common in Bantu languages like Lubukusu, cf. ((2a), (2b)): “Lubukusu displays a typologically interesting and theoretically challenging form of complementizer agreement where a declarative-embedding complementizer shows full phi-feature agreement (gender, number, and person) with the subject of the matrix clause” (Diercks 2013: 358). In (2a) the complementizer agrees with the matrix subject *babandu* ‘the people’, which is of class 2 (class marker *ba*), and in (2b) with the matrix subject *Alfredi*, which is of class 1 (class marker *a*) (Diercks 2013, van Koppen 2017).

- (2) Lubukusu (Diercks 2013, 358)
- a. baba-ndu ba-bol-el-a Alfredi ba-li a-kha-khil-e.
2-people 2s-said-AP-FV 1Alfred 2-that 1s-FUT-conquer
‘The people told Alfred that he will win.’

¹Note that we use the two terms purely descriptively in the defined sense (see below) and thus make no statement on the theoretical problem of the direction of the feature checking process (cf., among others, Bárány & van der Wal 2022, Bjorkman & Zeijlstra 2019).

²Cf. Poletto & Sanfelici (2018: 275): “We can conclude that in the XIV century [OI] texts, the complementizer in subject relatives is simultaneously marked for (nominative) case and for animacy.”

9 The phenomenon of Complementizer Agreement

- b. Alfredi ka-bol-el-a baba-ndu a-li ba-kha-khil-e.
 1Alfred 1s-said-AP-FV 2-person 1-that 2s-FUT-conquer
 ‘Alfred told the people that they will win.’

Note that although the complementizer introduces a subordinate clause, this clause is not embedded directly under the probe the complementizer is agreeing with.

Downward agreement is when the complementizer agrees with a lexical item that is lower in the structure. This is the case with complementizers that agree with the subject of the clause introduced by the complementizers. This type of CA is clause-bound; it uses the same morphology as verbal agreement (see sect. 2), and, as far as we currently know, it is restricted to Continental West Germanic dialects (Weiß 2005). (3a) is an example from North Bavarian (Schiepek 1899–1908: 76) and (3b) from Central Bavarian (Weiß 2005: 148).

- (3) a. dán d‘Láit häien
 that-3PL the people hear
 ‘that the people hear’
 b. I frog‘me, obsd ned du des mocha kannsd.
 I ask myself, whether-2SG not you that make could-2SG
 ‘I ask myself whether you can do that.’

In our paper, we are only concerned with CA of this type. What is special about this type is that the subject of the complementizer-introduced clause enters into a two-fold agreement relationship: with the verb and with the complementizer. A major difference between upward and downward agreement appears to be that the agreeing features are nominal in the former (e.g., noun class, animacy, case) and in the latter nominal-verbal in nature (i.e., person and number). We will not go into this further.

The paper is organized as follows: Section 2 discusses different types of CA, Section 3 the emergence of CA, and Section 4 presents accounts for CA that has been proposed in research so far; Section 5 is dedicated to presenting and discussing the new data collected for the South Bavarian variety of South Tyrol. Section 6 is a summary and the conclusion.

2 CA in Germanic: Typology of CA

As mentioned above, CA (of the Continental West Germanic type) uses the same morphology as verbal agreement (VA) (Weiß 2005, 2018; van Koppen 2017). With

respect to the forms of the morphemes, there are two types of agreement morphemes that occur in CA: They are of pronominal (4a), verbal origin (5), or both (6). Central Bavarian *-ma* used with complementizers (4a) and with verbs in second position (4b)³ has a pronominal origin, deriving from the pronoun *mia* ‘we’, whereas North Bavarian, *-n* in (5) did not result from reanalysis of a pronoun as an inflectional morpheme. Central Bavarian *-sd* (6) combines both since it emerged in OHG times by adding a *t*, going back to the pronoun *du* ‘you’ to the original inflectional marker *s* (cf. Weiß 2018 and the literature cited there).

(4) Central Bavarian

- a. *wemma mia aaf Minga foarn.*
when-1PL we to Munich drive
‘when we go to Munich.’
- b. *Mia foarma aaf Minga.*
we drive-1PL to Minga
‘we go to Munich.’

(5) North Bavarian (Schiepek 1899–1908: 76)

- dán d‘Láit häien*
that-3PL the people hear
‘that the people hear’

(6) Central Bavarian (Weiß 2005: 148)

- I frog‘me, obsd ned du des mocha kannsd.*
I ask myself, whether-2SG not you that make could-2SG
‘I ask myself whether you can do that.’

Note that we follow Zwart (1993), Weiß (2005, 2018) and others in using the term CA also for the inflectional morpheme that only occurs in C°/Fin° regardless of whether the position is occupied by a complementizer or a verb. That means that the verb in (4b) shows CA (just like the complementizer in (4a)), but the verb in (4a) VA. CA therefore means complementizer agreement in a narrower sense and C-agreement in a broader sense. The underlying assumption is that the syntactic position is inflected and not actually the respective lexeme that fills the position.

³In most Central Bavarian varieties, the *-ma*-ending of the verb only occurs on verbs raised to C°/Fin°, but there are some varieties where this ending also is used with auxiliaries and modals in end position. For more details on how *-ma* developed into an agreement marker cf. Weiß (2005).

9 *The phenomenon of Complementizer Agreement*

Concerning the completeness of the CA paradigms, we observe nearly complete as well as highly defective systems (Weiß 2005). Complete systems (in German dialects) show overt inflection in the 2Sg and in the complete plural (cf. (7), (8)), whereas 1 and 3Sg lack overt inflection. The same inflectional behavior is found in modals and strong preterites, so it is reasonable to assume a complete paradigm in these cases.

(7) Anhaltinish (Richter 1979)

- a. vu:sde ‘where-2SG-you’
- b. öpenmer ‘whether-1PL-we’
- c. venter ‘if-2PL-you’
- d. viinze ‘how-3PL-they’

(8) Erzgebirgish (Weise 1907, 200)

- a. wailst du ... ‘because-2SG-you’
- b. wosn mer ... ‘what-1PL-we’
- c. dastersch ... ‘that-2PL-you-it’
- d. ebnse ... ‘whether-3PL-the’

Defective systems often exhibit overt inflection only in the 2Sg or in the 2Pl (or in both). Frisian and Westphalian are dialects where CA only occurs in the 2Sg ((9a), (9b)), whereas CA was restricted to the 2Pl in two varieties of South-Hessian ((10a), (10b)).

(9) 2Sg

- a. dat-st (do) jûn komst (Frisian, Zwart 1993: 253)
that-2SG (you) tonight come-2SG
- b. vanste (Westphalian, Holthausen 1886: 89)
if/when-2SG

(10) 2Pl

- a. obd‘ə (South-Hessian, right of the Rhine, Bohn & Weiß
whether-2PL-you
2016, 2017)
- b. owən‘ə (South-Hessian, left of the Rhine, Bohn & Weiß
whether-2PL-you
2016, 2017)

Ermenegildo Bidese & Helmut Weiß

Central Bavarian is somewhere in the middle since it shows overt CA in the 2Sg, 1Pl, and 2Pl ((11a)–(11c)):

(11) Central Bavarian (Weiß 1998, 2005)

- a. wennst ... ‘when-2sg’
- b. wenma ... ‘when-1pl’
- c. wennds ... ‘when-2pl’

There is an interesting finding regarding the areal distribution (Weiß 2005): Although CA occurs in most Continental West Germanic dialects in one form or another, complete systems are mainly found at the edges. One finds complete CA-paradigm dialects in the eastern part of the German-speaking area in Upper Saxon, Thuringian, East Franconian, and North Bavarian, while West Flemish is known in the western part of the Continental West Germanic language area for the frequent occurrence of complete CA-systems. Defective/partial systems can be found everywhere in between.

3 The emergence of CA

There is little known about how and when CA emerged. Concerning the first question, Weiß (2018) proposed that the emergence of CA involved the three steps in (12):⁴

(12) Subject clitics > DA > CA

When subject clitics are reanalyzed as (part of the) inflection, they first produce a double agreement (DA); that is, agreement morphology that only occurs on verbs in C°/Fin°, but not on verbs in the end position, cf. in (13) the difference between *khumt* in end position and *seats* in second position: The latter exhibits an inflectional marker that consists of the original ending *-t* plus an *-s*, while the former lacks this addition that evolved by reanalyzing the subject pronoun *es* ‘you-Pl’ as part of the inflection.

(13) North Bavarian (Weiß 2005: 154)

- weits iwet pruk khumt, seatses wietshaus.
- when-2PL over-the bridge come-2PL, see-2PL-the pub
- ‘when you come over the bridge you will see the pub.’

⁴There are other explanations of the emergence of CA (e.g., Goeman 2000, Kathol 2001, Zwart 2006). See Weiß (2018: 148) for the shortcomings of these proposals.

9 *The phenomenon of Complementizer Agreement*

The most prominent case where this reanalysis occurred already in OHG times is the 2SG, whose original verbal ending consisted only of an -s, to which then a -t (< *thu, du* ‘you-SG’) was added, but only when the verb raised to C°/Fin°, cf. (14a) vs. b, quoted after [Weiß 2018: 146](#).

- (14) a. *daz thu thes kiscephes* (Christus und die Samariterin 13)
 that you that scoop
 ‘That you scoop it.’
 b. *biwaz kerost thu* (Christus und die Samariterin 7)
 why demand you
 ‘Why do you demand?’

Such reanalyzed forms led to DA. In the case of the 2SG, DA disappeared when the new inflectional marker also occurred on verbs in the final position – but there are other cases where DA still exists today (cf. (13)).

Children learning a DA dialect could probably get the impression that the special inflectional morphology appearing on verbs in C was connected with the syntactic slot as such, irrespective of whether the lexical item occupying the slot is a verb or a complementizer. They may have got this impression because the new inflection was restricted to the C-position, and only later the inflection was transferred via analogical extension onto complementizers, too. This explains why CA is often restricted to pronominal agreement, i.e., to cases where (a part of) the inflectional marker developed from subject pronouns (as with the 2SG in many dialects or the 2PL in Northern and Central Bavarian). However, as we have seen above (e.g., in (7) and (8)), CA also uses a non-pronominal agreement marker. Therefore, there must have been a further analogical expansion: In a second step, there was a general association between C° and agreement. The restriction to pronominal inflection was dropped.

There is little known about when exactly CA emerged (cf. [Axel & Weiß 2011](#), [Volodina & Weiß 2016](#), [Weiß 2018](#), [Weiß & Volodina 2018](#)). We can say with certainty that CA did not occur in the older Germanic languages because we have no evidence for it. Since it is now restricted to the CWG dialects, one must assume that it arose at a time when Old English was no longer part of West Germanic languages. For German, the first clear example occurs in a Bavarian Carnival play from the 17th century (15), but possible evidence for CA already appears a century earlier in the writings of Hans Sachs ((16a), (16b)) (cf. [Weiß & Volodina 2018](#)).⁵ However, similar examples are also attested in texts from the

⁵Hans Sachs was a famous *Meistersinger* from Nuremberg, a Bavarian town where an East Franconian dialect is spoken (cf. [Renn & König 2006: 18](#)).

13th century, so it is possible that CA already emerged in MHG times ((17a), (17b)).

- (15) Ja! wanst mir wolst varliegn? (*Ein kurtzweilig Faßnachtspill*, 1618)
 yes if-2SG me want-2SG to lie
 ‘Are you trying to lie to me?’
- (16) a. dast in habst verlorn (Sachs, FNS 172 V. 127)
 that-2SG him have lost
 ‘that you have lost him’
 b. weilt mir warst versprochwn (Sachs, ML 109 V. 126)
 because-2SG me were promised
 ‘because you were promised to me’
- (17) a. obd ie gewunnest mannes muot (Eckenlied, before 1260)⁶
 if-2SG ever won-2SG man’s courage
 ‘if you ever won manly courage’
 b. wan dazt aber else uncreftic bist (Tristan, ca. 1210)⁷
 when that-2SG but so unstrong are-2SG
 ‘but if you are not strong’

Note that while (15) is unambiguously an example of CA, the examples in (16) and (17) also allow an analysis with enclitic pronouns: The surface forms *dast/weilt/obd* can also be explained as consisting of a complementizer to which the clitic form of the subject pronoun *du* ‘you-2SG’ is rightattached. However, these forms can also be analyzed as inflected complementizers if *-t* is interpreted as an inflectional element, as is the case in Southern Bavarian and Alemannic dialects (cf. Weiß & Volodina 2018: 282 for further details). Therefore, a considerable age of CA can be proven, at least for German. This was to be expected because the first reanalysis of subject pronouns in the Wackernagel position as part of the verbal inflection had already occurred in OHG. Since this is the necessary precondition for the development of CA, it would be very astonishing if CA actually did not develop until the 16th or 17th century (cf. Weiß 2018 for further details). The 13th century, from which the first evidence comes, is chronologically much closer to the OHG period.

⁶The *Eckenlied* is an MHG verse epic and (17a) is quoted after Wierschin (1974: 40). According to Wierschin (1974: xxiv-xxvii), its dialect is Alemannic, and it was written before 1260.

⁷The author, Gottfried von Straßburg, wrote his epic in Alemannic. (17) is quoted after von Straßburg (1981: I: 470, l. 7801).

4 Theoretical accounts of CA

The phenomenon of CA has received considerable attention in theoretical syntax (cf. [van Koppen 2017](#) and the literature cited there). Roughly speaking, one can distinguish two types of approaches: syntactic and post-syntactic accounts.

Syntactic accounts have been proposed e.g., by [Zwart \(1993, 1997\)](#), [Carstens \(2003\)](#), or [van Koppen \(2015, 2017\)](#). They all have in common that they assume “that there is an agreement relation between the complementizer, or more accurately C°, and the subject” ([van Koppen 2017](#): 13). A central assumption is therefore that C° is equipped with phi-features that are spelled-out by CA and then enter in an agreement relation with the subject. While [Zwart \(1993, 1997\)](#) assumes that the phi-features get there through T-to-C raising, most researchers concerned with CA adopt [Chomsky’s \(2005\)](#) principle of feature inheritance, that is, that phi- (and tense) features are inserted at the phase head, i.e. C°, from where T inherits them. With such an approach, CA is what one would expect:⁸ CA just spells out the phi-features of C°. Since C° and T° have phi-features, both can (and do) enter in an Agree-relation with the subject of the clause (cf., e.g., [van Koppen 2005, 2017](#)).

On the other hand, there are also post-syntactic accounts, proposed, e.g., by [Fuß \(2004, 2005, 2008, 2014\)](#). Fuß assumes that phi-features are not present in C° from the beginning. AGR-in-C is rather “a morphological ornament that results from the post-syntactic insertion of phi-feature (a copy of T’s phi-set) under structural adjacency with T” ([Fuß 2014](#): 78). The presence of a finite verb is, therefore, an important piece of empirical evidence that Fuß claims to speak in favor for his post-syntactic account because without it, there is no agreement morphology present that could be copied to C°. This is seen in comparative deletion (([18a](#)), ([18b](#))):⁹

(18) Bavarian

- a. D’Resl is gresser als wia-st du bist.
the-Resl is taller than as-2sg you are
‘Resl is taller than you are.’
- b. D’Resl is gresser als wia(-*st) du.
the-Resl is taller than as-2sg you
‘Resl is taller than you.’

⁸One is rather surprised that it is such a typological rarity that only occurs in Continental West Germanic dialects ([Weiß 2018](#): 150).

⁹[van Koppen \(2017: 24\)](#), however, mentions that there are “examples with CA in phrasal comparatives are found in (a subset of) the Dutch dialects”.

We assume a kind of mixed approach. We remain agnostic with respect to Chomsky's (2005) feature inheritance hypothesis; that is, we do not assume that CA is a morphological expression of C°'s phi-features. As described in Section 3, we follow Weiß's (2018) explanation, according to which "the emergence of CA is more or less the result of a historical accident (some subject clitics were by chance reanalyzed as inflection markers)" (Weiß 2018: 149f.). Simply assuming, without further assumptions, that CA marks the phi-features of C° is in odd contrast to the astonishing fact that this phenomenon only exists in Continental West Germanic dialects. It should be much more common in the languages of the world.

5 CA in the South-Bavarian varieties of South Tyrol

In the South-Bavarian varieties spoken in South Tyrol, CA has hardly ever been investigated (see Birgit 1994 and, more recently, for South-Bavarian varieties spoken in Austria, Vergeiner & Bülow 2021). Recently, data concerning CA have been collected in some South-Tyrolean dialects as part of three BA theses on the base of standardized questionnaires, in which the informants had to translate the sentences from Italian into the local South German variety.¹⁰ The investigated areas in South Tyrol are the following:

- (a) Upper Vinschgau/Alta Venosta near Stilfserjoch/Passo dello Stelvio, more precisely the villages Trafoi and Sölden/Solda¹¹;
- (b) the dialects of the so-called German Nonsberg/Val di Non, in particular, of the villages Laurein/Lauregno und Proveis/Proves in Upper Fischbach-Valley/Valle della Pescara, and

¹⁰The questionnaires have been developed by the research teams of the Universities of Trento and Verona within the project "Advancing the European Multilingual Experience" (AthEME) (<https://cordis.europa.eu/project/id/613465>). The project focused on the various facets of multilingualism in Europe. The aims were (i) to raise the interest in and the awareness of multilingualism in Europe; (ii) to question existing public policies concerning the promotion of multilingualism and to collect best practices throughout Europe about it; and, (iii) to develop a model of academic public engagement around the same topic. The research was carried out from March 2014 to February 2019 and has been funded by the EU (Grant agreement ID: 613465) as part of the FP7-SSH (Specific Programme "Cooperation": Socio-economic Sciences and Humanities). It was coordinated by the Universiteit Leiden (P.I.: Lisa Cheng).

¹¹The data were collected by the bachelor students Anna Barera for the area (a), Giorgia Pigarella for the area (b) and Tamara Bassighini for the area (c) on the base of a standardized questionnaire. In their BA master thesis works dedicated to this topic, they analyzed the realization of the phenomenon in the different areas (see Barera 2016, Pigarella 2018 and Bassighini 2018).

9 The phenomenon of Complementizer Agreement

(c) the city dialect of Meran/Merano.

For all areas, we used the same sentences, which allowed us to compare the results consistently. The sentences were submitted in Italian as the German-speaking population of South Tyrol has usually a good passive command of Italian, whereas the active production can differ in particular from the cities of Bozen/Bolzano and Meran/Merano to the most mountain areas. Furthermore, the possibility of transfer from standard German to South Tyrolean was reduced by using Italian as the language of the questionnaire in order to avoid bias from the standard German form. For the city dialect of Merano/Meran, acceptability judgments were also collected, as will be explained in the following (cf. Section 5.3).

5.1 The first area: Upper-Vinschgau/Alta Venosta: Trafoi and Sölden/Solda

The first area to be investigated was the Upper Vinschgau/Alta Venosta, in particular the villages of Trafoi and Sölden/Solda.

Ten speakers of the local dialect were interviewed, five of them older than age 50, and five younger, along the following distribution:

(19) older generation:	younger generation:
E.P. – 83-years-old	S.R. – 26-years-old
L.G. – 76-years-old	C.P. – 19-years-old
A.T. – 57-years-old	F.G. – 19-years-old
M.R. – 57-years-old	T.T. – 17-years-old
H.G. – 56-years-old	T.Z. – 17-years-old

The task consisted of the translation of 38 sentences in Italian into the local dialect.

The first remarkable result is that CA is robustly attested in the South Bavarian varieties of Upper Vinschgau/Alta Venosta. However, these dialects clearly show a defective paradigm (see above, Section 2); in fact, CA is restricted to the 2Pl with the form $-[t]s$, while it is not attested in the other persons of the paradigm and in particular in the 2Sg (see above, the examples in (9) and (10)) (see [Vergeiner & Bülow 2021](#)). As with most varieties from South Tyrol, those from the Upper Vinschgau/Alta Val Venosta also have *eis* as a 2Pl pronoun (see [Wiesinger 1962-1969](#)). The pronoun *e(i)s* goes back to a dual form meaning ‘you both’ that was generalized as a plural form in Central Bavarian where it replaced the original plural form *iar* (i.e., German *ihr* ‘you’) ([Renn & König 2006](#): 86-87). Additionally,

this form was often grammaticalized as part of the verbal inflection, giving rise to the new ending *-ts* (or *-ds*) (see, among others, [Weiß 2005: 152](#); [Renn & König 2006: 87](#)). The original form *iar*, on the other hand, seems never to have been grammaticalized as part of the inflection.

The informants' group offers a very consistent picture with regard to CA. In example (19), we show a sentence representative for the 2Sg. No CA is realized. In example (21), we reproduce some sentences with the 2Pl (data from [Barera 2016](#)):

- (20) South-Bavarian of Upper Vinschgau (Trafoi and Sulden/Solda): 2Sg
 Konnsch (du) mir sogn, *wenn* (du) afn / zun Meer fohrsch?¹²
 can (you) me.DAT tell, when (you) at-the / to-the see go
 'Can you tell me when you are going to the beach?'
- (21) South-Bavarian of Upper Vinschgau (Trafoi and Sulden/Solda): 2Pl
- a. *Wenn-ts* (eis) ferte gessn hob, kemm-r mitnond noch Meran
 when-2PL (you) finished eat.INF have can-we together to Meran
 fohrn.
 go
 'When you have finished eating, we can go to Merano together.'
 - b. Er hot mi gfrog, *ob-s* (eis) ihm it helfn kennt.
 he has me asked whether-2PL (you) him.DAT not help can
 'He asked me if you could help him.'
 - c. Sie isch mit enk zournig, *weil-(t)s* eis ihr / sie nia
 she is with you angry because-2P you her.DAT / her.ACC never
 griaßt.
 greet
 'She is angry with you because you never greet her.'
 - d'. Dr Hund, *dein-ts* (eis) ihr gschenkt hob, isch a
 the dog that-2PL (you) her.DAT given have is a
 braver.
 well-behaved.MS
 'The dog you gave her is well-behaved.'

¹²Given the purpose of this article, we have transcribed the dialectal data in the most intuitive way, using the German orthography adapted to the dialects of South Tyrol, as it is used in works on dialectology that deal with syntactic phenomena.

9 The phenomenon of Complementizer Agreement

d`. Dr Hund, *dein* was-ts (eis) ihr gschenkt hob, isch a
the dog that wh.2PL (you) her.DAT given have is a
braver.
well-behaved.ms
‘The dog you gave her is well-behaved.’

The fact that *-(t)s* is obligatory with or without the full pronoun *eis* indicates that it is an agreement morpheme. Although the behavior of the speakers is very coherent, there are isolated sentences in which some younger speakers, in particular T.T. and T.Z. (both 17-years-old), do not realize the CA (cf. 22–25) (see Barera 2016: 25):

- (22) I bsinn mi, *wieso* eis in Meran a Haus kaft hob-s. (T.T., T.Z.)
I remember me why you in Meran a house bought have
‘I remember why you bought a house in Merano.’
- (23) Er hot mi gfrog, *ob* eis ihm helfn kenn-ts. (T.Z.)
he has me asked whether you him.DAT help can
‘He asked me if you could help him.’
- (24) Sie isch mit enk sierig, *weil* eis ihr nia griaß-ts. (T.T., T.Z.)
she is with you angry because you her.DAT never greet
‘She is angry with you because you never greet her.’
- (25) *Wenn* eis schun ferte gessn hob-s, kenn-mr mitnond Meran oi
when you already finished eat.INF have can-we together Meran down
fohrn. (S.R., T.Z.)
go
‘When you have finished eating, we can go down to Merano together.’

The second important result regards a diachronic effect between the older and the younger generation. In fact, while all participants show CA on verbs in C°, only the younger speakers realize it in the verb-end position. In this respect, the data show a clear-cut distinction (cf. 26–28):

- (26) Mi ricordo perché avete comprato casa a Merano
‘I remember why you bought a house in Meran’
a. I erinner mi nu guat, *brum-s* (eis) in Meran a Haus kaft *hob* (E.P., 83
y.o.)

Ermenegildo Bidese & Helmut Weiß

- b. I woafß nu, *wenn-ts* (eis) in Meran a Haus kaft *hob* (L.G., 76 y.o.)
 - c. I erinnern mi, *brum-s* (eis) in Meran a Haus kaft *hob* (A.T., 57 y.o.)
 - d. I erinner mi nu, *brum-s* (eis) in Meran a Haus kaft *hob* (M.R., 57 y.o.)
 - e. I konn mi erinnern, *brum-s* (eis) af Meran a Haus kaft *hob* (H.G., 56 y.o.)
 - f. Iaz erinner i mi, *wieso-s* (eis) in Meran a Haus kaft *hob-s* (S.R., 26 y.o.)
 - g. I woas nu, *brum-s* (eis) in Meran des Haus kaft *hob-s* (C.P., 19 y.o.)
 - h. I konn mi schun erinnern, *wieso-s* (eis) in Meran a Haus kaft *hob-s* (F.G., 19 y.o.)
 - i. I bsinn mi, *wieso* eis in Meran a Hitt kaft *hob-s* (T.T., 17 y.o.)
 - j. I erinner mi, *wieso* eis in Meran a Haus kaft *hob-s* (T.Z., 17 y.o.)
- (27) È arrabbiata con voi perché non la salutate mai
She is angry with you because you never greet her.
- a. Sie isch mit enk zournig, *weil-ts* eis ihr nia *griaßt* (E.P., 83 y.o.)
 - b. Sie isch mit enk zourne, *weil-ts* eis ihr nia *griaßt* (L.G., 76 y.o.)
 - c. Sie isch mit enk zourne, *weil-ts* eis nia *griaßt* (A.T., 57 y.o.)
 - d. Sie isch mit enk zourni, *weil-ts-es* eis nia *griaßt* (M.R., 57 y.o.)
 - e. Er isch mit enk zourni, *weil-ts-es* eis nia *griaßt* (H.G., 56 y.o.)
 - f. Sie isch mit enk zourne, *weil-ts* eis sie nia *griaßt* (S.R., 26 y.o.)
 - g. Sie isch mit enk zournig, *weil-ts* eis sie nia *griaßt* (C.P., 19 y.o.)
 - h. Sie isch sierig mit enk, *weil-ts* es sie nia *griaß-ts* (F.G., 19 y.o.)
 - i. Sie isch mit enk sierig, *weil* eis ihr nia *griaß-ts* (T.T., 17 y.o.)
 - j. Sie isch zournig mit enk, *weil-s* eis sie nia *griaß-ts* (T.Z., 17 y.o.)
- (28) So che avete commesso questo sbaglio, ...
I know that you have made this mistake, ...
- a. I woafß, *dass eis* an Fahlar gmocht *hob*, ... (E.P., 83 y.o.)
 - b. I woafß, *dass-ts* an Fahlar gmocht *hob*, ... (L.G., 76 y.o.)
 - c. I woafß, *dass-ts (es)* an Fahlar gmocht *hob*, ... (A.T., 57 y.o.)
 - d. I woafß, *dass-ts* an Fahlar gmocht *hob*, ... (M.R., 57 y.o.)
 - e. I woafß, *dass es* an Fahlar gmocht *hob*, ... (H.G., 56 y.o.)
 - f. I woafß, *dass-ts (eis)* an Fahlar gmocht *hob-s*, ... (S.R., 26 y.o.)
 - g. I woafß, *dass-ts* an Fahlar gmocht *hob-s*, ... (C.P., 19 y.o.)

9 The phenomenon of Complementizer Agreement

- h. I woäß, *dass* es an Fahlar gmocht *hob-s*, ... (F.G., 19 y.o.)
- i. I woäß, *dass-ts* es an Fahlar gmocht *hob-s*, ... (T.T., 17 y.o.)
- j. I woäß, *dass* eis an Fahlar gmocht *hob-s*, ... (T.Z., 17 y.o.)

With respect to adjacency effects between the agreement morpheme in C° and the overtly realized personal pronoun or DP-subject, the South Bavarian dialects of Obervinschau/Alta Venosta offer an interesting picture when compared to the other Bavarian dialects. As pointed out by Fuß (2014), the adjacency effect usually affects dialects that exhibit ‘double agreement’, i.e., in which the CA morpheme differs from the verbal inflection, as observed by Ackema & Neeleman (2003), Ackema & Neeleman (2004) for the East Dutch variety of Hellendoorn. On the contrary, single agreement dialects, such as Tegelen Dutch, show no adjacency effects (see van Koppen 2012). Furthermore, Bavarian is generally reported to show no adjacency effects even in double agreement dialects. However, there also seems to be a weak tendency, at least for some speakers, to prefer complementizers without inflection when the adjacency between the complementizers and the subject is broken by lexical material (see Fuß 2014). The Upper Vinschgau/Alta Venosta dialect clearly follows this last tendency. When adverbial phrases like ‘probably’ (cf. (29)) or ‘tomorrow’ (cf. (30)) intervene between the complementizer and the pronominal subject, most speakers prefer the uninflected complementizer; only a few informants, who usually belong to the older generation, randomly produce the inflected form:

(29) *Penso che probabilmente sarete voi ad essere interrogati*

I think it is probably you who will be questioned.

- a. I denk, *dass-ts* wohrscheinlich *eis* geprüft *weart* (E.P., 83 y.o.)
- b. I denk, *dass* wohrscheinlich *eis* prüft *weart* (L.G., 76 y.o.)
- c. I denk, *dass-ts* wohrscheinlich *eis* geprüft *weart* (A.T., 57 y.o.)
- d. I denk, *dass* wohrscheinla *eis* geprüft *weart* (M.R., 57 y.o.)
- e. I moan, *dass* ament *eis* prüft *weart* (H.G., 56 y.o.)
- f. I denk, *dass* wohrscheinla *eis* prüft *wear-ts* (C.P., 19 y.o.)
- g. I glab, *dass* eppr *eis* prüft *wear-ts* (F.G., 19 y.o.)
- h. I glab, *dass* wohrscheinlich *eis* geprüft *wear-ts* (T.T., 17 y.o.)
- i. I denk, *dass* wohrscheinlich *eis* prüft *wear-ts* (T.Z., 17 y.o.)

(30) *Sono sicuro che toccherà a voi domani*

I am sure that tomorrow will be your turn.

Ermenegildo Bidese & Helmut Weiß

- a. I bin mr sicher, *dass-ts* morgen *eis* oun dr Reih *kemmp* (L.G., 76 y.o.)
- b. I bin mr sicher, *dass* morgen *eis* oun dr Reih *kemmp* (A.T., 57 y.o.)
- c. I bin mr sicher, *dass* morgen *eis* oun dr Reih *kemmp* (M.R., 57 y.o.)
- d. I bin mr sicher, *dass* morgen *eis* oun dr Reih *kemm-s* (S.R., 26 y.o.)
- e. I bin mr sicher, *dass* morgen *eis* drun *kemm-s* (C.P., 19 y.o.)
- f. I bin mr sicher, *dass-ts* morgn *eis* oun dr Reih *sei-ts* (F.G., 19 y.o.)
- g. I bin mr sicher, *dass* morgn *eis* droun *kemm-s* (T.Z., 17 y.o.)

The situation does not really change with modal particles, which, according to Fuß (2014), do not break the prosodic domain. In fact, the data collected for the Upper Vinschgau/Alta Venosta dialect seem to indicate a slight preference of the older generation for the inflected form (cf. (31)), but they are too few to establish a correlation:

- (31) Voglio farvi presente che dovete essere voi stessi a ritirare il documento
Let me point out that you must collect the document yourselves.
- a. I will enk nu sogn, *dass-ts* jo *eis* selber s Dokument ohoul'n *miast*.
(L.G., 76 y.o.)
 - b. I moch enk drauf aufmerksam, *dass* jo *eis* selbst des Dokument
ohoul'n *miast*. (A.T., 57 y.o.)
 - c. I moch enk drauf aufmerksam, *dass-ts* jo *eis* selbst des Dokument
ohoul'n *kemmp*. (M.R., 57 y.o.)
 - d. Es isch jo wichtig, *dass* jo *eis* s Dokument ohoul'-*ts*. (C.P., 19 y.o.)
 - e. I moch enk drauf aufmerksam, *dass* jo *eis* selbst s Dokument ohoul'n
mias-ts. (F.G., 19 y.o.)
 - f. I muas enk nu sogn, *dass* jo *eis* selber es Dokument ohoul'n *kemm-s*.
(T.T., 17 y.o.)

This is, however, not the case when the finite verb is in the V2 position followed by the interrogative/modal particle *denn* (or in the reduced form *-n*), as in the following examples:

- (32) [I am at home. The doorbell rings. I open the door and to my surprise
there are three friends]
Cosa ci fate qui
What are you guys doing here?
- a. Wos well-*ts-n* *eis* heint? (E.P., 83 y.o.)

9 The phenomenon of Complementizer Agreement

- b. Jo was *tea-ts* denn *eis* do? (L.G., 76 y.o.)
- c. Jo hoi, was *tea-ts* denn *eis* do? (A.T., 57 y.o.)
- d. Jo was *tea-ts-n* *eis* do? (H.G., 56 y.o.)
- e. Was *tea-ts* denn *eis* do? (C.P., 19 y.o.)
- f. Na hoi, was *tea-ts-n* *eis* do? (F.G., 19 y.o.)
- g. Hoi, was *tea-ts-n* *eis* do? (T.T., 17 y.o.)
- h. Hoi, was *tea-ts-n* *eis* do? (T.Z., 17 y.o.)

These data clearly reveal that the adjacency effect is at work in this variety only when the CA morpheme concerns the complementizer and not the finite verb in C°. This points to a dissociation between CA and the T-related subject-verb agreement as indicated by other empirical facts, such as the First Conjunct Agreement, and suggests that CA is somehow a morphological ornament and should be considered separate from the verbal agreement itself (see [Richards 2012](#)). In fact, with respect to the first Conjunct Agreement, the Upper Vinschgau/Alta Venosta dialect shows the presence of CA to an increasing degree (cf. (33)), while it is completely absent when the pronominal subject is not overtly realized, although the phi-feature involved is the second plural person (cf. (34)):

- (33) Ho saputo che voi, tu e Anna, avete trovato un lavoro
I heard that you, you and Anna, found a job.

- a. I hon keart, *dass-ts eis*, du und die Anna, an Orbat gfundn *hob* (E.P., 83 y.o.)
- b. I hon keart, *dass-ts eis*, du und die Anna, an Orbat gfundn *hob* (A.T., 57 y.o.)
- c. I hon keart, *dass eis*, du und die Anna, an Orbat gfundn *hob* (M.R., 57 y.o.)
- d. I hon keart, *dass-ts eis*, die Anna und du, an Orbat gfundn *hob* (H.G., 56 y.o.)
- e. I hon keart, *dass-ts eis*, du und die Anna, an Orbat gfundn *hob-s* (S.R., 26 y.o.)
- f. I hon keart, *dass eis*, du und die Anna, an Orbat gfundn *hob-s* (C.P., 19 y.o.)
- g. I hon keart, *dass eis*, die Anna und du, an Orbat gfundn *hob-s* (F.G., 19 y.o.)

- h. I hon keart, *dass eis*, du und die Anna, an Orbat gfundn *hob-s* (T.T., 17 y.o.)
 - i. I hon keart, *dass eis*, du und die Anna, a nuie Orbat gfundn *hob-s* (T.Z., 17 y.o.)
- (34) textitHo sentito che Lisa e tu avete trovato un lavoro.
I heard that Lisa and you found a job.
- a. I hon keart, *dass die Lisa* und du an Orbat gfundn *hob* (E.P., 83 y.o.)
 - b. I hon keart, *dass die Lisa* und du a nuie Orbat gfundn *hob* (L.G., 76 y.o.)
 - c. I hon keart, *dass die Lisa* und du an Orbat gfundn *hob* (A.T., 57 y.o.)
 - d. I hon keart, *dass die Lisa* und du an Orbat gfundn *hob* (M.R., 57 y.o.)
 - e. I hon keart, *dass die Lisa* und du an Orbat gfundn *hob* (H.G., 56 y.o.)
 - f. I hon keart, *dass die Lisa* und du an Orbat gfundn *hob-s* (S.R., 26 y.o.)
 - g. I hon keart, *dass die Lisa* und du an Orbat gfundn *hob-s* (C.P., 19 y.o.)
 - h. I hon keart, *dass die Lisa* und du an Orbat gfundn *hob-s* (F.G., 19 y.o.)
 - i. I hon keart, *dass die Lisa* und du a Orbat gfundn *hob-s* (T.T., 17 y.o.)
 - j. I hon keart, *dass die Lisa* und du an Orbat gfundn *hob-s* (T.Z., 17 y.o.)

The data presented here confirm that the CA is an effect of the reanalysis of some clitics into bounded morphemes attracted to C°. CA does not satisfy the phi-feature of C°; nevertheless, it can be understood as a reflex of the power of C° in the V2 languages and its relation to the finite verb.

5.2 The second area: The dialects of Nonsberg/Val di Non

The second area of South Tyrol that was investigated was the Upper Fischbach-Valley/Val della Pescara at the Nonsberg/Val di Non (see [Pigarella 2018](#), as explained in footnote 12). The data of 9 speakers of the local dialect have been collected, 3 of the informants were younger than 50-years-old, 6 of them older. The task consists of translating 38 sentences from Italian into the local South-Bavarian varieties, the same ones as in the first area. The result is very consistent and shows no presence of CA at all. This seems to relate to the fact that in this area the dominant form for the second person plural is not *es*, but *ir* (with the reduced enclitic variants *er* or *ar* and the full ones *ier* and *iar*) (see [Kollmann 2008](#)). In ((35)–(37)), we reproduce the sentences with the finite verb in the verb-second position (cf. (35)) and in the final position (cf. (36) and (37)):

9 The phenomenon of Complementizer Agreement

(35) Cosa ci fate qui?

What are you doing here?

- a. *Wos tiat-er* do? (P.G., 67 y.o.)
- b. *Wos tiat-er* do? (A.G., 64 y.o.)
- c. *Wos tiat-er* do? (R.E., 57 y.o.)
- d. *Wos tiat-er* do? (U.G., 57 y.o.)
- e. *Wos tiat-ier* do? (F.K., 56 y.o.)
- f. *Wos tiart-er* do? (W.U., 52 y.o.)
- g. Na, *wos tiat-n es* do? (J.N., 28 y.o.)
- h. *Wos tiart-er* do? (M.U., 26 y.o.)
- i. *Wos tiasch-r* do? (J.U., 19 y.o.)

In (35), only J.N. uses the pronominal form *es* for the second person plural. All the other speakers show the pronoun *ir* without any CA.

(36) È arrabbiata con voi perché non la salutate mai.

She is angry with you because you never say hello to her.

- a. *Si isch mit enk zourni, weil ier sie nia griest.* (P.G., 67 y.o.)
- b. *Si isch zourni mit enk, weil ier sa nia griest.* (A.G., 64 y.o.)
- c. *Ier isch mit enk znicht, weil ier net griast.* (R.E., 57 y.o.)
- d. *Si isch zourni mit enk, weil-ier-sa nia gries tet.* (U.G., 57 y.o.)
- e. *Si isch zourni mit dier, weil da-se nia griest.* (F.K., 56 y.o.)
- f. *Si isch zourni mit enk, weil ier-sa nia griast.* (W.U., 52 y.o.)
- g. *Si isch siriag af enk, weil es sa nie grießt tent.* (J.N., 28 y.o.)
- h. *Si isch zourni mit enk, weil ier-sa nia griast.* (M.U., 26 y.o.)
- i. *Es isch sierig --- enk, weil es sa nie griest tent.* (J.U., 19 y.o.)

In (36), only two young speakers, J.N. and J.U., used *es* as a personal pronoun for the second person.¹³ According to Pigarella (2018: 39–40), who conducted the interviews (cf. footnote 12), both speakers are used to work outside of the investigated area in their everyday life, namely in linguistic environments where *es* is the prevalent form for the second person plural.

This result is consistent and confirmed by sentence (37):

¹³Note that in example (36), the oblique form of the 2Pl pronoun is *enk* that, just like *es* (see above), goes back to the dual form, while in the nominative the old form *iar* is preserved. The same split (old form in the nominative, new form in the oblique) occurs in (38) and (41). Note also that in (36e), the 2PL was mistakenly translated as 2SG.

(37) *Mi ricordo perché avete comprato casa a Merano.*

I remember why you bought a house in Meran.

- a. I bsinn mi nou, warumm-er in Meran a Haus gekauft *hob*. (P.G., 67 y.o.)
- b. I erinne mi, wiesou ier a Haus in Meran gkouft *hob*. (A.G., 64 y.o.)
- c. I bsinn mi, dass ier in Meran a Haus gkauft *hob*. (R.E., 57 y.o.)
- d. I woäß nu, pall/wall-ar (*hat*) dieser Hit gkaft g-Merò (U.G., 57 y.o.)
- e. I bsinn mi, dass ier *hab* a Haus gkauft in Meran. (F.K., 56 y.o.)
- f. I bsinn mi, worum-ier in Meran a Haus gekaft *hob*. (W.U., 52 y.o.)
- g. I bsinn mi, wiesou *es* in Meran a Haus gkauft *hob*. (J.N., 28 y.o.)
- h. I psinn mi, worum-ier in Meran a Haus gkaft *hob*. (M.U., 26 y.o.)
- i. I kann mi erinner, dass *es* in Meran a Haus gekauft *hab*. (J. U., 19 y.o.)

The fact that the only two speakers who produce *es* belong to the younger generation clearly suggests that *es* is a new feature that seems to have been imported from outside into a part of South Tyrol where the traditional form for the second person plural is *ir*, or, in the enclitic form, *ar*. We verify this hypothesis by looking at the traditional Wenker records collected in this area presumably in the 1940ies (see [Fleischer 2017](#): 127; [Rabanus 2020](#): 215).¹⁴ They all confirm that the pronoun *es* is a very recent innovation. In fact, the sentences with a second person plural only show the enclitic forms *-or* in the Wackernagelposition, as the translations of the original Wenker sentences 27. and 30. report (cf. here (38) and (39), respectively):

(38) Wenker sentence 27.:

Könnt ihr nicht noch ein Augenblickchen auf uns warten, dann gehn wir mit euch.

Can't you wait for us for a little while longer, then we'll go with you.

Kennt **or** nēit nou an augnblik wortn, nochr gea mor mit enk?

Can you not still a little while wait, than go we with you

¹⁴The Wenker catalog can be accessed and consulted online through the platform REDE (regionalsprache.de) <https://www.regionalsprache.de>. The Wenker records referring to the German part of Nonsberg/Val di Non, in particular to the village of Laurein/Lauregno have the sheet numbers 46324 and 46366, those referring to the village of Proveis/Proves the sheet number 46323. The sentences reported here have been extracted from the sheet 46324; the same sentences in the other sheets are fully consistent in not showing the personal pronoun *es* for the second person plural. We are very grateful to Jan Casalicchio (University of Siena) for bringing to our attention this period, in which the *Wenkersätze* were probably collected in South Tyrol.

9 The phenomenon of Complementizer Agreement

(39) Wenker sentence 30.:

Wieviel Pfund Wurst und wieviel Brot wollt ihr haben?

How many pounds of sausage and how much bread do you like?

Wiafil Pfunn wurscht unt broat wöillt or denn?

How-many pounds sausage and bread want you also

In addition, the tonic forms *ier* with the variant *iar* appear in front of the sentence, i.e., in the *Vorfeld* (cf. die Wenker sentences 28. and 31. reproduced here in (40) and (41)):

(40) Wenker sentence 28.:

Ihr dürft nicht solche Kindereien treiben!

You must not be so childish!

Ior törft net sölle Tumhaitn mochn.

You should not such stupid things do

(41) Wenker sentence 31.:

Ich verstehe euch nicht, ihr müßt ein bisschen lauter sprechen!

I don't understand you, you have to speak a little louder!

I vorschtea enk nēit, iar miast a bisl lautor rēidn!

I understand you not you must a little louder speak

The reason why CA does not appear in this area of South Tyrol is most likely related to the absence of the personal pronoun *es* for the second person plural. However, some of the younger speakers seem to be importing it from outside the area, laying the groundwork for the spread of this feature in the region and for the possible development of CA.

5.3 The third area: The town dialect of Meran / Merano

With regard to the phenomenon of CA, the city dialect of Merano/Meran shows an inconsistent picture, in contrast to what we have seen in Upper-Vinschgau/Alta Venosta (cf. Section 5.1, above) and Nonsberg/Val di Non (cf. Section 5.2, above) (see Bassighini 2018, as explained in footnote 12, above). In Merano/Meran, 10 speakers of the city dialect were interviewed, ranging in age from 9 to 79 years. The methodology used for the interviews was the same as in the other areas of South Tyrol, but in this case, grammaticality judgments were also requested. In fact, the phenomenon seems to be spreading, but the sociolinguistic dynamics of language use in a city, as opposed to rural areas, counteract the spread of CA: The older urban population rejects CA, at least for the most

part, as unnatural and “rural”, i.e., typical of people who live in the countryside, as opposed to those who live in the city. It is only among younger speakers that a higher acceptance of the phenomenon can be observed. Although they do not produce it actively, it belongs to their passive repertoire and judge it as possible or correct.

The example (42) displays the use of CA among the speakers interviewed in Merano/Meran.

(42) *Mi ha chiesto se potete aiutarlo a riparare la macchina.*

He asked me if you can help him to fix the car.

- a. Er hot mi gfrog, ob es kann-ts helfn, des Auto zu repariern. (L.S., 79 y.o.)
- b. Er hot mi gfrog, ob es ihm helfn kennt, es Auto repariern. (H.W., 54 y.o.)
- c. Er hot mi gfrog, ob es ihm kann-ts helfn, es Auto zu repariern. (B.M., 53 y.o.)
- d. Er hot mi gfrog, ob(-s) es ihm helfn kenn-ts, es Auto zu repariern. (M.S., 25 y.o.)
- e. Er hot mi gfrog, ob(-s) es ihm helfn kann-ts, es Auto zu repariern. (C.F., 24 y.o.)
- f. Er hot mi gfrog, ob(-s) es ihm helfn kenn-ts, es Auto zu repariern. (A.V., 22 y.o.)
- g. Er hot gfrog, ob(-s) es ihm helfn kenn-ts, es Auto zu repariern. (M.W., 22 y.o.)

It is interesting to note that, in contrast to what we observed in Upper Vinchgau/Alta Venosta, almost all of the participants realize the agreement morpheme in the verb end position. There is a clear difference between the younger and older generations regarding the acceptance of the CA on C°. The round brackets show that the speakers did not produce the CA on their own, but they accepted it as possible. On the contrary, the speakers of the older generation did not consider it as grammatical and judged it as “rural”.

A similar result, though less consistent, is shown in example (43), where all younger generation speakers produced the agreement marker on the final verb and three out of five accepted it (cf. (43e), (43f), (43i)) on the complementizer, while only one out of three older generation speakers realized it on the final verb and accepted it on the complementizer (cf. (43b)):

9 The phenomenon of Complementizer Agreement

- (43) *Se avete già finito di mangiare, possiamo andare a Merano assieme.*
 If you have already finished eating, we can go to Merano together.
- a. Wenn es schon fertig gessn hob, nor kennen mir zusammen noch Meran gian. (L.S., 79 y.o.)
 - b. Wenn(-s) (es) schon aufkeart hob-s zun Essn, kanntn mir mitnonder noch Meran fohrn. (W.S., 58 y.o.)
 - c. Wenn es schon fertig gessn hob, nor kemmr mitnondr Meran gian. (H.W., 54 y.o.)
 - d. Wenn es schon fertig gessn hob-s, kemmr mitnondr noch Meran gian. (B.M., 53 y.o.)
 - e. Wenn(-s) (es) schon fertig gessn hob-s, nor kennen mir a zusammen noch Meran gian. (M.S., 25 y.o.)
 - f. Wenn(-s) (es) schon fertig gessn hob-s, kemmr zusammen Meran innigian. (C.F., 24 y.o.)
 - g. Wenn es schon fertig sei-ts mit Essn, nor kemmr zusammen noch Meran gian. (A.V., 22 y.o.)
 - h. Wenn (es) schon fertig gessn hob-s, nor kemmr mitnondr noch Meran gian. (M.W., 22 y.o.)
 - i. Wenn(-s) (es) genuag hob, hear-ts auf zu Essn, terf-mr zusammen noch Meran gian? (J.W., 9 y.o.)

6 Conclusions: Two ways to develop CA

In conclusion, the data suggest that the phenomenon of CA is generally spreading in the South-Bavarian dialects of South Tyrol (see also [Vergeiner & Bülow 2021](#) for Nord-Tyrol), but in three very different ways and situations:

- (a) In the first area (Upper-Vinschgau/Alta Venosta: Trafoi and Sölden/Solda), the phenomenon is attested very robustly with a clear distinction between the older and the younger generation. Only in the data of the younger generation, CA has developed into an agreement marker, as it shows up in the final position of the sentence as well and not only on C°. Indeed, the presence of the CA morpheme on the verb, even outside the C domain, suggests that it became part of the verbal paradigm and lost any syntactic transparency as a subject clitic. This seems consistent with a diachronic path in which the subject clitic is realized first as an agreement morpheme and then as an inflectional marker.

- (b) In the second area (Upper Fischbach-Valley/Val della Pescara at the Nonsberg/Val di Non), the phenomenon is not attested, probably because the dual form *e(i)s* ‘you’ has never found its way into this dialect. The South-Bavarian variety spoken in this part of South Tyrol still preserves the original form *ir* ‘you’ and the enclitic reduced forms *-ar* and *-r*. Interestingly enough, the form *es* for the second person plural is going to be imported into this area by younger speakers who study or work elsewhere in South-Tyrol, but without any trace of CA yet.
- (c) As for the third area, the CA morphemes occurred consistently on verbs in the end position as well, but only the younger speakers ‘tolerated’ inflected complementizers, although they do not use them (yet?).

These data confirm Weiß, Weiß’s (2005, 2018) hypothesis on the diachronic path of the phenomenon, as they prove that the subject clitic has been reanalyzed as an agreement morpheme, thus becoming part of the inflectional paradigm for the speakers of the younger generation.

However, the data from South Tyrol suggest that there are two slightly different ways to inflected complementizers:

- I. DA (in C°) → CA → V/end (Upper Vinschgau)
- II. DA (in C°) → V/end → CA (Meran)

The first step is always the development of new inflectional morphology (DA) on verbs in the second position via reanalysis of subject clitics as (part of the) inflection. But the following extension into new contexts can proceed in two different ways:

- I. The new morphology can be transferred first on complementizers and then on verbs in the end position – this happened in the Upper Vinschgau – or,
- II. first on verbs in the end position and afterward on complementizers – this is what seems to be happening already in Meran.

Contributions

The present contribution was written by the authors in complete collaboration. For the formal definition of scholarly responsibility, as required by the Italian

9 *The phenomenon of Complementizer Agreement*

academic system, we declare that Ermenegildo Bidese was responsible for Sections 5 and 6, and Helmut Weiß was responsible for Sections 1, 2, 3, and 4, respectively. We would like to thank two anonymous reviewers for their comments and Maren Michel for proofreading. Heartfelt thanks to Anna Barera, Tamara Bassighini, and Giorgia Pigarella for collecting data in the three areas of South Tyrol.

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9 The phenomenon of Complementizer Agreement

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Chapter 10

Parasitic participles and ellipsis in VP-focus pseudoclefts

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1 Introduction

This article discusses pseudocleft constructions of the type in (1)-(2), found in Dutch and German, where the verb in the focused verb phrase (in square brackets) can be either an infinitive or a past participle.

(1) Dutch

Wat ik heb ge-daen is [keihard werk-en / ge-werk-t].
what I AUX:1SG GE-do:PTCP be:3SG real.hard work-INF GE-work-PTCP
‘What I did is work real hard.’

(2) German

Was ich ge-mach-t habe ist [einfach weitere Apps installier-en
what I GE-do-PTCP AUX:1SG be:3SG just more apps install-INF
/ installier-t].
install-PTCP
‘What I did is simply to install more apps.’

Past participles in Dutch and German normally occur with a temporal (*have/be*) or passive auxiliary, which is absent from the focus VP in (1)-(2). In the pseudocleft constructions in (1)-(2), the participle in the focus VP is parasitic on the past participle in the antecedent VP:

Jan-Wouter Zwart

(3) Dutch

Wat ik deed is [keihard werk-en / *ge-werk-t].
 what I do:PST be:3SG real.hard work-INF GE-work-PTCP
 ‘What I did is work real hard.’

Replacement of the antecedent VP past participle construction *heb gedaan* ‘did’ by a simple past form *deed* ‘did’ has the effect that the past participle can no longer be used in the focus VP.

In this article I present an analysis of the properties of parasitic past participles in VP-focus pseudoclefts, in the context of what we know about pseudoclefts more generally and about parasitic participle constructions crosslinguistically. The analysis leads me to consider an ellipsis analysis of the focus VP, which can be shown to have several advantages, especially in the variant of ellipsis espoused by Ott & de Vries (1981), where ellipsis is preceded by fronting of the focused material. However, my conclusion is that the relevant facts can be equally well understood in a ‘base-generation’ analysis, not involving ellipsis. Our discussion shows the importance of the feature ‘anterior’ (as distinct from ‘past’) in the morphosyntactic realization of Continental West Germanic verbs (cf. Zwart 2017).

2 Pseudoclefts

Pseudoclefts (Higgins 2003, Blom & Daalder 1979, Den Dikken 2005) are copular constructions of the type [A COP B] where A is a wh-clause and B a focused element:

(4) Dutch

[Wat ik<e> lees] is [een boek].
 what I read:1SG BE:3SG INDF book
 ‘What I’m reading is a book.’

In (4), A = *wat ik lees*, COP (the copula) = *is*, and B = *een boek*. The wh-clause is marked by a fronted wh-element (*wat* in (4)) associated with an empty position (a variable) inside the clause (indicated by <e> in (4); the variable precedes the finite verb, because of the verb-final character of embedded clauses in Dutch). B (the focused element) can be interpreted as providing the value for the variable, so that (4) entails (5).

(5) Dutch

10 *Parasitic participles and ellipsis in VP-focus pseudoclefts*

Ik lees een boek.
 I read:1SG INDF book
 'I'm reading a book.'

On this interpretation, the pseudocleft construction is called *specificational*.

Alternatively, A refers to a separate concept x , and B identifies or specifies a property of x . This is easier to see when B is not a noun phrase or not a suitable complement of the verb in A:

- (6) Dutch
 [Wat ik_{<e>} lees] is [interessant].
 what I read:1SG BE:3SG interesting
 'What I'm reading is interesting.'

This is a *predicational* pseudocleft construction. It does not allow a paraphrase like (7a) (cf. (5)), but requires the more involved paraphrase in (7b).

- (7) a. *I'm reading interesting.
 b. I'm reading x , and x is interesting.

When B is a noun phrase, as in (1), the pseudocleft construction is ambiguous between a specificational and a predicational interpretation, and we need diagnostic tests to tell the two readings apart.

For Dutch, the following tests can be applied:

- (i) Degree modification of the focus element brings out the predicational reading.

Such degree modification is illustrated in (8b), using *je reinste* 'utter':

- (8) Dutch
 a. Wat hij schrijf-t is porno.
 what he write-3SG be:3SG porn
 (i) 'He writes porn.' (specificational)
 (ii) 'The stuff he writes is porn.' (predicational)
 b. Wat hij schrijf-t is je reinste porno.
 what he write-3SG be:3SG utter porn
 (only) 'The stuff he writes is utter porn.' (predicational)

Jan-Wouter Zwart

The effect of the degree modification is that *porno* ‘porn’ can only be understood as predicating over whatever it is that he writes (i.e. the predication reading). The simple reading ‘He writes porn’ is not available.¹

(ii) Adjusting the copula brings out the predication reading.

In (9b), we change the copula from *zijn* ‘be’ to *worden* ‘become’:

(9) Dutch

- a. Wie jij <e> bedoel-t is de nieuw-e directeur.
 who you mean-2SG be:3SG DEF new-AGR director
 (i) ‘You are referring to the new director.’ (specificational)
 (ii) ‘The person you are referring to is the new director.’
 (predicational)
- b. Wie jij <e> bedoel-t word-t de nieuw-e directeur.
 who you mean-2SG become-3SG DEF new-AGR director
 (only) ‘The person you are referring to will be the new director.’
 (predicational)

In (9b), the specificational reading (‘You are referring to the new director’) is not available.

(iii) A pied-piped wh-phrase appears to bring out the specificational reading. This is illustrated in (10):

(10) Dutch

- a. Op wie jij<e> doel-t is de nieuw-e directeur.
 to who you refer-2SG be:3SG DEF new-AGR director
 (only) ‘You are referring to the new director.’ (specificational)
- b. *Op wie jij<e> doel-t word-t de nieuw-e directeur.
 to who you refer-2SG become-3SG DEF new-AGR director
 (intended) ‘The person you are referring to will be the new director.’

The contrast between (10a) and (10b) is reminiscent of the difference between embedded wh-questions and free relative clauses, the latter not allowing pied-piped wh-expressions:

¹A reviewer has different judgments on the use of ‘je reinste’ in these and following examples. I have no explanation for this variation in judgments. To me and native speakers I have questioned, the judgments are crystal clear.

10 Parasitic participles and ellipsis in VP-focus pseudoclefts

(11) Dutch

- a. Ik weet met wie hij praat.
 I know:SG with who he speak:SG
 ‘I know who he is talking to.’ (embedded question)
- b. *Ik ken met wie hij praat.
 I know:SG with who he speak:SG
 (intended) ‘I am familiar with the person who he is talking to.’ (free relative)

I take this to suggest that the *wh*-clause in specificational pseudoclefts is an embedded question, while the *wh*-clause in predicational pseudoclefts is a free relative clause (see [Den Dikken 2005](#) for extensive discussion of the status of the *wh*-clause in specificational pseudoclefts).

3 VP-focus pseudoclefts

Let us now return to pseudoclefts of the type in (1)-(2), where A and B are not noun phrases but verb phrases. These, too, can be specificational or predicational. Before turning to the type involving a parasitic participle, consider first the example in (12), where A contains a modal auxiliary and an empty complement to the modal auxiliary:

(12) Dutch

Wat ik_{<e>} wil is [keihard werk-en].
 what I want:SG be:3SG real.hard work-INF
 ‘What I want is to work real hard.’

In (12), the focus VP *keihard werken* ‘work real hard’ provides a value for the variable *<e>*, so that (12) can be paraphrased as (13):

(13) Dutch

Ik wil keihard werk-en.
 I want:SG real.hard work-INF
 ‘I want to work real hard.’

But (12) can also be interpreted as a predicational pseudocleft, as can be seen when we change the copula:

Jan-Wouter Zwart

(14) Dutch

Wat ik<e> wil word-t [keihard werk-en].
 what I want:SG become-3SG real.hard work-INF
 ‘What I want entails that I have to work real hard.’

Here the antecedent *wat ik wil* ‘what I want’ stands for some concept *x*, say finish your dissertation within four years, and what (14) means is that achieving *x* will involve a lot of hard work.

Another construction that brings out the predication reading of the pseudocleft is when the focus VP expresses a meta-comment, as in (15):

(15) Dutch

Wat ik<e> wil is [vrag-en om moeilijkhed-en].
 what I want:SG be:3SG ask-INF for trouble-PL
 ‘What I want is asking for trouble.’

Although (15) also has the specificational reading (‘I want to literally ask for trouble’), the more natural meta-comment reading as given in the translation is clearly predication: what I want is some concept *x*, say enforce the smoking ban on Dutch train platforms, and *x* (enforcing the smoking ban) means asking for trouble.

As expected, this predication reading of VP-focus pseudoclefts invites the use of degree modifiers such as *je reinste* ‘utter’:²

(16) Dutch

Wat ik<e> wil is je reinste [vrag-en om moeilijkhed-en].
 what I want:SG be:3SG utter ask-INF for trouble-PL
 ‘What I want is a clear case of asking for trouble.’ (predication)

VP-focus pseudoclefts in Dutch require the presence of verbal material in A. In (12)-(16), the verbal material is a modal auxiliary. Elsewhere, a dummy verb *do* must be employed, as we saw already in (1) and (3).

4 Parasitic participles in VP-focus pseudoclefts

Let us return now to the pseudocleft construction in (1), with a past participle in the focus VP:

²As suggested by a reviewer, infinitives like *keihard werken* ‘work real hard’ can arguably be viewed as ambiguously verbal and nominal, the former giving rise to the specificational reading, and the latter to the predication reading. This allows us to make the generalization that *je reinste* ‘utter’ can combine with nominal phrases only.

10 *Parasitic participles and ellipsis in VP-focus pseudoclefts*

(17) Dutch

Wat ik heb ge-daan is [keihard ge-werk-t].
 what I AUX:1SG GE-do:PTCP be:3SG real.hard GE-work-PTCP
 ‘What I did is work real hard.’

This participle, *gewerkt* ‘worked’ in (1)/(17), can only appear when the antecedent VP contains a past participle (cf. (3)):

(18) Dutch

- a. *Wat ik deed is [keihard ge-werk-t].
 what I do:PST be:3SG real.hard GE-work-PTCP
 ‘What I did is work real hard.’
- b. *Wat ik doe is [keihard ge-werk-t].
 what I do:1SG be:3SG real.hard GE-work-PTCP
 (intended) ‘What I do is work real hard.’
- c. *Wat ik wil is [keihard ge-werk-t].
 what I want:1SG be:3SG real.hard GE-work-PTCP
 ‘What I want is to work real hard.’

The condition has to refer to the morphology, not to the grammatical feature expressed by the morphology. The relevant feature is anteriority (relative tense), which is also present in the so-called IPP-construction, where a past participle that takes an infinitival complement is itself also realized with infinitive morphology (‘Infinitivus Pro Participio’, cf. [Lange 1981](#)). In that case, the focus VP must be infinitive as well. In (19), the infinitive *willen* ‘want’ replaces the expected participle *gewild* ‘wanted’, and the parasitic participle is now excluded:

(19) Dutch

- a. Wat ik heb wil-len doe-n is [keihard werk-en].
 what I AUX:1SG want-IPP do-INF be:3SG real.hard work-INF
 ‘What I wanted to do is work real hard.’
- b. *Wat ik heb wil-len doe-n is [keihard ge-werk-t].
 what I AUX:1SG want-IPP do-INF be:3SG real.hard GE-work-PTCP
 (intended: same as a.)

Descriptively, then, the participle of the focus VP is parasitic on the participle of the antecedent VP.

The triggering participle needs to be a dummy verb *do*:

Jan-Wouter Zwart

(20) Dutch

- a. Wat ik heb ge-wil-d is keihard { werk-en /
 what I AUX:1SG GE-want-PTCP be:3SG real.hard work-INF
 *ge-werk-t }.
 GE-work-PTCP
 ‘What I wanted is to work real hard.’
- b. Wat ik heb ge-probeer-d is keihard { werk-en /
 what I AUX:1SG GE-try-PTCP be:3SG real.hard work-INF
 *ge-werk-t }.
 GE-work-PTCP
 ‘What I tried is to work real hard.’

The IPP-effect remains in force in the parasitic participle. That is, when the verb in the focus VP has an infinitival complement, that verb cannot have participial morphology:

- (21) a. Ik heb keihard { lat-en / *ge-lat-en } werk-en.
 I AUX:1SG real.hard cause-INF GE-cause-PTCP work-INF
 ‘I made [them] work real hard.’
- b. Wat ik ge-daen heb is keihard { lat-en /
 what I GE-do:PART AUX:1SG be:3SG real.hard cause-INF
 *ge-lat-en } werk-en.
 GE-cause-PTCP work-INF
 ‘What I made [them] do is work real hard.’

Importantly, a pseudocleft construction with a parasitic participle lacks a predicational interpretation. Recall that a VP-focus pseudocleft like (22) can have both a specificational and a predicational interpretation:

(22) Dutch

Wat ik ge-daen heb is vrag-en om moeilijkheden.
 what I GE-do:PTCP AUX:1SG be:3SG ask-INF for trouble
 ‘What I did is ask for trouble.’

The two interpretations can be paraphrased as in (23):

- (23) a. specificational: I literally asked for trouble.
 b. predicational: The thing I did invited trouble for me.

10 *Parasitic participles and ellipsis in VP-focus pseudoclefts*

The second, predicational reading is lost when the parasitic participle appears:

(24) Dutch

Wat ik ge-daen heb is ge-vraag-d om moeilijkheden.
 what I GE-do:PTCP AUX:1SG be:3SG GE-ask-PTCP for trouble
 ‘What I did is ask for trouble.’
 ✓ specificational: I literally asked for trouble.
 ✗ predicational: The thing I did invited trouble for me.

This is a striking effect, given that the predicational reading is the more natural one, yet it is unavailable when the focus VP contains a parasitic participle.

As we now expect, the degree modifier (which calls out the predicational reading) cannot be used together with the parasitic participle:

(25) Dutch

Wat ik ge-daen heb is je reinste { vrag-en / *ge-vraag-d }
 what I GE-do:PTCP AUX:1SG be:3SG utter ask-INF GE-ask-PTCP
 om moeilijkheden.
 for trouble
 ‘What I did is a clear case of asking for trouble.’

Also, changing the copula from ‘be’ to ‘become’ (which forces the predicational reading) is impossible with the parasitic participle:

(26) Dutch

Wat ik ge-daen heb werd (op den duur) { vrag-en /
 what I GE-do:PTCP AUX:1SG become:PST.SG after some time ask-INF
 *ge-vraag-d } om moeilijkheden.
 GE-ask-PTCP for trouble
 ‘What I did became asking for trouble (after some time).’

We return to the obligatory specificational reading of the VP-focus pseudocleft construction with parasitic participle in Section 6, where we will see that this aspect of the construction gives initial support to an ellipsis analysis (as proposed for pseudoclefts as early as [Peters & Bach 1968](#)). First, however, we compare the pseudocleft parasitic participle to other parasitic participles crosslinguistically.

Jan-Wouter Zwart

5 Parasitic participles crosslinguistically

5.1 Germanic

Several cases of parasitic participles in Germanic have been discussed in the literature. At least four types may be distinguished (cf. [Wurmbrand 2012](#)):

A. Mainland Scandinavian parasitic supine constructions ([Wiklund 2001, 2007](#))

- (27) Swedish
 Han hade kunnat skrivit.
 he AUX:PST can:PTCP write:PTCP
 ‘He could have written.’

In (27), the verb selected by the modal shows participial rather than the expected infinitival morphology. As this does not happen when the modal is not itself a participle, the morphology seems parasitic.

B. Frisian Participium-pro-Infinitivo constructions ([Den Dikken & Hoekstra 1997](#))

- (28) West Frisian
 Hy sol it dien ha kinne-n.
 he AUX.MOD:3SG it do:PTCP AUX:INF can-PTCP
 ‘He would have been able to do it.’ (MOD > can > AUX > do)

Here, the verb *kinne* ‘can’ is selected by the modal auxiliary *sol*, and would normally show infinitival morphology.

C. The ‘scandalous’ construction in German ([Vogel 2009](#))

- (29) German
 ... ohne es verhinder-t hab-en zu könn-en.
 without it prevent-PTCP AUX-INF to can-INF
 ‘... without having been able to prevent it.’ (AUX > can > prevent)

Again, *verhindern* ‘prevent’ is selected by *können* ‘can’ and should have infinitival morphology. The scandalous monicker is due to the fact that the infinitival marker *zu* and the auxiliary *haben* are also wrongly placed (it should read *verhindern können zu haben*, with the IPP-effect on *können*).

10 *Parasitic participles and ellipsis in VP-focus pseudoclefts*

D. The perfect doubling construction of Dutch and German dialects (Koene-man et al. 2011, Brandner 2008, Wall 2018)

(30) Brabantish Dutch

Ik heb vandaag nog niet ge-rook-t ge-had.

I AUX:1SSG today yet NEG GE-smoke-PTCP GE-AUX:PTCP

‘I have not smoked today yet.’

(31) Alemannic

Er isch grad kum-me g-si.

he AUX:3SG just come-PTCP GE-AUX:PTCP

‘He had just arrived.’

The participial auxiliary *gehad* in (30) is absent outside circumscribed dialects, including Standard Dutch. In Alemannic (31), the doubling seems to be a device to express the relative past, an innovation brought about by the disappearance of the simple past.

These parasitic participle constructions share a property that is absent from the Dutch pseudocleft constructions with parasitic participles, namely that they appear in the context of verbal embedding or clustering. Consequently, the phenomenon may promisingly be approached as one of perseveration or ‘cross-wiring’ inside a verbal complex, essentially a (postsyntactic) morphological phenomenon if I am not mistaken (cf. Zwart 2017).³ By contrast, in the Dutch pseudocleft parasitic participle construction, with its copular structure ground plan, the parasitic participle cannot easily be explained as the result of a misconstrued dependency inside a verb cluster.

5.2 Beyond Germanic

Predicate doubling in connection with focus has been attested widely in so-called predicate clefting constructions (PCC), going back to Koopman (1984: chapter 6) on Vata. See Kandybowicz (2008: 80) for a survey. Example (32) is from Nupe (Kandybowicz 2008: 83):

³As pointed out by a reviewer, the perfect doubling construction type D is different still, in that the unexpected participial auxiliary acts as a copula (*gehad/gsi*), taking a complement that contains a participial predicate (*gerookt/kumme*). Strictly speaking, the construction does not involve doubling (cf. Poletto 2009: 47–48). This type of organization is also completely absent from the parasitic participle construction discussed here.

Jan-Wouter Zwart

(32) Nupe

bi-ba Musa à ba nakàn sasi èsun làzi yin o.

RED-cut Musa FUT cut meat some tomorrow morning PRT FOC

‘It is cutting that Musa will do to some meat tomorrow morning.’

Semantically the PCC seems to involve narrow focus on the verb (e.g. [Larson & Lefebvre 1991:256](#)), and the clefted predicate may not be accompanied by a complement (example from [Larson & Lefebvre 1991:248](#)):

(33) Haitian creole

Se manje (*pen an) Jan manje pen an.

IT.IS eat bread DEF Jean eat bread DEF

‘John ate the bread.’

This restriction does not apply to the Dutch pseudoclefts with parasitic participles:

(34) Dutch

Wat ik ge-daan heb is alle boek-en van Chomsky

what I GE-do:PTCP AUX:1SG be:3SG all book-PL of Chomsky

ge-lez-en.

GE-read-PART

‘What I did is read all books by Chomsky.’

Buli ([Hiraiwa 1999](#)), Yoruba ([Kandybowicz 2004](#)), Krachi ([Kandybowicz & Torrence 2004](#)), and doubtless other languages, do allow object pied-piping with predicate clefting. In Buli, pied-piping is limited to objects, so no adverbs or other adjuncts may be included ([Hiraiwa 1999: 555–556](#)), perhaps suggesting an incorporation analysis. No such restrictions apply to Dutch pseudoclefts with parasitic participles:

(35) Dutch

Wat ik ge-daan heb is gisteren uitgebreid met Chomsky

what I GE-do:PTCP AUX:1SG be:3SG yesterday extensively with Chomsky

ge-praat.

GE-talk:PART

‘What I did is talk extensively with Chomsky yesterday.’

Closer to home, Yiddish ([Cable 2004](#)), Russian ([Abels 2001](#)) and Hebrew ([Lan-dau 2006](#)) also allow predicate clefting with the object included in the fronted predicate, but the semantics is that of topicalization rather than focusing. (36) is an example from Yiddish ([Cable 2004: 2](#)).

10 *Parasitic participles and ellipsis in VP-focus pseudoclefts*

(36) Yiddish

- a. Ge-gess-en hot Maks ge-gess-en fish.
 GE-eat-PTCP AUX:3SG Max GE-eat-PTCP fish
 ‘As for eating, Max ate fish.’
- b. Ge-gess-en fish hot Maks ge-gess-en.
 GE-eat-PTCP fish AUX:3SG Max GE-eat-PTCP
 ‘As for eating fish, Max ate them.’

Similar constructions have been reported for Iberian Romance languages (Viente 2009, Cable 2004). Semantically this is quite different from the Dutch pseudocleft construction featuring parasitic participles, which express contrastive focus.⁴

I conclude that the existing literature on predicate clefting does not provide a model for the analysis of Dutch parasitic participle pseudo-clefts.

6 Ellipsis analysis

It has been observed as early as Peters & Bach (1968) that pseudocleft constructions alternate with an asyndetic construction where the focus NP is represented as part of a full clause:

(37) English

- a. What John reads <*e*> is BOOKS. (pseudocleft construction)
- b. What John reads <*e*> is he reads BOOKS. (asyndetic construction)

In (37b), where *books* is in focus, we may describe *he reads* as the focus related topic, a natural target for ellipsis (Tancredi 1992). This suggests that the pseudocleft construction (37a) may be derived from the asyndetic construction (37b) via deletion of the focus related topic *he reads*:

(38) Ellipsis analysis

What John reads <*e*> is he reads BOOKS.

⁴Interestingly, though, the participle in Yiddish alternates with an infinitive, suggesting that the participial morphology, when it appears, is again parasitic:

- (i) Ess-en hot Maks ge-gess-en a fish.
 eat-INF AUX:3SG Max GE-eat-PTCP INDEF fish
 ‘As for eating, Max ate a fish.’

Jan-Wouter Zwart

While the ellipsis analysis has been criticized and rejected early on (Higgins 2003, Blom & Daalder 1979), it has been revived in recent years (cf. Den Dikken et al. 2000), and has some immediate advantages for the analysis of pseudocleft constructions with parasitic participles.

First, a pseudocleft construction derived from an asyndetic construction via ellipsis can only be of the specificational type. Recall that in a specificational pseudocleft the focus NP provides a value for the variable $\langle e \rangle$ of the antecedent clause. This is different from a predicational pseudocleft, where the antecedent clause refers to a concept x , which is then predicated over by the focus NP. The status of the focus NP, then, is different in the two types of pseudocleft: in specificational pseudoclefts, it is a complement to the verb, while in predicational pseudoclefts, it is a predicate. In the ellipsis analysis (38), it is clear that the focus NP must be interpreted as a complement to the verb, so that only the specificational reading should be available.

Put differently, there is no source for the derivation of predicational pseudoclefts via ellipsis:

- (39) No ellipsis for predicational pseudoclefts
 What John reads $\langle e \rangle$ is he reads INTERESTING.

Recall now that pseudoclefts with parasitic participles are uniquely specificational. This is explained if they are derived via ellipsis from an asyndetic clausal source:

- (40) Ellipsis analysis for parasitic participle pseudoclefts (Dutch)
 Wat ik $\langle e \rangle$ ge-daan heb is ik heb [keihard
 what I GE-do:PTCP AUX:1SG be:3SG I AUX:1SG real.hard
 ge-werk-t].
 GE-work-PTCP
 ‘What I did is work real hard.’

In (40), the only interpretation that is available for the focus VP *keihard gewerkt* ‘worked real hard’ is that of complement to the temporal auxiliary *heb* ‘have’, leading us to interpret the VP as providing a value for the variable in the antecedent clause *wat ik gedaan heb* ‘what I did’. As a result, (40) can only be specificational.

Second, the ellipsis analysis immediately explains the distribution of the parasitic participle in pseudocleft constructions. Recall that the parasitic participle can only occur when the antecedent clause contains a past participle as well (see (18)–(19)). None of the examples in (18)–(19) can be derived via ellipsis from a grammatical source clause:

10 Parasitic participles and ellipsis in VP-focus pseudoclefts

- (41) No source for ellipsis
- a. *Ik deed keihard gewerkt (18a)
I do:PAST.SG real.hard GE-work-PTCP
- b. *Ik doe keihard gewerkt (18b)
do:1SG
- c. *Ik wil keihard gewerkt (18c)
want:SG
- d. *Ik heb willen keihard gewerkt (19b)
AUX:1SG want-INF

The ellipsis analysis also explains the observation illustrated in (20) above, that the participle in the antecedent clause must be a form of the dummy verb. All other verbs would destroy the parallelism between the antecedent clause and the asyndetically linked clause hosting the focus VP:

- (42) No source for ellipsis
- a. *Wat ik heb ge-wil-d is ik heb keihard
what I AUX:1SG GE-want-PTCP be:3SG I AUX:1SG real.hard
ge-werk-t.
GE-work-PTCP
(intended) ‘What I wanted is I worked real hard.’
- b. *Wat ik heb ge-probeer-d is ik heb keihard
what I AUX:1SG GE-try-PTCP be:3SG I AUX:1SG real.hard
ge-werk-t.
GE-work-PTCP
(intended) ‘What I tried is I worked real hard.’

The observation in (21), showing the IPP-effect in the focus VP, also follows naturally on an ellipsis analysis:

- (43) IPP in the focus VP with ellipsis
 Wat ik ge-daan heb is ik heb keihard { lat-en /
 what I GE-do:PART AUX:1SG be:3SG I AUX:1SG real.hard cause-INF
 *ge-lat-en } werk-en.
 GE-cause-PTCP work-INF
 'What I made [them] do is work real hard.'

Jan-Wouter Zwart

These two observations, the uniquely specificational interpretation and the limited distribution, provide immediate support for the ellipsis analysis of pseudocleft constructions with parasitic participles.

However, the ellipsis analysis of VP-focus pseudoclefts with parasitic participles also faces two immediate problems. First, the analysis cannot be generalized to regular VP-focus pseudoclefts, of the type in (3), repeated here:

- (44) Dutch
 Wat ik deed is keihard werk-en.
 what I do:PST be:3SG real.hard work-INF
 ‘What I did is work real hard.’

To derive (44) via ellipsis, we must posit a source clause like (45), which is ungrammatical.⁵

- (45) Dutch
 * Wat ik deed is || ik deed keihard werk-en.
 what I do:PST be:3SG I do:PST real.hard work-INF
 (intended) ‘What I did is I worked real hard.’

The ellipsis analysis now must either posit an ungrammatical source, or must apply to pseudoclefts with parasitic participles only. Neither option is very attractive.⁶

Second, while VP-focus pseudoclefts with parasitic participles may be considered somewhat marked, they lack the disjointed, anakolouthic character of the proposed asyndetic source for the ellipsis illustrated in (40). This raises the question how the asyndetic source construction for the ellipsis may have been derived. In the spirit of Zwart (2009) one might think that the derivation of a construction of this type requires derivation layering: an analysis in which part of the structure is derived in a separate derivation and included in the numeration for the current derivation as the single item [*ik heb keihard gewerkt*].⁷ But if that is the case, what is to exclude an analysis in which the separate derivation simply yields the sequence [*keihard gewerkt*], obviating the need for any ellipsis?

⁵The judgment in (45) applies to standard Dutch, and may be different in southern varieties of Dutch which allow *do*-support.

⁶This argument may be carried further by considering the inapplicability of the ellipsis analysis to it-clefts, which space prevents me from addressing here.

⁷Arguably, both the A and the B part in the A cop B construction making up the pseudocleft are best viewed as outputs of separate derivations.

10 *Parasitic participles and ellipsis in VP-focus pseudoclefts*

The first problem can be solved by assuming that ellipsis is preceded by VP-fronting (cf. Ott & de Vries 1981), as the next section discusses. But this analysis faces problems of its own, suggesting that an analysis without ellipsis might be preferable.

6.1 **Remnant ellipsis**

Ott & de Vries (1981) propose an analysis of right dislocation involving ellipsis from a biclausal source. Consider the example in (46).

- (46) Dutch
 Tasman heeft ze ge-zie-n de Maori-s.
 Tasman AUX:3SG them GE-see-PTCP DEF Maori-PL
 ‘Tasman saw them, the Maoris.’

In the analysis proposed by Ott and De Vries, *de Maori’s* ‘the Maoris’ is all that remains of a full clause (47), which is juxtaposed to the clause *Tasman heeft ze gezien* ‘Tasman saw them.’ Call this the dislocation clause.

- (47) Tasman heeft de Maori-s ge-zie-n.
 Tasman AUX:3SG DEF Maori-PL GE-see-PTCP
 ‘Tasman saw the Maoris.’

Inside the dislocation clause, *de Maori’s* is fronted before ellipsis takes place:

- (48) [De Maori-s]_i heeft Tasman *t_i* ge-zie-n
 DEF Maori-PL AUX:3SG Tasman GE-see-PTCP

Ellipsis then targets everything to the right of *de Maori’s*, so that the fronted noun phrase is all that remains of the dislocation clause. (46), then, is derived from a biclausal analysis via fronting and deletion in the second clause, as illustrated in (49).

- (49) Remnant ellipsis analysis of Right Dislocation
 Tasman heeft ze ge-zie-n || [[de Maori-s]_i heeft Tasman *t_i*
 Tasman AUX:3SG them GE-see-PTCP DEF Maori-PL AUX:3SG Tasman
 ge-zie-n].
 GE-see-PTCP
 ‘Tasman saw them, the Maoris.’

Jan-Wouter Zwart

Let us take this analysis of Ott & de Vries (1981) to provide the model for ellipsis in the deletion approach to pseudoclefts. That means that between (37b) and (37a), here repeated as (50a) and (50c), there is an additional step of fronting, illustrated in (50b).

(50) English

- a. What John reads $\langle e \rangle$ is he reads BOOKS. (asyndetic construction)
- b. What John reads $\langle e \rangle$ is BOOKS he reads. (fronting)
- c. What John reads $\langle e \rangle$ is BOOKS he reads. (pseudocleft construction)

We will now show that this approach to ellipsis (in general) strengthens the ellipsis analysis of pseudoclefts, in the sense that discrepancies that exist (i) between either the focus XP and the corresponding material in the asyndetic clause, or (ii) between the focus related topic and the corresponding material in the asyndetic clause (i.e. the material to be deleted under identity with non-focus material in the antecedent clause), are resolved after fronting of the focus XP inside the asyndetic clause.

To see what I mean by such ‘discrepancies’, consider again example (44)-(45), here repeated.

(51) Dutch

Wat ik deed is keihard werk-en.
 what I do:PST be:3SG real.hard work-INF
 ‘What I did is work real hard.’

(52) Dutch

* Wat ik deed is || ik deed keihard werk-en.
 what I do:PST be:3SG I do:PST real.hard work-INF
 (intended) ‘What I did is I worked real hard.’

In (45), the focus VP is *keihard werken* ‘work real hard’, and the focus related topic is *ik deed* ‘I did’. The problem is that the asyndetic clause *ik deed keihard werken* ‘I worked real hard’ is ungrammatical in Dutch. Yet the presence of the dummy verb is required by parallelism with the focus related topic material in the antecedent clause (roughly, *I did x*), hence the discrepancy. But after fronting of the focus VP (53), the dummy verb is allowed (in fact, obligatorily present), removing the discrepancy.

(53) Dutch

10 Parasitic participles and ellipsis in VP-focus pseudoclefts

[Keihard werk-en] *(deed) ik.
 real.hard work-INF do:PST I
 ‘I worked real hard.’

Cases like this, where VP-fronting removes discrepancies between the source construction (the asyndetic construction underlying ellipsis) and the target construction (the VP-focus pseudocleft) can be multiplied:

(54) Dutch

- a. VP-focus pseudocleft
 Wat ik hem heb zie-n doe-n is keihard werk-en.
 what I him AUX:1SG see-INF do-INF be:3SG real.hard work-INF
 ‘What I saw him do is work real hard.’
- b. asyndetic ellipsis source
 Ik heb hem keihard zien (*doen) werk-en.
 I AUX:1SG him real.hard see-INF do-INF work-INF
- c. asyndetic ellipsis source with VP-fronting
 Keihard werken heb ik hem zien doen.
 real.hard work-INF AUX:1SG I him see-INF do-INF

In (54), the asyndetic ellipsis source (54b), in addition to containing the ungrammatical dummy verb *doen* ‘do’, features a nonconstituent *keihard werken* ‘work real hard’ as the source for the focus VP in the VP-focus pseudocleft (54a). Fronting of *keihard werken* restores both the dummy verb and the constituent status of the VP.

(55) Dutch

- a. VP-focus pseudocleft
 Wat hij gisteren ge-daan heeft is ’m
 what he yesterday GE-do:PTCP AUX:3SG be:3SG OCL
 op-ge-lad-en.
 up-GE-charge-PTCP
 ‘What he did yesterday is charge it.’
- b. asyndetic ellipsis source
 Hij heeft ’m gisteren op-ge-lad-en.
 he AUX:3SG OCL yesterday up-GE-charge-PTCP
- c. asyndetic ellipsis source with VP-fronting
 ’m op-ge-lad-en heeft hij gisteren.
 OCL up-GE-charge-PTCP AUX:3SG he yesterday

Jan-Wouter Zwart

In (55), the third person object pronoun *'m* is part of the focus VP *'m opgeladen* ‘charged it’, but since the object pronoun is a clitic, it is separated from the verb in the asyndetic ellipsis source (55b), creating a non-constituent VP. Fronting of the VP again restores the constituency, creating a suitable source for the ellipsis (55c).

(56) Dutch

- a. VP-focus pseudocleft
 Wat hij niet ge-daan heeft is een boek ge-lez-en.
 what he NEG GE-do:PTCP AUX:3SG be:3SG INDF book GE-read-PTCP
 ‘What he didn’t do is read a book.’
- b. asyndetic ellipsis source
 Hij heeft geen boek ge-lez-en.
 he AUX:3SG no book GE-read-PTCP
- c. asyndetic ellipsis source with VP-fronting
 Een boek ge-lez-en heeft hij niet.
 INDF book GE-read-PTCP AUX:3SG he NEG

In (56), the asyndetic ellipsis source (56b) contains a negative article *geen* ‘no’, in which the negation *niet* and the indefinite article *een* have been fused. This fused negative article is absent from the focus VP in the target construction (56a). In the asyndetic ellipsis source with VP-fronting (56c), the fused negative article no longer appears.

(57) Dutch

- a. VP-focus pseudocleft
 Wat hij gaa-t prober-en is een boek schrijv-en.
 what he ASP.AUX-3SG try-INF be:3SG INDF book write-INF
 ‘What he is going to try to do is write a book.’
- b. asyndetic ellipsis source
 Hij gaa-t prober-en een boek *(te) schrijven.
 he ASP.AUX-3SG try-INF INDF book to write-INF
- c. asyndetic ellipsis source with VP-fronting
 Een boek schrijv-en gaa-t hij prober-en.
 INDF book write-INF ASP.AUX-3SG he try-INF

In (57), the asyndetic ellipsis source contains the infinitival marker *te* (57b), which is absent from both the target construction (57a) and the VP-fronting construction (57c).

10 *Parasitic participles and ellipsis in VP-focus pseudoclefts*

In all these cases fronting prepares the asyndetic clause for the ellipsis that yields the required focus VP in the pseudocleft construction. The approach to ellipsis proposed in Ott & de Vries (1981), then, improves the ellipsis analysis of VP-focus pseudoclefts with parasitic participles considerably, by removing the first of the two problems noted at the conclusion of Section 6.

However, the remnant ellipsis analysis is not without problems either. These can be listed as follows.

We saw in (54)-(53) that the material making up the focus VP in the pseudocleft construction is scattered (hence a non-constituent) in the asyndetic source clause, and only appears as a constituent after fronting. This creates the desired input for remnant ellipsis, but it raises the question how non-constituent material can be fronted as a single constituent. From this perspective, (54)-(53) are just as much of a problem for the ellipsis analysis as an argument in its favor.

Likewise, it is not clear how negation is defused (56) or the infinitival marker *te* is lost (57) under fronting. These facts showing a discrepancy between fronted and in situ VPs have in fact been taken to support a base-generation analysis of ‘topicalization’ (Weerman 1989). Of course, if a VP can be base-generated in topic position, it can also be base-generated in pseudocleft constructions, and the similarities between the two are in fact predicted.

We may also return to an argument against ellipsis advanced by Georgia (1971) and discussed in Higgins (2003: 56f) (under the rubric of ‘nonexistent deep structure sources’ for deletion).⁸ Here we adapt the argument to the domain of VP-focus pseudoclefts.

(58) Dutch

Wat we aan dat probleem ge-daen heb-ben is veel
 what we about that problem GE-do:PTCP AUX-PL be:3SG much
 vergader-d.
 meet-PTCP

‘What we did about that problem is hold a lot of meetings.’

As expected in the remnant ellipsis analysis, the *about*-constituent cannot appear together with the focus-VP without fronting (the PP is ungrammatical in any position):

⁸This argument questions the derivation via ellipsis of examples like (i), as the asyndetic source clause underlying the ellipsis (ii) is ungrammatical.

(i) What I like about John is his sense of humor

(ii) *I like his sense of humor about John

Jan-Wouter Zwart

- (59) Dutch
 We heb-ben (*aan dat probleem) veel vergader-d.
 we AUX-PL about that problem much meet-PTCP
 ‘We held a lot of meetings.’

But here fronting does not remedy the situation:

- (60) Dutch
 Veel vergader-d heb-ben we (*aan dat probleem).
 much meet-PTCP AUX-PL we about that problem
 ‘We held a lot of meetings.’

We have not explicitly addressed the derivation of the asyndetic source clause (‘elliptic clause’, in Ott & de Vries 1981), but the logic would seem to be that the material of the asyndetic source clause is copied from the antecedent clause, with no exceptions predicted.

An imperfect match between the antecedent clause and the asyndetic source clause is also in evidence with discourse particles:

- (61) Dutch
 Wat hij nou wil is componist word-en.
 what he PRT want:SG BE:3SG composer become-INF
 ‘Well, what he wants is to become a composer.’

Nou ‘now’ in (61) is a discourse particle that guides the flow of the argument, announcing the main point or a return to the main line of argumentation. Such discourse particles cannot very well appear in the asyndetic source clause, regardless of fronting of the focus-VP:⁹

- (62) Dutch
- a. Hij wil (??nou) componist word-en.
 he want:SG PRT composer become-INF
 - b. Componist word-en wil hij (??nou).
 composer become-INF want:SG he PRT

Another problem is that negative polarity items (NPIs) in Dutch do not like to be fronted. In (63), the NPI is *ook maar iets*, an indefinite marked by the minimizer *ook maar*:

⁹It is not clear to me what causes the degradation in these examples, but the point is that *nou* works well in pseudoclefts and less well in these regular constructions.

10 Parasitic participles and ellipsis in VP-focus pseudoclefts

- (63) a. *Geen* STUDENT wil ook maar IETS voorbereid-en.
 NEG.INDF student want:SG MINIM anything prepare-INF
 ‘No student wants to prepare the slightest thing.’
 b. * *Ook* maar IETS voorbereid-en wil geen STUDENT.
 MINIM anything prepare-INF want:SG NEG.INDF student

In (63), the NPI apparently needs to remain within the c-command domain of the negative subject *geen student* ‘no student’, so that fronting of the focus-VP *ook maar iets voorbereiden* ‘prepare anything’ is blocked. But the same focus-VP can appear in a pseudocleft without losing the negative polar (minimizing) interpretation:

- (64) Dutch
 Wat geen STUDENT wil is ook maar IETS
 what NEG.INDF student want:SG be:3SG MINIM anything
 voorbereid-en.
 prepare-INF
 ‘What no student wants is to prepare the slightest thing.’

The biclausal analysis, with fronting and remnant ellipsis in the asyndetic source clause, is unable to derive the pseudocleft in (64).

It appears, then, that while the remnant deletion analysis of ellipsis achieves some improvement over traditional approaches to ellipsis, it cannot derive all cases of VP-focus pseudocleft constructions, and it creates some new problems all its own. This adds to the other problems of the ellipsis analysis, already noted:

- (65) problems of the ellipsis analysis of VP-focus pseudoclefts with parasitic participles
- the ellipsis analysis cannot be generalized to other pseudoclefts, not involving parasitic participles.
 - the ellipsis analysis relies on an asyndetic source construction which has a marked character absent from the pseudocleft construction.
 - the asyndetic source construction must arguably be derived via derivation layering, removing much of the intuitive appeal of an ellipsis approach.

In the final section of the paper we elaborate on (65c).

Jan-Wouter Zwart

7 A ‘base-generation’ analysis

As an alternative to ellipsis, we may consider a base-generation analysis, in which the focus XP is created in a separate derivation layer (cf. Zwart 2009) and is merged as a single element in the B position (recalling the A copula B structure of pseudoclefts). This base-generation analysis can be applied to both specificational and predicational pseudoclefts (as well as *it*-clefts), thus generalizing over these constructions as far as the derivation of the B-part is concerned. Moreover, we may take the discrepancies noted above between in situ and fronted VPs to suggest that ‘topicalization’ of VPs is likewise not the result of movement but of base-generating in left-peripheral position the same constituent that is also merged in clefts and pseudoclefts.

The basic principle of layered derivations, a variant of the Multiple-Spell Out idea of Uriagereka (1999), is that a derivation is typically a network of derivations, each consisting of the three components of any minimalist derivation: (i) a Numeration (of elements to be merged), (ii) Merge (i.e. Narrow Syntax), (iii) Externalization (i.e. the interface components dealing with sound, including morphology, and meaning). Idiosyncratic properties of form and meaning are determined at the Externalization stage of each (sub-)derivation, which can lead to a clause or verb phrase being reanalyzed as a noun phrase before being included in the Numeration of another derivation (see Zwart 2009 for more discussion).

Both ellipsis and ‘base-generation’ of VP-focus pseudoclefts can be described in terms of a layered derivations architecture. In the case of ellipsis, a clause is derived in a separate derivation layer, and reinterpreted as a noun phrase (a nominal infinitive) before being included in the Numeration for the next derivation. In the case of ‘base-generation’, the separate derivation yields not a clause but a VP, which is likewise reinterpreted as a noun phrase before being included in the Numeration for the next derivation. The difference between ellipsis and ‘base-generation’, then, would be that ellipsis requires derivation of a clause *cum* fronting and ellipsis, whereas ‘base-generation’ requires derivation of a VP without any further operations.

From the base-generation perspective, the difference between VP-focus pseudoclefts with and without parasitic participles (cf. (1)) is purely morphological: the VP that is created in a separate derivation and included in the Numeration for the next derivation as a noun phrase gets spelled out as a participle in some situations, but not in others. The success of the analysis depends on how easily these situations can be defined.

Recall that past participle morphology in Dutch occurs only in the presence of a temporal auxiliary (*have/be*) or the passive auxiliary *worden* ‘become’. (I ig-

10 *Parasitic participles and ellipsis in VP-focus pseudoclefts*

nore the passive in what follows.) We may assume, for the time being, that the morphology reflects a dependency of the participle on the auxiliary. Let us say that the auxiliary assigns a feature to the verb which is spelled out as participial morphology. This is surely an oversimplification (cf. Zwart 2017), but it will probably do for now.¹⁰

We may also assume, uncontroversially, that the relevant dependency requires a relation of c-command, such that the auxiliary (or the functional head with which it is associated) c-commands the verb (and α c-commands δ iff δ is [contained in] the sister of α , making c-command a function of Merge; cf. Epstein 1999). As I've argued elsewhere (Zwart 2017), the feature relevant to past participle morphology in Dutch is anteriority (relative tense), and I adopt here the structure of the clause proposed by Wiltschko (2014), roughly as in (66), where ANCHORING and POINT OF VIEW are functional heads in the clausal spine, representing classical T(ense) and Asp(ect) heads, respectively.

(66) [subject ANCHORING (tense) [POINT OF VIEW (anteriority) [VP]]]

In the pseudocleft constructions we have seen so far, the *wh*-clause in the A-position includes the structure in (66), preceded by an additional projection hosting the *wh*-element (presumably CP), and with the variable $\langle e \rangle$ contained in VP:

(67) [CP *what* [subject ANCH [POV [VP V $\langle e \rangle$]]]]
[+ant]

It follows that POV c-commands $\langle e \rangle$ and may assign the feature [+anterior] to it. We may then hypothesize that the feature [+anterior] is replicated on the focus-VP by its association with the variable $\langle e \rangle$.

Obviously the feature [+anterior] is also assigned to V, as are the tense and agreement features, and this entire complex of features is spelled out as *heb gedaan* in (1). But as tense and agreement are never realized on the focus-VP in pseudoclefts, we have no reason to suppose that the features [tense] and [agreement] are assigned to $\langle e \rangle$ as well.

The analysis predicts that when the variable $\langle e \rangle$ is outside the scope of (i.e. not c-commanded by) POV, the focus-VP will never show parasitic participle morphology. This prediction is correct, as can be seen when the focus-VP is associated with a variable in subject position:

¹⁰As I argue in Zwart (2017), the auxiliary-participle combination is best viewed as a periphrastic morphological realization of a single terminal node V, spelling out the feature [ANTERIOR] (among others).

Jan-Wouter Zwart

(68) Dutch

Wat<e> hem ge-nek-t heeft is keihard { werk-en /
 what him GE-neck-PTCP AUX:3SG is:3SG realhard work-INF
 *ge-werk-t }.
 GE-work-PTCP
 ‘What killed him was to work real hard.’

It seems, then, that the parasitic participle in focus-VP pseudoclefts is limited to constructions where the variable <e> is in the scope domain of the functional element responsible for the participial morphology.

Note that this generalization linking parasitic participial morphology to the scope relation between POV and <e> cannot be achieved in an ellipsis analysis (without or without fronting). As (69) shows, formation of the asyndetic construction on which ellipsis supposedly operates to produce the pseudocleft is also possible when the variable appears in subject position:

(69) Dutch

Wat<e> hem ge-nek-t heeft is || hij heeft keihard
 what him GE-neck-PTCP AUX:3SG is:3SG he AUX:3SG real.hard
 ge-werk-t.
 GE-work-PTCP
 ‘What killed him was he worked real hard.’

Instead, on the ellipsis analysis the parasitic participle in (68) must be excluded by recourse to additional conditions on ellipsis, and it is unclear what those would be.

We now need to restrict the occurrence of parasitic participles in VP-focus pseudoclefts to situations where the VP in the antecedent clause contains a dummy verb. Recall that when V = *willen* ‘want’ or *proberen* ‘try’, the parasitic participle is excluded (20). In these constructions, the feature [+anterior] must be present, as *willen* and *proberen* show participial morphology. Yet, the parasitic participle cannot appear:

(70) Dutch

a. Wat ik heb ge-wil-d is keihard { werk-en /
 what I AUX:1SG GE-want-PTCP be:3SG real.hard work-INF
 *ge-werk-t }.
 GE-work-PTCP
 ‘What I wanted is to work real hard.’

10 *Parasitic participles and ellipsis in VP-focus pseudoclefts*

- b. Wat ik heb ge-probeer-d is keihard { werk-en /
 what I AUX:1SG GE-try-PTCP be:3SG real.hard work-INF
 *ge-werk-t }.
 GE-work-PTCP
 ‘What I tried is to work real hard.’

Apparently, the feature [+anterior] is only associated with the variable $\langle e \rangle$ when the feature derives from a dummy verb (cf. (18), where the dummy verb lacks the feature [+anterior]), and the parasitic participle does not appear).

This suggests that $\langle e \rangle$ gets the feature [+anterior] not from POV directly, but indirectly, via the dummy verb. I will assume that it is somehow in the nature of a dummy verb to (optionally) share relevant features with the empty category.¹¹

The base-generation analysis of parasitic participle VP-focus pseudoclefts, then, requires:

(71) Summary of analysis

- a. the focus VP is associated with $\langle e \rangle$.
- b. the dummy verb optionally shares the feature [+anterior] with $\langle e \rangle$.

This predicts that when the focus VP is not associated with $\langle e \rangle$, parasitic participle morphology will not show up. Precisely this situation obtains with predicational pseudoclefts, where the focus VP is not interpreted in the position of $\langle e \rangle$, but is interpreted as a predicate of the referent of the *wh*-clause.

The analysis posits a particular relation of ‘association’ between the variable $\langle e \rangle$ and the parasitic participle. Space precludes a further discussion of the nature of this association relation, but see Heycock & Kroch (1999), Sharvit (1999) for an analysis of this relation in semantic terms. Den Dikken (2005: 313f) discusses a number of tests suggesting that the focus XP in specificational pseudoclefts is interpreted in the position of the variable $\langle e \rangle$ in the antecedent clause. The relevant observations are often taken to support an ellipsis analysis of specificational pseudoclefts (e.g. Den Dikken et al. 2000), but they are also predicted by the association relation between the focus XP and the variable, provided we state:

(72) Specificational pseudocleft

The focus XP is interpreted at the position of the variable in the *wh*-clause.

¹¹The analysis would be simplified considerably were we to assume that the dummy verb is just an empty V-position in Narrow Syntax, creating a contrast with non-empty V positions such as those occupied by *want* and *try* in (20), which may be thought of as absorbing the [anterior] feature rather than sharing it with $\langle e \rangle$.

Jan-Wouter Zwart

As we have seen, it is a defining property of specificational pseudoclefts that the focus XP provides the value for the variable in the *wh*-clause. This property now follows directly from (72).

It remains to discuss the particulars of the morphological realization of the focus VP.

The tense system of Dutch can be characterized by the features *tense* (ANCHORING) and *anteriority* (POINT OF VIEW), such that [−anterior] expresses simultaneity with a temporal reference point, the location of which depends on the value for *tense* (present = the here and now, past = prior to the here and now). Likewise [+anterior] expresses that the event referred to takes place prior to the reference point, i.e. prior to the present or prior to the past (see Verkuyl 2008, Zwart 2017). The [−anterior] tenses are morphologically realized by the present (when *tense* = present) and the simple past (when *tense* = past), the [+anterior] tenses by the periphrastic past, i.e. the past participle in combination with a temporal auxiliary (*have/be*), the tense marking of which determines that the reference point is in the present (when *tense* on the auxiliary = present) or in the past (when *tense* on the auxiliary = past).

As I have argued in Zwart (2017), there is no need to assume that the auxiliary has a syntactic position: narrow syntax produces constituents and terminals with certain feature specifications, and the terminal V of the verb phrase VP can have any of eight feature combinations, if we limit ourselves to agreement, *tense*, and *anteriority*:

(73) Feature combinations of V

- | | |
|---------------------------------|---------------------------------|
| a. [+agr, +tense, −anteriority] | e. [−agr, +tense, −anteriority] |
| b. [+agr, +tense, +anteriority] | f. [−agr, +tense, +anteriority] |
| c. [+agr, −tense, −anteriority] | g. [−agr, −tense, −anteriority] |
| d. [+agr, −tense, +anteriority] | h. [−agr, −tense, +anteriority] |

Agreement in Dutch requires *tense*, so (70c,d) are not attested. But the remaining six feature combinations have morphological exponents (some even have many, depending on the values for *tense* and agreement—we will illustrate the two values for *tense* only):

(74) Dutch verbal morphology

verb = *werken* ‘work’, person = 3SG

- a. [+agr, +tense, −anteriority]

10 Parasitic participles and ellipsis in VP-focus pseudoclefts

tense = present	werk-t
tense = past	werk-te
b. [+agr, +tense, +anteriority]	
tense = present	heeft ge-werk-t
tense = past	had ge-werk-t
e. [-agr, +tense, -anteriority]	
tense = present	te werk-en
tense = past	te heb-ben ge-werk-t
f. [-agr, +tense, +anteriority]	
tense = present/past	te heb-ben ge-werk-t

The combination [-agr,+tense] yields the set of infinitives, which are expressed periphrastically as soon as tense = past or anteriority is positive. That tense is relevant in infinitives is clear from the morphological adjustment that takes place when a reference point in the past is made explicit, and the event is understood to be simultaneous with that reference point in the past. Compare finite (75) and nonfinite (76), where *toen ik binnenkwam* ‘when I came in’ provides the reference point in the past.

(75) Dutch, finite

- a. Hij beweert [dat hij werk-t].
 he claim-3SG c he work-3SG
 ‘He claims that he is working.’
- b. Hij beweert [dat hij werk-te toen ik binnen kwam].
 he claim-3SG c he work-PAST.3SG when I in come:PAST.SG
 ‘He claims that he was working when I came in.’

(76) Dutch, nonfinite

- a. Hij beweert [te werk-en].
 he claim-3SG INF work-INF
 ‘He claims to be working.’
- b. Hij beweert [ge-werk-t te heb-ben toen ik binnen kwam].
 he claim-3SG GE-work-PTCP INF AUX-INF when I in
 come:PST.SG
 ‘He claims to be have been working when I came in.’

Jan-Wouter Zwart

The explicit reference point in the past, with which the event is simultaneous, forces a morphological adjustment of the verb in both the finite and the nonfinite paradigm, yielding a simple past *werkte* ‘worked’ in the finite paradigm and a periphrastic infinitive *gewerkt te hebben* ‘to have worked’ in the nonfinite paradigm. (This shows that the characterization of infinitives as tenseless categories, as found e.g. in Wurmbrand 2001, needs to be refined; see also Stowell 1993, Ter Beek 2008.)

This leaves two feature combinations unaccounted for, but these, too, have their exponents:

(77) Dutch verbal morphology, continued

verb = *werken* ‘work’

- | | |
|---------------------------------|------------------|
| g. [–agr, –tense, –anteriority] | werk-en |
| h. [–agr, –tense, +anteriority] | ge-werk-t |

Werken (77g) is the unmarked verb form, the bare infinitive, not expressing any features. This leaves (77h) as the remaining logically possible feature combination, marked by pure anteriority. Given the interdependence between anteriority and tense one expects this feature combination to have only a limited use. But if we are correct in this article, the position inside the focus VP in specificational pseudoclefts is precisely where this morphological exponent can be attested.

Recall that our analysis contains the two elements in (71), repeated here:

(78) Summary of analysis

- a. the focus VP is associated with $\langle e \rangle$.
- b. the dummy verb optionally shares the feature [+anterior] with $\langle e \rangle$.

It follows that the focus VP, and hence its head V, is characterized by the feature combination (77h), i.e. pure anteriority. We suggest that this explains the optional occurrence of the parasitic participle in specificational VP-focus pseudoclefts, as well as its absence elsewhere.

7.1 Conclusion

I have argued that the main properties of VP-focus pseudoclefts (in Dutch) can be explained without recourse to ellipsis, if we assume a model of grammar in which derivations are layered (i.e. where the output of a derivation may enter the Numeration of the next derivation as a single item). Specificational pseudoclefts require that the focus VP (in VP-focus pseudoclefts) be associated with

10 Parasitic participles and ellipsis in VP-focus pseudoclefts

the variable in the antecedent clause. The parasitic participle in VP-focus pseudoclefts derives its morphology from a feature [+anterior] which the focus VP inherits from the variable in the antecedent clause (which in turn receives the feature [+anterior] from a functional head in the antecedent clause, mediated by the dummy verb. The analysis derives the fact that VP-focus pseudoclefts with parasitic participles are uniquely specificational, not predicational.

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Jan-Wouter Zwart

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10 *Parasitic participles and ellipsis in VP-focus pseudoclefts*

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Name index

- Abels, Klaus, 240, 301, 302, 364
Ackema, Pete, 145, 337
Ackema, Peter, 140, 337
Ackermann, Tanja, 136, 139
Adger, David, 13, 51
Alexiadou, Artemis, 163, 169, 315
Algeo, John, 237
Allan, Keith, 201
Allen, Cynthia, 79, 281
Anderson, Curt, 41
Anna, Roussou, 201
Anttila, Arto, 145, 146
Appel, Heinz-Winfried, 223
Arsenijević, Boban, 12
Askedal, John Ole, 157
Austin, John L., 119
Axel, Katrin, 212, 329
Axel-Tober, Katrin, 14, 16, 20, 33, 35
- Bach, Emmon, 361, 365
Bacsikai-Atkari, Julia, 44
Bader, Markus, 218
Baptista, Marlyse, 316
Bárány, András, 324
Barera, Anna, 332, 334, 335
Barwise, Jon, 119
Bassighini, Tamara, 332, 343
Baumgärtner, Klaus, 223
Baunaz, Lena, 14, 104
Bayer, Josef, 6, 201
Beekes, Robert S. P., 222
Behagel, Otto, 297
- Behaghel, Otto, 28, 52, 102
Belvin, Robert, 237
Benincà, Paola, 115
Bennis, Hans, 53
Berg, Kristian, 136
Berger, Claudia Bucheli, 13, 51
Berger, Dieter, 241
Bergh, Gunnar, 80
Bernolet, Sarah, 315
Bettelou, Los, 207
Betty, Birner, 185
Bhatt, Rakesh, 118
Biberauer, Theresa, 85, 185, 202, 203, 207, 210, 217
Bidese, Ermenegildo, 112, 113, 118
Bietenhard, Ruth, 288
Birgit, Alber, 332
Bjorkman, Bronwyn, 164
Bjorkman, Bronwyn M., 324
Blix, Hagen, 299
Blom, Alied, 354, 366
Blümel, Andreas, 215
Boeckx, Cedric, 72
Boef, Eefje, 15, 25, 314
Bohn, Isabella, 327
Bolinger, Dwight, 201
Bonet, Eulàlia, 141
Booth, Hannah, 132, 134, 136, 160, 184, 201, 284
Borer, Hagit, 298
Bowers, John, 13, 45
Boye, Kasper, 14

Name index

- Brandner, Ellen, 13, 20, 27, 51, 55, 132, 134, 363
- Brandt, Patrick, 182, 222
- Bräuning, Iris, 27
- Breckenridge, Janet, 201
- Breindl, Eva, 245, 246
- Breivik, Leiv, 182
- Breivik, Leiv Egil, 90, 91
- Bresnan, Joan, 162
- Brugmann, Karl, 193
- Buchholz, Oda, 40
- Bülow, Lars, 332, 333, 345
- Butler, Milton, 85
- Cable, Seth, 364, 365
- Calabrese, Andrea, 118
- Cardinaletti, Anna, 246
- Carstens, Vicki, 331
- Chafe, Wallace L., 220
- Charters, Helen, 162
- Chaves, Rui, 96
- Cheshire, Jenny, 58
- Chomsky, Noam, 2, 69–75, 78, 84, 201, 204–207, 215, 331, 332
- Cinque, Guglielmo, 38, 44, 45, 58, 299
- Cognola, Federica, 20
- Coniglio, Marco, 3, 102–107, 111, 113, 118, 207
- Corr, Alice, 201
- Corver, Norbert, 163
- Cowper, Elizabeth, 164
- Craenenbroeck, Jeroen van, 201
- Curme, George O., 187, 191
- D’Arcy, Alexandra, 58
- Daalder, Saskia, 354, 366
- Dagnac, Anne, 112
- Dahl, Östen, 135
- Dal, Ingerid, 28
- Dalrymple, Mary, 162
- Damköhler, Eduard, 286
- Damonte, Federico, 118
- Dayal, Veneta Srivastav, 55
- De Bastiani, Chiara, 103
- De Cuba, Carlos, 12, 30
- de Vries, Mark, 354, 369, 370, 373, 374
- Deal, Amy Rose, 72
- Déchaine, Rose-Marie, 302
- Dekeyser, Xavier, 17
- Delsing, Lars-Olof, 135, 285
- Demske, Ulrike, 132
- den Besten, Hans, 84
- Den Dikken, Marcel, 33, 51, 148, 354, 357, 362, 366, 379
- den Dikken, Marcel, 237
- Diercks, Michael, 324
- Diesing, Molly, 209
- Doherty, Cathal, 76
- Donhauser, Karin, 102
- Drubig, Bernhard H., 209
- Duden, 218
- Ebert, Robert P., 28–30
- Elliott, Patrick, 11, 12, 39–41, 43
- Elspaß, Stephan, 245
- Embick, David, 298
- Emonds, Joseph, 238
- Epstein, Samuel, 377
- Erben, Johannes, 191
- Erdmann, Oskar, 28, 49, 187
- Erlewine, Michael Yoshitaka, 72
- Esther, Torrego, 212, 213
- Faarlund, Jan T., 201
- Faarlund, Jan Terje, 14, 15, 34
- Fábregas, Antonio, 300
- Falk, Hjalmar, 285

Name index

- Fanselow, Gisbert, 201, 202, 207
 Faraci, Mary, 102
 Farkas, Donka, 3, 106
 Ferraresi, Gisela, 29
 Feyer, Ursula, 242
 Fischer, Olga, 85, 86, 283
 Fisher, Rose, 295, 300, 312, 313, 315, 316
 Fitzpatrick, Justin M., 263
 Fleischer, Jürg, 5, 20, 132, 136, 241–245, 284–287, 297, 306, 307, 342
 Fong, Vivienne, 145, 146
 Fort, Marron C., 285
 Francez, Itamar, 148
 Franco, Ludovico, 12, 15, 25, 131, 134, 147–149
 Freeze, Ray, 170
 Frey, Werner, 188, 197, 202, 207, 210, 213, 218, 220, 268, 269, 273, 274, 277
 Freywald, Ulrike, 245
 Fricke, Lea, 198, 199
 Fricke, Leah, 198
 Frings, Theodor, 33, 35
 Fukui, Naoki, 79
 Fuß, Eric, 152, 182, 193, 194, 196, 203, 206, 216, 222, 331, 337, 338
 Gallmann, Peter, 142, 240, 302
 Gámez, Perla B., 315
 Gärtner, Hans-Martin, 296
 Gavruseva, Elena, 163
 Georgi, Doreen, 131, 132, 163
 Georgia, Green, 373
 Gert-Jan, Schoenmakers, 212
 Ghomeshi, Jila, 147
 Giannakidou, Anastasia, 3, 106, 107, 118
 Gillian, Ramchand, 13, 51
 Gillon, Carrie S., 314
 Giorgi, Alessandra, 120
 Gippert, Jost, 102
 Goeman, Ton, 328
 Goldsmith, John, 140
 González i Planas, Francesco, 112
 Goto, Nobu, 2, 71–73, 79
 Götze, Lutz, 241
 Grewendorf, Günther, 118, 188, 201, 206, 207, 213, 269
 Greyerz, Otto von, 288
 Griffiths, James, 237, 238
 Grimshaw, Jane, 76
 Grohmann, Kleonthes, 72
 Gunkel, Lutz, 36
 Haag, Earl C., 21–23, 27, 37
 Hachem, Mirjam, 295, 302, 305, 314
 Haeberli, Eric, 87
 Haegeman, Liliane, 131, 132, 163, 172, 194, 201, 206
 Haider, Hubert, 82, 85, 147, 189, 204, 213, 217, 240, 268
 Halila, Hafedh, 79
 Harbert, Wayne, 14, 16, 24, 34, 49
 Harbour, Daniel, 210
 Härd, John Evert, 157, 158
 Harley, Heidi, 210, 238
 Hartmann, Jutta, 201
 Hartmann, M., 36
 Hartsuiker, Robert J., 315
 Harves, Stephanie, 238
 Haspelmath, Martin, 40
 Hazout, Ilan, 201
 Heck, Fabian, 201, 202, 212, 269
 Heim, Irene, 39, 44, 209, 314
 Heine, Bernd, 14, 25
 Heltoft, Lars, 14

- Henneberg, Luise Magdalena, 260
Herrmann, Tanja, 12
Hess-Lüttich, Ernest W. B., 241
Heycock, Caroline, 379
Higgins, Francis R., 354, 366, 373
Himmelreich, Anke, 269
Hinterhölzl, Roland, 84, 102–104, 106, 107, 111, 113, 118, 119, 186, 193, 194, 196, 203, 206, 216, 222, 223
Hinzelin, Marc-Olivier, 201
Hiraiwa, Ken, 143, 364
Hoekstra, Eric, 362
Hoekstra, Jarich, 286
Hoekstra, Jarich F., 285
Hohaus, Vera, 40
Höhle, Tilman, 196, 208, 213
Holmberg, Anders, 187, 201, 202, 206, 207
Holthausen, Ferdinand, 327
Hopper, Paul, 33
Horvath, Julia, 207
Ihden, Sarah, 29
Ishii, Toru, 2, 71–73, 79
Jacobs, Joachim, 220
Jacobs, Neil G., 24
Jäger, Agnes, 19, 28, 31
Jenkins, Lyle, 201
Jensen, Ella, 285, 286
Jespersen, Otto, 76
Johnk, Linn D., 33
Johnson, Kyle, 263
Jónsson, J. G., 203
Jung, Walter, 191
Jürgens, Carolin, 245
Kaiser, Elsi, 201
Kandybowicz, Jason, 363, 364
Kaplan, Ron, 162
Karimi, Yadgar, 148
Kasper, Simon, 131, 132, 136, 170
Kathol, Andreas, 328
Kayne, Richard, 77, 170, 299
Kayne, Richard S., 12, 14, 59, 238
Keiser, Steven Hartman, 296
Kiefer, Heinrich, 136, 170
Kimball, John P., 238
Kiparsky, Carol, 24
Kiparsky, Paul, 24
Kiss, Katalin É., 26
Kiziak, Tanja, 96
Klein, Thomas, 103
Klein, Wolfgang, 119
Köbler, Gerhard, 19
Kock, Ernst Albin, 29
Koeneman, Olaf, 363
Kokkelmans, Joachim, 246
Kollmann, Cristian, 340
Komen, Erwin, 103
Komen, Erwin R., 103
König, Werner, 329, 333, 334
Koontz-Garboden, Andrew, 148
Koopman, Hilda, 240, 298, 363
Kornacki, Pawel, 25
Kratzer, Angelika, 11, 39, 44, 119
Krause, Thomas, 102
Krifka, Manfred, 208
Kroch, Anthony, 379
Kuroda, Shige-Yuki, 79
Kuteva, Tania, 14, 25
Lakoff, George, 201
Landau, Idan, 364
Landemann, Willy, 246
Langacker, Ronald, 201
Lange, Klaus-Peter, 359

Name index

- Larson, Richard, 147–149
 Larson, Richard K., 364
 Larsson, Ida, 24, 25
 Lasch, Agathe, 157
 Lazard, Gilbert, 147
 Leben, William Ronald, 139, 140
 Ledgeway, Adam, 112, 118
 Lefebvre, Claire, 26, 364
 Legate, Julie A., 26
 Lehmann, Winfred P., 16
 Lekakou, Marika, 143
 Lenerz, Jürgen, 188, 190, 201, 208, 213
 Leu, Thomas, 132, 295, 301
 Leu, Tom, 20
 Liefke, Kristina, 26
 Lindauer, Thomas, 139, 163
 Lindberg, Conrad, 93
 Lindow, Wolfgang, 241
 Löbel, Elisabeth, 132
 Lockwood, William Burley, 283
 Lohndal, Terje, 298, 315
 Longbotham, James, 285
 Longobardi, Giuseppe, 16, 34
 López, Luis, 315
 Loranger, Virginie, 26
 Los, Bettelou, 33, 85, 87
 Los, Bettelou, 95
 Louden, Mark L., 296
 Lühr, Rosemarie, 14, 32, 33, 102, 108

 M. Teresa, Espinal, 201
 MacWhinney, Brian, 131, 140, 141, 143
 Mahajan, Anoop, 201
 Maling, Joan, 79
 Manzini, M. Rita, 12
 Marcin, Morzycki, 41
 Mascarenhas, Salvador, 112
 Massam, Dianne, 93

 Mayerthaler, Evi, 222
 Mayerthaler, Willi, 222
 McCawley, James, 201
 McDaniel, Dana, 51, 54
 McIntyre, Andrew, 237
 Meinunger, André, 209, 210
 Melinger, Alissa, 315
 Menn, Lise, 131, 140, 141, 143
 Meyerhoff, Miriam, 25
 Miller, D. Garry, 15, 16, 31, 34
 Miller, Kimberly A., 284
 Milsark, Gary, 185
 Mitchell, Bruce, 81, 82, 102, 108, 110
 Miyagawa, Shigeru, 205–207, 212, 213
 Mohanan, Karuvannur P, 140, 143
 Mohanan, Tara, 131, 140, 143, 145, 146
 Mohr, Sabine, 201, 202
 Möller, Robert, 245
 Moltmann, Friederike, 11, 42
 Moro, Andrea, 201
 Moulton, Keir, 11
 Müller, Gereon, 37, 201, 202, 212, 240, 241, 283, 310
 Müller, Gertraud, 33, 35
 Müller, Kalle, 16
 Muller, Robert, 287
 Munaro, Nicola, 112
 Myler, Neil, 238

 Natvig, David, 295, 300, 315, 316
 Neeleman, Ad, 134, 140, 141, 143–146, 337
 Negele, Michaela, 245, 297
 Nevalainen, Terttu, 86
 Nevins, Andrew, 4, 131, 134, 140–144, 146
 Newmeyer, Frederick J., 186
 Nicholas, Nick, 26

- Nikolas, Gisborne, 17
 Noonan, Máire, 240, 295, 298, 301, 303–305
 Norde, Muriel, 14, 132, 134
 Nordström, Jackie, 14
 Noyer, Rolf, 210, 298
 Nye, Rachel, 26

 Obata, Miki, 96
 Olsen, Susan, 132, 163
 Oppenrieder, Wilhelm, 245
 Östermann, Aune, 283
 Ott, Dennis, 354, 369, 370, 373, 374
 Otte-Ford, T. Clinton, 245, 297
 Ouali, Hamid, 212

 Padovan, Andrea, 112, 118
 Panieri, Luca, 113
 Pankau, Andreas, 56
 Paoli, Sandra, 112
 Paul, Hermann, 28, 30, 283, 284, 297
 Perez-Cortes, Silvia, 315
 Perlmutter, David, 140
 Perry, John, 119
 Perry, John R., 147
 Pesetsky, David, 212, 213, 264, 281
 Peter, Suchsland, 201
 Peters, Stanley, 361, 365
 Petrova, Svetlana, 102–104, 106, 107, 111, 113–115, 118
 Pianesi, Fabio, 120
 Piepers, Joske, 138
 Pigarella, Giorgia, 332, 340, 341
 Pintzuk, Susan, 87
 Pittner, Karin, 37, 50, 170
 Plag, Ingo, 140, 143
 Platzack, Christer, 85, 201
 Plesničar, Vesna, 112
 Poletto, Cecilia, 59, 118, 324, 363

 Polinsky, Maria, 301
 Postal, Paul M., 302
 Preminger, Omer, 210, 212, 214
 Pretorius, Erin, 298
 Puskás, Genoveva, 104
 Putnam, Michael T., 300
 Putnam, Michael T., 299, 315
 Putnam, Mike, 96
 Pütz, Herbert, 38, 183, 191, 192, 215, 220

 Quer, Josep, 106
 Quirk, Randolph, 76

 Rabanus, Stefan, 342
 Radford, Andrew, 76, 143
 Rauch, Irmengard, 102
 Rauth, Philipp, 152
 Redl, Theresa, 138
 Rehn, Alex, 136, 160
 Rehn, Alexandra, 143
 Reichenbach, Hans, 119
 Reis, Marga, 309
 ReN-Team, 132, 134
 Renn, Manfred, 329, 333, 334
 Rett, Jessica, 40
 Richards, Marc, 78, 85, 185, 202, 203, 217, 339
 Richards, Norvin, 131, 134, 140–143, 146, 147, 149, 167
 Richter, Helmut, 327
 Ritter, Elizabeth, 210, 238, 305
 Rivero, María Luisa, 118
 Rizzi, Luigi, 57, 74, 76, 84, 115, 213, 299
 Roberts, Ian, 14, 33, 35, 201, 206
 Robinson, Fred C., 102, 108, 110
 Roehrs, Dorian, 131, 132, 154, 160–163, 172

Name index

- Rögnvaldsson, Eiríkur, 188, 201
 Rosen, Sara Thomas, 238
 Ross, John, 201
 Rothmund, Walter, 223
 Roussou, Anna, 12, 14, 33, 35, 58, 118
 Rowley, Anthony, 20
 Rugna, Giuseppe, 56
 Russ, Charles V. J., 284
- Sabel, Joachim, 188, 213
 Sailor, Craig, 237, 238
 Saltveit, Laurits, 246
 Salzmänn, Martin, 13, 131, 132, 163
 Samiian, Vida, 148
 Samvelian, Pollet, 148
 Sánchez, Liliana, 315
 Sanfelici, Emanuela, 59, 324
 Sapp, Christopher D., 15, 49, 50
 Savoia, Leonardo M., 12
 Schallert, Oliver, 132, 136
 Schiepek, Josef, 325, 326
 Schippers, Annelien, 51, 54
 Schirer, Kasper, 263
 Schönhoff, Hermann, 223
 Schoorlemmer, Maaike, 132, 137
 Schrodt, Richard, 102
 Schütze, Carson T., 5, 238, 248, 267
 Scontras, Gregory, 315
 Sells, Peter, 201
 Seppänen, Aimo, 80
 Sharvit, Yael, 379
 Shlonsky, Uri, 76
 Sigurðsson, Halldór, 185, 188, 200–202, 206
 Simeonova, Vesela, 11
 Singhapreecha, Pornsiri, 148
 Sluckin, Benjamin L., 194, 222
 Speyer, Augustin, 188, 197, 212
 Spiekermann, Helmut, 297
- Sprouse, Jon, 255
 Starke, Michal, 298, 315
 Stassen, Leon, 170
 Sternefeld, Wolfgang, 37, 183, 191
 Stockwell, Richard, 5, 238, 246, 248, 267
 Stowell, Tim, 382
 Stowell, Timothy, 119, 120
 Streube, Anita, 37
 Stroh-Wollin, Ulla, 12
 Struik, Tara, 212
 Strunk, Jan, 152, 153, 155, 157, 159, 160, 162
 Stübs, Hugo, 242
 Sudhoff, Stefan, 37, 38, 183, 215
 Sugimoto, Yushi, 316
 Svenonius, Peter, 182, 239, 299
 Swan, Michael, 237
 Swantje, Tönnis, 198
 Szendrői, Kriszta, 143
- Tagliamonte, Sali, 58
 Tancredi, Christopher, 365
 Tanja, Milicev, 207
 Taraldsen, Knut Tarald, 24, 25
 Taube, Moshe, 24
 Taylor, Ann, 103
 Ter Beek, Janneke, 382
 Terzi, Arhonto, 298
 Thráinnsson, Höskuldur, 186, 201
 Thun, Harald, 240
 Tomaselli, Alessandra, 112, 118, 201, 223
 Toosarvandani, Maziar, 148, 149
 Torp, Alf, 285
 Torrence, Harold, 364
 Traugott, Elizabeth C., 33
 Traugott, Elizabeth Closs, 102
 Truswell, Robert, 17

- Tsiakmakis, Evripidis, 201
- Uriagereka, Juan, 376
- Urpo, Nikanne, 187, 202, 207
- Valentin, Paul, 108
- Van de Koot, Hans, 134, 140, 141, 143–146
- van der Wal, Jenneke, 324
- Van Gelderen, Elly, 50, 57, 102
- van Gelderen, Elly, 69, 82, 84, 85, 91, 93
- van Kemenade, Ans, 87, 207, 210, 281, 307
- van Koppen, Marjo, 324, 325, 331, 337
- Van Ness, Silke, 308
- van Riemsdijk, Henk, 131, 140
- van Riemsdijk, Henk C., 240, 287
- Van Urk, Coppe, 148, 149
- Vangsnes, Øystein Alexander, 184, 202, 203
- Vasilyeva, Marina, 315
- Vergeiner, Philip, 332, 333, 345
- Verkuyt, Henk, 380
- Vezzosi, Letizia, 102
- Vicente, Luis, 365
- Vikner, Sten, 181, 182, 184, 203
- Villa-García, Julio, 112
- Villata, Sandra, 255
- Visser, Frederik Theodor, 93, 95, 281
- Vogel, Ralf, 362
- Volodina, Anna, 329, 330
- von Straßburg, Gottfried, 330
- Wagener, Terje, 15
- Walker, Alastair G. H., 285
- Wall, Joanna, 363
- Ward, Gregory, 185
- Warner, Anthony, 93
- Weber, Albert, 222
- Weerman, Fred, 373
- Weinrich, Harald, 136, 152, 154
- Weise, Otto, 327
- Weiß, Helmut, 6, 29, 35, 131, 132, 136, 138, 160, 163, 172, 223, 325–332, 334, 346
- Weskott, Thomas, 103
- Westergard, Marit, 25
- Wierschin, Martin, 330
- Wiesinger, Peter, 155, 333
- Wiklund, Anna Lena, 362
- Williams, Edwin, 201
- Wiltshko, Martina, 131, 161, 167–170, 172, 299, 302, 305, 314, 377
- Windfuhr, Gernot, 147
- Wolfe, Sam, 84
- Woods, Rebecca, 84
- Wurff, Wim van der, 283
- Wurmbrand, Susi, 201, 362, 382
- Yamakido, Hiroko, 147–149
- Yip, Moira, 131, 134, 140, 143
- Yoon, James, 118
- Yvonne, Portele, 218
- Zaenen, Annie, 79
- Zeijlstra, Hedde, 118, 324
- Zeldes, Amir, 102
- Zeller, Jochen, 210
- Zifonun, Gisela, 36, 138, 155, 156
- Zimmermann, Ilse, 26, 37, 38
- Zimmermann, Malte, 40
- Zwaarts, Joost, 285
- Zwanziger, Ronald, 147
- Zwart, Jan-Wouter, 326–328, 331, 354, 363, 368, 376, 377, 380
- Zwarts, Frans, 106
- Zwicky, Arnold M, 141

Germanic syntax from a comparative perspective

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